Understanding the linear degree gap

Uniform polynomial consequences and the lifting principle

An AI-generated tutorial textbook

Making machine-generated proofs readable

Mathematical direction, supervision and editing by

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Preface: mathematics, machines and exposition

AI is changing how mathematics can be explored, checked and explained. This project suggests a corresponding way to present mathematical research: an interactive paper, with further explanations attached to its arguments, and a tutorial textbook which develops the ideas at a human pace. The two formats serve different purposes. The paper states the results and their proofs concisely; the textbook supplies the definitions, examples and intermediate reasoning needed to understand the whole route.

Machine-generated proofs can be hard to follow. They may combine ideas from several research communities without explaining why a construction was chosen or how its pieces fit together. The initial proofs generated by Astra and Claude also omitted several key references to prior human work. Their mathematical knowledge reflects a body of human research, although the outputs do not establish which particular sources were used in training or how they influenced the constructions. A formally checked argument therefore also needs a readable account of the mathematics and its antecedents. This book provides that account for the linear degree gap.

This is an *AI-generated tutorial textbook*, developed under my mathematical direction, supervision and editing. Astra supplied much of the drafting and restructuring. Astra and Claude also contributed substantially to the proof search and formalisation. I selected the programme, supplied my earlier work and mathematical input, challenged arguments, read successive chapters and requested clearer explanations and worked examples. This continuing exchange shaped both the proof development and its presentation.

The published mathematical starting point is my LICS 2008 paper [7]; the 53-page 2009 report supplies a longer account [8]. The conference paper established a four-way classification, or *quadrichotomy*, of uniform algebraic refutation complexity: $\mathrm{T}_1, \ldots, \mathrm{T}_4$, with periodicity, NS–PC type agreement and undecidability for every nontrivial union of types, including the no-finite-model promise. The logarithmic NS and double-logarithmic PC degree lower bounds used Krajíček's Theorem 3.5 [9], based on Ajtai's symmetric-system work [25]. My joint journal papers with Meera Sitharam develop uniform translations [23] and uniformly generated permutation submodules over fields of characteristic zero [24]. In a separate 1997 preprint with Meera Sitharam, we announced a nonconstant-to-linear degree gap. I acknowledged an unresolved gap in its inducing-down proof in the 2009 report [8, p. 50, footnote 3; p. 53, reference 33]; this concerns the proof and does not refute the asserted theorem. My later unpublished notes pursued stronger bounds and suggested lifting separating vectors between universe sizes, with important steps incomplete. The aim here was to obtain complete proofs of linear degree bounds in the universe size, in every fixed positive characteristic, and to treat uniform polynomial consequences as well as refutations.

The central proof search included, in my recollection, roughly two to three days of concentrated work with Astra and Claude, alongside my interactions. Checking, repairs, extensions and exposition continued afterwards. When the construction reached an $\Omega(\sqrt{n})$ bound in characteristic two, I pressed for an $\Omega(n)$ bound and an argument covering all positive characteristics. The difficult work was to build one lifting mechanism compatible with all the required operations.

The development and subsequent comparison with the literature brought together proof complexity, model theory, modular representation theory and algebraic combinatorics. Some of these connections were unfamiliar to me; the assistants helped bring them into the discussion. Krajíček's earlier work already treated uniform polynomial targets, Euler structures and bounded-degree NS–PC comparison, including the NS-span plateau argument [13][9, Theorem 5.5]. The wider context includes Church–Ellenberg–Farb's FI framework for representations as the number of labels grows, Harman's modular periodicity in a linear range, and Hone and coauthors' explicit periodic functors through semisimplification [15, 10, 11]. Harman's equivalences already respect induction, restriction, tensor products and duality [10, Propositions 2.7 and 2.12]. The construction here proves the required closure preservation and target detection on the actual polynomial maps with its own quantitative bounds. Integral centraliser and rook combinatorics provide another connection [12]: realising compatible linear maps as combinations of label permutations. These communities study stability as the number of labels grows and the construction of maps between representations. Chapter 5 explains these antecedents; the finite constructions required for our polynomial proof rules are proved here.

The principal construction developed in this project is in Chapters 11–13: compatible corrections reuse one retained pool of labels, and the odd-characteristic argument completes the same linear budget. Chapter 14 transports the actual axiom, multiplication and indexed-target maps. Chapter 15 independently supplies the comparison's uniform chain bound. The four qualitative types are inherited from my earlier work; the present quantitative bounds are linear in T_2, linear on the hard residues in T_3, and give finite linear peaks in T_4. Chapter 19 gives explicit reductions separating any two types under the no-finite-model promise already stated in LICS 2008, refining its every-union formulation. This treatment was developed interactively by me and Astra from my earlier undecidability arguments. The NS proof system is due to Beame, Impagliazzo, Krajíček, Pitassi and Pudlák [21]; PC is due to Clegg, Edmonds and Impagliazzo [22]. The induction-chain separation used in Chapter 4 is due to Buss and Pitassi [2]; the design method is developed in Buss [26], and the modular bijection antecedent is Beame–Riis [27]. The compression discussion belongs to separate companion work.

I developed this tutorial interactively by reading drafts, identifying steps I could not follow, proposing abstractions and requesting examples that would carry the reader through the general argument. Alongside supplying my earlier manuscripts and guiding the proof search, an important part of my contribution was to identify relevant prior human work, check the connections against the literature, and ensure that these contributions received proper credit. My intention is to make the textbook available on GitHub alongside the interactive version of the paper, and potentially use it for lectures and presentations. The contribution to exposition is this coordinated combination: a scientific paper with expandable explanations and a longer, progressively developed tutorial.

I stated and proved the following undecidability theorem: for every nonempty proper subset $A \subsetneq \{1, 2, 3, 4\}$, it is undecidable whether a uniform family arising from a first-order sentence belongs to one of the types T_i with $i \in A$. This theorem appears in my LICS 2008 paper, with further details in the 2009 report [7, 8]. The no-finite-model promise is explicitly part of its published statement [7, Theorem 1]. Its Lemma 8 gives a T_1-versus-prescribed-type switch; the controlled-join construction in Chapter 19 supplies the remaining pairwise reductions. This comparison concerns that published formulation and inspected construction. The core degree, comparison and undecidability results now have Lean formalisations and independent source, statement and axiom audits. Newly added teaching examples and the topological reformulation are written deductions; they are not additional Lean theorem endpoints. The book explains the mathematics and its provenance, while making explicit how its text was generated and how I directed and revised it.

How to read this book

This book develops the mathematics behind *Degree profiles of uniform polynomial consequences*. It begins with actual algebraic proof lines and ends with a self-contained account of the linear degree gap. The intended reader is comfortable with polynomials, coefficient vectors, linear combinations and spans. We recall kernels and quotient spaces and introduce tensor products through explicit calculations. Nullstellensatz proofs, Polynomial Calculus and the required label actions are introduced here. No prior knowledge of modular representation theory, homological algebra or of the earlier complexity-gap paper is assumed. Understanding the conclusion is easier than checking the complete proof: the common-witness construction in Chapters 11–13 requires sustained work with the displayed maps.

The central problem is

$$\Gamma_n \vdash G^{(n)},$$

where both the axioms and the polynomial expression to be derived come from fixed rules as the universe size n changes. A target with a bounded-support index can be included as well. Deriving 1 is the important refutation special case, but it is not the only case. Here Γ_n is the set of input polynomials, each interpreted as an equation equal to zero; $G^{(n)}$ is the target polynomial, whose equation we want to derive. The symbol $\vdash$ means that an algebraic proof exists. The two proof systems and their degree conventions are defined in Chapter 1.

The question which organises the whole proof

Suppose a uniform system at size n has a proof with degree at most d. Can that degree budget also work at $n + q$, for a suitable prime-power block q? For an invariant uniform target $G^{(n)}$, the picture is

$$\Gamma_n \vdash_{\le d} G^{(n)} \quad\Longrightarrow\quad \Gamma_{n+q} \vdash_{\le d} G^{(n+q)} \quad\Longrightarrow\quad \Gamma_{n+2q} \vdash_{\le d} G^{(n+2q)} \quad\Longrightarrow \cdots .$$

Here the proof system is fixed as either NS or ordinary PC. The axioms and target follow their prescribed uniform rules at each size; the larger proof may have a different list of lines. The resource preserved by lifting is the degree bound for the whole proof. A *degree budget* means an upper bound on the degrees allowed in the proof; it is not a bound on its length or on the universe size. In this picture, *invariant* means that every permutation of the universe labels fixes the target polynomial. A target with distinguished indices has a different invariance condition, defined in Chapter 2.

This is the main question of the book. First we describe exactly what has to be transported. Then we construct progressively stronger lifts, measuring how large n must be in relation to d. Finally we reverse that numerical condition to obtain degree lower bounds. General indexed targets use the same membership principle, with an additional argument to detect their images; their individual vectors need not be literal images under the invariant-target lift.

The linear-algebra picture behind lifting

At a fixed degree budget d, ordinary PC is a calculation in a finite vector space. Start from usable axioms, close under linear combinations, and multiply a vector by a variable whenever that vector has degree at most $d-1$. This gives a subspace $C_d(n)$. NS uses a different subspace $I_d(n)$, spanned by the permitted monomial–axiom products. The target question is membership in one of these spaces.

The difficult theorem relates the membership calculations at n and $n+q$, where $q=p^a$ is a suitably large power of the characteristic. One exact operation deletes the extra block of labels. A related linear operator lifts smaller consequences into the larger universe. The construction has to preserve actual maps, intersections and images; a count of dimensions would not suffice.

For a first reading of this transition, begin with the worked multiplication calculation in Section 5.3. Then follow the four operations making one closure stage in Section 5.4. Section 5.5 explains why preserving this calculation still leaves a separate target question. Chapter 7 introduces the objects in Theorem 7.1 one by one and gives a guide to each clause. At that point the theorem is a precise statement of what the later construction will prove, not a fact to accept without proof.

There are two different operations which should never be confused. Multiplication by a variable takes degree-$d-1$ vectors to degree-d vectors *within one universe*. The lifting operator takes polynomials from the smaller universe to the larger one *without increasing degree*. It is linear, but is not generally a homomorphism of polynomial rings.

Why one proof gives a constant bound on a progression

Fix one degree d and one eligible starting size n. Lifting gives proofs with that same budget at $n+q, n+2q, \ldots$. The absolute budget stays d. Once that starting proof has been selected, d is a *constant* for the whole progression, even if it was originally described as, say, a small multiple of $\log n$. At the later universe sizes we keep its numerical value; we do not recompute the logarithm. Its ratio to universe size tends to zero:

$$\frac{d}{n+\ell q} \longrightarrow 0.$$

The constant bound initially belongs to one arithmetic progression. It is not yet a constant bound over all universe sizes. Two-way transfer at every lower threshold shows exactly which finite minimum degree recurs on that progression. The four types describe how those progressions fit together.

The lower-bound argument reverses this observation. If a finite degree cannot recur, it must have occurred before transfer became available. The theorem makes that availability threshold linear in d. Solving this inequality for d gives the linear lower bound. In the fluctuating case, a separate bound on the length of the period locates finite linear peaks. These are different uses of the same transfer theorem.

The stages are improvements in how far we can lift

The earlier refutation classification gives logarithmic NS and double-logarithmic PC lower bounds [7, 8]. The underlying transfer estimates summarised in the LICS paper have exponential and double-exponential starting universe sizes as functions of the fixed degree. These are historical refutation estimates, obtained there from substantial earlier work; they are not elementary lemmas reconstructed in this book.

The constructions developed here make the starting universe size much smaller:

Lifting stage	Sufficient starting size	Degree lower bound
Earlier NS refutation result	at most 2^{cd}	$\Omega(\log n)$
Earlier PC refutation result	at most $2^{2^{cd}}$	$\Omega(\log \log n)$
Characteristic-two construction with fresh labels	$O(d^2)$	$\Omega(\sqrt{n})$
One retained pool, every positive characteristic	$O(d)$	$\Omega(n)$

These are degree lower bounds on the hard sizes specified by the classification, not universal degree upper bounds. The constants depend on the fixed family and field. A small degree means small enough to meet the relevant starting-size inequality. Chapter 8 gives an elementary counting explanation for the prime-power steps. Chapter 10 supplies the first fully worked lifting construction. The central advance in Chapters 11–13 is to replace its quadratic label cost by a linear one, and to make that construction work also in odd characteristic. The T4 conclusion uses an additional linear bound on the period and gives finite linear peaks.

This development connects proof complexity with modular representation theory and algebraic combinatorics. Those communities already study representations built from bounded numbers of labels and periodic behaviour as the universe grows [15, 10, 11, 12]. Section 5.8 identifies the close antecedents. The task here is to prove transport for the actual polynomial proof rules and targets, with all the required finite constructions supplied in the book.

A layered route through the construction

The first part replaces proof syntax by consequence spaces, then works through a Boolean first-order example over a fixed finite field: a strict order in which every element chooses a smaller predecessor. Chapter 4 proves the elementary growing-degree obstruction and shows exactly how comparison and lifting strengthen it to a linear bound. The predecessor may be many-to-one; the example uses no matching or injection constraint. The second part states the precise transfer contract and derives the four profiles from it; it also identifies the desired lifting map. The third part constructs the common algebraic projector which fulfils that contract, first in characteristic two and then in every positive characteristic. The fourth part puts the target and consequence maps into the same argument and proves the independent NS–PC comparison. The final part works through examples, dual witnesses, matching obstructions and the logical undecidability consequence.

Within a technical chapter the order is always the same: say which property is needed, explain why it is enough, prove the structural lemma, and then inspect the calculation that realises it. The full normalisation, zero-marginal, rook and odd-characteristic arguments remain in the main chapters. Exercises have worked solutions at the end, but the proofs of the main results do not depend on doing them.

A first reading can follow the contracts, examples and checkpoints. A second reading can fill in each finite algebraic construction in turn. This is a way of learning the complete proof, not permission to regard its hard inputs as unproved assumptions.

When returning to a technical section, keep track of the space on which an operator acts and which parameters are fixed. Powers of a linear operator mean repeated composition. A polynomial degree, a tuple length and a support budget measure different things, even when the argument relates them by an inequality. Local reminders at the main changes of construction, together with the notation guide in the final chapter, identify these roles.

The most direct first route through the opening is to read the basic template definition in Chapter 2, then the target request and support budget in Sections 2.5–2.6. The orbit-sum and compression discussion can be read later. In Chapter 3, follow the closure recurrence and its maps; the coefficient-list size interpretation is a separate application. Then follow the running order example in Chapter 4. Its final steps specify the two general inputs still to be constructed: uniform comparison and degree-preserving lifting. Chapter 5 then names the three obligations which organise the rest of the proof: identify the data, preserve the calculation, and reflect target vanishing. The chapters below refer back to these obligations as each construction is supplied.

Specialist tools and a route through the proof

The proof uses several specialist languages. Its representation-theoretic part concerns permutation modules, group algebras, annihilator ideals and nilpotent chains. Its transport language consists of tensor products, quotients, naturality and exactness. Its finite constructions use common partial inverses, ordered projectors, marginals, sections and rook coordinates. Dual functionals and matching designs supply obstructions; definability and effective reductions explain the logical consequences. The background tools are developed at their first use and recalled when their role changes. The following map identifies the main transitions between mathematical subjects. It also helps a reader returning to the proof locate the relevant definitions.

Tool	Concrete use in this book	First development
Ideals and degree cut-offs	Polynomial combinations, with the allowed products constrained by degree	Chapters 1–3
Orbits and stabilisers	Grouping coefficient patterns under relabelling	Chapter 2
Scalar extension	Solving the same coordinate problem with auxiliary field coefficients	Chapter 3
Tensor products	Coordinates for paired inputs, with a diagonal label action	Chapter 5
Naturality and exactness	Commuting maps and preserving the actual kernel, image and quotient calculations	Chapter 5
Nilpotent chains	Reading which directions a block operator kills or leaves in its quotient	Chapter 7
Frobenius and sections	Prime-power counting and a chosen lift of each retained vector	Chapter 8
Common partial inverses	One group-algebra expression certifying the required projector on several spaces	Chapters 8–9
Marginals and layer sections	Compatible coordinate sums and corrections using one retained label pool	Chapter 11
Rook coordinates	Encoding overlap patterns and solving a triangular coefficient system	Chapter 12
Dual functionals and pairings	Coefficient tests separating targets from consequence spaces	Chapters 4–5
Matching designs	Bounded-degree coefficient tests and their extension rules	Chapter 17

Definability and reductions	The model-theoretic interpretation and the transfer of an undecidable computation problem	Chapters 18–19

Background reading. For ordinary algebra, Lang [18] has chapters on modules, algebraic extensions, a single endomorphism, bilinear forms and tensor products. Alperin [19] introduces the modular representation theory of finite groups. The exactness and reflection vocabulary is described in the Stacks Project [17]; here we develop it through kernels and images of the displayed linear maps. Cutland [20], especially the chapters on computable functions, universal programs, undecidability and reducibility, gives background for the counter-program reductions. The references in Section 5.8 locates the representation-stability connection. Chapter 18 explains the paper's Section 11 topological classification, including a worked residue-tree example. Section 18.5 gives optional historical background on Euler structures, not a criterion characterising T1–T4. These are reading routes for the terminology. The common witness, its natural map compatibility, the rook correction and the ordinary-PC extension invariant are established explicitly in this book.

The opening route above is for learning the problem: it reaches the running example and the transport contract quickly. The book's chapter order is for learning the construction: it explains the destination before constructing the engine, including the simpler quadratic stage. The following dependency order is for checking that every input of the linear theorem has been discharged; it returns to classification only after transfer is proved. These purposes account for their different orders. Each item states a concrete conclusion to check, rather than asking the reader to accept a general stability slogan.

1. **Replace proofs by linear algebra.** In Chapters 1–3, check that I_d and C_d describe *exactly* the legal degree-d proofs. In particular, PC intersects with $P_{\leq d-1}$ after cancellation. Check stable coefficients and the separate degree-zero convention.
2. **Identify the certificate for exactness.** In Chapter 9, verify that one common identity $zwz = z$ gives the projector $(1 - zw)(1 - wz)$ and exact deletion on the entire common annihilator class. The two-point failure of exactness explains why the moved-basis calculation alone is insufficient.
3. **Construct that common certificate in characteristic two.** In Chapter 11, check the partition of unity, the signed layer sections and the correction interface (Lemma 11.5). In Chapter 12, check strengthened forcing and triangular realisation. These prove that one correction repairs length b while annihilating every shorter length, on the same pool of retained labels.
4. **Check the changes for an arbitrary prime.** Chapter 13 replaces z by its top power $u = z^{p-1}$ in the witness identity. Check purity on tuple generators, the ordered projector factors, the p-level sections and the actual transpose relabelling. Purity is not assumed for arbitrary nilpotent representations.
5. **Return to the prescribed polynomials.** In Chapter 14, check the axiom and multiplication maps entry by entry, then the full target map J. For indexed targets, check the nonzero-subquotient pairing (Lemma 14.2); exactness alone does not reflect vanishing. The pairing doubles the original monomial support kd.
6. **Read off the lower bounds.** Now check Chapter 6. The linear onset controls nonrecurring values in T2 and on T3 hard residues. The period bound has the additional job of locating *finite* T4 peaks. This step is an argument about integer sequences after transfer has been proved.

7. **Check the separate comparison theorem.** Chapter 15 uses a strict-chain bound and an NS plateau to contain the whole PC closure. It proves same-type agreement; it is not the source of the linear PC lower bound.

Chapter 10 is a complete alternative construction in characteristic two, useful for seeing why spending fresh labels gives only a square-root bound. The linear route does not depend on that construction or its normalisation lemma. Chapters 7 and 8 state and explain the operation before its coefficients are constructed; their promised input is discharged by Chapters 9–13. The later example, matching and undecidability chapters have their own additional arguments.

The core formal results have independently checked Lean source builds, theorem statements, axioms and dependencies, as recorded with the paper. That evidence does not certify every new pedagogical formulation in this book. A human check can follow the finite calculations above without treating either the formalisation or an external representation-stability theorem as a substitute for a missing step.

Notation which keeps the roles separate

Symbol	Meaning
p	Fixed prime characteristic of the coefficient field
n	Retained, smaller universe size
$q = p^a$	Size of the added/deleted block; a is an exponent
$N = n + q$	Larger universe size
d	Degree bound for the whole proof
$G_b^{(n)}, G^{(n)}$	Indexed target and fully invariant target at size n
D_G	Fixed upper bound on the target polynomial degree
k, s, t	Supports of a variable, axiom index, target index
$r = R(d)$	One support budget once d has been fixed
$N(d)$	Onset beyond which a degree-d threshold transfers
$Q(d)$	A suitable prime-power period for that threshold
$I_d(n), C_d(n)$	NS and ordinary-PC consequence subspaces
L	Degree-preserving lift from n to N
$\mathcal{D}$	Exact deletion operation from larger modules to smaller ones

In particular, the support budget r is *not* the exponent a in p^a, and the variable support k is neither of them. The number of polynomial variables can be of order n^k and is not itself n.

Reading alongside the current paper

This edition aligns terminology and target notation with the 78-page paper revised on 4 November 2026 (fixed GitHub version). The paper is the concise statement of the research; this book supplies the slower route through its arguments. In particular $N(d)$ now denotes the same periodicity onset in both texts. Ordinary $N = n + q$ still denotes a larger universe size. The older refutation-only tutorials are historical guides, rather than additional prerequisites.

Question while reading the paper	Where to turn here
What is uniform, and what does degree count?	Chapters 1–3; distinguish a full polynomial from its bounded support of distinguished labels.
How does transfer imply the four cases?	Chapter 6; compare *thresholds*, then recurring exact values.
Why does deletion preserve the proof?	Chapter 5; transport maps and intersections, not just dimensions.
Where is the linear improvement?	Chapters 10–13; compare fresh labels with reuse of one common pool.
Why is the prescribed target preserved?	Chapter 14; the quotient obstruction must survive deletion.
How does topology separate T1–T4?	Section 18.1; the residue tree, compactness, and the binary worked example.
What is formalised?	The accompanying Lean packet's scope and endpoint tables; the new topological reformulation has a written proof.

A short first pass is Chapters 1 and 3, the contract in Chapter 5, Chapter 6, and Section 18.1. Then return to the common witness and rook constructions. Their purpose is to justify the contract with a *linear* label budget; the classification proof itself is considerably simpler.

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Part I

From proof lines to a finite linear closure

Chapter 1

Polynomial consequences and proof degree

A polynomial equation can be used in two different ways. One can evaluate it at a proposed solution, or one can combine it algebraically with other equations. This book studies the second activity. Given equations $f = 0$ for f in a set Γ, how complicated is a derivation of a specified equation $G = 0$? Our measure of complexity is the largest total degree needed in the derivation.

The distinction between an equation and its left-hand polynomial is worth keeping in view. We usually write f instead of $f = 0$, and write $\Gamma \vdash G$ for deriving $G = 0$. Thus a derivation of the polynomial 1 is a contradiction, since it asserts $1 = 0$. A derivation of $x + y$ need not be a contradiction: it may simply express a consequence shared by all solutions of the original equations.

1.1 The ambient polynomial ring

Let K be a field and let V be a finite set of variable names. We work in the ordinary polynomial ring $K[x_v : v \in V]$. A monomial is a product $\prod_v x_v^{a_v}$ with nonnegative integer exponents. Its total degree is $\sum_v a_v$. The total degree of a nonzero polynomial is the largest degree of a monomial with nonzero coefficient, after equal monomials have been collected. We use the convention $\deg 0 = 0$.

For example, over $\mathbb{F}_2$ the polynomial

$$(xy + x) + (xy + 1) = x + 1$$

has degree one. The two degree-two terms have cancelled. In a polynomial ring over a field, nonzero products satisfy $\deg(fg) = \deg f + \deg g$. We will use this fact when we define the degree of NS certificates and PC derivations in the next two sections.

A system Γ consists of polynomials regarded as equations equal to zero. It may include Boolean equations $x_v^2 - x_v$, but their presence does not change the ambient ring. In particular, x_v^2 is still a polynomial of degree two. Replacing it by x_v requires a justified algebraic step using $x_v^2 - x_v$.

1.2 Nullstellensatz certificates

The algebraic proof system was introduced by Beame, Impagliazzo, Krajíček, Pitassi and Pudlák [21]. We use its polynomial-identity form, with degree measured on the complete products.

Definition 1.1 (NS consequence)**.** An NS derivation of G from Γ is a finite polynomial identity

$$G = \sum_{f \in F} a_f f, \qquad F \subseteq \Gamma \text{ finite.} \tag{1.1}$$

Its degree is the largest degree of a nonzero product $a_f f$. Thus it has degree at most d if $\deg a_f + \deg f \leq d$ for every nonzero summand. Zero summands can be removed. The empty identity is a degree-zero derivation of $G = 0$.

The letters NS refer to Nullstellensatz, but the definition is simply an identity which can be checked by multiplication and addition. There is no appeal to an existence theorem hidden in a certificate. The multiplier a_f records how the input equation $f = 0$ is used.

Example 1.2 (A first certificate)**.** From $f_1 = x + y$ and $f_2 = y - z$, over any field, we derive $G = x + z$ by

$$x + z = f_1 - f_2.$$

Both products have degree one. From $f_1 = x$ and $f_2 = 1 - x$ we instead obtain $1 = f_1 + f_2$. The first identity expresses a consequence; the second is a refutation.

The relevant degree is the degree of each product before the products are added. A large-degree contribution cannot be ignored merely because it cancels against another summand in the final identity. Otherwise every refutation would appear to have degree zero, since its final polynomial is the constant 1.

1.3 Ordinary Polynomial Calculus

Polynomial Calculus was introduced by Clegg, Edmonds and Impagliazzo [22]. The rules below fix the ordinary polynomial convention used throughout this book.

Definition 1.3 (Ordinary-PC consequence)**.** A PC derivation of G is a finite sequence $P_1, \ldots, P_M$ ending in G. A line may be an input polynomial from Γ, a linear combination $\alpha P_i + \beta P_j$ of earlier lines, or a variable multiple $x_v P_i$ of an earlier line. Coefficients α, β belong to K. The zero polynomial is allowed as the empty linear combination. The degree of the derivation is $\max_i \deg P_i$, with each line collected as an actual polynomial.

The word *ordinary* stresses that there is no automatic Boolean reduction and no rule which multiplies by an arbitrary polynomial in a single step. A monomial multiple can be obtained by repeated variable multiplication, but every intermediate line counts towards the degree. The number of lines is unrestricted.

Example 1.4 (A cancellation which enables the next step)**.** Over $\mathbb{F}_2$, consider the inputs

$$f_1 = xy + x, \qquad f_2 = xy + 1, \qquad f_3 = xz.$$

We want a PC derivation of z in which every line has degree at most two. This is a restriction on the whole proof, including its inputs and intermediate lines. After listing these inputs, PC may continue with

$$\begin{aligned} f_1 + f_2 &= x + 1, \\ z(x + 1) &= xz + z, \\ (xz + z) + f_3 &= z. \end{aligned}$$

Every line stays within the chosen degree-two budget. Multiplication by z is always a PC rule, but here we must also check that its result fits that budget. The preceding line $x + 1$ has

degree one, so $z(x+1) = xz + z$ has degree two and is an admissible line in this degree-two derivation. Multiplying one of the original quadratic inputs by z would instead produce a degree-three line. Expanding the same calculation into one NS identity gives

$$z = zf_1 + zf_2 + f_3,$$

whose first two products have degree three. This is a concrete reason to keep track of cancellations inside a PC proof.

The axioms are consistent: $x = y = 1$ and $z = 0$ is a solution. We have proved the nonconstant consequence $z = 0$, not a contradiction. Exercise 1.9 checks that the two minimum degrees in this example really are different.

More generally, suppose we require every line to have degree at most d, and the proof has derived a nonzero polynomial P. A variable multiple stays within this degree budget only if $\deg P \le d - 1$. This follows from $\deg(x_v P) = \deg P + 1$. Even if P is represented as a linear combination of large-degree earlier lines, it is its actual degree *after cancellation* that controls this next step. This observation is the source of the intersection operation in our later linear-algebra model.

1.4 Least degrees, zero, and infinity

Definition 1.5 (Least target degree)**.** For $S \in \{\mathrm{NS}, \mathrm{PC}\}$, let $h_S(\Gamma \vdash G)$ be the least degree of an S derivation of G, or ∞ if there is no derivation. We order $\mathbb{N} \cup \{\infty\}$ by placing ∞ above every integer. For refutations write

$$d_S(\Gamma) = h_S(\Gamma \vdash 1).$$

Thus $h_{\mathrm{PC}}(\Gamma \vdash G) = 2$ says both that a PC derivation with every line of degree at most two exists and that no derivation confined to degree at most one exists. The symbol S selects the proof system; h_S then records the smallest degree achievable in that system.

These conventions separate three situations. The zero target has degree zero, even from empty input. A nonzero constant target is equivalent to the target 1, by rescaling with a field element. A nonconstant target has a positive degree floor, but its minimum proof degree can be much larger or infinite.

Proposition 1.6 (Degree floor and simulation)**.** *For every target G,*

$$\deg G \le h_{\mathrm{PC}}(\Gamma \vdash G) \le h_{\mathrm{NS}}(\Gamma \vdash G). \tag{1.2}$$

If a finite proof exists, the first inequality is an ordinary integer inequality. With an infinite value it is read in the extended order.

Proof. The last PC line is G, so its degree is at least $\deg G$. For NS, a sum of products of degree at most d is also of degree at most d. To simulate an NS certificate, expand each multiplier into monomials. Derive a permitted product μf by listing f and multiplying by one variable factor of μ at a time. None of these intermediate products exceeds $\deg \mu + \deg f$. Rescale and add all products to obtain G. Every step remains within the original NS degree bound. □

The algebraic object which organises unrestricted derivability is an *ideal*. An ideal in a polynomial ring is a subspace closed under multiplication by every polynomial. The ideal generated by Γ, written $\langle \Gamma \rangle$, consists of all finite sums $\sum_f a_f f$ with $f \in \Gamma$ and polynomial coefficients a_f. It is the smallest ideal containing the inputs. Thus unrestricted NS derivability

is precisely membership in $\langle \Gamma \rangle$. At a fixed degree, the restriction on each product is an additional condition; an arbitrary ideal representation need not respect it.

There is a converse without a degree bound. Annotate each PC input f with its expression as a single input multiple. Linear combination of two lines combines their annotations, and multiplication by x_v multiplies every coefficient in the annotation by x_v. At the last line this produces an NS identity. It may have a higher product degree than the largest actual PC line, because cancellations have been expanded rather than reused. Consequently NS and PC derive precisely the same polynomials when degree is unrestricted: the ideal generated by Γ. The difficult comparison later in the book concerns their *bounded-degree* behaviour uniformly in the universe size.

1.5 Algebraic consequence and solutions

Every derivation is sound under evaluation. If a tuple a satisfies $f(a) = 0$ for every input, an NS identity gives $G(a) = 0$. The PC rules preserve this assertion line by line. The same is true when a lies in an extension field of K.

The converse need not hold for the proof systems defined here. From $x^2 = 0$, every field-valued solution has $x = 0$, but x is not a polynomial multiple of x^2. It therefore has no NS or PC derivation. So there are two different questions: does G vanish at every solution, and does an NS or PC derivation of G exist? We call the first *semantic implication* and the second *algebraic derivability*. For the proof systems defined here, algebraic derivability is ideal membership. The example shows that semantic implication alone does not guarantee it.

Hilbert's Nullstellensatz describes the general connection precisely [18]. If $I = \langle \Gamma \rangle$, a polynomial G vanishes on all common roots in an algebraic closure $\overline{K}$ exactly when $G \in \sqrt{I}$, the radical of I: some power G^r with $r \geq 1$ belongs to I. A proof of the target G itself requires $G \in I$. For refutation these distinctions coincide: there is no common root over $\overline{K}$ if and only if $1 \in I$. The resulting ideal-membership certificate can be taken over K, by the coefficient-field argument in Chapter 3. Absence of roots in K alone is insufficient: $x^2 + 1$ has no root in $\mathbb{F}_3$ but has roots over its algebraic closure.

With all equations $x_v^2 - x_v = 0$ included, every root is Boolean. The quotient by these equations is the algebra of functions on the finite Boolean cube, a product of copies of K. Every ideal in that quotient is radical. Thus a polynomial target vanishes on all roots of a Boolean system exactly when it has an NS derivation.

The following standard Boolean completeness construction also keeps track of the degree [21]. Here a *Boolean assignment* assigns each variable either zero or one, and an *inconsistent Boolean system* means that no such assignment satisfies all its equations. The construction below explains the degree bound in Section 2.4 of the paper, using ordinary polynomial products throughout.

Proposition 1.7 (Completeness for Boolean refutation)**.** *Let Γ be a finite set of polynomials in finitely many variables, over any field K, containing $x_v^2 - x_v$ for every variable x_v. If no Boolean assignment satisfies Γ, there is an NS identity deriving 1 from Γ, and hence a PC refutation. If there are $v = |V|$ variables and all inputs have degree at most $D \geq 2$, then*

$$d_{\mathrm{PC}}(\Gamma) \leq d_{\mathrm{NS}}(\Gamma) \leq v + D.$$

Proof. The Boolean axiom gives $x_v(x_v - 1) = 0$ at every solution. Since a field has no zero divisors, each coordinate is zero or one; this remains true over an extension field. For each Boolean assignment a, put

$$\delta_a = \prod_{a_v = 1} x_v \prod_{a_v = 0} (1 - x_v).$$

This is its indicator polynomial: $\delta_a(a) = 1$, while $\delta_a(b) = 0$ for every other Boolean assignment b. Expanding $\prod_v (x_v + (1 - x_v))$ gives $\sum_a \delta_a = 1$.

There are only $2^{|V|}$ Boolean assignments. Because none satisfies the system, for each a choose one input f_a with $f_a(a) \neq 0$. Its nonzero value has an inverse in K. Form the explicit polynomial

$$H = 1 - \sum_a \frac{\delta_a}{f_a(a)} f_a.$$

At any Boolean assignment b, all summands except the one indexed by b vanish, and that summand has value one. Thus $H(b) = 0$.

We claim that a polynomial vanishing on every Boolean assignment is an explicit sum of multiples of the Boolean axioms. To see this, repeatedly reduce powers using the identity

$$x_v^e - x_v = (x_v^2 - x_v)(1 + x_v + \cdots + x_v^{e-2}) \qquad (e \geq 2).$$

Each replacement records a polynomial multiple of $x_v^2 - x_v$. After finitely many replacements every exponent is at most one, so

$$H = R + \sum_v b_v(x)(x_v^2 - x_v),$$

where R is multilinear. The replacements do not change values on Boolean assignments, so R also vanishes at all of them. A multilinear polynomial is determined by these values. For a direct induction, write $R = A + x_v B$: evaluating at $x_v = 0$ gives that A vanishes on the smaller Boolean cube, and evaluating at $x_v = 1$ then gives the same for B. Starting with constant polynomials, induction proves $R = 0$. Therefore rearranging the definition of H gives the actual identity

$$1 = \sum_a \frac{\delta_a}{f_a(a)} f_a + \sum_v b_v(x)(x_v^2 - x_v).$$

Every polynomial on the right which is used as an input belongs to Γ. Each indicator has degree at most v, so each product $\delta_a f_a$ has degree at most $v + D$. The power reductions never increase degree: each recorded Boolean-equation multiple has degree at most that of the term being reduced. Thus every product in the final identity has degree at most $v + D$. The degree-preserving simulation proved earlier gives the PC bound as well. □

This is an upper bound in the number v of polynomial variables, not the universe size n. A uniform family with variable support at most k can have $v = O(n^k)$, even though its input-degree bound D is fixed independently of n. The Boolean argument therefore does not give a universal linear bound in n, and it does not apply to arbitrary non-Boolean systems. The injection example in Section 16.3 has a separate certificate of degree at most $n + 1$. Its matching linear lower bound establishes sharpness of the general lower bound, not a universal upper bound. The interpolation construction may use exponentially many assignments; a degree bound is not a bound on the number of terms or proof lines.

1.6 The next question: can a proof survive a change of universe?

The definitions so far concern one fixed polynomial system. In the next chapter the axioms and targets will come from fixed rules indexed by the universe size n. We shall then ask whether a degree-d proof at one size can be lifted to a larger size with the same degree budget.

The earlier proof may use many lines; our aim is to preserve the degree bound without imposing a bound on their number.

If this can be repeated at $n, n+q, n+2q, \ldots$, the one chosen proof degree becomes a constant bound on that progression. If the family is known to have no such bounded progression, an eligible starting proof cannot exist. Measuring which degrees are eligible is how the lifting construction will become a lower-bound argument.

1.7 Exercises

Exercise 1.8. Find the least NS and PC degrees of the targets 0, 1, and x from $\Gamma = \varnothing$ and from $\Gamma = \{1\}$. Explain the role of the last-line degree floor.

Model answer (p. 163).

Exercise 1.9. For Example 1.4, prove $h_{\mathrm{PC}}(\Gamma \vdash z) = 2$ and $h_{\mathrm{NS}}(\Gamma \vdash z) = 3$. For the NS lower bound, list the monomials which can occur in a degree-two certificate.

Model answer (p. 163).

Exercise 1.10. A proposed degree-two PC calculation multiplies the derived line xy by x and replaces x^2y by xy, using a Boolean equation. Explain why this is not an ordinary degree-two step, even if $x^2 - x$ is an input.

Model answer (p. 164).

Chapter 2

What it means for a family to be uniform

The number of variables in our examples grows, but the rule describing them does not. At universe size n there may be a variable x_{ij} for every pair of labels $i, j < n$, an equation for every row, and a target for every distinguished label. The classification theorem requires a precise version of this fixed-rule condition. Symmetry at each size, by itself, does not imply the four complexity profiles: the equations could be chosen differently at each size, according to an arbitrary rule. We need symmetry together with one coefficient rule which remains the same as the universe grows.

For the main proof, this chapter has three jobs: specify the labelled variables, fix the coefficient rule for the axioms, and do the same for the requested targets. The intervening orbit-sum and compression results explain the geometry of supported expressions, but the linear lower bound does not use compression. On a first reading, one may pass from Definition 2.3 and its examples directly to Sections 2.5–2.6, then return to the structural results later.

2.1 Labels, support, and relabelling

The universe size n counts the elements of the finite universe. A label is simply a name for one such element. The paper sometimes calls n the rank of the labelled family; here we normally say universe size. Neither meaning should be confused with polynomial degree, the dimension of a polynomial space, or the rank of a matrix.

Write $[n] = \{0, \dots, n-1\}$. Think of its members as names rather than numerical values. A variable x_{ij} is indexed by the ordered pair (i, j), and relabelling by a permutation π sends it to $x_{\pi(i),\pi(j)}$. The labels needed to specify that index are $\{i, j\}$; this is the support of the pair, and therefore of the variable indexed by it. When $i = j$, its support has size one, not two. Permutations fixing those labels leave that variable unchanged. We start with individual indices because monomials and whole polynomial expressions can have different kinds of support, as the row-sum example below will show.

Definition 2.1 (Bounded basis system)**.** A bounded basis system is a collection of objects labelled from one infinite pool $\mathbb{N}$, with an action of label permutations. There is one fixed integer b, independent of n and of the object, and each object v is assigned a set $\operatorname{supp}(v) \subseteq \mathbb{N}$ with $|\operatorname{supp}(v)| \leq b$. Supports are equivariant: $\operatorname{supp}(\pi v) = \pi(\operatorname{supp}(v))$. Every permutation fixing $\operatorname{supp}(v)$ pointwise fixes v. At universe size n, we keep precisely the objects whose supports lie in $[n]$; we require that this set be finite. We call b a support bound for the system.

Examples include ordered pairs, tuples with specified equality patterns, and finitely many tagged sorts of such tuples. There may also be objects with empty support: a distinguished variable y fixed by every label permutation is an example. A support bound is a bound on the labels needed by one object, not on the number of objects present at universe size n. The important requirement is the *same bound at every universe size*. Finiteness at one universe size would not provide that: a bound of n would grow with the universe and would not be a bounded-support system in this sense.

A variable basis $\mathcal{V}$ gives the ring $K[x_v : v \in \mathcal{V}_n]$. We write $P_{n,\le D}$ for its vector space of polynomials of total degree at most D. If each variable has support at most k, a monomial of degree at most d involves at most kd labels: it has at most d variable factors, each involving at most k labels. Powers and overlapping supports may make the true number smaller.

Example 2.2 (The row sum mentions every label)**.** With variables x_{ij}, consider

$$f_{i,n} = \sum_{j<n} x_{ij} - 1.$$

Its index is i, of support one. Every permutation of $[n]$ fixing i fixes the whole polynomial, because it merely permutes the summands. Nevertheless the union of labels mentioned in the polynomial is generally all of $[n]$. Each monomial uses at most two labels, and the index uses one; neither assertion says that the whole polynomial uses only one or two labels. This distinction permits totality sums in the theory.

2.2 Stable coefficient rules

Definition 2.3 (Uniform axiom template)**.** A uniform template consists of a variable basis $\mathcal{V}$ with support bound $k \ge 1$, an axiom-index basis $\mathcal{A}$ with support bound s, and coefficients $c(a, \mu) \in K$. Here a is an axiom index and μ is a monomial in the labelled variables. The rule is independent of universe size and invariant under simultaneous relabelling:

$$c(\pi a, \pi\mu) = c(a, \mu).$$

For a fixed D, it vanishes whenever $\deg \mu > D$. At universe size n the inputs are

$$\Gamma_n = \{f_{a,n} : a \in \mathcal{A}_n\}, \qquad f_{a,n} = \sum_{\mu \text{ supported in } [n]} c(a, \mu)\mu. \tag{2.1}$$

Fix $n_0 \ge \max\{1, kD + s\}$ as part of the template data.

Because there are finitely many variables at each universe size and the degree is bounded, each displayed sum is finite. The same coefficients are used when the universe grows; there are simply more monomials available. The system consists of all equations indexed by $\mathcal{A}_n$, not one equation: a fixed list of patterns can generate many equations. For example, the n row equations $\sum_{j<n} x_{ij} - 1 = 0$ together with $x_{ij}^2 - x_{ij} = 0$ for every pair give $n + n^2$ indexed equations. For the row sum, the coefficient of x_{ij} is one when its row index is i, the constant coefficient is minus one, and other coefficients are zero. None of these rules asks how large n is.

For a group acting on indices, the *orbit* of a is the set of all its relabellings πa. Its *stabiliser* consists of the permutations which fix a. These familiar group-action notions have a specific use here: indices in the same orbit describe the same coefficient pattern with different label names. Fixing a support *pointwise* means fixing each of its labels, not merely preserving the set of labels as a whole. Later, H_S denotes the subgroup fixing the distinguished set S pointwise.

The stable range allows us to recognise the degree of an axiom from its index orbit. A monomial witnessing the largest nonzero coefficient uses at most kD labels. Together with the axiom index it uses at most $kD + s$. A permutation fixing the index can move all the additional labels into $[n]$ when $n \geq kD + s$. Hence, beyond n_0, each index orbit has one fixed polynomial degree. This matters when deciding which multiplier–axiom products are legal at a proposed proof degree.

Example 2.4 (Symmetric but not uniform)**.** Choose an arbitrary set A of positive integers. At universe size $n \in A$ let the inputs be $\{1\}$, and at other universe sizes let there be no inputs. Each instance is invariant under every label permutation. Yet the refutation degree is zero on A and infinite elsewhere. Arbitrary choices of A cannot satisfy the periodic conclusions of this book. They are excluded because the axiom rule changes with universe size.

2.3 Supported expressions are orbit-sum expressions

The row sum suggests a converse question. Suppose we are given a polynomial and know that every permutation outside a short list of labels fixes it. Can we describe the polynomial by a uniform rule? At one universe size the answer is yes. To represent a whole family by one rule, we also need compatibility as universe size changes.

Here is the support notion for a whole expression. At universe size n, a set $S \subseteq [n]$ *supports a polynomial* F if every permutation of $[n]$ fixing S pointwise fixes F. The permutation may move every other label. This is different from collecting every label mentioned in its monomials. For example, $\sum_{j<n} x_j$ has empty support as an expression, although its monomials collectively mention all labels. The row sum $f_{i,n}$ is supported by $\{i\}$ in this sense.

Proposition 2.5 (Orbit-sum description)**.** *Let F have degree at most D, and let H_S be the permutations of $[n]$ fixing S pointwise. The polynomial F is supported by S exactly when it has a unique expression*

$$F = \sum_{O} a_O \, \Omega_{O,n}, \qquad \Omega_{O,n} = \sum_{\mu \in O} \mu,$$

where O runs over the H_S-orbits of degree-at-most-D monomials.

Proof. Relabelling permutes the monomial basis. Thus $hF = F$ for all $h \in H_S$ exactly when monomials in the same orbit have the same coefficient. Collect these equal coefficients to obtain the displayed sum. Different orbits have disjoint monomial sets, so the orbit sums are linearly independent and their coefficients are unique. □

This is a grouping of distinct monomials, not an average over a group. In particular, it is valid in characteristic p even when the order of the stabiliser is divisible by p.

Example 2.6 (All supported linear expressions in one-variable indices)**.** For variables x_j and a distinguished label i, every linear polynomial fixed by the permutations fixing i has the form

$$ax_i + b\sum_{j \neq i} x_j + c.$$

For $n \geq 2$ there are three monomial orbits: the constant monomial, x_i, and all other variables. At $n = 1$ the last orbit is absent, so its coefficient cannot yet be recovered from the polynomial. At $n \geq 2$ all three orbits are present. Keeping a, b, c fixed gives a uniform expression at every larger universe size.

2.3.1 From one supported expression to one uniform template

Return to a variable basis of support at most k. Each monomial of degree at most D uses at most $r = kD$ labels. Let $s = |S|$ and choose an ordering a_0 of S. Relative to those s distinguished labels, every global monomial orbit can be represented using at most $s + r$ labels in total. Hence all its orbit types are visible at any universe size $n \geq s + r$ containing S.

Proposition 2.7 (Uniform extension of one supported polynomial)**.** *Suppose $n \geq s + kD$ and $F \in P_{n,\leq D}$ is supported by an s-element set S. Using ordered distinct s-tuples as indices, there is a unique size-independent, simultaneously relabelling-invariant coefficient rule of degree at most D whose polynomial at index a_0 and universe size n is F. It gives a uniform axiom or target orbit with index-support bound s and no increase in polynomial degree.*

Proof. Consider a pair (a, μ) consisting of an ordered distinct s-tuple and a monomial of degree at most D. First relabel a to a_0. The remaining monomial labels can then be moved into $[n]$ by a permutation fixing S, because there are at most kD of them and $n - s \geq kD$. Thus every global orbit of pairs has a representative (a_0, μ_0) at the chosen universe size.

Assign to that orbit the coefficient of μ_0 in F. This value does not depend on the representative. Indeed, a global relabelling between two such representatives fixes a_0. Its restriction to the labels used by the tuple and monomial is a finite injection within $[n]$, which extends to a permutation of $[n]$ fixing S. It therefore relates the two monomials by H_S, and their coefficients agree by Proposition 2.5.

The assigned value defines $c(a, \mu)$ on all pairs, with zero coefficients above degree D. Being constant on simultaneous pair orbits makes it relabelling invariant. Summing its coefficients over size-m monomials gives the required polynomial for every index available at universe size m. At the original universe size and index, every coefficient agrees with F. All pair orbits already occurred there, so the coefficient rule is unique. □

Thus one supported polynomial at a sufficiently large universe size determines its uniform continuation. This does not say that a separately chosen polynomial at every universe size necessarily follows that continuation.

2.3.2 The additional condition for a whole family

For $m \leq n$, let $\rho_{n,m}$ retain the coefficients of monomials supported in $[m]$ and discard the other monomials. This is the coefficient restriction map, rather than a transport rule for proof steps. A stable uniform family satisfies

$$\rho_{n,m}(F_{a,n}) = F_{a,m} \quad \text{whenever the index } a \text{ is supported in } [m]. \tag{2.2}$$

Proposition 2.8 (Characterisation of a stable coefficient rule)**.** *Fix bounded variable and index bases and a degree bound D. A family $F_{a,n} \in P_{n,\leq D}$, defined for every $n \geq 1$ and every index a available at that universe size, comes from one uniform coefficient rule if and only if it is equivariant under permutations at each universe size and satisfies (2.2).*

Proof. A uniform rule gives equivariance by simultaneous relabelling of indices and monomials. Restriction preserves exactly its coefficients on the retained monomials, proving coherence.

Conversely, for a global pair (a, μ) choose a universe $[n]$ containing all its labels and define its coefficient to be the coefficient of μ in $F_{a,n}$. Coherence makes this independent of the chosen universe size. To compare it with the coefficient of $(\pi a, \pi\mu)$, choose a larger universe containing the labels of both pairs. The action of π on the finite support of the pair extends to a permutation of that universe. Equivariance there gives equality of the two coefficients.

The resulting rule has all the required properties, and reproduces the family coefficient by coefficient. □

For example, $F_n = a_n \sum_{i<n} x_i$ is invariant at every universe size for *any* sequence of scalars a_n. If the scalars change, the coefficient of a retained variable changes, and coherence fails. Small support and bounded degree control the form of an expression; coherence controls its behaviour across different universe sizes.

2.4 Smaller-support generators: what can be compressed?

The bound kD applies to one monomial. It need not bound the number of distinguished labels needed to specify a polynomial. A further, different possibility is to replace a polynomial's entire permutation orbit by generators with smaller support. First we state a sufficient condition for replacing axioms without changing proof degrees.

Proposition 2.9 (Degree-preserving axiom replacement)**.** *Suppose two axiom sets* Γ *and* Γ' *in the same polynomial ring satisfy, for every integer* $e \geq 0$*,*

$$\operatorname{span}_K\{f \in \Gamma : \deg f \leq e\} = \operatorname{span}_K\{g \in \Gamma' : \deg g \leq e\}.$$

Then every target has the same least NS degree and the same least PC degree from the two sets, including degree zero and infinity.

Proof. An input $f \in \Gamma$ of degree e is a constant-coefficient linear combination of inputs from Γ' of degree at most e. In an NS certificate replace f by that combination. Every new multiplier–input product has degree no larger than the product it replaces, since its new input degree is at most e. This transforms any certificate without increasing its degree. In a PC derivation, derive each old input by these linear combinations of new inputs, within its old degree, and then reuse the original inference steps. This also preserves the degree budget. Interchanging the two sets gives the converse transformations and proves equality of their least degrees. □

The increasing spaces $A_e = \operatorname{span}_K\{f \in \Gamma : \deg f \leq e\}$ form a *filtration*: $A_0 \subseteq A_1 \subseteq \cdots$, organised by the usable input degree. The replacement criterion compares this entire filtration. The unrestricted span forgets which inputs were usable at each cutoff.

An arbitrary change of basis for the full axiom span need not meet this condition. In Example 1.4, replacing f_2 by $f_1 + f_2 = x + 1$ leaves the axiom span unchanged, but now $z = z(x + 1) + f_3$ is an NS certificate of degree two. From the original axioms the least NS degree is three, as Exercise 1.9 shows. Equality of the unrestricted span alone therefore does not justify a claim about exact proof degrees. The proposition supplies a sufficient condition by comparing the inputs usable at *each* degree.

The next example satisfies the criterion and supplies a successful support compression.

Proposition 2.10 (Compression of a linear subset sum)**.** *Let* $\varnothing \neq S \subsetneq [n]$*, put* $s = |S|$*, and consider* $F_S = \sum_{i \in S} x_i$*. Its smallest support as an expression has size* $\min\{s, n - s\}$*. Nevertheless its* S_n*-orbit span has generators supported on at most two labels. More precisely, with* s *interpreted as a scalar in* K*,*

$$\operatorname{span}_K\{\pi F_S : \pi \in S_n\} = \begin{cases} \operatorname{span}_K\{x_i : i < n\}, & s \neq 0 \text{ in } K, \\ \operatorname{span}_K\{x_i - x_j : i, j < n\}, & s = 0 \text{ in } K. \end{cases}$$

Proof. The coefficient of x_i is one on S and zero off S. If labels outside a proposed support include one label of each kind, swapping those two labels changes the polynomial. Thus a support must contain all of S or all of its complement, and either choice is sufficient. This proves the minimum-support formula.

Take $i \in S$ and $j \notin S$. Then

$$F_S - (ij)F_S = x_i - x_j.$$

Relabelling shows that the orbit span contains every difference. Those differences span the subspace $A = \{\sum_i a_i x_i : \sum_i a_i = 0\}$: fix a label j and write a vector of A as $\sum_{i \neq j} a_i(x_i - x_j)$. Each orbit vector has coefficient sum s. If $s = 0$ in K, the orbit span is contained in A and is therefore exactly A. If $s \neq 0$, it also contains $F_S \notin A$; since A has codimension one, it is the whole linear polynomial space. □

For example, $\sum_{i \in S} x_i$ can need many labels to specify even though each monomial uses only one. The proposition replaces its orbit span, not the polynomial itself, by one- or two-label generators. All replacements are linear combinations of the original orbit polynomials and conversely. Thus making this replacement in an axiom set changes neither the NS nor the PC consequence spaces at any positive degree budget: the starting linear spans agree, and all replacement equations have degree one.

The two-label bound can really be necessary. Suppose the characteristic p divides both n and s, with $n \geq 3$. A vector of A supported by at most one label i has the form $ax_i + b\sum_{j \neq i} x_j$. Its coefficient sum is $a + (n-1)b = a - b$, so $a = b$. Such vectors are scalar multiples of the fully invariant sum $\sum_j x_j$. They span only one dimension, whereas A has dimension $n - 1$. Consequently one-label generators cannot replace the whole orbit span in this case.

Example 2.11 (Four-label axioms replaced by two-label axioms)**.** In characteristic two, let Γ_n consist of all sums $\sum_{i \in S} x_i$ over four-element subsets $S \subseteq [n]$. For every $n \geq 5$, their span is exactly the span of the differences $x_i - x_j$. Thus we can replace all the four-label equations by the two-label equations $x_i - x_j = 0$, preserving every NS and PC consequence degree. Both descriptions have fixed coefficient rules across universe sizes. The starting-size condition matters: at $n = 4$ there is only one four-element subset, so its single sum does not span all differences.

At $n = 12$, an individual four-label sum has smallest support four, whereas its orbit span has two-label generators. Since the characteristic divides both 12 and 4, the preceding argument shows that one-label generators would not suffice. This gives a concrete reason to distinguish the polynomial's own support from the support of generators for its orbit span.

The general compression question has been resolved in separate companion work [16]; its proof is not developed in this book. In the stable-template setting over a field, put $r = kD$. There is one coherent replacement template with the same variables and degree bound, and with axiom-index support at most $2r$, such that for every

$$n \geq \max\{n_0, \max(s, r) + 2r\}$$

the old and new axiom spans agree at *every input-degree cutoff*. The replacement criterion above then preserves every target's least NS degree and its least ordinary-PC degree separately, including zero and infinity. This is an optional normal form, not an assertion about arbitrary submodules without a bound on the starting supports, and not an executable compression algorithm. No equivalence at the exceptional small universe sizes is claimed.

The monomial bound kD alone does not establish that compression; the companion proof supplies an additional structural argument. The lifting and classification proofs below retain the given index support s in their budget $2kd + s + t$; they do not assume that it has already been compressed to kD or $2kD$.

2.5 Targets also need a fixed rule

Definition 2.12 (Uniform target request)**.** A uniform target family for a fixed axiom template has a target-index basis $\mathcal{B}$ with support bound t, a fixed degree bound D_G, and coefficients $\gamma(b,\mu) \in K$. They are independent of universe size, invariant under simultaneous relabelling of b and μ, and zero above degree D_G. The target at universe size n, indexed by $b \in \mathcal{B}_n$, is

$$G_b^{(n)} = \sum_{\mu \text{ supported in } [n]} \gamma(b,\mu)\mu.$$

For $S = \mathrm{NS}$ or PC, define the aggregate profile

$$h_S(n) = \max_{b \in \mathcal{B}_n} h_S(\Gamma_n \vdash G_b^{(n)}). \tag{2.3}$$

Here the superscript (n) records the universe size, not a power; the subscript b is the target index, with support of size at most t. For a fully invariant target we omit b and write $G^{(n)}$. The empty maximum is zero. If any requested target is underivable, the maximum is infinity.

The statement $h_S(n) \le d$ means that *every* requested target has some derivation within degree d. Those derivations need not share lines, and their total length is not bounded. This is a maximum of individual minimum degrees, not the degree of the sum of the targets. Nor are the targets inserted into the axiom system.

There are three scopes which we will keep distinct.

1. A fully invariant target has a single index of empty support. Its polynomial $G^{(n)}$ is fixed by every permutation of $[n]$. For example, $G^{(n)} = \sum_{i<n} x_i$ is invariant and has degree one, although it mentions all labels.
2. A fixed target orbit can contain many indices at a universe size. For $G_i^{(n)} = x_i$, the indices i form one orbit. Any one representative has the same least degree as every other representative, since relabelling transports proofs.
3. An all-target request may contain several orbits. Its profile is the maximum of their individual profiles.

Every global index orbit has a representative at universe size $n \ge t$. Indeed, move its support into $[t]$. There are only finitely many global orbits because $\mathcal{B}_t$ is finite. Within one orbit, two size-n indices can be related by a permutation of $[n]$: first use the given relabelling on their supports and then extend that finite bijection. Thus choosing a representative which varies with n is harmless *provided it stays in one fixed orbit.* Switching between different orbits according to an arbitrary rule is not a uniform request.

2.6 Stable target degrees and the label budgets

The same support argument used for axioms applies to targets, now with at most $kD_G + t$ labels. We therefore use

$$n_* = \max\{n_0, 1, kD_G + t\}. \tag{2.4}$$

Beyond this universe size, every target orbit has a fixed degree. Their largest degree is a floor for both aggregate profiles. A proof budget below that floor fails; we do not replace the target by its lower-degree part in order to make it fit.

For an axiom multiplier, the combined support of a monomial of degree at most d and an axiom index is at most $kd + s$. For multiplication inside a PC proof, a variable and a monomial of degree at most $d-1$ use at most $k + k(d-1) = kd$ labels. These elementary counts explain why proof degree enters the transfer theorem linearly.

General indexed targets require a further detection argument. The resulting safe budget is

$$R(d) = 2kd + s + t = \kappa d + \sigma, \qquad \kappa = 2k, \quad \sigma = s + t. \tag{2.5}$$

For one fully invariant target the sharper budget is $R(d) = kd + s$. The factor of two in the general case is not an extra degree cost in a proof. It is a support cost in the finite-dimensional argument which shows that target membership is reflected, as well as preserved, when labels are deleted.

2.7 Where first-order systems enter

A universal clause is a disjunction of literals required to hold for every assignment of its logical variables to universe labels. A *literal* is an atom or its negation. A *grounding* replaces all those logical variables by particular labels; one clause pattern therefore produces many polynomial equations at universe size n. For example, grounding $R(u) \vee \neg R(v)$ at $u = i, v = j$ gives the clause whose polynomial translation appears below.

The normalised syntax is fixed before proof degrees are measured. *Flattened* function-graph atoms name an output explicitly, as in $f(i) = j$, instead of putting nested function terms inside an atom. They can therefore be indexed by bounded tuples of labels. Equality between two grounded labels is evaluated as equality of those labels; it is not an extra unknown relation.

A fixed first-order sentence can supply a uniform template. Relation atoms receive Boolean variables. A function f is represented by its graph variables $x^f_{\bar{i},j}$, with equations

$$\sum_{j<n} x^f_{\bar{i},j} - 1 = 0, \qquad x^f_{\bar{i},j} x^f_{\bar{i},j'} = 0 \quad (j \neq j').$$

Boolean equations make each value zero or one; collision equations give at most one output; the sum gives at least one. The collisions cannot be dropped in characteristic p, where $p + 1$ ones also sum to one. A clause is translated as a product of its literal-falsity indicators. For example $R(i) \vee \neg R(j)$ gives $(1 - r_i)r_j = 0$.

Fixed arities and a fixed finite list of clause patterns give bounded variable and axiom-index supports, and expansion gives a stable coefficient rule. Functions of arity zero represent constants. There is no built-in numerical order on the labels: an order, if used in a sentence, is a relation symbol with its own axioms. Later examples keep this graph-and-sum encoding visible, since an arbitrary change of encoding need not preserve minimum degree.

Proposition 2.13 (Normalised first-order compiler)**.** *A fixed finite list of universal clauses in flattened relation, function graph, and equality atoms gives a uniform template by the displayed translation. At every $n \geq 1$, its roots over K correspond exactly to models on $[n]$ of that normalised sentence.*

Proof. Use atom tuples as variable indices, and grounding tuples as clause indices. Totality and collision indices are fixed tuples as well. Expansion of a clause product depends only on the fixed syntax and on equalities among the grounding labels; it therefore gives stable, simultaneously invariant coefficients, including repetitions and cancellations. If a_R, a_f are the maximum relation and function arities, safe parameters are

$$k = a_R + a_f + 1, \qquad s = k + 1 + \max_C \operatorname{nvars}(C), \qquad D = 2 + \max_C |C|, \qquad n_0 = kD + s + 1.$$

Here $\operatorname{nvars}(C)$ is one more than the largest variable index occurring in C, and is one if none occurs; an unused variable slot is harmless. Maxima over empty symbol or clause lists may be taken as zero where appropriate.

At a root, every relation and graph variable is zero or one. Totality and collisions select exactly one output for each function input, and a clause product vanishes precisely when one of its literals is true. Equality atoms use their actual label-equality values. This constructs a model. Conversely a model supplies the corresponding Boolean values and satisfies all translated equations. □

Normalisation is part of the input convention. This proposition does not say that different normalisations, or arbitrary introductions of witness functions, preserve exact proof degree. The later examples use specific normalised sentences so that their equations are unambiguous.

2.8 Exercises

Exercise 2.14. For variables x_{ij}, find the label supports and total degrees of x_{ij}^3 and $x_{ij}x_{jk}$ when i, j, k are distinct. Compare the actual supports with the general bound $2d$. Explain why the row sum in Example 2.2 has an index of support one.

Model answer (p. 164).

Exercise 2.15. Take no axioms, a single variable x of empty support, and two target indices of empty support requesting 0 and x. Determine the two fixed-orbit profiles and the aggregate profile. What happens if one chooses the first target at sizes in an arbitrary set and the second elsewhere?

Model answer (p. 165).

Exercise 2.16. Take variables x_i, no axioms, and targets $x_i - x_j$ indexed by ordered distinct pairs. Use $k = 1, s = 0, t = 2, D_G = 1, n_0 = 1$ to compute n_*. At which positive universe sizes does this request actually contain a target? Why is the stated stability threshold allowed to be larger than necessary?

Model answer (p. 165).

Chapter 3

All proofs inside one finite-dimensional space

A proof can be arbitrarily long even when its degree is fixed. To study all such proofs, we need not enumerate their possible lengths. There are only finitely many monomials of degree at most d in a fixed finite set of variables. Every line of a degree-d proof is a vector in the space on those monomials. The proof rules describe which vectors can be reached.

This chapter identifies the reachable space exactly. The distinction between NS and PC becomes the distinction between one linear span and an iterated closure. In particular, the closure must recognise a low-degree vector created by cancellation among higher-degree vectors.

3.1 Coordinates for bounded-degree polynomials

Let $\mathcal{X}$ be the finite set of variable names and put $v = |\mathcal{X}|$. Work over the fixed field K. The space $P_{\leq d}$ consists of polynomials in x_u, $u \in \mathcal{X}$, of total degree at most d. Its monomial basis has size

$$\dim P_{\leq d} = \binom{v+d}{d}.$$

Indeed, a monomial has exponents $a_1, \ldots, a_v$ with $\sum a_i \leq d$. Add the slack coordinate $a_{v+1} = d - \sum a_i$. This gives $v+1$ nonnegative integers summing to d. Placing v separators among $d+v$ positions counts those choices by $\binom{v+d}{v} = \binom{v+d}{d}$.

For two variables and $d = 2$, one possible ordering is $1, x, y, x^2, xy, y^2$. A polynomial is its vector of coefficients in this basis. Addition and scalar multiplication are the usual vector-space operations; polynomial multiplication is a different operation which may leave this space.

Fix a finite system Γ. We omit its zero inputs when forming spanning sets, since zero contributes no new vector. For NS set

$$I_d = \operatorname{span}_K\{\mu f : f \in \Gamma,\ \mu \text{ a monomial}, \quad \deg \mu + \deg f \leq d\}. \tag{3.1}$$

There are only finitely many permitted pairs (μ, f). If desired, put one column μf in a matrix for each such pair. Deriving a target by NS in degree d will mean that its coefficient vector belongs to the column span of this matrix.

3.2 The closure for ordinary PC

For $d \geq 1$ define

$$\begin{aligned} V_0 &= \operatorname{span}_K\{f \in \Gamma : \deg f \leq d\}, \\ V_{j+1} &= V_j + \sum_{u \in \mathcal{X}} x_u (V_j \cap P_{\leq d-1}), \\ C_d &= V_\infty := \bigcup_{j \geq 0} V_j. \end{aligned} \tag{3.2}$$

For $d = 0$, define $C_0 = V_0$ without a multiplication term. The notation $x_u U$ means the image of a subspace U under the linear map $f \mapsto x_u f$. The sum of subspaces consists of finite sums of vectors from them.

The intersection in this formula is crucial. It asks which vectors already derivable in degree d happen to have actual degree at most $d-1$. It does not merely select input axioms of degree at most $d-1$, nor basis vectors that were short when first listed. A linear combination may cancel its highest terms and enter the intersection.

Example 3.1 (The first closure step sees cancellation)**.** Return to $f_1 = xy + x$, $f_2 = xy + 1$, $f_3 = xz$ over $\mathbb{F}_2$, with $d = 2$. A combination

$$af_1 + bf_2 + cf_3 = (a+b)xy + cxz + ax + b$$

is linear precisely when $a = b$ and $c = 0$. Thus

$$V_0 \cap P_{\leq 1} = \operatorname{span}\{x + 1\}.$$

The vector $z(x+1)$ lies in the new multiplication space, and adding f_3 produces $z \in V_1$. The degree-two NS space, in contrast, is just the span of the three quadratic inputs and contains no z term.

Proposition 3.2 (Exact description of consequences)**.** *For every polynomial G and every $d \geq 0$,*

$$\begin{aligned} h_{\mathrm{NS}}(\Gamma \vdash G) \leq d &\iff G \in I_d, \\ h_{\mathrm{PC}}(\Gamma \vdash G) \leq d &\iff G \in C_d. \end{aligned}$$

The chain V_j stabilises after finitely many strict increases, and $I_d \subseteq C_d \subseteq P_{\leq d}$.

Proof. Expand every coefficient a_f in an NS identity into monomials. Every resulting product belongs to the spanning set in (3.1). Conversely, group a linear combination of such products by input f; the resulting multiplier has degree at most $d - \deg f$. This proves the NS equivalence, including the empty identity for $G = 0$.

For PC, first prove by induction on a derivation that every line is in C_d. Usable input axioms belong to V_0. Two vectors in stages V_i and V_j both belong to $V_{\max(i,j)}$, which is closed under linear combinations. If a nonzero line f is multiplied by a variable while staying within degree d, then $\deg f \leq d - 1$. If $f \in V_j$, its multiple belongs to V_{j+1} by definition. Multiples of zero are zero.

For the converse, induct on the closure stage. Every vector of V_0 is a finite linear combination of usable axioms, hence has a legal proof. A vector of V_{j+1} is a finite sum of an old vector and terms $x_v f_v$ with $f_v \in V_j \cap P_{\leq d-1}$. Concatenate proofs of the finitely many f_v, append the permitted multiplications, and add to obtain the desired vector as the last line. Every partial sum still has degree at most d.

The chain is increasing inside $P_{\leq d}$. A strict increase raises its dimension, so there are at most $\dim P_{\leq d}$ such increases. If $V_{j+1} = V_j$, the recurrence gives equality at every later stage. Thus one stage equals the whole union. Finally each NS generator μf has a PC derivation within the same bound by successive variable multiplication, giving $I_d \subseteq C_d$. □

The proposition includes two different finiteness statements. Every particular vector in the closure has a finite proof, by the constructive induction. The space containing *all* degree-d proofs is also reached after finitely many stages. Neither statement assumes in advance an upper bound on the length of proofs presented to us.

3.3 The linear maps behind the recurrence

The closure formula can be expressed entirely through fixed ambient spaces and linear maps. This is the form needed for transferring it between universe sizes.

The immediate idea is elementary. To describe all permitted variable multiplications, keep a separate low-degree input polynomial f_v for each variable x_v, and send this list to $\sum_v x_v f_v$. This is a linear map on lists. Later, in Section 5.3, we call the space of these lists $X(n) \otimes L_d(n)$ and the map m_n. The tensor notation packages the separate registers; it does not introduce a new PC inference rule. All these operations occur at one fixed universe size. Lifting to another size is a separate task.

The first map sends an axiom index to its coefficient vector. Restrict to axioms of degree at most d; its image is V_0. The NS map instead sends a permitted pair (μ, a) to μf_a; its image is I_d. Both use the actual input coefficients.

Let X be the vector space with one basis element for each polynomial variable, and put $L = P_{\leq d-1}$. Multiplication is represented by a single linear map

$$m : X \otimes L \longrightarrow P_{\leq d}, \qquad x_v \otimes f \longmapsto x_v f.$$

Here the tensor product can be understood concretely: its basis is made of pairs (x_v, μ) with $\deg \mu \leq d-1$, and the map sends such a pair to the monomial $x_v\mu$. Bilinearity is built into the vector space on these pairs.

For any current stage V_j, its low-degree part is the intersection $V_j \cap L$. It can be described as the kernel of

$$V_j \oplus L \longrightarrow P_{\leq d}, \qquad (f, g) \longmapsto f - g,$$

identified with either equal coordinate. Multiplication then gives an image, and adjoining old vectors gives a sum:

$$V_{j+1} = V_j + \operatorname{im}\big(m|_{X\otimes(V_j\cap L)}\big).$$

Thus the construction uses a small vocabulary: coefficient maps, kernels, intersections, images, sums, and a finite iteration. A transport operation preserving this vocabulary, and agreeing with the actual maps, transports the closure. Merely matching dimensions would not identify which target vectors lie in it.

3.4 Changing the coefficient field

Extending scalars enlarges the allowed coefficients while keeping the same coordinate equations. If $U \subseteq K^M$ has basis $u_1, \ldots, u_a$ and $K \subseteq E$, its scalar extension is $U_E = \operatorname{span}_E\{u_1, \ldots, u_a\} \subseteq E^M$. It is also written $E \otimes_K U$. The subscript K records which scalars may be moved between the two tensor factors. For example, the kernel of $(a, b) \mapsto a + b$ is $K(1, -1)$ over K and $E(1, -1)$ over E. The same basis and relation remain, although there are more choices of scalar multiples. The next lemma applies this coordinate principle to every stage of the proof calculation.

Lemma 3.3 (Field extension preserves minimum degree)**.** *If $K \subseteq E$ are fields and Γ, G have coefficients in K, then the least NS and ordinary-PC degrees of $\Gamma \vdash G$ are the same over K and over E.*

Proof. At a fixed degree, NS membership is solvability of a finite linear system with coefficients in K. Row reduction over K already determines whether it is consistent; extending scalars does not change that answer. Equivalently, a matrix and its augmented matrix have the same ranks over the two fields.

For PC, extending scalars commutes with sums and images of finite matrices, and also with their kernels: row reduction describes these spaces over K and gives bases after extension. Intersections are kernels of the difference map just displayed. Induction on (3.2) therefore identifies each extended stage with the stage computed over E. Once both chains have stabilised, membership of the original vector G is unchanged. This proves equivalence at every integer degree, including degree zero, and hence equality of the minimum degrees, including infinity. □

This lemma will let us use convenient auxiliary coefficients in the finite-dimensional construction. It does not change the target to an arbitrary polynomial over the larger field. The target whose membership is tested remains the original K-coefficient vector.

3.5 When degree gives a proof-size bound

This section is a separate application of the consequence-space view. It explains the size interpretation under full coefficient lists; the construction of the lifting operator and the degree lower bounds do not depend on this encoding. A reader following the degree proof can continue to Chapter 5 and return here for the size consequences.

Degree alone is not a count of the nonzero monomials in a proof. We use one particular, deliberately explicit representation in which a degree bound has a direct size interpretation.

Definition 3.4 (Full coefficient-list size)**.** In an ambient ring with v variables, store a nonzero polynomial of degree e by the full list of coefficients of all monomials of degree at most e, including zeros. The number of entries is $B_v(e) = \binom{v+e}{e}$. Store zero by one entry. PC size is the sum of these lengths over all lines, including the final target. NS size is the sum over its nonzero multipliers; the inputs and target are supplied separately. The empty NS certificate for zero has size zero.

Write S_{NS} and S_{PC} for the least coefficient-list sizes of proofs of the fixed target G from Γ. These are minimum sizes over all proofs in the specified system; they are infinity when no proof exists.

Lemma 3.5 (Degree versus coefficient lists)**.** *With v variables and input degree at most D, a degree-d NS proof has a representative certificate of size at most $B_v(d)^2$. A degree-d PC proof has a representative proof of size $O(B_v(d)^3)$. Conversely, if the least degrees are finite, minimum sizes satisfy*

$$S_{\mathrm{PC}} \geq B_v(h_{\mathrm{PC}}), \qquad S_{\mathrm{NS}} \geq B_v(\max\{0, h_{\mathrm{NS}} - D\}) \quad (G \neq 0).$$

Furthermore $B_v(d) \geq 2^{\min(v,d)}$.

Proof. Put $M = B_v(d)$. A vector in I_d is a linear combination of at most M generator columns. Grouping by input gives at most M nonzero multipliers, each representable by at most M entries.

For PC retain a basis of usable axioms. If its span W is not closed under the permitted multiplications, choose $f \in W \cap P_{\leq d-1}$ with $x_v f \notin W$. Derive f as a combination of the at most M retained basis vectors, append $x_v f$, and enlarge the basis. Each enlargement raises dimension, so there are at most M enlargements. At the end, the closed space contains every

degree-d consequence. Append a combination giving the desired target. This uses $O(M^2)$ lines, each with at most M entries.

Any PC proof has a line of degree at least the least possible proof degree. An NS certificate for a nonzero target has a nonzero product of degree at least its least degree; its multiplier has degree at least $h_{\mathrm{NS}} - D$. These give the lower bounds. Finally the table contains all squarefree monomials in any $\min(v, d)$ variables, of which there are $2^{\min(v,d)}$. □

All variables of the ambient ring count, including those absent from nonzero terms. Over a fixed finite field, coefficient entries have bounded bit length. The results do not automatically become lower bounds for sparse representations, where zero coefficients are omitted. This distinction will remain explicit in the size classification.

3.6 Exercises

Exercise 3.6. Describe I_0 and C_0 for an arbitrary system. Prove directly that a nonzero constant target has a degree-zero proof exactly when an input is a nonzero constant.

Model answer (p. 166).

Exercise 3.7. With input $\Gamma = \{x^2\}$, show that every vector in every closure C_d has zero coefficient of x. Deduce that x is underivable, although it vanishes at every field-valued solution of Γ.

Model answer (p. 166).

Exercise 3.8. List the monomials counted by $B_2(2)$, and compare the full coefficient list of x^2 with its sparse representation. For $v \geq d$, identify explicitly the 2^d entries used in the final lower bound of Lemma 3.5.

Model answer (p. 167).

Chapter 4

A running example: an order with no first element

We now apply the consequence-space viewpoint to one concrete family. The equations describe a finite total order in which every element has a selected predecessor. A finite order has a first element, so each instance is inconsistent. The question is how much polynomial degree is needed to expose that contradiction.

For the running example fix a finite field K of characteristic p. The chain and restriction arguments below work over every field; the final linear bound also holds over any fixed field of positive characteristic.

The proof has three stages. First, a restriction to an elementary induction chain shows that NS degree tends to infinity. Second, the uniform NS–PC comparison proved in Chapter 15 shows that PC degree also tends to infinity. Third, the degree-preserving lifting theorem, constructed in the intervening chapters and assembled in Chapter 14, upgrades this qualitative information to a linear lower bound. The chain obstruction and the restriction are proved here in full. The two general results are stated at their point of use, so the reader can see exactly what their later proofs must deliver.

4.1 The equations and their finite proofs

For this chapter write $[n] = \{1, \dots, n\}$; this is just a renaming of the book's universe. There are two sorts of variables:

L_{ij} means that i precedes j, F_{ij} means that the selected predecessor of i is j.

Let $\mathcal{O}_n$ consist of the following polynomials, all interpreted as equations equal to zero:

$$L_{ij}^2 - L_{ij}, \quad F_{ij}^2 - F_{ij} \qquad (i, j \in [n]), \tag{4.1a}$$

$$L_{ii} \qquad (i \in [n]), \tag{4.1b}$$

$$L_{ij}L_{ji}, \quad (1 - L_{ij})(1 - L_{ji}) \qquad (i \neq j), \tag{4.1c}$$

$$L_{ij}L_{jk}(1 - L_{ik}) \qquad (i, j, k \in [n]), \tag{4.1d}$$

$$\sum_{j=1}^{n} F_{ij} - 1 \qquad (i \in [n]), \tag{4.1e}$$

$$F_{ij}F_{ik} \qquad (j \neq k), \tag{4.1f}$$

$$F_{ij}(1 - L_{ji}) \qquad (i, j \in [n]). \tag{4.1g}$$

Here a selected predecessor means any strictly earlier element; it need not be the immediately preceding element. Different elements may choose the same predecessor.

The first four lines describe a strict total order. The next two select exactly one predecessor in each row. The last requires that this predecessor really come earlier. This is the graph-and-sum encoding of the first-order assertion that L is a strict total order and $L(F(i), i)$ holds for every i.

The functionality collisions matter in positive characteristic. Boolean values and a sum equal to one alone could permit $p + 1$ selected entries. The collisions allow at most one, and the sum then forces one.

Proposition 4.1 (Finite inconsistency and finite proof degree)**.** *For every $n \geq 1$ and every field K, the system $\mathcal{O}_n$ has an NS refutation and an ordinary-PC refutation. Their minimum degrees are finite and positive.*

Proof. At a root, all variables are zero or one. The order equations give a strict total order on the labels. Starting from any label, move to an earlier one while possible. Transitivity and irreflexivity forbid a repeated label, so finiteness forces this process to stop at a first element i. The function equations select a label j with $F_{ij} = 1$. Equation (4.1g) then gives $L_{ji} = 1$, contradicting the choice of i. The same argument applies over every extension field.

Boolean refutation completeness (Proposition 1.7) constructs a finite NS certificate, and its simulation gives a finite PC proof. No input is a nonzero constant, so the degree-zero consequence space is zero. Neither refutation can have degree zero. □

At $n = 1$ the contradiction is already visible in six ordinary-PC lines:

Line	Polynomial	Reason	Degree
1	L_{11}	input	1
2	$F_{11} - 1$	input	1
3	$F_{11}(1 - L_{11})$	input	2
4	$F_{11}L_{11}$	multiply line 1 by F_{11}	2
5	F_{11}	add lines 3 and 4	1
6	1	subtract line 2 from line 5	0

For larger n there is no designated first label. The order variables must determine it, while every row insists on a predecessor.

4.1.1 Why this is one uniform family

Relabelling sends L_{ij} and F_{ij} to the variables with the relabelled ordered pair. Every input has an index using at most three labels, and all coefficients follow one fixed rule. For example, the row with index i has coefficient one on every F_{ij} and constant coefficient minus one, regardless of n.

Safe template parameters are

$$k = 2, \qquad s = 3, \qquad D = 3, \qquad n_0 = 9. \tag{4.2}$$

There are $2n^2$ variables, but one variable uses at most two labels. The target 1 is fully invariant and has degree zero. At proof degree d the invariant-target support budget is therefore

$$r = kd + s = 2d + 3. \tag{4.3}$$

This budget bounds the bases and maps of the consequence calculation; it does not require each proof line to mention only $2d + 3$ labels.

4.2 A smaller chain problem with two different proof degrees

Here the input comes with a *fixed successor chain*: the numerical chain $1, 2, \ldots, n$, its successor relation and its two endpoints are fixed in advance. They are part of the input rule, rather than relations to be determined by a satisfying assignment. For variables $x_1, \ldots, x_n$, define the induction system

$$\mathrm{IND}_n = \{1 - x_1,\ x_i(1 - x_{i+1})\ (1 \le i < n),\ x_n,\ x_i^2 - x_i\ (1 \le i \le n)\}. \tag{4.4}$$

Its meaning is sequential: the first variable is one, a one forces the next variable to be one, and the last variable is zero.

There is an ordinary-PC refutation of degree at most two. Starting from a derived line $P = 1 - x_i$, first multiply it by the single variable x_{i+1}. Then form

$$P - x_{i+1}P + x_i(1 - x_{i+1}) = 1 - x_{i+1}$$

by two linear combinations. Every intermediate line has degree at most two. Iterate and add the final input x_n to obtain 1. For $n = 1$, the two endpoint inputs already sum to one.

NS behaves differently: its minimum degree is $\Theta(\log n)$. We prove both a logarithmic lower bound and a logarithmic upper bound in the next section. Thus the fixed chain gives a constant versus logarithmic degree separation. The linear NS and PC bounds belong to the original uniform order family $\mathcal{O}_n$, whose order is an unknown relation. The distinction is possible because this chain family is *not* a uniform template of the kind defined in Chapter 2. It singles out numerical adjacency and the last label. An arbitrary label permutation does not preserve its input rule. Thus the uniform NS–PC comparison cannot be applied to IND_n itself.

The induction system is a useful restriction of the uniform order family. We next prove its obstruction using a finite graph of monomials. The classical NS degree behaviour of induction is discussed by Buss and Pitassi [2]; the needed lower bound is proved below rather than used as an external theorem.

4.3 A finite graph that witnesses the NS obstruction

Fix an integer degree budget $b \ge 0$. A multilinear monomial $x^S = \prod_{i \in S} x_i$ is indexed by a subset $S \subseteq [n]$. Make a graph whose vertices are the sets with $|S| \le b$. Its edges allow the following changes:

1. Insert or delete label 1.
2. Insert or delete label $i > 1$ when label $i - 1$ is present.

The number of occupied labels must remain at most b. Call an occupied label a pebble. The moves are reversible: the predecessor is still present after the move, so the reverse move is legal.

4.3.1 The clearance invariant

We need a bound on how far a sequence starting with no pebbles can go. The following auxiliary test supplies it. Write a configuration as $a_1 < \cdots < a_k$, put $a_0 = 0$, and call pebble a_j *eligible* for removal when

$$a_j - a_{j-1} \le 2^{b-k}. \tag{4.5}$$

A configuration is *clearable* if successive eligible removals can empty it, recalculating the threshold after each removal. These auxiliary removals need not be legal graph moves. They

give an invariant for the actual graph moves, whose predecessor gaps are one. In this clearance argument, k counts the currently occupied pebbles; it is independent of the variable-support parameter $k = 2$ for $\mathcal{O}_n$.

Lemma 4.2 (Any eligible removal can come first)**.** *Removing any eligible pebble from a clearable configuration leaves a clearable configuration.*

Proof. Induct on the number k of pebbles. The assertion for $k \leq 1$ is immediate. Put $h = 2^{b-k}$, choose an eligible pebble a, and let c be the first removal in one clearing order. If $a = c$, use the remainder of that order.

Otherwise both are initially eligible. Removing either doubles the threshold to $2h$. The other's gap stays the same unless the two pebbles are adjacent in the configuration; in that case its gap can become the sum of two gaps at most h, hence is at most $2h$. Thus either order can remove both. The configuration with c removed is clearable; by induction it can remove a next and then clear the remaining configuration. Starting instead by removing a, remove c next and use that same remainder. This gives a clearing order after removing a. □

Lemma 4.3 (The reachable frontier)**.** *Every configuration reachable from the empty set using at most b pebbles is clearable. In every nonempty clearable configuration of k pebbles,*

$$a_k \leq 2^b - 2^{b-k} \leq 2^b - 1. \tag{4.6}$$

Proof. The empty configuration is clearable. For a legal deletion, the deleted pebble has gap one, so is eligible; the exchange lemma preserves clearability. For a legal insertion, the new pebble has gap one and can be removed immediately by the auxiliary test. This returns to the previous clearable configuration, proving that the new one is also clearable. This proves the invariant along every graph path.

For the bound, induct on k, assigning maximum zero to the empty configuration. A clearing order first removes an eligible pebble with gap at most $h = 2^{b-k}$. The remaining configuration is clearable and has $k - 1$ pebbles, so its maximum is at most $2^b - 2h$. If the deleted pebble was not the largest, this bounds the original maximum as well. If it was the largest, restoring it adds at most its gap h to that maximum. Either way, $a_k \leq 2^b - h$. Since $k \leq b$, one has $h \geq 1$, giving the final inequality. □

For example, two pebbles can reach label three:

$$\varnothing,\ \{1\},\ \{1,2\},\ \{2\},\ \{2,3\}.$$

They cannot reach label four. The clearance test explains the limitation without bounding the number of moves: with two occupied labels the threshold is one, and after an eligible removal it becomes two.

4.3.2 From graph components to a linear functional

A *linear functional* assigns a scalar to each coefficient vector and respects addition and scalar multiplication. Specifying its value on every monomial specifies it on every polynomial by linearity. To exclude an NS identity for 1, it suffices to find such a functional which is zero on every permitted multiplier–input product but nonzero on 1. Applying it to the identity would give a contradiction. This is a dual certificate for nonmembership in a span.

The graph supplies its weights. A difference of two adjacent monomial coordinates receives value zero if the weights are constant along edges, hence on connected components. We give the empty component weight one and the other components weight zero. The proof below checks that this assignment also handles powers and endpoint inputs.

Proposition 4.4 (An elementary logarithmic NS obstruction)**.** *If $n \geq 2^b$, the system* IND_n *has no total-product degree-b NS refutation, over any field.*

Proof. Let $\mathcal{E}$ be the component of the empty vertex in the graph. By Lemma 4.3, no vertex of $\mathcal{E}$ contains n. For a monomial μ of degree at most b, let $S(\mu)$ be the labels with positive exponent and define

$$\Lambda(\mu) = \begin{cases} 1, & S(\mu) \in \mathcal{E}, \\ 0, & S(\mu) \notin \mathcal{E}. \end{cases}$$

Extend Λ linearly to $P_{\leq b}$. In particular $\Lambda(1) = 1$. We check that it vanishes on every permitted monomial–input product.

The two monomials of $\mu(1 - x_1)$ reduce to the same subset or to vertices joined by a label-one edge. Their Λ values are equal, so their difference has value zero. For $\mu x_i(1 - x_{i+1})$, the subsets are $S(\mu) \cup \{i\}$ and $S(\mu) \cup \{i, i+1\}$. They are identical or adjacent, because i is present. They both have size at most b by the product-degree restriction, so this difference also has value zero. The product μx_n corresponds to a vertex containing n, hence has value zero. Finally $\mu(x_i^2 - x_i)$ has the same set of positive exponents in both monomials, so again has value zero.

Every allowed NS identity is a linear combination of these products. Applying Λ to a supposed identity for 1 would give $0 = 1$. This contradiction proves the proposition. □

The functional used for the lower bound is a test on coefficient vectors, not a truth assignment. Replacing a positive exponent by one in its definition does not grant a free Boolean-reduction rule to a PC proof. Indeed, the functional need not respect multiplication of derived low-degree lines; the degree-two PC refutation above demonstrates precisely why an NS witness need not be an ordinary-PC witness.

For a small calculation, take $b = 2$ and $n \geq 4$. The empty component contains $\{2\}$ and $\{2, 3\}$ but not $\{3\}$: the singleton's gap three fails its clearance threshold two. Consequently

$$\Lambda(1 - x_2) = 0, \qquad \Lambda\big(x_3(1 - x_2)\big) = 0 - 1 = -1 \neq 0.$$

The line $1 - x_2$ has a degree-two PC derivation. Its permitted multiplication by x_3 is exactly an operation this NS functional does not preserve.

4.3.3 Why the chain also has logarithmic NS certificates

The chain obstruction cannot be strengthened to a linear bound. There is a short divide-and-conquer construction that makes this clear without appealing to an external upper-bound theorem.

Proposition 4.5 (A logarithmic NS upper bound for the fixed chain)**.** *For $n \geq 2$, the system* IND_n *has a total-product NS refutation of degree at most $2 + \lceil \log_2(n-1) \rceil$, over every field.*

Proof. For $a < b$ write $T_{ab} = x_a(1 - x_b)$. When $b = a + 1$, this is an input. For any $a < c < b$, the polynomial identity

$$T_{ab} = (1 - x_b)T_{ac} + x_a T_{cb}$$

combines two shorter intervals. Choose c so that their lengths differ by at most one, and recursively expand both terms until only the original adjacent-transition inputs remain. Each recursive level multiplies their certificates by a polynomial of degree one. An interval of length $b - a$ needs at most $\lceil \log_2(b - a) \rceil$ levels, so every resulting multiplier–input product has degree at most $2 + \lceil \log_2(b - a) \rceil$. This is an NS identity in the original inputs, obtained by expanding all intermediate transitions.

Finally use the endpoint identity

$$1 = (1 - x_1) + T_{1n} + x_1 x_n.$$

The first term is an input, the last is a degree-two multiple of the input x_n, and the middle has the certificate just constructed. The displayed degree bound follows. For $n = 1$, the endpoint inputs already sum to one with total-product degree one. □

Together with Proposition 4.4, this proves the claimed $\Theta(\log n)$ NS degree. The fixed successor chain allows PC to carry a single degree-one conclusion down the chain; an NS certificate records the original inputs, and balanced composition keeps its growing degree logarithmic.

4.4 Restricting the uniform order problem to the chain

In this argument, a *substitution* assigns a polynomial to each variable and extends by addition and multiplication to the whole polynomial ring. It is a ring homomorphism: $\sigma(af + bg) = a\sigma(f) + b\sigma(g)$ and $\sigma(fg) = \sigma(f)\sigma(g)$, where $a, b \in K$. Thus it sends an identity $1 = \sum a_f f$ to $1 = \sum \sigma(a_f)\sigma(f)$. To obtain a certificate from the chain inputs, we must additionally express each substituted input $\sigma(f)$ using those inputs. The degree accounting below controls both steps. Substitution here is a fixed-size reduction between two systems, whereas the later lifting operator is a linear map between universe sizes.

We now relate $\mathcal{O}_n$ to IND_n, for $n \geq 2$. The substitution need not respect relabelling: it will be used only to transform an NS identity at one fixed size. Set

$$\sigma(F_{ij}) = \begin{cases} 1, & i > 1 \text{ and } j = i - 1, \text{ or } i = 1 \text{ and } j = n, \\ 0, & \text{otherwise}, \end{cases} \tag{4.7}$$

and put

$$\sigma(L_{ij}) = \begin{cases} 0, & i = j, \\ 1 - x_i(1 - x_j), & i < j, \\ (1 - x_i)x_j, & i > j. \end{cases} \tag{4.8}$$

Every substituted variable has degree at most two. On a Boolean assignment, the relation in (4.8) orders labels first by their Boolean value, with zero before one, and then by their numerical label. It is therefore always a strict total order.

The selected predecessors run around one fixed cycle. For $i > 1$, the nonzero guard becomes exactly

$$1 - \sigma(L_{i-1,i}) = x_{i-1}(1 - x_i).$$

The wraparound guard is

$$1 - \sigma(L_{n,1}) = (1 - x_1) + x_1 x_n. \tag{4.9}$$

This has a degree-two NS identity from the two endpoint inputs of IND_n. All unselected guards become zero, and the function Boolean, totality and collision equations become zero identically.

Lemma 4.6 (Boolean reduction retains a product-degree bound)**.** *If a polynomial P of degree at most m vanishes on every Boolean assignment, it is a sum of multiples of $x_i^2 - x_i$ with each nonzero product of degree at most m.*

Proof. Reduce powers one monomial at a time using

$$x_i^a - x_i = (x_i^2 - x_i)(1 + x_i + \cdots + x_i^{a-2}) \qquad (a \geq 2).$$

Multiplying this identity by the other factors of that monomial does not exceed its original degree. Continue until every monomial is multilinear, recording the Boolean-axiom products. The final multilinear remainder still vanishes on the Boolean cube, so is zero by the induction proved in Chapter 1. □

Apply this lemma to the substituted order equations, which vanish on every Boolean assignment. A Boolean or two-label order axiom has degree two before substitution and degree at most four afterwards. A transitivity axiom has degree three and becomes a polynomial of degree at most six. Irreflexivity becomes zero. Thus each source axiom f of actual degree e has a substituted NS identity from IND_n with product degree at most $2e$; the guards have the stronger bound two, as just computed.

Lemma 4.7 (Degree accounting for the restriction)**.** *A degree-d NS refutation of $\mathcal{O}_n$ gives a degree-at-most-$2d$ NS refutation of* IND_n*.*

Proof. Write the original identity as $1 = \sum_f a_f f$, omitting zero products. If $e = \deg f$, then $\deg a_f \leq d - e$. Hence $\deg \sigma(a_f) \leq 2(d - e)$. Replace $\sigma(f)$ by the identity just established, whose individual products have degree at most $2e$. After multiplication by $\sigma(a_f)$, every new product has degree at most $2(d-e)+2e = 2d$. Summing gives the required NS refutation. □

Corollary 4.8 (The qualitative NS seed)**.** *For every $n \geq 2$ and every field,*

$$d_{\mathrm{NS}}(\mathcal{O}_n) > \tfrac{1}{2} \log_2 n. \tag{4.10}$$

In particular, the NS minimum degrees tend to infinity.

Proof. If a degree-d refutation existed with $n \geq 2^{2d}$, Lemma 4.7 would produce a chain refutation forbidden by Proposition 4.4 at budget $b = 2d$. Therefore $n < 2^{2d}$ for the finite minimum degree d, which is the displayed inequality. □

4.5 Why PC degree also tends to infinity

We now use the fixed positive characteristic. The uniform comparison theorem, Theorem 15.2, supplies the following input for the original order family: for each fixed PC budget d there is a finite NS budget $E(d)$, independent of n, such that

$$d_{\mathrm{PC}}(\mathcal{O}_n) \leq d \quad \Longrightarrow \quad d_{\mathrm{NS}}(\mathcal{O}_n) \leq E(d). \tag{4.11}$$

The function may grow quickly; only independence from n is needed.

For a fixed d, inequality (4.10) eventually exceeds $E(d)$. Consequently a degree-d PC refutation is impossible at every sufficiently large size. Since this holds for each fixed d, the PC minimum degree tends to infinity too. No finite degree can recur infinitely often in either proof system.

This uses comparison on $\mathcal{O}_n$, where the uniform hypothesis holds. Applying it to the restricted chain would be invalid: that nonuniform family has the degree-two PC proofs already displayed. The comparison transfers *qualitative unboundedness*; the linear PC bound will come from lifting with its original PC degree unchanged.

4.6 The same lifting mechanism gives the linear bound

For this family, the invariant-target case of Theorem 14.1 gives a particularly simple statement. For either $S = \mathrm{NS}$ or $S = \mathrm{PC}$, if

$$q = p^a > 2d + 3, \qquad n \geq \max\{9, 4d + 5\}, \tag{4.12}$$

then a degree-d refutation at n gives one at $n + q$. These numbers come from $r = 2d + 3$ and the retained-label requirement $n \geq 2r - 1$. We use the actual range of the transfer theorem; the profile chapter also defines a slightly more conservative onset for general bookkeeping.

Theorem 4.9 (A finite linear lower bound for the order family)**.** *Over every fixed field of prime characteristic, the NS and ordinary-PC refutation degrees of $\mathcal{O}_n$ are finite at every positive size, tend to infinity, and satisfy*

$$d_{\mathrm{PC}}(\mathcal{O}_n) > \frac{n-5}{4}, \qquad d_{\mathrm{NS}}(\mathcal{O}_n) > \frac{n-5}{4} \qquad (n \geq 9). \tag{4.13}$$

Both profiles are of type T2 in Chapter 6.

Proof. Finiteness and positivity were proved in Proposition 4.1. The preceding two sections show that both minimum degrees tend to infinity.

Suppose a degree-d refutation existed at a size satisfying (4.12). Keep this d fixed and choose one admissible prime power q. Iterated forward lifting gives refutations of degree at most d at

$$n, n + q, n + 2q, \ldots.$$

Every step remains eligible: the retained universe size grows, while d, r, q stay fixed. This contradicts the fact that the minimum degree tends to infinity. The contradiction works separately for NS and PC.

Now take any $n \geq 9$ and let d be its finite minimum degree in the chosen system. If $n \geq 4d + 5$, the preceding contradiction applies. Therefore $n < 4d + 5$, which rearranges to $d > (n - 5)/4$. Absence of any recurring finite degree is exactly the T2 case of the later classification. □

The conclusion concerns unavoidable proof degree at every sufficiently large size. It uses an infinite progression, not a drop in the degree: the lifted proofs would all have the same budget d. A proof of arbitrary length cannot evade the bound, because lifting transports the complete degree-d consequence space.

4.7 What the later common projector must do for this example

The application has now isolated the jobs of the algebraic construction. The common projector will be a linear projection obtained from one shared combination of label permutations. Write L for the resulting linear map from smaller-universe polynomials to larger-universe ones. The variable basis has two-label support; monomials of degree at most d use at most $2d$ labels. The usable-axiom source and the NS multiplier–axiom source are covered by $r = 2d + 3$. Every closure stage stays inside these fixed ambient spaces.

Consider the row input $\sum_{j \leq n} F_{ij} - 1$. Plain inclusion into a larger universe leaves out its added summands. Setting all variables involving deleted labels to zero instead turns a deleted row into -1. The common projector must repair the actual axiom coefficient map. Its axiom-lifting formula gives a linear combination of genuine larger-instance inputs, rather than assuming that the same retained polynomial remains an axiom.

For PC it must also preserve $V_j \cap P_{\leq d-1}$ after cancellation, and the joint tensor multiplication map. This ensures that every lifted vector in the larger closure has a legal ordinary-PC derivation. The target here is 1, so the identity $\mathsf{L}(1) = 1$ turns consequence lifting into refutation lifting. The retained-label cost $2r - 1 = 4d + 5$ is what can be inverted to obtain the linear bound.

The separate comparison theorem uses a uniformly bounded chain of invariant subspaces and an NS plateau to contain the whole PC closure. In this example it rules out bounded PC degree before lifting supplies the numerical lower bound. The restriction to the induction chain, the clearance invariant, comparison, and the common projector each have one task. Together they turn a sequential finite-order contradiction into the same linear-degree mechanism used throughout the book.

Part II

The lifting principle and the four profiles

Chapter 5

The lifting problem: transport a closure and its target

The syntax of an NS or PC proof has now been replaced by a finite membership calculation. We can therefore ask a more general linear question before constructing any operator: which parts of that calculation must survive a change of universe?

We separate three tasks: identify the actual polynomial maps, preserve the subspace calculation, and detect whether a target remains outside the consequence space. The finite constructions later in the book will fulfil this single transport contract.

5.1 From one eligible proof to an infinite progression

It helps to see the whole argument before its linear-algebra interfaces. Recall that k bounds the labels needed by one variable, s bounds the labels in one axiom index, and t bounds those in one target index. We work over a fixed field of positive characteristic p. Fix one proof degree $d \geq 1$, and choose a support budget r covering the degree-d data. A fully invariant target uses $r = kd + s$; the general indexed-target calculation uses $r = 2kd + s + t$. The completed lifting theorem will say that, for

$$q = p^a > r, \qquad n \geq \max\{n_*, 2r - 1\},$$

a prescribed target request has proofs of degree at most d at size n exactly when it does at size $n + q$. Here n_* is a fixed starting universe size after which the template and target data are stable; its precise value is given in Chapter 6. The number p is the characteristic, and q is the size of the added block, not necessarily the size of the coefficient field.

The first implication can be repeated with the same d, r, q, because every larger starting size still satisfies the displayed inequality. Thus one proof gives a constant degree bound on an infinite progression. The reverse implication, applied also at lower degree thresholds, identifies exact recurring minimum degrees. This is the bridge to the four profiles.

Now suppose a finite value d cannot recur. It cannot occur at an eligible starting size, so its occurrence forces

$$n < \max\{n_*, 2r - 1\}.$$

Since r is affine in d, this is a linear lower bound on d once n is beyond the fixed starting universe size. In T4 a value does recur; the two-way comparison and a bound on q instead move an occurrence to a representative of size $O(d)$, creating a finite linear peak. The remaining mathematical task is to prove the displayed lifting statement with its linear retained-label budget. The definitions below split that task into explicit obligations.

5.2 Label actions in linear-algebra language

We need some language from linear algebra and representation theory. It is useful language, but none of the words in this section should stand for an unexplained theorem. We introduce them through the polynomial spaces already familiar from Chapter 3.

5.2.1 A vector space together with relabelling

Consider the degree-one space

$$M = Kx_1 \oplus Kx_2 \oplus Kx_3.$$

The permutation (1 2) exchanges x_1 and x_2 and fixes x_3. It therefore sends $ax_1 + bx_2 + cx_3$ to $bx_1 + ax_2 + cx_3$. This is an invertible linear map on coefficient vectors. Other permutations give other linear maps, and doing two relabellings in succession has the same effect as their composite.

In general, a *representation* of S_n is a vector space M equipped with these compatible linear actions of the label permutations. A *module* is another name for such a space here; later the same action is extended to linear combinations of permutations. The vectors remain ordinary coefficient vectors; the representation records how relabelling acts on them.

For a finite labelled set $\mathcal{U}_n$, the notation $K[\mathcal{U}_n]$ means the vector space with one independent basis vector $[u]$ for each $u \in \mathcal{U}_n$. Its vectors are $\sum_{u\in\mathcal{U}_n} a_u[u]$. If permutations relabel the objects u, they permute these basis vectors. This is a *permutation module*. Monomials of degree at most d give precisely such a basis for our polynomial ambient space.

An *invariant subspace*, also called a *submodule*, is a subspace carried to itself by every label permutation. In the displayed example, $K(x_1 + x_2 + x_3)$ is invariant, whereas Kx_1 is not. The consequence spaces are invariant because relabelling an entire legal proof gives another legal proof from the uniform axioms. An individual proof line need not be invariant.

5.2.2 Maps that respect the relabelling

A linear map $f : M \to M'$ is *equivariant* when

$$f(\pi v) = \pi f(v)$$

for every label permutation π. The equation says that it makes no difference whether we relabel first or apply the map first. For example, sending an axiom index $[a]$ to its polynomial $f_{a,n}$ is equivariant by uniformity. Ordinary variable multiplication is equivariant when the *same* permutation acts on the variable and the polynomial being multiplied.

Equivariant maps encode rules that are compatible with renaming labels. Their symmetry lets one common expression in permutations interact with all the maps at once. Chapter 7 gives the elementary calculation behind this compatibility.

5.2.3 Kernel, image and quotient: three different questions

For a linear map $f : A \to B$, its *kernel* is $\ker f = \{a : f(a) = 0\}$; its *image* is $\operatorname{im} f = \{f(a) : a \in A\}$. They record respectively what the map kills and what the map can produce. If f is equivariant, both are invariant subspaces. Indeed, $f(a) = 0$ implies $f(\pi a) = \pi f(a) = 0$, and $\pi f(a) = f(\pi a)$ is again an image vector.

For a subspace $H \subseteq W$, the quotient W/H identifies two vectors whose difference belongs to H. The quotient map is

$$\rho_H : W \to W/H, \qquad g \mapsto [g] = g + H.$$

Its kernel is exactly H. Thus

$$[g] = 0 \text{ in } W/H \quad \Longleftrightarrow \quad g \in H.$$

This will express the question of whether a target is derivable. The zero *class* does not mean that the original polynomial is zero: a nonzero derived polynomial has zero class modulo the consequence space.

For example, let $W = Kx \oplus Ky$ and $H = K(x + y)$. Then $[x + y] = 0$, while $[x] \neq 0$ and $[y] \neq 0$, with $[x] = -[y]$. To see that $[x] \neq 0$, an identity $x = a(x + y)$ would force both $a = 1$ and $a = 0$ by comparing the two coefficients. This calculation works in characteristic two as well. If H is invariant, relabelling the representatives defines an action on the quotient: $\pi[g] = [\pi g]$. A *subquotient* of W is a quotient U/H with $H \subseteq U \subseteq W$ invariant. It records the directions in U that are still missing from H.

5.2.4 What the word exact will mean

Suppose $A \xrightarrow{f} B \xrightarrow{g} C$ are linear maps. The sequence is *exact at* B if $\operatorname{im} f = \ker g$: the vectors arriving from A are exactly those killed by the map to C. This is a statement about two specified subspaces of B. For instance,

$$0 \longrightarrow H \longrightarrow W \xrightarrow{\rho_H} W/H \longrightarrow 0$$

says that inclusion is injective, its image is the kernel of the quotient map, and the quotient map is surjective.

Later we will construct an operation $\mathcal{D}$ taking suitable spaces and maps at universe size $N = n + q$ to spaces and maps at universe size n. To say that this operation is *exact* means that it preserves these image–kernel relationships, with their actual inclusions. It must carry an injection to an injection, and an onto map to an onto map, on the specified class. It must also identify the transported kernel and image *inside the transported ambient space*. Matching their dimensions would not do this.

The one-dimensional space K, with every permutation acting trivially, is called the *trivial module*. It is *simple*: it has no proper nonzero subspace. We will use this particular observation for invariant targets, rather than assume a general structure theorem about representations.

5.2.5 Background for the transport construction

There are four layers of background here. The first is ordinary linear algebra: spans, linear maps, kernels and quotient spaces. The second is representation theory: we keep track of relabelling while doing that linear algebra. Since the coefficient field has positive characteristic, this is *modular* representation theory. One familiar shortcut is unavailable: averaging over all permutations would divide by $n!$, which is zero in the field once $n \geq p$. The later construction supplies a different operator rather than using this averaging argument.

A representation is *semisimple* if it is a direct sum of simple representations. Here *simple* means that there is no proper nonzero invariant subspace. In positive characteristic, an invariant subspace need not have an invariant complementary subspace. For example, the swap of two coordinates over a characteristic-two field fixes $K(x_1 + x_2)$. Any invariant complementary line would be an eigenline for the swap, but its only eigenvalue is one and its fixed space is precisely that same line. Thus no invariant complement exists. This elementary example explains why a decomposition into invariant pieces cannot simply be assumed in the later construction.

The third layer is the language of categories and functors: a way to say that one operation changes many spaces and their connecting maps consistently. We spell out that requirement in Section 5.4. The fourth is the language of exactness, usually taught in homological algebra. Here we use just its linear-algebra content: preserving kernels, images and quotient relations. No knowledge of chain complexes or homology groups is needed. For optional background on this terminology, see [17]; the required calculation is developed in this book.

The tensor product introduced next belongs to ordinary multilinear algebra. In our application it has a particularly concrete meaning: one register for each possible variable multiplication.

5.3 The data at one universe size

We now put names on the ingredients of one ordinary PC step. *Nothing in this section changes the universe size.* In particular, the map m_n below is ordinary polynomial multiplication at one fixed universe size; it is not the lifting operator. We first understand this familiar operation, then ask how to transport it. The letter z in the worked polynomial example below is just a polynomial variable. In Chapter 7, z will instead denote a linear operator made from block permutations; that operator will be defined there.

5.3.1 Three spaces with three different jobs

Fix a universe size n and a proof-degree budget $d \geq 1$. The variable names are the members v of the finite set $\mathcal{V}_n$; write x_v for the corresponding polynomial variable. There may be many more variables than universe labels. For example, variables x_{ij} give n^2 names. The two numbers must not be identified.

The spaces we shall use are

$P_d(n) = P_{n,\leq d}$	all polynomials allowed as proof lines,
$L_d(n) = P_{n,\leq d-1}$	all polynomials small enough to multiply once,
$X(n) = K[\mathcal{V}_n]$	one independent basis vector $[v]$ per variable name.

$P_d(n)$ contains both derivable and underivable polynomials. $L_d(n)$ also contains both kinds; it imposes only a degree restriction. If $V_j(n)$ is the space reached after j closure stages, the part we may actually multiply is $V_j(n) \cap L_d(n)$. The space $X(n)$ records which variable we use. We do *not* require that variable itself to have a proof.

For an explicit coordinate picture, take variables x, y, z and $d = 2$. Then

$$\begin{aligned} P_2(n) &= \operatorname{span}\{1, x, y, z, x^2, xy, xz, y^2, yz, z^2\}, \\ L_2(n) &= \operatorname{span}\{1, x, y, z\}, \\ X(n) &= \operatorname{span}\{[x], [y], [z]\}. \end{aligned}$$

Here the brackets indicate variable-name basis vectors, not quotient classes. This is a fixed-instance illustration; it does not specify a uniform axiom family or compute a lower bound.

5.3.2 Multiplication by one specified variable

For each variable x_v, consider the map

$$M_v : L_d(n) \to P_d(n), \qquad M_v(f) = x_v f.$$

It is linear: $x_v(af + bg) = a\,x_v f + b\,x_v g$. It stays inside $P_d(n)$ because $\deg f \le d-1$. For $d = 2$, multiplication by z sends the coefficient vector $a + bx + cy + ez$ to $az + bxz + cyz + ez^2$. This is an ordinary matrix calculation.

Now we want one map that packages *all* the maps M_v. Its input should allow a separate polynomial f_v for each variable: one slot for what is multiplied by x, one for what is multiplied by y, and so on. On these slots the output should be $\sum_v x_v f_v$. A direct sum of one copy of $L_d(n)$ for each variable does exactly this.

5.3.3 Variable-labelled copies and tensor products

The space of variable-labelled inputs is $X(n) \otimes L_d(n)$. Give it one basis vector $[v] \otimes \mu$ for each pair consisting of a variable name v and a monomial μ of degree at most $d-1$. Extend by linear combinations. Grouping these basis vectors by v identifies the space with those variable-labelled copies of $L_d(n)$:

$$X(n) \otimes L_d(n) \;\simeq\; \bigoplus_{v \in \mathcal{V}_n} L_d(n).$$

Every vector has a unique expression $\sum_v [v] \otimes f_v$. The notation $[v] \otimes f$ means: put the polynomial f in the slot labelled by v, with zero in every other slot.

For the three-variable example this space has dimension $3 \cdot 4 = 12$. It remembers inputs such as $[x] \otimes y$ and $[y] \otimes x$ as distinct vectors, even though their products are the same polynomial xy. It also allows sums such as $[x] \otimes f + [z] \otimes g$, with two different multiplication requests.

The tensor symbol is different from an ordinary Cartesian product. Polynomial multiplication on pairs $(a, f) \in X(n) \times L_d(n)$, after identifying a with its linear polynomial, is *bilinear*: it is linear in either argument when the other is fixed. It is not a linear map on the vector space of pairs. For example,

$$([x], 0) + (0, y) = ([x], y),$$

whereas multiplication is zero on the two summands and is xy on their sum. The tensor product makes all these bilinear requests into one *linear* map. In a basis it is just a matrix on the pair coordinates described above.

In particular, $([x] + [y]) \otimes f = [x] \otimes f + [y] \otimes f$ and $[x] \otimes (f + g) = [x] \otimes f + [x] \otimes g$. These rules distribute sums in either input; the monomial-pair description already implements them. The variable-labelled-copy interpretation uses the specified basis of $X(n)$. It is not a claim that a general tensor product comes with a preferred decomposition into copies.

The same coordinate construction applies to arbitrary finite-dimensional spaces A and B. After choosing bases a_i and b_j, their tensor product has basis $a_i \otimes b_j$. A bilinear map $\beta : A \times B \to C$ extends to a linear map on these coordinates by sending $a_i \otimes b_j$ to $\beta(a_i, b_j)$. For linear maps $f : A \to A'$ and $g : B \to B'$, the tensor map is defined by $(f \otimes g)(a \otimes b) = f(a) \otimes g(b)$ and extended linearly. Changing either basis changes the coordinates by this same rule; the construction does not depend on which bases were chosen. If labels act on both spaces, their *diagonal action* is $\pi(a \otimes b) = (\pi a) \otimes (\pi b)$. Each summand uses the same permutation in both positions. This is the tensor action used when multiplication and later detection pairings are transported.

5.3.4 The single multiplication map

Combining the variable multiplications gives the linear map

$$\boxed{m_n : X(n) \otimes L_d(n) \longrightarrow P_d(n), \qquad m_n\left(\sum_v [v] \otimes f_v\right) = \sum_v x_v f_v.}$$

The subscript n only tells us at which universe size the variables and polynomials are being used. On a basis pair it sends $[v] \otimes \mu$ to $x_v\mu$. This determines the whole map by linearity. Some later formulas write x_v itself for the basis vector of $X(n)$; at its output, x_v always means the actual polynomial variable.

This map need not be injective. For instance, $[x] \otimes y - [y] \otimes x$ is a nonzero input vector with output zero. Different multiplication requests can produce the same polynomial; their difference belongs to the kernel. The map is not being asked to recover the request from its output. We use its *image* to collect the polynomials obtained by legal multiplications.

5.3.5 A complete closure step with cancellation

Return to the three axioms

$$f_1 = xy + x, \qquad f_2 = xy + 1, \qquad f_3 = xz \quad \text{over } \mathbb{F}_2,$$

and keep the degree budget $d = 2$. We first take all their linear combinations, $V_0 = \operatorname{span}\{f_1, f_2, f_3\}$. Comparing the coefficients of xy and xz, as in Chapter 3, gives

$$V_0 \cap L_2 = \operatorname{span}\{x + 1\}.$$

Both f_1 and f_2 were quadratic. Their sum $x+1$ is linear and is now eligible for multiplication. The input $[z] \otimes (x + 1)$ lies in $X \otimes (V_0 \cap L_2)$, and

$$m_n([z] \otimes (x + 1)) = xz + z.$$

Adjoining the old vector $f_3 = xz$ gives the target z. The corresponding PC lines are completely explicit:

line	polynomial	reason
1	$xy + x$	input f_1
2	$xy + 1$	input f_2
3	xz	input f_3
4	$x + 1$	add lines 1 and 2
5	$xz + z$	multiply line 4 by z
6	z	add lines 5 and 3.

Every line has degree at most two. Multiplying line 1 or line 2 directly by z would instead give a cubic and exceed the budget. This is why the operation must first identify the low-degree *combinations*, rather than select low-degree inputs.

For a general current stage the same calculation reads

$$\underbrace{V_{j+1}(n)}_{\text{next stage}} = \underbrace{V_j(n)}_{\text{old consequences}} + \underbrace{m_n\big(X(n) \otimes (V_j(n) \cap L_d(n))\big)}_{\text{allowed new multiples}}.$$

The plus sign is a sum of subspaces: take all sums of one vector from each displayed space. The image term is itself a subspace because m_n is linear.

5.3.6 The input, inclusion and target maps

There are two other PC maps besides multiplication. Let $\mathcal{A}_{n,\le d}$ be the axiom indices whose polynomials fit the degree budget. A vector in $K[\mathcal{A}_{n,\le d}]$ stores one coefficient for each such index. The map

$$\alpha_n : K[\mathcal{A}_{n,\le d}] \to P_d(n), \qquad [a] \mapsto f_{a,n}$$

forms the resulting linear combination of usable axioms. Thus $V_0(n) = \operatorname{im} \alpha_n$. Distinct input columns may be dependent; the image, not the number of columns, is the initial

consequence space. The map $\ell_n : L_d(n) \hookrightarrow P_d(n)$ is simply inclusion: the same polynomial is now regarded as a possible degree-d line. Remembering this inclusion is necessary when forming an intersection.

For NS, instead put one coordinate on each permitted pair (μ, a), where μ is a monomial and $\deg \mu + \deg f_{a,n} \leq d$. Calling this set $\mathcal{C}_d(n)$, the certificate map is

$$\phi_n : K[\mathcal{C}_d(n)] \to P_d(n), \qquad [\mu, a] \mapsto \mu f_{a,n}.$$

Its image $I_d(n)$ is one span, with no PC iteration. The set $\mathcal{C}_d(n)$ here consists of certificate *indices*; it is not the representation class $\mathcal{C}_r(N)$ introduced in Chapter 7.

Finally, the requested targets have a separate column map

$$J_n : K[\mathcal{B}_n] \to P_d(n), \qquad [b] \mapsto G_b^{(n)}.$$

We use this map into $P_d(n)$ only when every full target degree is at most d. A budget below a stabilised target degree cannot derive that target; we do not truncate the target to make it fit. The map records what we ask to derive, not extra input axioms. Adding its columns to α_n would change the problem.

> **Checkpoint.** P_d contains possible proof lines; V_j contains the lines already reachable at stage j; L_d is the degree test; X labels the permitted variables. The map m_n assembles ordinary variable multiples. Only after understanding these fixed-universe ingredients do we ask for an operation relating them at two different universe sizes.

5.4 Building a consequence space by linear operations

We build each next consequence space by four familiar tasks: find an intersection, make variable-labelled copies, apply the multiplication matrix, and add the old space. We now spell out that recipe and why kernels and images describe it.

5.4.1 A direct sum stores two vectors; a sum combines them

If U, V are subspaces of the same ambient P, then $U \oplus V$ is the space of *pairs* (u, v). In contrast, $U + V \subseteq P$ is the space of outputs $u + v$. The linear map

$$\sigma : U \oplus V \to P, \qquad (u, v) \mapsto u + v$$

has image $U + V$. These spaces need not have the same dimension. For $U = V = Kx$, the input has two independent coordinates, while every output is a multiple of x. The pair $(x, -x)$ belongs to the kernel. The direct sum remembers the two requests; the sum forgets how an output was divided between them.

5.4.2 An intersection is found by solving a difference equation

Keep the same ambient space P and define

$$\delta : U \oplus V \to P, \qquad (u, v) \mapsto u - v.$$

A pair belongs to its kernel exactly when $u = v$. The common vector then belongs to both U and V. Thus the map

$$U \cap V \longrightarrow \ker \delta, \qquad h \mapsto (h, h)$$

is an isomorphism, with inverse $(u, u) \mapsto u$. Strictly, the kernel is a space of pairs, whereas the intersection is a subspace of P. The inverse just given, followed by inclusion in P,

specifies how to identify them. This small distinction matters when the whole calculation is transported.

In the cancellation example, $U = V_0$ and $V = L_2$. A kernel vector records a linear combination of the three axioms which equals a degree-one polynomial. Solving the difference equation gives the common line $K(x + 1)$. It finds a cancelled combination, not merely a subset of the original input columns.

5.4.3 Read the closure recurrence as a four-step algorithm

Suppose the current embedded space $V_j(n) \subseteq P_d(n)$ is known.

1. Find $U_j(n) = V_j(n) \cap L_d(n)$ by the difference-map kernel above. These are the already derived polynomials eligible for one further multiplication.
2. Make one copy of $U_j(n)$ for each variable name. This is $X(n)\otimes U_j(n)$, sitting in $X(n)\otimes L_d(n)$ by the actual inclusion $U_j(n) \subseteq L_d(n)$.
3. Apply the restriction of m_n to that input space, and take its image. Call the image $M_j(n)$. It consists of all sums $\sum_v x_v f_v$ with $f_v \in U_j(n)$.
4. Apply the sum map from $V_j(n) \oplus M_j(n)$ to $P_d(n)$. Its image is the next space $V_{j+1}(n)$.

All these operations are finite linear algebra at one universe size. Finding kernels or images is Gaussian elimination; making the variable-labelled copies is just organising coordinates. Each vector thereby obtained has a legal proof, by the constructive argument in Proposition 3.2.

An unchanged dimension would not tell us that the right space had been obtained. For example, Kx and Ky are isomorphic one-dimensional subspaces of $Kx \oplus Ky$, but only the first contains the target x. We must keep the embeddings and the specified maps, not just abstract isomorphism types.

5.4.4 Diagram and functor: the language behind the recipe

A *diagram* in this book is a collection of specified vector spaces and linear maps between them. Calling the data of Section 5.3 the *degree-d polynomial diagram* simply says that we remember the ambient, low-degree, variable, input and target spaces together with their maps. The inclusions are part of the data.

The vocabulary comes from category theory. A *category* organises objects and the maps allowed between them, with identity maps and composition. Here the objects are spaces with label actions and the allowed maps are equivariant linear maps. A *functor* coherently assigns a new object to each object and a new map to each allowed map: it preserves identities and composition. Thus, for maps $A \xrightarrow{f} B \xrightarrow{g} C$,

$$\mathcal{D}(gf) = \mathcal{D}(g)\mathcal{D}(f).$$

This equation ensures that a composite calculation can be transported either all at once or step by step. The operation used here is linear on the allowed maps and must also preserve the finite direct sums needed in the recipe.

An identification between transported and smaller-universe spaces must also respect their connecting maps. For fixed-data maps f_N and f_n, write these identifications as η_A and η_B. The required square is

$$\begin{array}{ccc} \mathcal{D}(A_N) & \xrightarrow{\mathcal{D}(f_N)} & \mathcal{D}(B_N) \\ {\scriptstyle \eta_A}\downarrow & & \downarrow{\scriptstyle \eta_B} \\ A_n & \xrightarrow{f_n} & B_n. \end{array}$$

It *commutes* when the two paths have the same value: $\eta_B \mathcal{D}(f_N) = f_n \eta_A$. A *natural* identification satisfies this compatibility for the allowed maps. This is the content of naturality used here; it is not a claim that auxiliary witness coefficients have a unique choice.

We have *not* constructed this operation yet. Write $\mathcal{D}$ for the operation we want, from a larger universe size $N = n + q$ to the retained universe size n. We must first identify the transported fixed data with the data at n, so that the transported multiplication map really is m_n, rather than some unrelated matrix of the same size. We must then show that it preserves the kernels, images and sums used by the four-step recipe.

If those statements hold, the induction is straightforward. Transport the initial image $\operatorname{im} \alpha_N$ to $\operatorname{im} \alpha_n$. If the current space has been transported, preserve the difference-map kernel, the variable-labelled copies with their inclusion, the multiplication image, and finally the sum image. The next stage is therefore transported too. At a stage past both stabilisation points, this gives transport of the entire degree-d consequence space. No individual proof length has to be bounded or followed through the change of universe.

This is why exactness is useful here. It is the preservation of a concrete linear-algebra recipe. The formal contract below records the three things which will make this reasoning valid; its existence is the subject of the later construction.

Definition 5.1 (Transport contract for a polynomial diagram)**.** At a fixed degree $d \geq 1$, a transport from universe size N to universe size n has three obligations.

1. **Identify the data.** Identify the transported ambient, lower-degree and basis spaces with their smaller-universe counterparts, carrying α, ℓ, m, ϕ, J to the actual smaller-universe maps. For multiplication, the identification $\mathcal{D}(X(N) \otimes Y) \simeq X(n) \otimes \mathcal{D}(Y)$ must respect the inclusions of the subspaces $Y \subseteq L_d(N)$ used in the recurrence.
2. **Preserve the calculation.** Preserve kernels, images, finite direct sums and their defining maps throughout the closure and target calculation. This transports intersections, sums and every closure stage as embedded subspaces.
3. **Reflect target vanishing.** For $H_N = I_d(N)$ or $C_d(N)$, put $Y_N = (H_N + \operatorname{im} J_N)/H_N$. Require $\mathcal{D}(Y_N) = 0 \Longrightarrow Y_N = 0$. This last condition says that transport cannot hide a requested target outside the consequence space.

The next two lemmas separate the work: the first two obligations transport the closure, while the third turns that transport into a two-way target-membership test. Throughout this discussion, transport includes the coherent maps and finite direct sums just described. The tensor identification is compatible with the actual subspace inclusions, not an independently chosen isomorphism for each space.

Lemma 5.2 (Transporting a closure recurrence)**.** *Suppose an operation $\mathcal{D}$ takes the larger-universe ambient, lower-degree, variable and usable-axiom spaces and maps to their actual smaller-universe counterparts. Suppose it preserves kernels and images on all the spaces used by the recurrence, and has the natural permutation-first-factor identification*

$$\mathcal{D}(X(N) \otimes Y) \simeq X(n) \otimes \mathcal{D}(Y).$$

Then, as embedded subspaces of the identified smaller ambient space,

$$\mathcal{D}(V_j(N)) = V_j(n) \quad \textit{for every } j.$$

It also transports the stable PC closure. Transport of the certificate map gives $\mathcal{D}(I_d(N)) = I_d(n)$.

Proof. The initial equality is preservation of the actual axiom-map image. Given the equality at stage j, express the intersection as the kernel of $(v, l) \mapsto v - l$. Preserve that kernel and

its inclusion. Apply the natural tensor identification and the actual multiplication map. Preserve its image and then its sum with the old stage. This gives the next equality. Both finite-dimensional chains eventually stabilise; choose a stage past both stabilisation points to obtain the closure statement. NS is the single certificate-map image calculation. □

> **Checkpoint.** The content is compatibility of the diagrams, not equality of dimensions. An isomorphism chosen separately for every space need not carry the intersection, multiplication map or target vector to the right one.

5.5 Preserving consequences and detecting missing targets

We now reach a separate question. Suppose the calculation of the consequence space has been transported correctly. Does that already tell us, in both directions, whether the prescribed target is derivable? The answer depends on whether transport can erase the obstruction to deriving it. We start with the ordinary quotient test.

5.5.1 A target leaves a remainder modulo the consequences

Let H_N be $I_d(N)$ for NS or $C_d(N)$ for PC, inside $W_N = P_d(N)$. For a single target g, its class $[g] \in W_N/H_N$ is zero exactly when g has a proof within the degree budget. Think of the quotient as recording only directions not yet proved.

For several targets the composite map

$$K[\mathcal{B}_N] \xrightarrow{J_N} W_N \xrightarrow{\rho_{H_N}} W_N/H_N$$

sends each target index to that target's unproved class. Every target is derived exactly when this map is zero. Its image is naturally the space

$$Y_N = (H_N + \operatorname{im} J_N)/H_N.$$

To check the formula, map $H_N + \operatorname{im} J_N$ into W_N/H_N. Its kernel is H_N, and its image is the target classes just described. Thus Y_N is a space of missing target directions, not a new polynomial or a new axiom. As before, the full target degrees must fit d before we use J_N into $P_d(N)$; there is no degree truncation.

For the small example $W = Kx \oplus Ky$, $H = K(x + y)$, a request for $x + y$ has zero obstruction. A request for both x and y has obstruction $W/H \neq 0$. Knowing that their sum is a consequence does not derive the two summands individually.

5.5.2 What exact transport establishes, and what is still missing

If all the actual maps are identified correctly, exactness takes the target image and its quotient to the corresponding calculation at n:

$$\mathcal{D}(Y_N) \simeq Y_n.$$

If $Y_N = 0$, then $Y_n = 0$. A derived request therefore remains derived after this downward transport. But from $Y_n = 0$ we know only $\mathcal{D}(Y_N) = 0$. To conclude $Y_N = 0$ we need to know that this operation cannot send a *nonzero* missing-target space to zero. That additional implication is called *reflection of vanishing*. In ordinary language: a missing consequence must still be detectable after transport.

There is no conflict with exactness. Even if an injection is transported to an injection, both its transported source and its transported destination might be zero. The map $0 \to 0$ is injective and surjective. Preserving relations among surviving spaces does not by itself require every original space to survive.

5.5.3 An exact operation which really loses an obstruction

Here is a concrete illustration. Work with pairs of vector spaces (A, B), with maps given by pairs of linear maps (f_A, f_B). Maps act separately on the two components and are not allowed to mix them. Define $F(A, B) = A$: retain the first component and forget the second.

Kernels, images, quotients and direct sums are computed componentwise. For example, $\ker(f_A, f_B) = (\ker f_A, \ker f_B)$, and forgetting the second component leaves $\ker f_A$. The same check works for images. Consequently F is an exact operation on these paired spaces. This is not a claim that ordinary projection preserves every intersection of arbitrary subspaces of K^2; our objects and allowed maps are the componentwise ones just specified.

Now take

$$W = (K, K), \qquad H = (K, 0), \qquad Q = (0, K), \qquad J : Q \hookrightarrow W.$$

The target map includes the second component. Its missing direction has obstruction $Y = (H + \operatorname{im} J)/H = (0, K) \neq 0$. After transport,

$$F(W) = K, \qquad F(H) = K, \qquad F(Q) = 0, \qquad F(Y) = 0.$$

All the image–kernel calculations have been preserved, but the target obstruction was forgotten. This example explains the logical issue only; it is not the polynomial deletion construction or a counterexample to our final theorem. The additional survival lemma is precisely what prevents this phenomenon on the relevant polynomial subquotients.

5.5.4 How the actual proof will keep missing directions visible

If a polynomial g is outside H, elementary linear algebra supplies a functional λ with $\lambda(H) = 0$ and $\lambda(g) = 1$: extend a basis of H by g and prescribe these values. For $H = K(x + y)$, $g = x$, the example is $\lambda(ax + by) = a - b$. Such a functional certifies nonmembership without requiring an invariant representative of a proof.

The permutation basis of the polynomial space has a dot product, so a functional can be represented by a coefficient vector. All functionals annihilating H form $H^\perp$. Concretely, this pairing is $\langle a, b\rangle = \sum_\mu a_\mu b_\mu$ on monomial coordinates. It is nondegenerate because each coordinate can be tested separately. There is no positivity requirement over the coefficient field. In characteristic two, for example, $(1, 1)$ pairs to zero with itself while pairing nontrivially with $(1, 0)$. Accordingly $H^\perp$ means the *annihilator* of H under this pairing, not a chosen complementary subspace. Permutations preserve the pairing because they permute its coordinates simultaneously. The induced action on a functional is $(\pi\lambda)(v) = \lambda(\pi^{-1}v)$; this is the usual dual action. Pairing them with the missing classes gives a map onto K whenever the obstruction is nonzero. The pairing is compatible with relabelings as a whole; an individually chosen separating functional need not be invariant.

The formal argument in Chapter 14 uses exactness and a tensor compatibility to show that, if deletion erased the obstruction, it would erase the source of this onto pairing. It would then produce a map from zero onto the surviving line $\mathcal{D}(K) \simeq K$, which is impossible. This is the reason for a doubled ambient support budget. One monomial coordinate uses at most kd labels; a pair of such coordinates uses at most $2kd$. We are bounding the coordinates of the pairing's ambient basis, not the union of labels in an arbitrary polynomial or functional.

We have described the idea here, not assumed the survival theorem. The next lemma records the implication once survival is proved.

Lemma 5.3 (The target-reflection contract)**.** *In addition to the closure transport, suppose the actual target map is transported and every nonzero subquotient of $P_d(N)$ survives deletion. Then*

$$\operatorname{im} J_N \subseteq H_N \quad \Longleftrightarrow \quad \operatorname{im} J_n \subseteq H_n.$$

Proof. The two membership conditions are $Y_N = 0$ and $Y_n = 0$. Exactness identifies $\mathcal{D}(Y_N)$ with Y_n. Since Y_N is a subquotient of the ambient polynomial space, survival of nonzero subquotients reflects its vanishing. This proves both directions. □

The survival lemma will come from a nondegenerate permutation-basis pairing, with doubled support. It is proved in Chapter 14; it is not an extra unproved premise of the final theorem.

5.6 The easier invariant target

The simplest important request is refutation: derive the polynomial 1. More generally, let $G^{(N)}$ be fixed by every label permutation. The input space of its target map has only one coordinate, recording the scalar coefficient with which that target is requested. For a fully invariant polynomial $G^{(N)}$, the source is just K:

$$j_N : K \to P_d(N), \qquad a \mapsto aG^{(N)}.$$

Its image in $P_d(N)/H_N$ is zero or a copy of the simple trivial module. The deletion of Chapter 7 fixes the trivial module K, so exactness already reflects whether this composite is zero. This explains why the invariant-target budget is smaller than the general indexed-target budget. The constant one is only one example of an invariant target.

Here is the argument without the module terminology. A map from a one-dimensional space has either zero image or a one-dimensional image. If the missing class $[G^{(N)}]$ is nonzero, its line has trivial label action because $G^{(N)}$ was invariant. Deletion preserves the image map and keeps that trivial line nonzero. It therefore cannot hide this particular obstruction. An indexed request can instead have a higher-dimensional missing space, so the doubled-support pairing argument is needed there.

5.7 The budget table

The fixed variable support is at most k, axiom-index support at most s, and target-index support at most t. The spaces have the following safe bounds.

Basis or ambient space	Support bound
Degree-d monomials	kd
Multiplier–axiom pairs	$kd + s$
Variable with degree-$d-1$ monomial	$k + k(d-1) = kd$
Target index	t
Two degree-d monomial coordinates	$2kd$

One affine choice covering the whole diagram and the survival argument is

$$R(d) = 2kd + s + t.$$

For an invariant target the pairing is unnecessary, so use $R(d) = kd + s$. Nothing in the PC recurrence increases the bound because its intermediate spaces are subspaces of those fixed ambient spaces. In particular, the support budget does not grow with proof length.

5.8 The connection with representation stability

The contract belongs to a familiar mathematical pattern: hold the complexity of a labelled object fixed and let the number of available labels grow. Here the complexity parameter is proof degree d, and one monomial uses at most kd labels. The ambient spaces grow, but the allowed equality patterns among the labels in a basis object come from a finite list. Representation stability studies how such families of symmetric-group representations are related across universe sizes. Three references help locate our argument within that setting. The terminology in this section concerns the surrounding literature. A *categorical equivalence* matches objects and their allowed maps, with an inverse up to compatible isomorphisms. An *abelian category* is a setting in which kernels, images and quotients have the formal properties familiar from modules. Spaces with equivariant linear maps are an example. Our construction below will establish the needed maps directly.

5.8.1 Church–Ellenberg–Farb: the broad fixed-degree picture

An FI-module is a functor from finite sets and injections to vector spaces: every injection has an assigned linear map, and compositions are respected. Church–Ellenberg–Farb [15] turn representation stability into a finite-generation condition. Their characteristic-zero results give eventual character polynomials and polynomial dimensions; fixed-degree configuration-space cohomology is one motivating example. This provides a useful comparison with fixing polynomial degree here. A character is the trace of a permutation acting on a representation. On a permutation basis it counts fixed basis objects. For example, the single-label space has character equal to the number of fixed labels; the space on ordered pairs of distinct labels has character $a(a-1)$ when a permutation fixes a labels. A *character polynomial* describes such traces using counts of cycles of specified lengths. The cited cohomology examples are background applications of that framework, not prerequisites for this book's proof.

Our consequence spaces need not form an FI-submodule under ordinary inclusion. For example, the uniform axiom $f_n = \sum_{i<n} x_i$ has degree-one consequence space Kf_n. Under $[n] \hookrightarrow [n+1]$, ordinary inclusion takes it to $K\sum_{i<n} x_i$, which is not contained in Kf_{n+1}. The new coefficient of x_n proves the failure directly. Thus fixed-degree FI intuition does not supply the consequence-transfer theorem automatically. We have to construct the maps which preserve the actual uniform equations.

5.8.2 Harman: the close modular-periodicity antecedent

Harman [10, Theorem 2.5 and Corollary 2.11] constructs periodic equivalences for bounded-weight permutation-module categories and extends them to the relevant abelian categories. The stable range is linear: his permutation result compares $n, m > 2r$. In positive characteristic, the suitable residue of the universe size matters, rather than one characteristic-zero eventual-stability statement. The structural principle of linear-range modular periodicity therefore predates this proof-complexity application. His Proposition 2.7 includes compatibility with natural operations on permutation modules; Proposition 2.12 includes induction, restriction, tensor products and duality on the abelian envelope. Compatibility with these operations is therefore also an antecedent. Here we construct the actual polynomial diagrams independently and prove that their closure operations and prescribed-target obstructions survive deletion, with the support bounds and universe-size conditions needed for the degree application. The bounded parameter in this comparison controls the small non-growing part of a permutation-module index. In the tuple models used here, the corresponding concrete parameter is the number of labelled positions $b \leq r$.

Keep the period convention precise. This book uses a prime power *strictly greater* than the support budget. Harman's printed $p^{\lceil \log_p r \rceil}$ needs care when r is itself a power of p; the strict choice avoids that boundary. For example, $\binom{5}{2} = 0$ and $\binom{7}{2} = 1$ modulo two, although 5 and 7 agree modulo two. Counts of degree r can detect a digit beyond that non-strict modulus. The present argument proves its strict-period statement directly.

5.8.3 Hone and coauthors: an explicit functor viewpoint

Hone, Klein, Pauwels, Sherman, Yacobi and Zhang [11] construct explicit periodic functors through semisimplification, motivated by the Duflo–Serganova functor in Lie superalgebra theory. Their symmetric-group result in Theorem 8.5.3 gives a closely related categorical setting. It helps explain why an operation which removes a prime-power block can have more structure than a numerical coincidence of dimensions. Its hypotheses and restricted categories matter; this is not global exactness on all representations. In that literature, *semisimplification* quotients a category by maps invisible to trace tests: $f : A \to B$ is negligible when $\operatorname{tr}(gf) = 0$ for every allowed $g : B \to A$. This is an operation on maps, not merely replacing each representation by a list of its simple factors. The Duflo–Serganova analogy starts with a parity-changing square-zero operator on a graded representation and takes its kernel modulo image. For the present proof the concrete quotient below is sufficient; none of the superalgebra machinery is assumed.

The concrete operation here is $\mathcal{D}(M) = \ker z_M / (\ker z_M \cap \operatorname{im} z_M)$. Here z_M is a linear operator on a representation M, formed from permutations of the added block; Chapter 7 gives its definition. Its powers mean repeated composition of that operator. In characteristic two, $z^2 = 0$ makes this look like the homology of a square-zero operator. Homology generally fails to preserve injections, and the two-point example in Chapter 7 shows that failure directly. For a square-zero map $d : M \to M$, the relation $d^2 = 0$ implies $\operatorname{im} d \subseteq \ker d$. In homological language, kernel vectors are *cycles*, image vectors are *boundaries*, and $\ker d / \operatorname{im} d$ is the *homology*: the cycles remaining after boundaries are identified with zero. This vocabulary describes a quotient of two concrete subspaces. Our characteristic-two deletion has this form. In odd characteristic, $z^p = 0$ need not give $\operatorname{im} z \subseteq \ker z$, so the intersection in the displayed quotient remains essential. Our common group-algebra projector proves exactness on the required bounded-support class, with the actual coefficient maps. In odd characteristic the quotient is defined exactly as displayed, not silently replaced by $\ker z / \operatorname{im} z^{p-1}$.

The proof which follows is self-contained: it constructs the witness, proves exactness and transports the whole NS or ordinary-PC diagram. The references explain the mathematical setting and the antecedents; they are not unproved inputs to the argument. The application requires prescribed-target reflection, the separate degree-zero case and the onset/period estimates used in Chapter 6. No claim that modular periodicity itself originates here is needed.

5.9 How the rest of the proof fulfils the contract

We add $q = p^a > R(d)$ labels and retain at least $2R(d) - 1$. The prime-power block defines a nilpotent operator z. On permutation bases a quotient of its kernel deletes exactly the block labels, but that quotient is not automatically exact on subspaces. One common group-algebra construction produces an idempotent e which models the quotient on every needed space at once: $e^2 = e$, so e acts as a projection. That common expression supplies exactness and map compatibility. The only hard quantitative question is how many retained labels are needed to build it.

In characteristic two, the preliminary construction uses $1 + 2 + \cdots + r$ labels for a support budget r. The improved construction reuses one pool and requires only $\max_{b \le r}(2b - 1)$. Once these bounds are proved, this chapter's contract becomes the actual transfer theorem rather than a conditional plan. The next chapter first shows exactly how that theorem yields the four types, so we know why the improvement from a sum to a maximum matters.

Chapter 6

From fixed-degree transfer to four global profiles

We now extract the global conclusions from the transfer theorem. This chapter does not yet construct the label-deletion operator; the later algebraic chapters do that. Its purpose is to show exactly what the transfer statement buys us, which direction of transfer is used, and where the linear lower bounds enter.

Fix a field of prime characteristic p, a uniform axiom template, and a uniform target request. Let $h(n)$ be either its NS or ordinary-PC profile. It may be an aggregate profile or the profile of one fixed target orbit. Recall the fixed bounds: k, s and t bound the variable-label, axiom-index and target-index supports, respectively; D_G bounds target degree, and n_0 is the template's starting universe size. For general indexed targets put

$$\kappa = 2k, \qquad \sigma = s + t, \qquad R(d) = \kappa d + \sigma, \qquad n_* = \max\{n_0, 1, kD_G + t\}.$$

For a single invariant target one can instead use $\kappa = k$, $\sigma = s$, and $n_* = \max\{n_0, 1, kD_G\}$. These are fixed constants of the problem, while d is a proposed proof degree.

6.1 What lifting does, and what it does not do

The transfer contract of Theorem 14.1 compares universe sizes which differ by a p-power block $q > R(d)$. Once the retained universe size is large enough relative to $R(d)$ and the stability threshold, it gives

$$h(n) \le d \quad \Longleftrightarrow \quad h(n+q) \le d.$$

The degree d is the same on both sides. We do not divide the degree by the new universe size, lower it by one, or change the target to its low-degree part. The operation transports the consequence space and the prescribed target-membership test.

Suppose a forward lifting argument alone starts from a degree-d proof at n. Iterating with the same admissible q gives proofs of degree at most d at

$$n, n+q, n+2q, \dots.$$

Their absolute degree bound remains d. The ratio $d/(n+\ell q)$ tends to zero, but that is not a statement that their absolute degrees fall. Nor does this progression cover all sufficiently large universe sizes. One cannot conclude that the entire profile is eventually bounded.

Forward lifting also need not preserve the exact minimum degree. For a simple sequence, let $g(1) = 2$ and $g(n) = 1$ for $n \ge 2$. Every threshold predicate $g(n) \le d$ persists under the

step $n \mapsto n+1$, but the initial exact value two never reappears. Forward persistence of a finite bound does force *some* finite value to recur, by the pigeonhole principle. The stronger conclusion that a particular value repeats, and the exact periodic statements below, come from two-way transfer at all the relevant thresholds.

6.2 A period and a starting universe size for each threshold

A *threshold predicate* answers a yes-or-no question at each size: does the request have degree at most d? Its eventual periodicity means that this answer repeats after the starting universe size. The period and the starting universe size may both depend on d. There need not be one period which works for the unbounded profile as a whole. Truncating h at $d+1$ retains the exact values $0, \dots, d$ and records every larger value, including infinity, by the single symbol $d+1$. This allows several threshold comparisons to be combined.

Use $N(d)$ for the periodicity onset, as in the paper. The argument (d) distinguishes this threshold from a larger universe size $N = n + q$. Define

$$\begin{aligned} N(0) &= n_*, \qquad Q(0) = 1, \\ N(d) &= \max\{n_*, 2(\kappa d + \sigma) + 1\}, \qquad Q(d) = \min\{p^a : p^a > \kappa d + \sigma\} \quad (d \geq 1). \end{aligned} \tag{6.1}$$

The conservative onset makes the retained-universe condition of the transfer theorem automatic. The degree-zero predicate is handled separately: beyond n_* the constant-target and constant-axiom data are stable, so that predicate is constant. In particular its period is one, even if σ is positive.

Proposition 6.1 (Threshold and truncated periodicity)**.** *For $n, m \geq N(d)$ with $n \equiv m \pmod{Q(d)}$, one has*

$$\min\{h(n), d+1\} = \min\{h(m), d+1\}. \tag{6.2}$$

The convention is $\min\{\infty, d+1\} = d+1$. *For $d \geq 1$,*

$$Q(d) \leq pR(d). \tag{6.3}$$

Proof. Repeated one-block transfer gives equivalence of the threshold $h \leq d$ at universe sizes in the same residue class, since every intermediate universe size remains beyond the onset. The smaller thresholds $e \leq d$ can also be compared there: $N(e) \leq N(d)$, and $Q(e)$ divides $Q(d)$ because both are powers of p and the budgets are nondecreasing. The zero threshold is constant in this range.

If $h(n) = e \leq d$, threshold e holds at m, while threshold $e-1$ fails there when $e > 0$. Hence $h(m) = e$; for $e = 0$ the first condition alone suffices. If $h(n) > d$, threshold d fails at both universe sizes. This proves the truncated equality.

Write $Q(d) = p^a$. Since $R(d) \geq 1$, the preceding power exists and minimality gives $p^{a-1} \leq R(d)$. Multiplication by p proves (6.3). □

Example 6.2 (A numerical budget)**.** Take $p = 2$, $\kappa = 4$, $\sigma = 3$, and $n_* = 7$. For degree $d = 3$,

$$R(3) = 15, \qquad Q(3) = 16, \qquad N(3) = 31.$$

Thus the degree-at-most-three predicate agrees at $31, 47, 63, \dots$. If the exact minimum at 31 is three, the minimum is exactly three at every size of that progression: the comparison of the lower thresholds rules out a drop. This conclusion says nothing about a different residue class modulo sixteen.

The inequality in the definition of Q is strict. If a budget equals 16 in characteristic two, the next permitted power is 32, not 16.

6.3 Which finite degrees come back infinitely often?

Definition 6.3 (Recurring finite degrees)**.** Let

$$E(h) = \{d \in \mathbb{N} : \text{there exists } n \geq N(d) \text{ with } h(n) = d\}.$$

Lemma 6.4 (An occurrence beyond its onset recurs)**.** *The set $E(h)$ consists exactly of the finite values assumed at arbitrarily large universe sizes. If $h(n) = d$ and $n \geq N(d)$, then*

$$h(n + \ell Q(d)) = d \qquad (\ell \geq 0).$$

A nonrecurring finite value can occur only before its own onset.

Proof. Apply truncated periodicity at threshold d to the displayed progression. A truncated value equal to d represents the exact value d, not a value above the cutoff. Conversely, any value appearing at arbitrarily large universe sizes appears beyond the fixed integer $N(d)$. □

The lemma is the central restriction on possible profiles. An arbitrary integer sequence might take the value 100 for the last time at an astronomically large universe size. Here, if 100 will not recur, its final appearance must precede the prescribed linear onset $N(100)$. Inverting that restriction gives a linear degree lower bound.

6.4 Inverting the onset

Define the nonnegative integer

$$\beta(n) = \left\lfloor \frac{(n - 2\sigma - 1)_+}{2\kappa} \right\rfloor, \qquad x_+ = \max\{x, 0\}. \tag{6.4}$$

For every $n \geq n_*$, one has $N(\beta(n)) \leq n$. If $\beta(n) = 0$, this is the special definition $N(0) = n_*$. Otherwise the floor inequality gives $2\kappa\beta(n) \leq n - 2\sigma - 1$, so the positive-degree formula for A applies. This separate zero case avoids subtracting natural numbers as if negative values were allowed.

Consequently,

$$n \geq n_*, \quad h(n) \leq \beta(n) \quad \Longrightarrow \quad h(n) \in E(h). \tag{6.5}$$

Indeed, the actual value $e = h(n)$ is finite and $N(e) \leq N(\beta(n)) \leq n$, so the recurrence lemma applies.

6.5 The four alternatives

We use limits in the extended order $\mathbb{N} \cup \{\infty\}$. The two tail bounds are

$$\liminf_{n\to\infty} h(n) = \sup_{m\geq 1} \inf_{n\geq m} h(n),$$
$$\limsup_{n\to\infty} h(n) = \inf_{m\geq 1} \sup_{n\geq m} h(n).$$

The liminf measures the best degree level still visible arbitrarily far out; the limsup measures the largest levels that keep returning. For an integer-valued profile with a recurring finite minimum e, the liminf is e once the finitely many smaller values have stopped occurring. A finite liminf therefore does not mean convergence. In T4 below, bounded subsequences coexist with arbitrarily large finite peaks.

Theorem 6.5 (Four degree profiles)**.** *For each fixed uniform request, and separately for NS and ordinary PC, exactly one of the following holds.*

T1. *The profile is bounded by a finite constant at every sufficiently large universe size.*

T2. *There is no recurring finite value. For every $n \geq n_*$,*

$$h(n) > \beta(n). \tag{6.6}$$

Infinity is allowed in this inequality; finite values satisfy a linear lower bound.

T3. *The set $E(h)$ is finite and nonempty, but the profile is not eventually bounded. Put $D = \max E(h)$. A nonempty proper set $U \subseteq \{0, \dots, Q(D) - 1\}$ has the following properties from universe size $N(D)$ onward. On every class $c \in U$, the profile has one constant finite value $v_c \leq D$. Off U, inequality (6.6) holds.*

T4. *The set $E(h)$ is infinite. For every prescribed bound B, there are integers $n, d \geq B$, with $d \in E(h)$ and $h(n) = d < \infty$, such that*

$$n \leq (p+2)(\kappa d + \sigma), \qquad h(n) \geq \frac{n}{(p+2)\kappa} - \frac{\sigma}{\kappa}. \tag{6.7}$$

Moreover $\liminf_{n\to\infty} h(n) = \min E(h) < \infty$.

In T1 and T3 the liminf also equals the least recurring finite value; in T2 it is infinity. Initial universe sizes below the specified onsets remain unconstrained.

Proof. If $E(h)$ is empty, (6.5) excludes $h(n) \leq \beta(n)$ and proves T2. Since $\beta(n)$ grows linearly, this is an $\Omega(n)$ lower bound wherever the minimum degree is finite. It is stronger to state the inequality in the extended order: an underivable target is not an exceptional counterexample.

Suppose next that $E(h)$ is finite and nonempty, with maximum D. Truncated periodicity at D divides the residue classes modulo $Q(D)$ into two kinds. On one kind the profile has a fixed finite value at most D; on the other it is always greater than D. Let U be the classes of the first kind. A recurring degree appears beyond $N(D)$, so U is nonempty. If U is the entire residue set, the profile is eventually bounded and belongs to T1. Otherwise, at a size off U, an inequality $h(n) \leq \beta(n)$ would force a recurring value, hence a value at most D, a contradiction. This proves T3. The easy values v_c need not be equal on different classes.

If $E(h)$ is infinite, choose recurring values d arbitrarily large, in particular so large that $N(d) = 2R(d) + 1$. An occurrence of d beyond that onset determines a residue class modulo $Q(d)$. There is a representative n in the interval

$$N(d) \leq n < N(d) + Q(d).$$

Two-way truncated periodicity ensures $h(n) = d$, even if the original occurrence was much later. Integrality and (6.3) now give

$$n \leq N(d) + Q(d) - 1 \leq 2R(d) + pR(d) = (p+2)R(d).$$

Rearrange for d to obtain the second inequality in (6.7). Taking d larger also makes $n \geq N(d)$ larger, so both may exceed the prescribed bound B.

To finish the exhaustion, eventual boundedness forces $E(h)$ to be finite and nonempty: only finitely many degrees can occur eventually, and at least one occurs infinitely often. Thus a profile either is eventually bounded, or falls into exactly one of the other three cases according as $E(h)$ is empty, finite nonempty, or infinite.

If $e = \min E(h)$ exists, every value below e occurs only finitely often. There are finitely many such values, so eventually $h(n) \geq e$. The value e keeps recurring, proving the liminf assertion. When $E(h)$ is empty, the same argument applies below each finite bound, and $h(n)$ tends to infinity in the extended order. □

Notice the two different numerical inputs. The linear onset $N(d)$ bounds the last occurrence of a nonrecurring small degree; this yields the T2 and hard-residue T3 bounds. The upper bound on $Q(d)$ additionally puts a recurring large degree at a universe size proportional to that degree; this yields the *finite* T4 peaks. Infinitely many infinite values would not by themselves prove the latter assertion.

If the proof only gave $N(d) = O(d^2)$, inverting the onset would give a square-root lower bound. If it gave no useful upper bound on $Q(d)$, a recurring value might first appear very far along its progression, and the finite linear peak argument would not follow. These observations explain why the later construction spends effort on a linear retained label budget rather than merely proving some eventual periodicity.

6.6 Reading the four cases correctly

In the examples below, $v_p(n)$ is the largest integer $a \geq 0$ such that p^a divides the positive integer n. Thus $p^{v_p(n)}$ is the largest power of p dividing n.

The following sequences illustrate the shapes. They are abstract profiles, not claims that these formulas are exact degrees of particular encodings. For fixed prime p and $n \geq 1$ consider

shape	$h(n)$	$E(h)$
T_1	1	$\{1\}$
T_2	n	$\varnothing$
T_3	1 if $p \nmid n$, $\quad n$ if $p \mid n$	$\{1\}$
T_4	$p^{v_p(n)}$	$\{1, p, p^2, \ldots\}$.

They satisfy the abstract threshold contract with, for example, $\kappa = 1$, $\sigma = 0$, and $n_* = 1$. For the last row, the condition $p^{v_p(n)} \leq d$ says that n is not divisible by the least p-power strictly larger than d. This makes its threshold periodicity explicit.

T1 does not mean eventually constant; bounded periodic variation is allowed. T2 can include infinite values, including the constant-infinity profile. T3 is linear only on its hard residues. T4 has a finite liminf, but that number is not an eventual value: infinitely many other finite levels recur as well. None of these descriptions removes finite initial exceptions. A sharp statement about every small universe size must be proved separately.

Proposition 6.6 (T4 admits no fixed easy/hard periodic partition)**.** *In T4 there do not exist a period q_0, a set U of its residues, a finite bound B, and $c > 0$ such that, at every sufficiently large universe size, $h(n) \leq B$ on U and $h(n) \geq cn$ off U.*

Proof. Choose a recurring finite value $e > B$. At sufficiently late occurrences of e, the universe size cannot be on an easy class. But once $n > e/c$, it cannot be on a hard class either. This contradicts recurrence. □

For a fully invariant target the same deductions use the smaller parameters $\kappa = k$, $\sigma = s$. For the target 1, $D_G = 0$ and $n_* = n_0$, giving the refutation corollary. The later comparison theorem shows that NS and PC assign the same type to each fixed request. It does not assert equal exact minimum degrees or equal recurring sets.

6.7 The size consequences under full coefficient lists

Recall that a full coefficient list for degree e in v variables has $B_v(e) = \binom{v+e}{e}$ entries, including zeros. In this section D again denotes the fixed axiom-degree bound, rather than the largest recurring degree used in the T3 statement. For an indexed request define $S_S(n)$ as the

maximum of the minimum coefficient-list sizes of the individual target proofs. Use zero for an empty request and infinity if any target is underivable. A bundle of proofs for all targets differs from this maximum by a factor between one and $|\mathcal{B}_n|$ when the request is nonempty. That factor is polynomial in n and does not change the cases below.

Corollary 6.7 (Polynomial bounds and exponential lower bounds)**.** *For full coefficient lists, the four degree types have these consequences.*

T1. *The minimum sizes are polynomial in n eventually.*
T2. *They are at least 2^{cn} eventually, for some $c > 0$, with infinity interpreted in the extended order.*
T3. *They have a polynomial upper bound on the easy residue classes and an exponential lower bound on the hard ones, beyond fixed onsets.*
T4. *They have a polynomial upper bound at infinitely many sizes and finite exponential peaks at arbitrarily large sizes. In addition, for every $e > 0$ there is b such that $n^e < S_S(n) \le n^b$ at infinitely many sizes.*

There is no eventual fixed periodic partition into one uniformly polynomial part and one exponentially hard part in T4. These are exponential lower bounds, not universal matching upper bounds, and they concern full coefficient lists rather than sparse monomial size.

Proof. Bounded supports give polynomially many variables and indices. If some variable has support size $j > 0$, its relabellings have every j-element support, giving at least $\binom{n}{j} \ge n$ variables for all sufficiently large n. A fixed degree therefore gives a polynomial coefficient-list upper bound by Lemma 3.5. A finite linear degree lower bound gives an exponential size lower bound, since both systems have size at least $B_{v(n)}(d - D)$ for a target attaining a minimum degree $d > D$, where D bounds input degree. Such a target is nonzero. The zero-target empty certificate causes no exception. Underivability gives infinite size directly.

For the extra T4 assertion choose a recurring degree d with $\ell = d - D > e$. At its sufficiently late occurrences,

$$S_S(n) \ge B_{v(n)}(\ell) \ge n^\ell/\ell! > n^e,$$

while the fixed degree d gives $S_S(n) \le n^b$ for some b. These intermediate polynomial subsequences cannot fit a proposed partition with one fixed polynomial exponent and an exponential lower bound on the other part.

If every variable has empty support, there are only finitely many variables and relabelling fixes them. Each axiom or target index orbit gives a fixed polynomial, and all finitely many orbits appear simultaneously beyond a fixed universe size. The systems and requests then stabilise. Their degree is a finite constant, or infinity, producing T1 or the infinite-size case of T2. This handles the degenerate case without assuming a growing number of variables. □

6.8 Exercises

Exercise 6.8. Let $p = 2$, $\kappa = 4$, $\sigma = 4$, and $n_* = 7$. Compute $N(0), Q(0)$, $N(3), Q(3)$ and $\beta(62)$. Check $N(\beta(62)) \le 62$ and explain why the strict inequality defining Q matters.

Model answer (p. 167).

Exercise 6.9. For $g(1) = 2$ and $g(n) = 1$ for $n \ge 2$, verify forward persistence of every threshold under $n \mapsto n + 1$. Identify a reverse threshold implication that fails. Explain why exact recurrence of the starting value cannot be deduced from forward persistence alone.

Model answer (p. 168).

Exercise 6.10. For $h(n) = p^{v_p(n)}$, determine every recurring value, its liminf, and a sequence of finite linear peaks. Explain why no eventual linear lower bound holds at all sizes.

Model answer (p. 168).

Exercise 6.11. Show that T3 cannot have $E(h) = \{0\}$. State exactly where stability of the degree-zero predicate is used.

Model answer (p. 169).

Chapter 7

What it means to delete a block of labels

We now describe the operation that the preceding contract requires. It acts on spaces with label permutations, not by setting some polynomial variables to zero. Its construction will justify the word "deletion" on bases, and then explain why extra work is needed on subspaces.

The construction has three ingredients: spaces on labelled tuples, one nilpotent operator built from block translations, and a quotient of its kernel. We develop these tools before stating the exactness theorem. Its clauses identify the surviving basis, preserve the actual coefficient maps, and transport the subspace calculation of a proof. The common-operator construction in the later chapters establishes these properties.

7.1 Why substitution is the wrong operation

Consider the totality axiom $\sum_j x_{ij} - 1$. If the row label i is deleted and all its variables are set to zero, this polynomial becomes -1. Such a substitution refutes the remaining system immediately. It does not produce the intended smaller uniform instance.

Restriction of coefficient matrices to retained rows and columns is correct on uniform data, but arbitrary subspaces do not restrict exactly. A linear relation supported partly on deleted coordinates may change its meaning. The proof therefore uses a quotient operation on modules with label action, together with a theorem that this operation preserves the required subspace calculations.

7.2 Permutation modules and their subspaces

Recall the label-action language of Section 5.2. The monomial space is a permutation module, and the NS/PC consequence spaces are invariant subspaces: relabelling a legal proof gives another legal proof from the uniform axioms.

A vector space with named basis vectors

For a finite set of labelled objects $\mathcal{U}_N$, the notation $K[\mathcal{U}_N]$ means the vector space with one basis vector e_u for each object $u \in \mathcal{U}_N$. A vector is a coefficient list

$$v = \sum_{u \in \mathcal{U}_N} a_u e_u.$$

The basis objects might be variables, monomials, axiom indices or ordered tuples. A label permutation $g \in S_N$ acts by $ge_u = e_{gu}$ and extends linearly. In matrix language it permutes

the basis coordinates. The phrase *permutation module* means exactly this vector space together with these permutation matrices.

For instance, the space on single labels has basis $e_1, \dots, e_N$, and the swap (12) sends e_1 to e_2, e_2 to e_1, and fixes the other basis vectors. More general spaces below need not have a basis permuted by labels: an invariant subspace or a quotient still carries linear permutation actions. Such a space is called a representation, or module. No additional algebraic operation on its vectors is intended by that word.

Subspaces, quotients and maps which respect relabelling

A subspace $U \subseteq M$ is *invariant* if every label permutation maps it to itself. Its quotient M/U consists of classes $[v]$ with v and v' representing the same class exactly when $v - v' \in U$. Relabelling acts by $g[v] = [gv]$, which is well defined because $gU = U$.

As a small example, in the space on three labels the line $U = K(e_1 + e_2 + e_3)$ is invariant. In the quotient the sum $e_1 + e_2 + e_3$ becomes zero, while $[e_1]$ is still nonzero. Thus a quotient does not discard a designated subset of coordinates: it imposes the specified linear relations. A *subquotient* is a quotient of an invariant subspace, such as U/V with $V \subseteq U \subseteq M$ invariant. These are precisely the objects which arise when we look at targets modulo consequences.

A linear map $f : M \to M'$ is *equivariant* if $f(gv) = gf(v)$ for every label permutation g. This just says that the two ways of relabelling and applying f give the same answer. For example, on the space E_N with basis $e_1, \dots, e_N$, the coordinate-sum map

$$\epsilon : E_N \longrightarrow K, \qquad \epsilon\Big(\sum_i a_i e_i\Big) = \sum_i a_i$$

is equivariant when labels act as the identity on the output K. Forgetting one position of a tuple is another example. The axiom, target and multiplication maps in the uniform proof calculation are equivariant too; their actual coefficients specify which maps they are.

One expression in permutations, acting on many spaces

The group algebra $K[S_N]$ consists of finite sums $\sum_g a_g g$. Its multiplication is composition, extended bilinearly. It acts by the corresponding linear combination of renamings. If a map f respects label permutations, then

$$f((\sum_g a_g g)v) = (\sum_g a_g g)f(v).$$

This elementary identity is the reason a common group-algebra expression is so valuable: the same expression automatically commutes with every map in the uniform diagram.

For example, $h = 2g - \tau$ acts on M as $v \mapsto 2gv - \tau v$. The same expression acts on M' and commutes with f whenever f is equivariant. We do not choose unrelated matrices separately on the two spaces. Nor do we claim that h commutes with every label permutation: the statement is that every equivariant map commutes with the common expression. In positive characteristic, the coefficient 2 in this example is interpreted in K, so it is zero when the characteristic is two.

Why ordered tuples are the test spaces

For $b \geq 0$ let

$$P_b(N) = K[\mathrm{Inj}([b], [N])]$$

be the space on ordered tuples of distinct labels. Every orbit of a support-b index is a quotient of such a space. Choose an ordering of its support; an injective tuple specifies where that

support is sent. Two extensions agreeing on the support act identically on the index. Thus ordered tuples provide a finite universal list of spaces on which to construct an operator valid for all support-bounded data. Here $P_b(N)$ denotes a tuple permutation space; the bounded-degree polynomial space is written $P_{N,\leq d}$.

Here $[b] = \{1, \ldots, b\}$ names the *positions*; an injection assigns distinct universe labels to them. In particular:

- $P_0(N)$ is the one-dimensional space on the empty tuple. Every permutation fixes its basis vector.
- $P_1(N)$ has basis $e_1, \ldots, e_N$.
- $P_2(N)$ has basis $e_{(i,j)}$ for $i \neq j$. It has $N(N-1)$ basis vectors, and $ge_{(i,j)} = e_{(g(i),g(j))}$.

There is also an action permuting tuple *positions*: swapping the two positions sends (i, j) to (j, i). It is different from swapping two particular universe labels. The two actions commute. Throughout the construction, b counts positions and r bounds their number; neither is a polynomial degree.

The universality claim in the preceding paragraph is concrete. If an object u has support $\{u_1, \ldots, u_b\}$, send $e_{(t_1,\ldots,t_b)}$ to e_{gu}, where g sends each u_i to t_i. Any two such permutations agree on the support and hence give the same object. The map is linear, equivariant and onto the space on the orbit of u. Therefore any operator which vanishes on $P_b(N)$ also vanishes on that orbit space.

The common-annihilator class

Let $J_r(N)$ be the ideal of group-algebra elements acting as zero on $P_b(N)$ for every $b \leq r$. Write $\mathcal{C}_r(N)$ for the finite-dimensional modules annihilated by $J_r(N)$. Every support-r permutation basis, and every submodule or quotient of it, belongs to this class. It is also closed under finite direct sums. We will not assume that it is closed under every tensor product: the actual tensor products used in the proof are placed inside known support-bounded permutation ambients.

In formulas, the definition is

$$J_r(N) = \{h \in K[S_N] : hv = 0 \text{ for every } v \in P_b(N) \text{ and every } 0 \leq b \leq r\}.$$

Calling it an *ideal* records that sums and scalar multiples remain in it, as do products on either side by group-algebra elements. The latter fact follows because each $P_b(N)$ is preserved by all those operators.

Membership of M in $\mathcal{C}_r(N)$ means that *every* operator in this set acts as zero on M too. Equivalently, all the operator relations shared by the tuple spaces up to length r hold on M. This is a condition on the permutation action on a space, not a claim that each of its vectors mentions only r labels. For example, $\sum_{i=1}^{N} a_i e_i$ can use every label, while its single-label permutation space belongs to $\mathcal{C}_1(N)$. Likewise the degree-d monomial space has basis support at most kd, although a polynomial in it can involve many different labels across its monomials. The symbol $J_r(N)$ denotes this set of operators, not the target map J_N from the previous chapter.

The closure assertions now have elementary proofs. If an operator kills every vector of M, it kills every vector of an invariant subspace and acts as zero on every quotient class. It also kills a finite direct sum of such spaces. Hence the class includes the subspaces and subquotients needed by a proof. It need not include an arbitrary tensor product: combining two basis objects can add their support sizes. We will check the relevant *ambient* support each time a tensor product is used.

7.3 The regular prime-power block

The parameters have different roles. We retain n labels, add $q = p^a$ block labels, and obtain $N = n + q$ universe labels in total. Here p is the characteristic, a is a block exponent, and r is the support budget for the spaces in the proof diagram. Choosing $q > r$ does not mean choosing q larger than n.

Write $N = n + q$ and identify the last $q = p^a$ labels with $A = (\mathbb{F}_p)^a$. Translation by a coordinate vector gives a permutation σ_i of the block, fixing the retained labels. The translations commute and have order p. Over a suitable coefficient extension choose $c_1, \dots, c_a$ independent over $\mathbb{F}_p$ and put

$$z = \sum_{i=1}^{a} c_i(\sigma_i - 1).$$

Here 1 is the identity permutation. On a representation M, this group-algebra element acts as a linear operator z_M. A power of z means repeated composition: $z^p = 0$ will say that p successive applications send every vector to zero. The characteristic-p binomial formula gives $z^p = 0$. The retained permutations commute with z, so its kernel and the quotient below retain an action of S_n. In operator formulas, I denotes the identity linear map, the action of this identity permutation. Independence of the coefficients is over the prime field $\mathbb{F}_p$: a relation $\sum_i t_i c_i = 0$ with $t_i \in \mathbb{F}_p$ must have all $t_i = 0$. It is not independence over the larger working field.

The word *regular* means that the translations move any block label to any other in exactly one way. For $p = 2, a = 2$, write the four block labels as $00, 10, 01, 11$. Translation in the first coordinate swaps 00 with 10 and 01 with 11; translation in the second swaps 00 with 01 and 10 with 11. They leave all retained labels fixed. In general, the identity

$$z^p = \sum_i c_i^p(\sigma_i - I)^p = \sum_i c_i^p(\sigma_i^p - I) = 0$$

uses the commutation of these translations and the characteristic-p binomial formula. It is a statement about matrices or composed operators, not about a scalar whose power is zero.

The independence of the coefficients prevents a nontrivial translation orbit from being accidentally invisible. It can require extending the field. Membership of a target with original-field coefficients descends by Lemma 3.3; no proof over an arbitrary extension is silently substituted for a proof over the original field.

A quotient made from a kernel and an image

Define

$$\mathcal{D}(M) = \ker z_M / (\ker z_M \cap \operatorname{im} z_M).$$

For $p = 2$ this is $\ker z / \operatorname{im} z$. For odd p that shorter formula is not meaningful in general, because z^2 need not be zero.

The kernel consists of vectors v with $z_M v = 0$. The image consists of vectors $z_M w$. We first keep the kernel vectors and then identify two of them when their difference is also an image vector. Thus an element of $\mathcal{D}(M)$ is a class $[v]$ of a vector already in the kernel. An arbitrary basis vector need not define such a class.

For a fully explicit case, work in characteristic two, retain one label a, and swap two block labels b, c. Choose $z = (bc) - I$ on the space with basis e_a, e_b, e_c. Then

$$ze_a = 0, \qquad ze_b = e_b + e_c, \qquad ze_c = e_b + e_c.$$

Consequently

$$\ker z = Ke_a \oplus K(e_b + e_c), \qquad \operatorname{im} z = K(e_b + e_c), \qquad \mathcal{D}(M) \simeq Ke_a.$$

The class of $\alpha e_a + \beta(e_b + e_c)$ is represented by αe_a. The combination $e_b + e_c$ lies in the kernel, so it qualifies as a representative; it also lies in the image, so it represents the zero class. The vector e_a survives. The individual vector e_b is not in the kernel and hence is not a representative for this quotient. This distinction is why deletion is not a coordinate restriction applied to every vector.

To see what the quotient keeps, take a nilpotent chain with basis $v_0, \ldots, v_{\ell-1}$ and

$$zv_0 = 0, \qquad zv_j = v_{j-1} \quad (j > 0).$$

This is a nilpotent *Jordan block*: applying z moves each vector one position down the chain and eventually sends it to zero. The chain basis describes the linear operator z. It need not be a permutation basis or an invariant decomposition for the full symmetric group. This distinction will matter when we later build an operator which works on many representations at once.

For $\ell = 1$, the kernel is the whole line and the image is zero, so the line survives. For $\ell \geq 2$, the kernel is Kv_0 and $v_0 = zv_1$ already lies in the image, so the quotient is zero. Thus deletion retains length-one chains and discards every longer chain. The equation $z^p = 0$ permits every length $1 \leq \ell \leq p$. The later odd-characteristic construction separately proves that the relevant tuple generators have only lengths one and p; neither that purity statement nor exactness follows from nilpotence alone.

How a linear map induces a map of the quotients

Let $f : M \to M'$ commute with label permutations. The common group-algebra definition gives $fz_M = z_{M'}f$. If $v \in \ker z_M$, then $f(v) \in \ker z_{M'}$. If two kernel vectors differ by $z_M w$, their images differ by $z_{M'}f(w)$, which is again an image vector and remains in the kernel. Therefore the rule

$$\mathcal{D}(f) : \mathcal{D}(M) \longrightarrow \mathcal{D}(M'), \qquad [v] \longmapsto [f(v)]$$

is well defined. It is the actual induced map, not an independently chosen map between spaces of the same dimension. Applying two maps in succession before or after taking classes gives the same answer.

In the three-label example, the coordinate-sum map $\epsilon : M \to K$ is equivariant. On a kernel vector it gives

$$\epsilon\big(\alpha e_a + \beta(e_b + e_c)\big) = \alpha + \beta + \beta = \alpha.$$

Thus its induced map sends the surviving class $[e_a]$ to 1 and the zero class $[e_b + e_c]$ to 0. On the trivial output space K every label permutation is the identity, so $z = 0$ and $\mathcal{D}(K) = K$.

Finally, permutations of the retained labels commute with each block translation. They preserve the kernel and image and act on the quotient by $[v] \mapsto [gv]$. This is why the output carries an action of the *smaller* group S_n. The deleted block does not become a set of new polynomial variables in that output.

7.4 Why small supports make a basis object retained

The support bound in the next argument belongs to an individual *basis object*, not to every vector in its span. A basis variable x_i uses label i; the monomial $x_i x_j$ uses labels i, j; a sum of many such monomials may mention most of the universe.

Suppose $\mathcal{U}$ has support bound $u < q$. If an object is fixed by all block translations, its support is a block-translation-invariant set. The regular action on the block is transitive. Hence a support meeting the block would contain all q block labels, contrary to $u < q$. The fixed objects are exactly those supported on retained labels.

This is the first structural use of the inequality $q > r$. It has nothing to do with choosing q larger than the universe itself. It must exceed the support needed by the degree-d calculation. A one-label change usually has $q = 1 \leq r$ and cannot satisfy this argument. This explains why the natural shifts are prime powers rather than $n \mapsto n + 1$.

The strict inequality is visible in an example. With a four-label block and unary variables, the product of all four block variables is a basis monomial fixed by every translation. It uses the whole block, so it is excluded by a support bound $u < 4$. A monomial meeting the block in one, two or three labels cannot be fixed by all translations. The support argument identifies which basis objects are fixed; the later orbit calculation must still prove that the moved part has zero deletion quotient.

The theorem in plain language

Choose a block larger than the supports we need to track, and keep enough labels outside it. The theorem asserts that the block coefficients can be chosen so that the quotient just defined does three things at once. It retains the expected smaller basis; it carries the actual uniform coefficient maps to the smaller maps; and, on the class $\mathcal{C}_r$, it preserves the subspace operations used by a proof. The last assertion is the substantial one: neither the definition of the quotient nor $z^p = 0$ proves it.

Here is the precise statement. The phrases “exact”, “natural” and “trivial module” are read clause by clause immediately below.

Theorem 7.1 (The exact deletion input)**.** *Work first over an algebraically closed field of characteristic p. For $q = p^a > r$ and retained universe size $n \geq \max\{0, 2r - 1\}$, the block coefficients can be chosen so that $\mathcal{D}$ is exact on $\mathcal{C}_r(n + q)$. It fixes the trivial module. For every support-u permutation basis with $u < q$ there are natural identifications*

$$\mathcal{D}(K[\mathcal{U}_{n+q}]) \simeq K[\mathcal{U}_n], \tag{7.1}$$

$$\mathcal{D}(K[\mathcal{U}_{n+q}] \otimes Y) \simeq K[\mathcal{U}_n] \otimes \mathcal{D}(Y), \tag{7.2}$$

with a permutation module as the first tensor factor. Stable invariant coefficient matrices are carried to their actual retained-universe matrices.

This finite-algebra theorem is proved by the chapters on square-zero blocks, linear layer repair, rook realisation and top-power projection. No representation-theoretic equivalence is invoked as an external assumption.

Reading the clauses

The working field. An algebraically closed field has the coefficients needed for the independent block directions. This is where the construction is carried out, not a change of the original input equations. Lemma 3.3 brings membership of original-coefficient targets back to the original field. The block size $q = p^a$ is a number of universe labels, not a prescribed cardinality for that field.

The support budget and the range. The integer r bounds the tuple test spaces and the supports of the proof-diagram ambients which must undergo exact transport. The inequalities are $q > r$ and $n \geq \max\{0, 2r - 1\}$. They concern block labels and retained labels, respectively. For $r \geq 1$, the latter is the sufficient retained-label budget $2r - 1$; for $r = 0$, it is simply $n \geq 0$. The proof will construct one compatible witness for all lengths $0, \ldots, r$ on that one retained pool. Template stabilisation and target-degree thresholds are additional requirements when this finite-algebra input is applied to polynomial families.

Exactness on the specified class. Take an equivariant map $f : M \to M'$ between spaces in $\mathcal{C}_r(N)$. Its kernel and image are invariant subspaces. Exactness says that their deleted spaces are the actual kernel and image of $\mathcal{D}(f)$, with the induced inclusions. In particular, if $U \subseteq V$ are invariant spaces in this class, the induced map $\mathcal{D}(U) \to \mathcal{D}(V)$ is injective, and

$$\mathcal{D}(V/U) \simeq \mathcal{D}(V)/\mathcal{D}(U).$$

This is a claim about maps and their linear relations, not only about dimensions. It is restricted to $\mathcal{C}_r(N)$; arbitrary label representations are not being declared exact.

Fixing the trivial module. On the one-dimensional space K with all label permutations acting as the identity, every $\sigma_i - I$ acts as zero. Hence $z = 0$, the kernel is K, the image is zero and $\mathcal{D}(K) = K$. The constant line does not disappear. This clause will be needed both for invariant targets and for the detection pairing.

The first basis identification. The first displayed isomorphism says that the surviving directions in the space on $\mathcal{U}_{n+q}$ are precisely the basis objects supported on retained labels. A retained object u gives the class of its basis vector e_u, identified with the same object in $\mathcal{U}_n$. Moved basis vectors need not themselves be kernel vectors; the assertion is that their entire moved-orbit summand contributes zero to the quotient. For a single-label basis, this turns the large label space into the space on the n retained labels.

The tensor identification and its naturality. Here Y is any space with the larger label action. The first factor has a permutation basis. Regard a tensor $\sum_u e_u \otimes y_u$ as a list of vectors from Y, one packet for each basis object u. A label permutation acts on *both* the packet index and its contents: $g(e_u \otimes y) = e_{gu} \otimes gy$. The theorem keeps retained packet indices and applies deletion to their contents. More precisely, for retained u and $y \in \ker z_Y$, it matches

$$e_u \otimes [y] \quad \text{with} \quad [e_u \otimes y].$$

It also proves that the moved packet summand disappears even though translations act on its Y contents. There is no separate bounded-support assumption on Y in this basis identification. When exactness is used on a tensor source, we separately check that source's ambient support budget.

"Natural" means that applying any equivariant map $Y \to Y'$ to the packet contents before or after this identification gives the same result. This is what lets the rule apply to a consequence subspace $Y \subseteq L_d$ with its actual inclusion, rather than to a separately chosen basis for Y. It is also why the tensor clause specifically states which factor has a permutation basis.

The coefficient matrices are the actual maps. If F_N sends a retained input basis vector e_u to $\sum_v a_N(v,u)e_v$, equivariance puts this image in the kernel of z. Its retained coefficients are exactly $a_N(v,u)$ for retained v. A stable rule gives $a_N(v,u) = a_n(v,u)$ on those objects, and the moved kernel component vanishes in the quotient. Thus the induced map is the prescribed F_n entry by entry. The statement does not allow us to replace it by an unrelated matrix with the same rank. It applies in particular to the uniform axiom, target and multiplication maps from Chapter 5.

How the later proof makes exactness possible

The later chapters construct one group-algebra element w with

$$z^{p-1}wz^{p-1} = z^{p-1} \quad \text{on every } P_b(N),\ b \le r.$$

This is a partial-inverse condition: on the image of z^{p-1}, w supplies a preimage. It does not invert the nilpotent operator on the whole space. From z and w the proof constructs another common operator e, using also the proved chain property of the tuple generators. This e is a projector with $e^2 = e$, and identifies $\mathcal{D}(M)$ with its image eM as a vector space. The letter e here names an operator, unlike the basis-vector notation e_u.

Every actual equivariant map commutes with this same expression in permutations. For example, if $f : V \to W$ is onto and $y \in eW$, choose an ordinary preimage v of y. Then $f(ev) = ef(v) = ey = y$, so the restricted map $eV \to eW$ is still onto. The kernel and injection checks use the same commutation. This is the source of exactness. Its hard part is constructing one such expression which works on the entire annihilator class; separate projectors on separate spaces would not justify these equations. The quotient itself carries the retained label action; the projector image is its vector-space model.

An admissible numerical instance

Take characteristic two, $r = 2$, $q = 4$, $n = 3$, and $N = 7$. Work in an algebraic closure of $\mathbb{F}_2$, choose $\omega^2 + \omega + 1 = 0$, and use coefficients $1, \omega$. Retain labels $1, 2, 3$ and identify labels $4, 5, 6, 7$ with $00, 10, 01, 11$, respectively. The two translations and their common operator are

$$\sigma_1 = (45)(67), \qquad \sigma_2 = (46)(57), \qquad z = (\sigma_1 - I) + \omega(\sigma_2 - I).$$

The numbers satisfy $q > r$ and $n = 2r - 1$. The tuple test spaces are $P_0(7)$, $P_1(7)$ and $P_2(7)$, of dimensions $1, 7, 42$. The moved-orbit calculation, which uses the independence of $1, \omega$ over $\mathbb{F}_2$, identifies their deletion quotients with retained tuple spaces of dimensions $1, 3, 6$. This illustrates the basis calculation for the displayed operator.

The all-class exactness assertion is separate. For these same parameters, Theorem 7.1 provides a suitable choice of block coefficients for which deletion is exact on $\mathcal{C}_2(7)$. The present numerical illustration does not certify that its specific choice $1, \omega$ has that additional property or supply a common witness for it. The later construction establishes the exactness input with its required coefficient choice.

For comparison with polynomial notation, the ordinary degree-at-most-two monomial space in seven unary variables has 36 basis monomials: one constant, seven linear monomials, seven squares and twenty-one products of distinct variables. Each monomial uses at most two labels. Its deletion is the corresponding space in three unary variables, with 10 monomials. This is a consequence of the basis identification, not a computation of any minimum NS or PC degree. Squares are separate coordinates because this is the ordinary polynomial ambient, not the space of functions on Boolean assignments. An application to a proof must additionally cover the actual axiom and target maps, multiplication domains and, when needed, the doubled-support detector. Those data may require a larger r.

The instance illustrates the theorem's parameters and conclusions; its concrete operator illustrates basis deletion, and the theorem's existential coefficient choice supplies exactness. It does not independently construct a common witness or prove general lifting by finite experiment.

7.5 Why deletion on bases is not enough

In characteristic two, let two block labels be swapped. Their permutation space has basis e_1, e_2 and $ze_1 = ze_2 = e_1 + e_2$. Its kernel equals its image, so its deletion is zero. The invariant line $T = K(e_1 + e_2)$ has $z = 0$ and survives. Therefore deletion of $T \hookrightarrow Ke_1 \oplus Ke_2$ is the zero map on a nonzero space. An injection was not preserved.

Here there are no retained labels: $n = 0$, $q = 2$ and the single-label basis has support budget $r = 1$. The retained-universe hypothesis would require $n \geq 1$. The example therefore explains the need for a range restriction rather than contradicting Theorem 7.1. The vector $e_1 + e_2$ represents a nonzero quotient class when the operation is applied to T alone, but represents the zero class after inclusion into the larger two-point space. Inducing maps of quotients is always possible; preserving injectivity is an additional theorem.

The extra retained labels in Theorem 7.1 rule out this problem in the relevant class by providing a common projector. The basis orbit calculation and the exactness construction have distinct jobs. Forgetting the second job would leave the proof unable to handle intersections and hence ordinary-PC degree falls.

7.6 What exactness will buy

In the contract of Definition 5.1, matrix naturality identifies the actual axiom, multiplication and target maps. Exactness preserves the kernels, images, intersections and sums of the closure calculation. The doubled-support pairing supplies the remaining target-vanishing reflection for general indexed targets. The roles should remain distinct. Exactness carries a consequence space and its actual inclusion to the smaller universe. The target-map compatibility then identifies the deleted request. To reflect underivability, we still need to know that a nonzero target class in the relevant quotient cannot disappear. That last step is the nonzero-subquotient detector, not a consequence of the word “exact” alone.

The next chapter gives the related upward lift and explains how it interacts with multiplication without pretending to be a ring map.

Chapter 8

How to lift a proof without increasing its degree

The deletion operation goes from a larger universe to a smaller one. The lifting operation goes in the other direction: take a low-degree argument, enlarge the universe by a suitable block, and construct a larger-universe consequence. These are related viewpoints. Deletion is an operation on entire invariant spaces and their maps. The lift is an ordinary linear map on polynomial vectors.

We first specify and verify the lift from the identities of a common projector. The following technical chapters construct that projector. Thus the present chapter gives the formula and its mathematical role, while the later construction proves that an operator with the required identities exists.

8.1 An elementary reason for prime-power steps

Before the operator construction, consider the simplest counting phenomenon behind it. Adding an arbitrary number of labels changes counting expressions. Adding a sufficiently large prime-power block leaves small binomial counts unchanged in the field.

The basic tool is the *Frobenius map*, $u \mapsto u^p$, in a commutative ring of characteristic p. Since the intermediate binomial coefficients $\binom{p}{i}$ vanish, $(u+v)^p = u^p + v^p$; multiplication is preserved as well. Repeating this identity gives the analogous formula for $q = p^a$. It applies in the polynomial ring $K[T]$ used below, and to the commuting block-translation operators used later. Commutation is part of the latter application; no such formula is being asserted for arbitrary noncommuting matrices. Integer counts are read in K through the map $m \mapsto m\,1_K$.

Lemma 8.1 (Counts below a prime-power block)**.** *Let K have characteristic p, let $q = p^a$, and let $n \geq 0$. For every $0 \leq j < q$, the integer binomial coefficients have the same image in K:*

$$\binom{n+q}{j} = \binom{n}{j}.$$

Proof. In $K[T]$, repeated application of the characteristic-p binomial identity gives $(1+T)^q = 1 + T^q$. Consequently

$$(1+T)^{n+q} = (1+T)^n (1+T^q).$$

Multiplication by T^q contributes no coefficient in degrees below q. Comparing the coefficient of T^j proves the assertion. □

For example, in characteristic two, adding four labels preserves $\binom{n}{j}$ for $j = 0, 1, 2, 3$. At $n = 7$ and $n + 4 = 11$ all four values are one in the field. Degree four can see the change:

$\binom{7}{4} = 35$ is odd, whereas $\binom{11}{4} = 330$ is even. This explains why the inequality between block size and complexity budget is strict.

The count has a polynomial interpretation. For unary variables put

$$E_{j,n} = \sum_{1 \le i_1 < \cdots < i_j \le n} x_{i_1} \cdots x_{i_j}.$$

Use $E_{0,n} = 1$ and $E_{j,n} = 0$ for $j > n$. Its value at the all-one assignment is $\binom{n}{j}$. More strongly, setting the q added variables to one gives, for $j < q$,

$$E_{j,n+q}(x_1, \ldots, x_n, 1, \ldots, 1) = \sum_{h=0}^{j} \binom{q}{h} E_{j-h,n} = E_{j,n}.$$

The intermediate coefficients $\binom{q}{h}$, $0 < h < q$, vanish in K.

This is a complete elementary calculation, but not a lifting theorem for arbitrary axioms. Setting new variables to one need not preserve their equations. Even stable coefficient counts do not yet preserve the intersections which allow PC to multiply a polynomial after cancellation. The full lift must preserve that subspace calculation. The lemma explains the choice of step; the common projector below provides the extra structure needed for proofs.

8.2 Low degree, a large starting universe size, and repeated lifting

Write $N(d)$ for a sufficient starting universe size for degree-d transfer. If $n \ge N(d)$, one proof with that budget gives proofs at every size $n + \ell q$, $\ell \ge 0$, with one fixed degree bound d. This is already enough to contradict a separate proof that the minimum degree tends to infinity along that progression. It is the argument used in Chapter 4.

The earlier LICS refutation result [7, p. 276] summarised estimates $A_{\mathrm{NS}}(d) \le 2^{cd}$ and $A_{\mathrm{PC}}(d) \le 2^{2^{cd}}$, extracted there from earlier work of Krajíček. A degree bounded by a sufficiently small multiple of $\log n$ or $\log \log n$, respectively, therefore lies in that historical transfer range. Inverting the starting-size estimates gives the earlier logarithmic and double-logarithmic gaps. These estimates concern the historical first-order refutation setting; we do not use them as inputs to the general target theorem here.

A small budget is easier to fit inside a given universe, but does not by itself supply a simple construction of the lift. The first complete construction in this book is the characteristic-two argument of Chapter 10. Its starting universe size is quadratic in d. The later constructions improve this to a linear starting universe size and cover every positive characteristic. Each stage answers the same question: how many retained labels are sufficient to transport the entire degree-d proof calculation?

8.3 What we are allowed to keep fixed

We now specify the operator which makes those repeated lifts possible.

Recall the three support bounds: one variable uses at most k labels, one axiom index at most s, and one target index at most t. A degree-d monomial therefore uses at most kd labels. The number r below covers all coordinates of the required diagram, rather than just one monomial; this is why indexed-target detection has a larger budget than closure transport alone.

Fix a proof degree d and a support budget r. The consequence-space calculation needs $r \ge kd + s$. To combine it with general indexed target reflection we will use $r = 2kd + s + t$.

Fix $q = p^a > r$, retain enough labels for exactness, and set $N = n + q$. Every later iteration may use the same d, r, q. Only the universe size and the particular witness coefficients change.

8.3.1 A numerical trace of the retained-label test

Take characteristic two, variable support $k = 2$, axiom-index support $s = 1$, input-degree bound $D = 1$, input stability threshold $n_0 = 3$, target-degree bound $D_G = 2$, and proposed proof degree $d = 2$. Compare a fully invariant target with an indexed request whose target-index support is $t = 1$. The final transfer theorem in Chapter 14 gives the following sufficient ranges:

Request	Budget r	Least 2-power $q > r$	Retained universe size n
Invariant	$kd + s = 5$	8	$n \geq \max\{3, 1, 4, 9\} = 9$
Indexed	$2kd + s + t = 10$	16	$n \geq \max\{3, 1, 5, 19\} = 19$

For the invariant request, a degree-two proof at size 9 lifts to sizes $17, 25, \ldots$. For the indexed request, the corresponding progression starts at 19 and runs through $35, 51, \ldots$. These are implications from an existing proof, not assertions that either example request has such a proof. At each step, n is the *smaller* universe size and $N = n + q$ the larger one. A starting universe size below the stated range supplies no conclusion from this test.

The count $2r - 1$ is one retained pool sufficient for every tuple length $b \leq r$: the individual requirements $2b - 1$ are bounded by $2r - 1$. They are not added together as the witness is repaired. The block size q is additional to that pool. If $r \geq 1$, the least p-power strictly above r satisfies $r < q \leq pr$; at a p-power budget the upper equality can occur. Once d is fixed, all three numbers $r, q, 2r - 1$ stay fixed through the progression.

The lift should not be thought of as replacing every variable by a new polynomial and then substituting into every product. Its coefficients are shared across all relevant spaces through a single group-algebra expression. This gives linear and diagrammatic compatibility, not a multiplicative substitution law.

8.4 The proof can be reconstructed from the closure

We spell out reconstruction for PC, whose consequence space is the iterated closure C_d. For NS the same construction transports the single certificate-map image I_d described in Chapter 3. Suppose an original PC proof produces f. The exact consequence-space characterisation says $f \in C_d(n)$. If a linear lift sends that vector into $C_d(N)$, there is a legal PC proof at degree d in the larger universe. One obtains it by storing derivations for bases of closure stages, expressing the required lifted vectors in those bases, and concatenating the permitted inference steps.

This reconstruction is finite, but no efficient algorithm or bound on proof length is asserted. Degree and length are different resources. The closure lets us explain why a proof exists without forcing every larger proof to have the same list of intermediate lines as the original.

8.5 A lifting operator for bounded-degree consequences

Fix a proposed proof degree $d \geq 1$. Degree zero uses the separate constant-space calculation from Chapter 3. Write $W_n = P_{n,\leq d}$, $H_n = P_{n,\leq d-1}$ and $X_n = K[\mathcal{V}_n]$. Let $V_j(n)$ be the closure stages from (3.2) and write $C_d(n)$ for their stable union. A proof ending in a polynomial f says exactly that $f \in C_d(n)$. For the uniform-consequence theorem we must additionally compare the prescribed targets at different universe sizes.

Choose a support budget $r \geq kd + s$ and add a block of $q = p^a > r$ labels and write $N = n + q$. The required properties of the linear operator are:

$$\mathsf{L} : W_n \longrightarrow W_N, \qquad \deg \mathsf{L}(f) \leq \deg f, \qquad \mathsf{L}(1) = 1, \qquad \mathsf{L}(V_j(n)) \subseteq V_j(N) \quad \text{for every } j. \tag{8.1}$$

The degree assertion for zero is interpreted using its harmless zero image. This contract carries every degree-d consequence to its specified projector lift and, since $\mathsf{L}(1) = 1$, carries a refutation upwards. A legal proof is recovered from the larger closure. Iteration reaches $n + \ell q$ for every $\ell \geq 0$. The maps at different steps need not have the same coefficients.

For refutations, full periodicity needs a little more than the forward contract. It needs

$$1 \in C_d(n) \quad \Longleftrightarrow \quad 1 \in C_d(n + q), \tag{8.2}$$

and it needs this at each lower degree budget too. The exact deletion construction will supply both directions. A forward lift by itself would not justify all the arguments below about exact recurring values and representatives of T4 peaks. General targets require the same two-way conclusion for their actual coefficient map, obtained by the image-subquotient detection argument in Chapter 14.

Let $\iota : W_n \to W_N$ include each retained monomial, and let $\rho : W_N \to W_n$ discard all other monomials. We also seek $\rho\mathsf{L} = \mathrm{id}$. Plain inclusion has this property, but usually does not carry smaller axioms to consequences of larger axioms. The sum-axiom example below will show the failure explicitly. The remedy is to compose inclusion with a common projector.

A map L with $\rho\mathsf{L} = \mathrm{id}$ is a *section*, or right inverse, of ρ. It chooses one larger representative for each smaller vector, and is injective because $\mathsf{L}(f) = 0$ implies $f = \rho\mathsf{L}(f) = 0$. The lifting problem asks for a section which also respects the consequence spaces. The term will recur later for sections of coordinate-deletion maps.

The following verification uses Theorem 7.1 and the common projector constructed in Chapters 9–13. It describes the actual operator and its compatibility with the PC inference rules.

8.6 The lift from a common projector

We now give the map, leaving the construction of its certificate to the next sections. Temporarily allow a coefficient extension E/K and use the same notation for the extended polynomial spaces. Section 8.9 returns the lifting statement to K.

Identify the added block B with $A = \mathbb{F}_p^a$, acting on it by translations and fixing $[n]$. Let $\sigma_1, \ldots, \sigma_a$ be the coordinate translations. Choose $c_1, \ldots, c_a \in E$ linearly independent over $\mathbb{F}_p$ and set

$$z = \sum_{i=1}^{a} c_i(\sigma_i - 1) \in E[S_N], \qquad z^p = 0.$$

The group algebra $E[S_N]$ consists of linear combinations of label permutations; it acts on polynomials by the corresponding linear combinations of variable renamings. The permutations commute on the block, and $\sigma_i^p = 1$, which proves $z^p = 0$.

The finite algebraic target is one element $w \in E[S_N]$ for which

$$uwu = u \quad \text{on every } P_b(N) = E[\mathrm{Inj}([b], [N])], \quad b \leq r, \qquad u = z^{p-1}. \tag{8.3}$$

Thus the same expression must work on every tuple length up to the support budget. The identity makes w a partial inverse for the top power $u = z^{p-1}$ on these spaces; it does not invert the nilpotent operator z. Here $P_b(N)$ is a tuple permutation space; the polynomial ambient space remains $W_N = P_{N,\leq d}$. For $y = uv$ in $\operatorname{im} u$, the witness identity

says $uwy = uwuv = uv = y$. Thus wy is one preimage of y under u; the word *partial* specifies that this inversion is required on $\operatorname{im} u$ only. The top power sends vectors to the bottom of length-p nilpotent chains. Its common witness will let the construction remove those chains while retaining the length-one directions.

For characteristic two put

$$e = (1 - zw)(1 - wz).$$

For general p, put $m = p - 1$ and

$$S_0 = 1, \qquad S_{j+1} = (1 - z^j w z^{m-j}) S_j \ (0 \le j < m), \qquad e = (1 - z^m w) S_m. \tag{8.4}$$

The two formulas agree at $p = 2$. For odd characteristic, the witness identity is used together with a property of these particular tuple spaces: their nilpotent chains have length one or p. The projector lemma in Chapter 13 proves the formula under that property; nilpotence $z^p = 0$ alone is not sufficient. Once these inputs are established, the formulas are explicit finite algebra expressions. The lifting operator is

$$\boxed{\mathsf{L}_{n,N,d}(f) = e\,\iota(f).} \tag{8.5}$$

Equivalently, writing $e = \sum_{g \in S_N} a_g g$,

$$\mathsf{L}(f) = \sum_{g \in S_N} a_g \, g(\iota(f)).$$

This explains why degree does not increase: inclusion and every renaming preserve degree. The difficult point is to choose the *common* coefficients a_g so that all consequence spaces are preserved. An arbitrary sum of renamings does not have that property.

8.6.1 The two facts that make the formula work

Let J_r be the ideal of group-algebra elements acting as zero on all $P_b(N)$ for $b \le r$, and let $\mathcal{M}_r$ consist of the finite dimensional representations annihilated by J_r. This class contains all the bounded-support permutation spaces needed by PC, and their invariant subspaces and quotients. It is the class $\mathcal{C}_r(N)$ of Chapter 7, now over the auxiliary field E. Define

$$\mathcal{D}(M) = \ker z_M / (\ker z_M \cap \operatorname{im} z_M).$$

The construction establishes two facts.

1. On permutation bases, $\mathcal{D}$ retains exactly the objects supported on $[n]$. On polynomial spaces its identification is induced by ρ:

   $$\ker z_{W_N} / (\ker z_{W_N} \cap \operatorname{im} z_{W_N}) \simeq W_n.$$

   Stable axiom matrices and the actual multiplication maps are transported to their smaller-universe counterparts. With a permutation space as first tensor factor, the identification is natural on the other factor.
2. The common e is an idempotent with $ze = ez = 0$ on $\mathcal{M}_r$, and for $v \in \ker z$,

   $$ev = v - z^{p-1} w v. \tag{8.6}$$

   It identifies $\mathcal{D}(M)$ naturally, as a vector space, with eM. Consequently $\mathcal{D}$ preserves kernels, images and exact sequences in this class.

These facts are proved in Chapters 9–13 from the common witness and the block action. They are stronger than a count of dimensions. They refer to the actual maps used in the consequence calculation. The idempotent property means $e^2 = e$: applying the common projection twice has the same effect as applying it once.

8.7 Compatibility with the entire PC closure

Apply the two facts just stated to the recurrence (3.2). The usable axioms form the image of a stable coefficient matrix, so deletion takes $V_0(N)$ to $V_0(n)$. Suppose it has already taken $V_j(N)$, with its inclusion in W_N, to $V_j(n)$. The intersection $V_j(N) \cap H_N$ is the kernel of

$$V_j(N) \oplus H_N \longrightarrow W_N, \qquad (f, h) \longmapsto f - h.$$

Exactness preserves it. The tensor identification and the actual multiplication map then preserve the multiplication image, and exactness preserves its sum with $V_j(N)$. Induction gives

$$\rho(V_j(N) \cap \ker z_{W_N}) = V_j(n) \qquad \text{for every } j. \tag{8.7}$$

The tensor domain is a subspace of $X_N \otimes H_N$, whose support bound is kd. No closure of $\mathcal{M}_r$ under arbitrary tensor products is assumed.

Proposition 8.2 (The lifting formula preserves and reflects consequences)**.** *Assume the two facts in Section 8.6. Then* $\mathsf{L} = e\iota$ *preserves degree,* $\rho\mathsf{L} = \mathrm{id}$*,* $\mathsf{L}(1) = 1$*, and*

$$\mathsf{L}(f) \in V_j(N) \quad \Longleftrightarrow \quad f \in V_j(n).$$

The same equivalence holds for the stable spaces C_d*.*

Proof. Every retained monomial is fixed by the block action, so $z\iota(f) = 0$. Equation (8.6) gives

$$\mathsf{L}(f) = \iota(f) - z^{p-1} w \iota(f).$$

The image of z has no retained component: z kills retained basis vectors and preserves the span of moved ones. Hence $\rho\mathsf{L}(f) = f$. On constants, every permutation acts trivially, so the correction term vanishes and $\mathsf{L}(1) = 1$.

If $f \in V_j(n)$, equation (8.7) supplies $v \in V_j(N) \cap \ker z$ with $\rho v = f$. The permutation-space identification says that $v - \iota(f) \in \ker z \cap \operatorname{im} z$. Since $e \operatorname{im} z = 0$, we have $\mathsf{L}(f) = ev$. The space $V_j(N)$ is invariant under all label permutations, hence under e, and $ev \in V_j(N)$. Conversely, $\mathsf{L}(f) \in V_j(N)$ is killed by z, so (8.7) and $\rho\mathsf{L}(f) = f$ give $f \in V_j(n)$. Degree preservation follows from the renaming formula. Pass to the unions for the stable statement. □

This proves both the forward lifting implication and (8.2). The image of the lift need not be all of $C_d(N)$: the larger universe has many additional consequences. Every smaller consequence has this projector lift. For $n \geq n_*$ and a fully invariant target whose stabilised degree is at most d, naturality of its map from K gives $\mathsf{L}(G^{(n)}) = G^{(N)}$: the target map commutes with the common projector, which acts as the identity on K. More concretely, invariance gives $zG^{(N)} = 0$ and $eG^{(N)} = G^{(N)}$, while stable coefficients give $\rho G^{(N)} = G^{(n)}$. The retained-vector identification and the argument in Proposition 8.2 therefore yield $e\iota(G^{(n)}) = G^{(N)}$. Hence invariant targets, including 1, are exactly preserved. For indexed targets this equality need not hold. The image-subquotient argument in Chapter 14 proves transfer of prescribed indexed targets without assuming it.

8.7.1 Axioms and joint tensor multiplication

For an axiom index a supported in $[n]$, let F_n and F_N be the usable-axiom coefficient maps. Write ι_A for inclusion of retained axiom indices and e_A for the action of the same group-algebra element on their source space. Naturality gives

$$\mathsf{L}(f_{a,n}) = F_N(e_A \iota_A(a)). \tag{8.8}$$

The right side is a linear combination of actual usable axioms of Γ_N. This is the precise sense in which axioms lift.

For the order example, a retained predecessor row has axiom $\sum_{j<n} F_{ij} - 1$. Plain inclusion omits the new summands with j in the added block, so it is not the larger row axiom. Equation (8.8) repairs precisely this mismatch: the lifted smaller row is a linear combination of the actual larger axioms. The order, Boolean, functionality and predecessor-guard equations are handled by the same usable-axiom map. We need no injectivity of the predecessor function. The common projector fixes the last line 1, and joint tensor compatibility handles every multiplication step, including one made possible by cancellation.

Now suppose $f \in V_j(n) \cap H_n$, so the smaller proof may derive $x_v f$. Apply the same e to the *joint* tensor input and expand by its variable coordinate:

$$e_{X_N \otimes H_N}\big(\iota(x_v) \otimes \iota(f)\big) = \sum_{y \in \mathcal{V}_N} x_y \otimes g_y. \tag{8.9}$$

Exactness and the natural tensor identification show that this vector lies in $X_N \otimes (V_j(N) \cap H_N)$. Thus each g_y belongs to $V_j(N)$ and has degree at most $d-1$. Multiplication commutes with the common group-algebra action, so

$$\mathsf{L}(x_v f) = \sum_y x_y g_y. \tag{8.10}$$

Each term is a permitted multiplication of a derivable polynomial.

There is even an explicit expression for these coefficients:

$$g_y = \sum_{g \in S_N \colon g(v) = y} a_g \, g\big(\iota(f)\big).$$

Individual summands need not be known consequences; their grouped sum g_y is a consequence by the tensor argument. One must not replace the joint operator in (8.9) by $e_X \otimes e_H$. That would discard the correlations supplied by the common permutation g.

To produce a literal proof sequence, compute bases of the closure stages while storing derivations for those bases. Express each g_y in the appropriate basis by linear algebra, append the multiplication steps in (8.10), and add. This is a finite procedure, with no efficiency or proof-length claim. It explains exactly how consequence-space lifting supplies a legal PC derivation.

8.7.2 A multiplication step with all its proof lines

The distinction between a joint action and two independent actions is visible in a very small calculation. Work over any field, with three labels $0, 1, 2$, variables x_0, x_1, x_2, and the uniform axioms $\Gamma_3 = \{x_0, x_1, x_2\}$. These mean $x_i = 0$. Set $d = 2$, $X = \operatorname{span}\{x_0, x_1, x_2\}$ and $H = P_{3,\le 1}$. Take $f = x_0 + x_1$, let $\tau = (0\ 2)$ rename the labels, and put $T = 1 - \tau$ in the group algebra. Applying T to the joint input gives

$$\begin{aligned} T_{X \otimes H}(x_0 \otimes f) &= x_0 \otimes (x_0 + x_1) - x_2 \otimes (x_2 + x_1) \\ &= x_0 \otimes g_0 + x_2 \otimes g_2, \qquad g_0 = x_0 + x_1, \quad g_2 = -x_2 - x_1. \end{aligned}$$

Both grouped coefficients are derivable linear combinations of the axioms. Multiplying the two tensor coordinates produces a literal degree-two PC derivation:

Line	Polynomial	Reason
1	x_0	axiom
2	x_1	axiom
3	x_2	axiom
4	$x_0 + x_1$	lines 1 and 2
5	$-x_2 - x_1$	lines 3 and 2
6	$x_0^2 + x_0x_1$	multiply line 4 by x_0
7	$-x_2^2 - x_2x_1$	multiply line 5 by x_2
8	$x_0^2 + x_0x_1 - x_2^2 - x_2x_1$	add lines 6 and 7

Lines 4 and 5 have degree one, so their multiplication is legal within the degree-two budget. The last line is $T(x_0 f)$. There has been no reduction modulo Boolean equations and no multiplication by a polynomial of degree greater than one.

By contrast, acting independently on the two factors gives

$$(T_X \otimes T_H)(x_0 \otimes f) = (x_0 - x_2) \otimes (x_0 - x_2),$$

whose product is $(x_0 - x_2)^2$. This is a different polynomial, also in characteristic two: the joint result still has the terms involving x_1. The common permutation in each summand is therefore essential.

This calculation illustrates the inference mechanism at one universe size. The element T is not a deletion projector and does not fix 1. For the actual larger-universe lift, replace T by the common e and use (8.9). The grouped g_y are then certified consequences by exactness and the tensor identification, rather than by the two-line calculation above. Once that certificate is available, the final multiplication and addition steps are exactly the same.

8.8 A one-label lifting example

Work in characteristic two. Retain the label a and add b, c. Let $z = 1 + (bc)$ and $w = (ab)$ act on the linear variables x_a, x_b, x_c. On their three-dimensional span,

$$z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives $zwz = z$ and

$$e = (1 - zw)(1 - wz) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Consequently

$$\mathsf{L}(\alpha + \beta x_a) = \alpha + \beta(x_a + x_b + x_c). \tag{8.11}$$

Its retained coordinate is the original polynomial, and $\mathsf{L}(1) = 1$.

Take the uniform degree-one axiom $\sum_{i<\ell} x_i = 0$ at every size ℓ. The all-zero assignment satisfies this system at every size. At size one the axiom is x_a; at size three it is $x_a + x_b + x_c$. Formula (8.11) sends the former exactly to the latter and lifts every degree-one consequence. Plain inclusion would leave x_a, which is not a consequence of the larger system: the assignment $(x_a, x_b, x_c) = (1, 1, 0)$ satisfies its sum but not $x_a = 0$.

Here the invariant target $G_\ell = \sum_{i<\ell} x_i$ is transported to the prescribed larger target. In contrast, the indexed expression $G_a^{(\ell)} = x_a$ has projector lift $x_a + x_b + x_c$, not x_a. The map

is linear but is not a ring homomorphism and does not preserve every indexed expression literally. This calculation illustrates the distinction but lies outside the indexed transfer range: for $k = d = t = 1$ and $s = 0$, where $t = 1$ bounds the support of the target index, that theorem uses $R(d) = 3$, whereas here $q = 2$ and $n = 1$. It is therefore no counterexample to indexed periodicity.

The failure of a literal indexed lift requirement also has a general explanation. At an admissible universe size, naturality for the invariant target $\sum_{i<n} x_i$ gives $\mathsf{L}(\sum_{i<n} x_i) = \sum_{i<N} x_i$. If every retained indexed target x_i had lift exactly x_i, linearity would instead give $\sum_{i<n} x_i$, a different polynomial. Thus at least one indexed target must change under the projector lift. This is why the proof uses actual target-map images and nonzero-subquotient detection. For the displayed invariant example, $r = 1$ and $q = 2 > r$, so a two-label block suffices. Larger support budgets require larger blocks. The sufficient witness threshold $n \geq 2r - 1$ is attained in this example; the general template theorem uses a slightly more conservative onset.

8.9 Descent to the coefficient field

Here *descent* means returning a consequence established with auxiliary coefficients to the original coefficient field. The auxiliary choices have to meet the conditions of the block and common-witness constructions; they are not probabilistic assumptions. Their existence is supplied by those constructions.

The generic block coefficients may require an extension field. This does not change the derivation degree of any target with coefficients in K: at fixed degree, scalar extension commutes with the finite-dimensional spans, intersections and images in (3.2). In particular,

$$V_j(n; E) = E \otimes_K V_j(n; K),$$

with the analogous equality for C_d. Membership of every K-coefficient target is therefore unchanged, so each target-membership equivalence proved over E holds over K.

There is also a concrete K-linear lifting map satisfying the forward requirements (8.1). All coefficients used in the construction lie in a finite extension E/K inside the algebraic closure. Choose a K-linear functional $\lambda : E \to K$ with $\lambda(1) = 1$, for example the coefficient of 1 in a vector-space basis containing 1. Apply it coefficientwise to polynomials and put

$$\widehat{\mathsf{L}}(f) = \lambda_*(\mathsf{L}_E(f)) \qquad (f \in W_n(K)). \tag{8.12}$$

The finite-extension assertion uses only finitely many coefficients of the block operator and group-algebra witness. Each is algebraic over K when chosen in its algebraic closure; adjoining this finite list gives a finite-dimensional extension.

The functional λ is a linear projection, not a field map. For example, over $\mathbb{F}_2 \subseteq \mathbb{F}_4$ with $\omega^2 + \omega + 1 = 0$, take $\lambda(a + b\omega) = a$. Then $\lambda(\omega) = 0$, but $\lambda(\omega^2) = 1$. This explains why descent works through linear consequence spaces: it is not substitution by a multiplicative map of coefficients.

This map preserves degree, satisfies $\rho\widehat{\mathsf{L}} = \mathrm{id}$ and fixes 1. If $f \in V_j(n; K)$, then $\mathsf{L}_E(f)$ is an E-linear combination of a K-basis of $V_j(N; K)$. Applying λ_* leaves a K-linear combination of that same basis. Thus $\widehat{\mathsf{L}}(f) \in V_j(N; K)$, as required.

This is descent of consequence-space membership, not coefficientwise projection of an arbitrary proof with extension-field scalars. Such a projection need not preserve each scalar inference. A proof over K is recovered from the K-closure bases. We use only the forward lifting properties for $\widehat{\mathsf{L}}$; the stronger reflection of individual consequences in Proposition 8.2 refers to the projector lift over E. The two-way prescribed-target statement over K follows

separately from target-map transport, nonzero-subquotient detection and field invariance, as above.

8.10 A summary of the map's three different uses

For any smaller consequence f, the projector lift gives a larger consequence $\mathsf{L}(f)$ of no greater degree. For a full stable invariant target fitting the budget, this vector is exactly the prescribed larger target $G^{(N)}$. For general indexed targets, literal pointwise equality is not available: the actual target-map image and the quotient $(C_d(N) + \operatorname{im} J_N)/C_d(N)$ supply the transfer instead. For NS, use the analogous quotient by $I_d(N)$. These quotients test whether a target lies outside the relevant consequence space.

These uses share one construction but have different logical inputs. Keeping them separate is the simplest way to avoid proving only a statement about an unspecified lifted vector while claiming a theorem about the specified target.

> **Checkpoint.** A single common projector makes the entire closure calculation compatible. It does not replace ordinary-PC multiplication by a new proof rule. The larger consequence remains derivable by the original legal rules.

Part III

Constructing a lift: from square-root to linear bounds

Chapter 9

Nilpotent blocks and a common projector

The property we need

The lifting contract asks for one operation which works on an entire proof diagram. It must transport the input-axiom map, the maps used for multiplication, the inclusions of consequence spaces and the actual target map. The first problem is to find an operation which has the right effect on bases. The second, more difficult problem is to make that operation exact on all the subspaces arising from a proof.

This chapter separates those problems. In characteristic two, a square-zero operator gives a very concrete quotient: vectors killed by the operator, modulo vectors in its image. On a permutation basis, this quotient deletes the block labels. That calculation supplies actual basis and tensor maps. It does *not* prove exactness. For exactness we will derive a projector from one group-algebra witness, common to every tuple length up to the chosen support budget. The next three chapters construct that witness rather than leaving its existence as a hypothesis.

Keep the five roles separate:

n	number of retained universe labels,
$q = 2^a$	number of block labels to be deleted,
$N = n + q$	size of the larger universe,
r	support budget for all spaces in the proof diagram,
$b \le r$	current tuple length in the witness construction.

Neither b nor r is the maximum degree of a polynomial proof. For an application to degree-d consequences, that degree determines r: $r = kd+s$ for invariant targets and $r = 2kd+s+t$ for the general indexed request. We first solve the finite algebra problem with r as an independent integer.

The construction proves Exact Deletion (Theorem 7.1) independently of the equations and targets. We begin with the square-zero operator in characteristic two, then build a witness with a quadratic retained-label cost, replace that cost by the linear threshold, and finally handle odd characteristic.

9.1 A block of labels and a square-zero operator

For this part $\operatorname{char} K = 2$. Choose $q = 2^a > r$ and split the labels into retained labels $R = [n]$ and a block B with q labels. Identify B with the vector space $A = \mathbb{F}_2^a$. Translation by the ith

unit vector gives a permutation σ_i of B, extended by the identity on R. These permutations commute and square to the identity.

Extend the coefficient field if necessary so that it contains a elements linearly independent over $\mathbb{F}_2$. In this finite-algebra construction, K denotes that working field; Lemma 3.3 later returns degree-bounded derivability to the original field. Choose $c_1, \dots, c_a$ linearly independent over $\mathbb{F}_2$ in K, and put

$$z = \sum_{i=1}^{a} c_i(\sigma_i - 1). \tag{9.1}$$

This notation means an operator on every vector space on which label permutations act. Products of operators mean composition, so $z^2 = 0$ means that applying z twice sends every vector to zero; it does not mean that z itself is zero. Since the summands commute, cross terms cancel in characteristic two, and

$$z^2 = \sum_i c_i^2(\sigma_i - 1)^2 = 0.$$

A linear operator is called *nilpotent* when some positive power is zero. Here that power is already two, so every vector in $\operatorname{im} z$ lies in $\ker z$. The nonzero two-point example below shows that nilpotence need not make the operator itself zero. For $q = 2$, z can be the swap of the two block labels plus the identity. For $q = 4$, work over $\mathbb{F}_4 = \mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$ and use $(c_1, c_2) = (1, \omega)$.

The independence is essential. On a two-point orbit where σ_1 and σ_2 induce the same swap, choosing $c_1 = c_2 = 1$ makes $z = 0$. The orbit is moved, yet would survive deletion. Independent coefficients ensure that every nonzero binary combination of the c_i remains nonzero.

For a space M with label action define

$$\mathcal{D}(M) = \ker z / \operatorname{im} z. \tag{9.2}$$

This quotient is legal because $z^2 = 0$ implies $\operatorname{im} z \subseteq \ker z$. Two vectors killed by z represent the same class if their difference is zw for some w. A map commuting with label permutations commutes with z and induces a map between these quotients. Permutations of the retained labels commute with z, so they still act on the quotient.

9.1.1 Two actions which must not be confused

The coordinates of a tuple have *positions*, whereas its entries are *universe labels*. For example, the tuple (a, b) has two positions and uses two labels. A label permutation g sends it to $(g(a), g(b))$. A position swap sends it to (b, a). These actions commute, but they play different roles. Group-algebra elements in $K[S_N]$ combine label permutations. The auxiliary inverses used later must commute with position permutations so that their matrix entries can be expressed as functions of an unordered matching between input and output labels.

For a concrete check, take labels $1, 2, 3$ and the tuple $(1, 3)$. The label swap (12) sends it to $(2, 3)$; the position swap sends it to $(3, 1)$. Doing both, in either order, gives $(3, 2)$. It is this commuting pair of actions which allows the same correction to be discussed both as an operator on ordered tuples and as a function on rook placements.

The block subgroup A acts on labels, never on tuple positions. The operator z therefore commutes with every position permutation. An auxiliary operator chosen by the underlying *set* of labels, instead of by a preferred ordering of a tuple, has the same property. That observation will matter when constructing an inverse without averaging over S_b.

The usual averaging recipe would replace an operator h by $(b!)^{-1}\sum_{\tau\in S_b}\tau h\tau^{-1}$, thereby forcing it to commute with positions. In characteristic two, $b!$ is zero in the field when $b \geq 2$, so this recipe is unavailable. Instead we construct h from the unordered label set of a tuple. That choice is already unchanged by every reordering of its positions.

9.1.2 The two-point orbit calculation

For the illustrative two-label choice $z = \sigma - I$, on basis vectors e_β, e_γ swapped by σ,

$$ze_\beta = ze_\gamma = e_\beta + e_\gamma.$$

The kernel and image are both the one-dimensional span of $e_\beta + e_\gamma$, so the quotient is zero. On a fixed basis vector e_i, $z = 0$, and its line survives. This is the prototype of deletion: moved directions disappear, fixed directions remain.

A useful failure to commute. On this two-point orbit let S be the swap and let D multiply the first basis vector by 0 and the second by 1. Their matrices are

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad DS - SD = S \quad (\operatorname{char} K = 2).$$

The difference $DS - SD$ is called a *commutator*; it measures the failure of the two operators to commute. Here that difference is invertible. The next calculation constructs the same useful failure to commute on every moved orbit, and uses it to write any kernel vector as an image vector.

9.1.3 The calculation for every moved orbit

An A-set is a set of basis objects on which the translations in $A = \mathbb{F}_2^a$ act. The orbit of x_0 consists of all vx_0, $v \in A$; its stabiliser H consists of translations fixing x_0. Since the orbit is moved, H is a proper subspace. A basis of H can be extended to a basis of A, so a nonzero linear functional ℓ can be chosen to vanish on H. It assigns a binary coordinate to the orbit, independent of the translation used to reach an object. The coordinate-reading operator D below is diagonal, just as in the two-point illustration.

Let X be a finite A-set with no point fixed by all of A. On each orbit choose x_0. Its stabiliser H is a proper binary subspace. Choose a nonzero linear functional $\ell : A \to \mathbb{F}_2$ vanishing on H. The function

$$\pi(vx_0) = \ell(v)$$

is well defined: two choices of v differ by an element of H. Let D multiply e_x by $\pi(x)$, and put $e_i = \ell(\varepsilon_i)$. These are constants on that orbit. If S_i denotes the action of σ_i, direct evaluation at a basis vector gives

$$DS_i - S_iD = e_iS_i, \qquad Dz - zD = W := \sum_i c_ie_iS_i.$$

Moreover $Wz = zW$ and

$$W^2 = \Big(\sum_i c_ie_i\Big)^2 I = \alpha^2 I, \qquad \alpha \neq 0.$$

In particular $W^{-1} = \alpha^{-2}W$. Thus the inverse used here is explicit; no choice of a complement or inverse matrix is hidden in the notation. The coordinate vector ε_i is the ith unit vector in A, and $e_i = \ell(\varepsilon_i)$ is a scalar, distinct from a basis vector e_x. Thus $\Theta = W^{-1}D$ satisfies

$\Theta z - z\Theta = I$. For $zm = 0$ this gives $m = -z\Theta m$, hence $\ker z \subseteq \operatorname{im} z$. The reverse inclusion was automatic, so the quotient vanishes.

The calculation has three ingredients, each serving a different purpose. The orbit coordinate π produces a diagonal operator whose commutator with translation is computable. Independence of the c_i makes the commutator invertible. The commutator identity then turns a kernel vector into an image vector. We do not choose a complement of $\ker z$ by hand and hope it will respect later maps.

For a two-point orbit of $A = \mathbb{F}_2^2$ with stabiliser $H = \{(0,0),(1,1)\}$, both coordinate translations induce the same swap S. Choose $\ell(v_1, v_2) = v_1 + v_2$. With $(c_1, c_2) = (1, \omega)$, the operator is $z = (1+\omega)(S - I)$, and $\alpha = 1 + \omega \neq 0$. Thus the one-dimensional kernel is exactly the image. If both coefficients had instead been 1, z would be zero on this moved orbit. This explicit orbit is why using one coefficient for every translation is not a harmless simplification.

This calculation also works on $K[X] \otimes Y$ for *any* space Y carrying the label action. Let D inspect only the X coordinate and let S_i act on both factors. We still have $S_i^2 = I$ and the same commutator calculation. This stronger statement will justify multiplication, whose domain contains a variable factor tensored with a consequence subspace.

9.2 Natural deletion on permutation bases

Let $\mathcal{U}$ be a basis system whose objects have support less than q. An object fixed by all of A cannot have support meeting B. Otherwise invariance of its support under the regular action would force its support to contain all q block labels. Thus its fixed objects are precisely $\mathcal{U}_n$.

Split the basis into fixed and moved objects. The preceding calculation kills the moved summand and fixes the retained one. Hence

$$\mathcal{D}(K[\mathcal{U}_{n+q}]) \cong K[\mathcal{U}_n], \tag{9.3}$$

$$\mathcal{D}(K[\mathcal{U}_{n+q}] \otimes Y) \cong K[\mathcal{U}_n] \otimes \mathcal{D}(Y). \tag{9.4}$$

The second identification sends $e_u \otimes [y]$ to $[e_u \otimes y]$ for retained u. It is natural in Y: applying a map to y before or after taking its class gives the same result. For a product of two permutation bases, it simply keeps pairs of retained objects. There is no choice of unrelated isomorphisms here.

Suppose a matrix from one permutation basis to another has stable relabelling-invariant coefficients. On a retained input vector, its image is killed by z. Its retained component has exactly the smaller-universe coefficients; its moved component is in $\ker z = \operatorname{im} z$ on the moved summand, so vanishes in the quotient. Thus the *actual matrix*, not merely its rank, is transported correctly.

9.2.1 What naturality says about actual matrix entries

Let $F_N : K[\mathcal{U}_N] \to K[\mathcal{V}_N]$ be an equivariant coefficient map. On a retained basis vector e_u, its image is

$$F_N(e_u) = \sum_v a_N(v,u) e_v.$$

It is killed by z because e_u is killed by z and $F_N z = z F_N$. Split this image into its retained and moved parts. Both parts are killed separately because z preserves those two summands. The moved part is in $\operatorname{im} z$ by the orbit calculation, and so contributes zero to the quotient. The remaining coefficients are $a_N(v,u)$ for retained u, v. A stable coefficient rule identifies each of these with $a_n(v,u)$. The smaller matrix is therefore obtained entry by entry; we have not merely proved equality of two ranks or dimensions.

For a tensor map, apply the same argument to the permutation first factor. A retained basis vector in that factor is fixed by the block, so the quotient sees precisely the class of the second factor. A moved first-factor orbit has zero quotient even when the block acts nontrivially on the second factor. This is why the strengthened orbit calculation, with a diagonal action on $K[X] \otimes Y$, was needed. The tensor rule subsequently applies to a variable factor tensored with a consequence subspace.

9.2.2 A small failure of exactness

Take the two-point permutation space $E = Ke_\beta \oplus Ke_\gamma$ and its invariant line $T = K(e_\beta + e_\gamma)$. Inclusion $T \hookrightarrow E$ is injective. But $\mathcal{D}(T) = T$ whereas $\mathcal{D}(E) = 0$. Deletion takes an injection to the zero map on a nonzero space. Therefore the basis calculation alone does not prove that deletion preserves kernels, images or intersections.

This failure is precisely why a proof cannot stop at "the monomials delete correctly." We must show that, with enough retained labels, every subspace produced by the PC calculation lies in a class where the quotient operation is exact.

9.3 A common group-algebra witness

The group algebra $K[S_N]$ consists of finite formal sums $\sum_g a_g g$ of label permutations. It acts on every space with an S_N action by the corresponding linear combination of operators. Its multiplication is composition. This provides a common language for certificates valid on many spaces at once.

For $b \geq 0$ let

$$P_b(N) = K[\mathrm{Inj}([b], [N])],$$

the space on ordered injective b-tuples. In these construction chapters $P_b(N)$ denotes this tuple space, not the polynomial ambient space $P_{N,\leq d}$ from the consequence calculation. The integer b counts tuple positions, whereas d bounds polynomial degree. Label permutations act on entries; a second group S_b permutes tuple positions. These actions commute. Define

$$J_r(N) = \{a \in K[S_N] : aP_b(N) = 0 \text{ for every } b \leq r\}.$$

It is a two-sided ideal. Write $\mathcal{C}_r(N)$ for the finite-dimensional spaces with label action annihilated by this ideal: every element of $J_r(N)$ acts as zero on each such space. This is the same annihilator class called $\mathcal{M}_r$ in Chapter 8. Subspaces invariant under labels, quotients and finite direct sums remain in this class.

Every bounded-support permutation basis of support at most r belongs to the class. For an orbit of an object, order its support of size $b \leq r$. Sending an injective tuple to the corresponding relabelling of that object gives a surjective equivariant map from $P_b(N)$ onto the orbit space. The support condition makes this well defined. Elements of J_r kill that quotient as well.

The required witness is a *partial inverse*, not an inverse of the nilpotent operator z. If $zwz = z$ and $v = zt$ is an image vector, then $z(wv) = zwzt = zt = v$: applying w supplies a preimage under z. The witness condition below requires this one expression to work on all tuple lengths simultaneously. Its error need only lie in J_r, since such an error acts as zero on the entire class.

Definition 9.1. A support-r witness is one element $w \in K[S_N]$ satisfying

$$zwz - z \in J_r(N). \tag{9.5}$$

Such a w gives $zwz = z$ on *every* object of $\mathcal{C}_r(N)$. Choosing a separate matrix inverse on each space would not give this compatibility. A module map here is a linear map commuting with all label permutations. It therefore satisfies $f(wv) = wf(v)$ for a common group-algebra element, which is exactly what is needed.

9.3.1 The projector calculation, without an exactness assumption

A projector is an operator e with $e^2 = e$. It fixes every vector in its image eM and sends every vector in $\ker e$ to zero; it need not be a coordinate projection. We seek one which selects representatives of the quotient $\ker z/\operatorname{im} z$ in a way compatible with every equivariant map. The factors in the formula below act from right to left. They are not assumed to commute.

Suppose $z^2 = 0$ and $zwz = z$. Set

$$e = (I - zw)(I - wz).$$

Then

$$ze = z(I - wz) = z - zwz = 0, \qquad ez = (I - zw)z = 0.$$

If $zv = 0$, then $ev = v - zwv$. Also

$$(zw)e = (zw - zwzw)(I - wz) = 0,$$

since $zwzw = zw$. Applying the formula on $\ker z$ to ev gives $e^2v = ev$. Thus e is a projector.

The map $[v] \mapsto ev$ is an isomorphism $\ker z/\operatorname{im} z \to eM$: it is well defined because $ez = 0$, its image lies in $\ker z$, and $ev - v = -zwv$ is in $\operatorname{im} z$ when $zv = 0$. Its inverse sends a vector of eM to its quotient class.

Now let $0 \to U \to V \to W \to 0$ be an exact sequence in $\mathcal{C}_r(N)$. All maps commute with the same e. Restricting to eU, eV, eW preserves injectivity. A vector of eW has a lift $v \in V$; ev is a lift in eV, proving surjectivity. A vector ev mapping to zero lies in U and equals its own projection, so lies in eU. This proves exactness explicitly. By the natural isomorphism, $\mathcal{D}$ is exact.

The projector need not commute with permutations of the retained labels. We use its image as a vector-space model to prove exactness, and keep the original quotient for the residual label action. Nothing here asserts an unwarranted equivariant direct-sum decomposition for that residual group.

Example 9.2 (One retained label)**.** Let the retained label be a and the two block labels be b, c. Put $z = I + (bc)$ and $w = (ab)$. On $P_1(3)$,

$$zwz(e_b) = zw(e_b + e_c) = z(e_a + e_c) = e_b + e_c = ze_b,$$

and similarly for e_c; both sides vanish on e_a. The projector is the all-ones 3×3 matrix. Its image is the constant line, naturally representing the retained one-dimensional quotient. With no retained label, every element of $K[S_2]$ satisfies $zwz = 0$, so there is no such witness. The spare label makes a real difference.

9.4 Reading the projector as an exactness certificate

In a short exact sequence the first map i is injective, the last map f is onto, and $\ker f = \operatorname{im} i$. Thus U is a specified subspace of V, and W records the quotient by it. A *diagram chase* checks these assertions after transport by following an individual vector and its preimages through the maps. Here the common projector makes those vector checks possible.

The exactness proof is worth restating as a diagram chase, because it is the point at which a basis calculation becomes a tool for ordinary PC. Suppose

$$0 \longrightarrow U \xrightarrow{i} V \xrightarrow{f} W \longrightarrow 0$$

is exact and all three spaces are in the common annihilator class. The equations $ei = ie$ and $ef = fe$ hold because i, f commute with every label permutation and hence with their common linear combination e. A separate projector selected independently on each space would not justify these equations.

If $u \in eU$ maps to zero, its original image is zero and injectivity of i gives $u = 0$. If $w \in eW$, take any lift $v \in V$ with $f(v) = w$. Then $f(ev) = ef(v) = ew = w$. Finally, if $v \in eV$ maps to zero, exactness of the original sequence gives $v = i(u)$ for some $u \in U$. Applying e gives $v = ev = i(eu)$. These are respectively injectivity, surjectivity and equality of image with kernel for the projected sequence. The natural quotient-to-projector isomorphism transfers all three conclusions to deletion.

This argument proves more than the existence of some complement to $\operatorname{im} z$. It proves that every actual map in the consequence diagram uses compatible complements. Intersections, sums and restricted multiplication images can therefore be transported. The joint tensor operation in (8.9) uses the same group-algebra expression on both factors together; it is not the tensor product of two independently selected projectors.

9.4.1 The three-label matrix in full

With retained label a and block labels b, c, use the ordered basis (e_a, e_b, e_c). Then

$$z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Matrix multiplication gives $zwz = z$ and

$$e = (I - zw)(I - wz) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Its square is itself: every entry of the square is $1 + 1 + 1 = 1$ in characteristic two. The quotient can be represented by the retained coordinate, while the projector realises that coordinate as $e_a + e_b + e_c$. Thus the lift of a retained vector can involve all labels. This is already the mechanism that turns a smaller totality sum into its larger counterpart.

The example also shows why deletion is not substitution of zero for block variables. The projector lift of e_a is a sum of three basis vectors. Its block terms are required to preserve the relations of the larger system. A support budget does not assert that lifted proof lines mention only retained labels; it limits the labels in each *basis object* used by the transport construction.

What has been proved, and what remains

We have proved natural deletion of permutation bases and its permutation-first-factor tensor rule. We have also proved that the single identity $zwz = z$ on all tuple lengths up to r gives a common projector and therefore exact deletion on the full annihilator class. The identity has not been assumed as an external theorem: its construction is the subject of the next chapters. The preliminary construction spends fresh labels at each length; the improved construction realises a smaller, specially shaped error on one reused pool.

Chapter 10

Lifting in characteristic two: a quadratic label budget

The property we need

We remain in characteristic two. The block has $q = 2^a$ labels, and $z = \sum_i c_i(\sigma_i - I)$ is its square-zero translation operator. A common witness is a group-algebra element w satisfying $zwz = z$ on every injective-tuple space P_b with $b \le r$. The block stays fixed throughout the following induction; the retained labels are the resource we count.

We now construct the witness used in the previous chapter. The first construction is intentionally generous with labels: at tuple length b, it allocates b labels which have not been used before. It therefore proves exact deletion with a quadratic retained-label budget. This gives a complete route to a square-root degree lower bound and isolates exactly where that route loses a factor of r.

Keep the progression picture in view. Given a degree-d proof at one size eligible for this construction, we obtain a degree-d proof at every later size separated by the same block q. Thus d is constant along that progression. The quadratic cost below governs which starting proofs are eligible; its inverse is the square-root degree lower bound.

The induction maintains one group-algebra element, not a list of independently chosen inverses. At the start of stage b, this element satisfies $zwz = z$ on every shorter tuple module. We must repair length b while preserving all those previous identities. Three steps do that: normalise the old element, localise its residual to tuples without the fresh labels, and simulate a position-compatible inverse by permutations involving those fresh labels.

Keep the order of those steps. A correct partial inverse of z on one vector space does not automatically give a partial inverse for the old residual, and an operator which works after projecting back to old labels need not work on all larger tuples. Normalisation settles the first issue; localisation settles the second.

10.1 Fresh-label construction

We now prove that a support-r witness exists in characteristic two when

$$n \ge T(r) := 1 + 2 + \cdots + r = \frac{r(r+1)}{2}, \qquad r < q.$$

This proof is independent of the rook theorem used later. Its ingredients are finite algebra, a position-compatible auxiliary inverse, and an explicit way to simulate that inverse using fresh labels.

The phrase "generic coefficients" below refers to the independent block coefficients from Chapter 9. We choose them in a finite field so that the subsequent ring calculation takes place in a finite set. If Ω is the current finite label set, $F[S_\Omega]$ has only $|F|^{|S_\Omega|}$ elements: each permutation has one coefficient in F. The finite field is a construction device, not a restriction on the field of the original polynomial request.

Choose the generic coefficients in a finite field F of characteristic two, large enough to contain a independent elements. We may work inside the algebraic closure of the original field; the degree question descends by the field-extension argument. The finiteness of F will be used only to normalise an element of a finite group algebra.

10.1.1 Normalisation of the residual

Suppose w_0 works on shorter tuples but not yet on length b. Put $y = z - zw_0z$. An auxiliary linear map h may satisfy $zhz = z$ on the larger tuple space. It is tempting to conclude $yhy = y$. In general the expansion contains an extra term zw_0zw_0z, and this need not collapse. Before correcting the error we arrange the additional identity $w_0zw_0 = w_0$ in the actual group algebra. This normalisation was a genuine mathematical step in the earlier construction.

Here *normalisation* is an algebraic replacement: it modifies a ring element to satisfy the additional algebraic identity needed for the residual calculation. A unital ring has addition, associative multiplication and an identity 1; its multiplication need not commute. A ring representation applies its elements as operators, respecting addition, composition and the identity. Thus an identity proved in the ring holds in all its representations at once. In the lemma, v is a ring element and the letters i, ℓ, h in the proof are integers recording powers; this local h is not the auxiliary linear operator used elsewhere.

Lemma 10.1 (Normalisation in a finite ring)**.** *Let v, z belong to a finite unital ring. There is an element $v^\natural$ in the same ring with $v^\natural z v^\natural = v^\natural$, such that every ring representation in which $zvz = z$ also satisfies $zv^\natural z = z$.*

Proof. Put $x = vz$. Finiteness gives $x^{i+\ell} = x^i$ for some $i \geq 0$ and $\ell > 0$. Multiplication by powers of x gives $x^{j+\ell} = x^j$ whenever $j \geq i$. Choose $h \geq i$ such that $h + 1$ is divisible by ℓ. Then $2h + 1 - h = h + 1$ is a multiple of ℓ, so $x^{2h+1} = x^h$. Define $v^\natural = x^h v$. Associativity, without commutation of factors, gives

$$v^\natural z v^\natural = x^h (vz) x^h v = x^{2h+1} v = x^h v = v^\natural.$$

In a representation with $zvz = z$, induction gives $z(vz)^j = z$ for every $j \geq 0$. Hence

$$zv^\natural z = z(vz)^{h+1} = z.$$

The exponent was selected inside the original finite ring. The normalisation therefore preserves the witness relation simultaneously in every representation where it was already true. □

In the construction below we denote the normalised old witness by b_0. It is one ring element shared by every shorter tuple representation. The lemma preserves $zvz = z$ wherever it was already valid. It does not assert that a formerly unsuccessful tuple length has been repaired. The new identity $b_0zb_0 = b_0$, obtained in the ring itself, is what makes the following sandwich of the residual work.

Lemma 10.2 (Normalised residual)**.** *If $b_0zb_0 = b_0$ and $zhz = z$, then, for $y = z - zb_0z$,*

$$yhy = y.$$

Proof. Expand without commuting any factors:

$$\begin{aligned} yhy &= zhz - zhzb_0z - zb_0zhz + zb_0zhzb_0z \\ &= z - 2zb_0z + zb_0zb_0z \\ &= z - zb_0z = y. \end{aligned}$$

The last equality uses the normalisation. This identity works in any ring; characteristic two is not needed for this particular calculation. □

10.1.2 A residual calculation over $\mathbb{F}_4$

The notation E_{ij} denotes a matrix unit: it sends basis vector e_j to e_i and sends the other basis vectors to zero. Thus $E_{12}(s,t) = (t,0)$ and $E_{21}(s,t) = (0,s)$ on a two-dimensional coordinate space. Applying E_{12}, E_{21}, E_{12} successively returns $E_{12}(s,t)$, which makes the partial-inverse identity in the calculation below transparent.

The normalisation is not optional algebraic tidying. Let $\omega^2 + \omega + 1 = 0$, let $z = E_{12}$ on $\mathbb{F}_4^2$, and let $h = E_{21}$. Then $zhz = z$. If we take the unnormalised old operator $w_0 = \omega E_{21}$, its residual is

$$y = z - zw_0z = (1-\omega)z = \omega^2 z.$$

But $yhy = \omega^4 z = \omega z \neq \omega^2 z = y$. Thus a partial inverse for z does not automatically serve as an inverse for y. The finite-ring lemma repairs exactly the missing identity used in the expansion of yhy.

This example is a two-dimensional operator calculation explaining one algebraic step. It is not being used as a label-permutation construction or as an asymptotic counterexample. The group-algebra construction obtains its normalisation in the finite ring $F[S_\Omega]$ itself, which is why the corrected identity holds in all the required modules at once.

10.1.3 An auxiliary inverse compatible with tuple positions

Let Ω be the current label set, consisting of the block B and the retained labels already allocated. On $Q = P_b(\Omega)$, partition the tuples according to the A-orbit of their *underlying label set*. A set meeting B has proper stabiliser because it has size $b < q$. Choose the functional ℓ and coordinate π as in Section 9.1, now on these sets. Let D read π from the underlying set, and use the same W and nonzero α as before. The choices and the scalar α are made separately on each orbit; the resulting operators are combined on their direct sum. On each moved orbit,

$$Dz - zD = W, \quad Wz = zW, \quad W^2 = \alpha^2 I, \quad z^2 = 0.$$

Multiplying the commutator by z shows $zDz = zW$. Thus

$$h = \alpha^{-2} WD \quad \text{satisfies} \quad zhz = z.$$

On tuples wholly outside B, put $h = 0$, since $z = 0$ there. Because D depends on the underlying set and translations act on labels, h commutes with every permutation of tuple positions. We have not claimed that h is a group-algebra element. It is just an explicitly constructed linear operator. Here and in this section, "inverse" is shorthand for the partial identity $zhz = z$, not for $zh = hz = I$.

10.1.4 Fresh labels simulate any position-compatible operator

Write $C = \{\kappa_1, \ldots, \kappa_b\}$ for the fresh labels. The old tuple space sits inside the larger one by including its basis vectors; projection back discards the basis vectors using a fresh label. The composite of that inclusion and projection on the old space is the identity. We want only the old-to-old part of a larger permutation combination, which is the meaning of the compressed operator below.

Adjoin b new labels $\kappa_1, \ldots, \kappa_b$ to Ω. Let $i : P_b(\Omega) \to P_b(\Omega \cup C)$ include the old tuples and let p discard every tuple using a fresh label. We prove that any operator h commuting with position permutations has the form

$$h = pci \tag{10.1}$$

for some $c \in F[S_{\Omega \cup C}]$. The order of operations in pci is: include the old input, apply c, then project the result to old outputs. The symbol p here names this projection, not the prime characteristic. The formula does not say that c preserves the old space or agrees with h on every larger tuple; it identifies its old-to-old matrix corner.

Fix a b-element set $U = \{u_1, \ldots, u_b\} \subseteq \Omega$, with one chosen ordering. Form

$$E_U = \prod_{j=1}^{b} (1 - (u_j \ \kappa_j)).$$

The transpositions are disjoint. On an old tuple missing some u_j, the corresponding factor kills the tuple, and the other factors cannot introduce u_j. On a tuple using all the u_j, the identity term leaves the tuple old, while every nonidentity term introduces a fresh label. Therefore $pE_U i$ is precisely the projector onto the orderings of U.

The $b!$ orderings of U form a *free orbit* for position permutations: exactly one reordering carries a chosen ordering to any other. Consequently a position-compatible map is determined on that orbit by its value at one ordering. All other values are obtained by the same position reordering of the output. This is why one can prescribe the map without averaging or dividing by $b!$.

For an ordered output tuple v, choose a permutation $g \in S_\Omega$ taking $(u_1, \ldots, u_b)$ to v. Extend g by fixing fresh labels. Then $pgE_U i$ takes that input ordering to v, takes its position permutations to the corresponding permutations of v, and kills all other underlying sets. The orderings of each U form a free orbit for S_b. Consequently an operator commuting with S_b is completely determined by its value on one ordering for each U. Expanding those values as linear combinations of output tuples and using the operators $pgE_U i$ proves (10.1). There is no division by $b!$.

Example 10.3 (Two registers)**.** For $U = \{u_1, u_2\}$,

$$E_U = 1 - (u_1\kappa_1) - (u_2\kappa_2) + (u_1\kappa_1)(u_2\kappa_2).$$

On (u_1, u_2), projection p kills the last three terms. On (u_1, v) with $v \neq u_2$, the identity contribution is cancelled by the second swap factor, which fixes that tuple. The detector recognises the set U, while leaving the two possible orderings available for a position-compatible output rule.

10.1.5 Why compression is enough only after localisation

Equation (10.1) asserts $h = pci$ on the old tuple space. It does not assert that c acts like h on every tuple in the larger universe. Terms of c may introduce fresh labels. This would normally be a serious problem: a proof must preserve all basis objects, not just an arbitrarily selected corner of a matrix.

The residual has a special support in the larger tuple space. It is zero on every summand with a fresh label, because such a summand uses fewer than b old labels and the old witness is already valid at those shorter lengths. Therefore the two residual factors in ycy project to the old corner before and after the middle operator. The identity $h = pci$ is exactly the information the sandwich needs. The next calculation verifies this point as an equality of maps, rather than assuming that compression itself is a global realisation.

In matrix terms, order the basis with old tuples first and tuples using fresh labels second. Under the induction hypothesis the calculation below gives

$$y = \begin{pmatrix} y_Q & 0 \\ 0 & 0 \end{pmatrix}.$$

In ycy the rightmost y removes fresh input directions, and the leftmost y removes fresh output directions. Only the old-to-old corner $pci = h$ of the middle operator matters. The zeros here come from validity of the old witness at shorter lengths, not from a claim that an arbitrary permutation combination has this block form.

10.1.6 The induction, including localisation of the error

For support zero take $w_0 = 0$: z is zero on constants. Suppose a witness exists on the block and $T(b-1)$ retained labels, collectively Ω. Normalise it in the finite ring $F[S_\Omega]$ to b_0. It still works on all tuple lengths below b and satisfies $b_0 z b_0 = b_0$ in the ring itself. Set $y = z - zb_0z$.

Adjoin b fresh labels. Decompose $P_b(\Omega \cup C)$ according to the positions occupied by fresh labels and their actual values. The group S_Ω fixes each fresh label, so every summand with at least one fresh label is a copy of $P_j(\Omega)$ for $j < b$. On it $y = 0$ by the old witness relation. The summand with no fresh labels is $Q = P_b(\Omega)$. Thus, as actual maps,

$$y = iy_Qp.$$

This localisation is what allows a compressed inverse to suffice. Construct the auxiliary h on Q above. Lemma 10.2 gives $y_Qhy_Q = y_Q$. Choose c with $pci = h$. Then

$$ycy = iy_Q(pci)y_Qp = iy_Qp = y.$$

Define the new group-algebra element

$$w_b = b_0 + (1 - b_0z)c(1 - zb_0).$$

Since $z(1 - b_0z) = (1 - zb_0)z = y$,

$$zw_bz = zb_0z + ycy = zb_0z + y = z$$

on length b. On every shorter tuple space the same fresh-label decomposition makes $y = 0$, so the relation remains valid there too. The new cost is exactly b retained labels, giving

$$T(b) = T(b-1) + b.$$

If still more retained labels are present, extend the constructed permutations by fixing them. Decomposing according to those outside labels reduces every tuple of length at most b to a shorter tuple on the constructed labels. Thus the same witness works for all $n \geq T(b)$.

Theorem 10.4 (Quadratic retained-label budget)**.** *For every $r < q = 2^a$ and $n \geq r(r+1)/2$, there is a common witness w with $zwz = z$ on all tuple modules of length at most r. Therefore deletion is exact on the support-r class. It transports the entire degree-d PC consequence space with $r = kd + s$, beyond the fixed template threshold. For prescribed indexed targets use the budget $r = 2kd + s + t$ to include the target map and the doubled-support detector.*

Every construction needed for this theorem has now been supplied. In particular, the proof is not an inference from witnesses found for small tuple lengths.

Remark 10.5 (The square-root degree bound). For either prescribed-target budget, the construction gives an onset $O(d^2)$ for degree-d transfer. Inverting that onset gives an $\Omega(\sqrt{n})$ lower bound when no finite degree recurs, and on the hard residues of the periodic case. The loss is precisely the cumulative allocation $T(r) = r(r+1)/2$. The following construction replaces this sum by the maximum $2r-1$, using one retained pool at every layer.

10.2 The first three induction stages

At length zero there is one basis vector, the empty tuple. Every permutation fixes it, so $z = 0$ and $w_0 = 0$ is a valid witness. No retained label is needed.

At length one, the auxiliary operator recognises the moved label orbits. One fresh retained label realises its compressed inverse. The witness can therefore be built on the block plus one retained label. For a block of two labels, the previous chapter exhibited this witness explicitly as a transposition between one block label and the retained label.

At length two, normalise that same witness first. Allocate two additional retained labels κ_1, κ_2. A tuple containing either fresh label has at most one old entry; the residual is zero on it. Only the tuples with both entries old require repair. The detector

$$(1 - (u_1\ \kappa_1))(1 - (u_2\ \kappa_2))$$

selects an old two-element set. Combining its translates realises any position-compatible inverse on the old corner. The total number of retained labels is now $1 + 2 = 3$.

At length three, the same argument spends three further labels. Tuples with one, two or three of those fresh labels reduce to old tuple lengths two, one or zero, respectively. All three pieces already have zero residual. The only unsolved piece consists of old three-tuples, and a detector with three disjoint swaps compresses its auxiliary inverse. The retained total is $1 + 2 + 3 = 6$.

The block size must exceed the largest tuple length being treated. For example, one may choose a block of size four when explaining stages one through three. A two-label block illustrates stage one, but cannot be reused as a valid block for stage three. Retained labels and block labels have different jobs in this count.

Tuple length r	1	2	3	4
Fresh-label total $T(r)$	1	3	6	10
Linear total proved later $2r-1$	1	3	5	7

The first strict improvement is at length three. The linear method will not allocate the third batch separately: it will exploit the residual's zero marginals and reuse a common retained pool.

10.3 From quadratic label cost to square-root degree hardness

Return now from tuple spaces to one fixed polynomial request and one proof system, NS or PC. Its profile $h(n)$ is the least degree for one fixed target orbit, or the maximum of the least degrees for an all-target request. It is infinity if any requested target is underivable; an empty request has value zero. A finite value recurs if that exact value is attained at infinitely many universe sizes. The block construction will force recurrence whenever such a value occurs beyond its eligible starting universe size.

Let $r(d) = \alpha d + \beta$, where $(\alpha, \beta) = (k, s)$ for an invariant target or $(\alpha, \beta) = (2k, s + t)$ for an indexed request. The fixed starting universe size n_* includes degree stabilisation of the targets. A convenient safe onset for this preliminary construction is

$$N_{\mathrm{quad}}(d) = \max\{n_*,\, 2r(d) + 1,\, T(r(d))\} \qquad (d \geq 1).$$

The extra linear term is harmless and makes this onset comparable with the eventual linear statement. Exactness of the quadratic construction, together with the target-map and detection arguments, gives degree-threshold periodicity beyond this onset. Take the least power of two $Q(d) > r(d)$; then $Q(d) \leq 2r(d)$.

Here α, β are integer coefficients of the support budget, unrelated to the nonzero field scalar α in the orbit calculation. The onset estimates concern positive degree budgets. Degree zero uses its previously established fixed predicate beyond n_*; it therefore cannot be a nonrecurring value on a hard tail. The hard-value inversion below concerns positive finite values, and the unbounded recurring values chosen for peaks are positive too.

Suppose a profile has no recurring finite degree. If $h(n) = e < \infty$ and $n \geq N_{\mathrm{quad}}(e)$, threshold periodicity at all budgets through e repeats that exact value on an infinite arithmetic progression. This is a contradiction. Thus every finite value must satisfy $n < N_{\mathrm{quad}}(e)$. Since $N_{\mathrm{quad}}(e) \leq A(e + 1)^2$ for a constant A depending only on the fixed request, it follows that

$$h(n) > \sqrt{n/A} - 1$$

whenever n is beyond the fixed initial range. The assertion also holds when $h(n) = \infty$. The same argument works on hard residues in a fixed periodic easy/hard split: a value attained beyond its onset would recur and would therefore be one of the bounded easy values.

For a fluctuating profile, choose recurring values e tending to infinity. A representative of its recurring residue can be selected in

$$N_{\mathrm{quad}}(e) \leq n < N_{\mathrm{quad}}(e) + Q(e).$$

Both the onset and the period are bounded by a constant times $(e+1)^2$, so these representatives are finite degree peaks with a square-root lower bound, $e \geq c\sqrt{n} - O(1)$, for a fixed $c > 0$. They are not an all-size lower bound; low recurring values remain at arbitrarily large sizes.

Nothing in this argument makes the degree of an individual proof shrink. At fixed d we compare derivability at two universe sizes. The lower bound follows by comparing that *fixed budget* with its allowed universe-size onset. The preliminary route just proved is in characteristic two. The later prime-characteristic chapter will give the full linear result directly over every fixed positive characteristic.

Chapter 11

Towards linear lifting: reuse one pool of labels

The property we need

Keep the characteristic-two block from the previous chapters: $N = n + q$, with n retained labels, $q = 2^a > r$ block labels and the square-zero operator z. We seek one w with $zwz = z$ on all ordered-tuple spaces of lengths at most r. The final retained label requirement will be $n \geq 2r - 1$; the block and the budget r stay fixed while the tuple length b increases.

The quadratic construction repaired an arbitrary position-compatible operator on the whole old b-tuple space. That is more freedom than we need. The old witness error already annihilates every direction coming from shorter tuples and has zero marginal in every output position. We will use that shape to construct a correction on the same retained label pool.

There are two separate contracts. The present chapter constructs a position-compatible operator φ with $z\varphi z = -\varepsilon$, whose image has zero marginals and which kills the incoming shorter layers. The next chapter proves that every operator with exactly this shape is the action of a group-algebra element annihilating the shorter tuple modules. It uses no additional labels. Combining the two contracts yields one common witness at retained universe size $2r - 1$.

This is the improvement which changes the degree gap. With fresh labels, the retained-label cost was a sum over all tuple lengths. Reusing one pool makes it a maximum. A proof with degree d can therefore be lifted once the universe is only linearly large in d. If that degree cannot recur on an infinite progression, every occurrence must have been before this linear starting universe size.

Semisimplicity would supply invariant complements to every invariant subspace (Section 5.2); here we construct the particular compatible complements needed by the lift. The construction builds sections using label-avoiding decompositions, then makes those sections commute with the block operator. In positive characteristic, a request for an equivariant section cannot be justified by averaging. Every such section below is explicit.

A complement to a subspace $U \subseteq V$ is a subspace C with $V = U \oplus C$. A basis extension always supplies a vector-space complement, but it need not respect a permutation action or the operator z. A compatible complement therefore needs a construction. For a surjection $f : E \to Q$, a *section* is a linear map $\sigma : Q \to E$ with $f\sigma = I$: it chooses a preimage of each vector, consistently with linear combinations. Such a section gives $E = \ker f \oplus \operatorname{im} \sigma$. Requiring it to commute with an action is an additional condition, not a consequence of this decomposition.

11.1 The zero-marginal layer

We still work in characteristic two. Fix a tuple length $b \geq 1$, distinct from the proof-degree budget d. Let $V = P_b(N)$. For each position i there is a deletion map

$$\partial_i : P_b(N) \longrightarrow P_{[b]\setminus\{i\}}(N)$$

which forgets the entry in position i. On a vector of coefficients it sums over that entry, so we call it a marginal. The target is a copy of $P_{b-1}(N)$ with the surviving positions named. Give tuple spaces the dot product making their bases orthonormal, and put

$$V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}, \qquad \mathcal{Z}_b(N) = \bigcap_i \ker \partial_i = (V')^{\perp}, \qquad T_b(N) = V/V'. \tag{11.1}$$

Here $\mathcal{Z}_b$ is the zero-marginal space, and T_b is the top quotient. This notation distinguishes the tuple zero-marginal space from the polynomial consequence space $C_d(n)$. All statements below use the displayed definitions.

These are marginal sums, rather than projections onto some of the original coordinates. At length two, for example,

$$\partial_1\Big(\sum_{i\neq j} a_{ij}e_{ij}\Big) = \sum_j \Big(\sum_{i\neq j} a_{ij}\Big)e_j.$$

All entries with the same surviving label contribute to one output coefficient. A zero marginal says that these contributions cancel.

The dot product here is the bilinear coefficient pairing $\langle v, w\rangle = \sum_t v_t w_t$, without conjugating coefficients. It is nondegenerate because the tuple basis is orthonormal. It is not a positive inner product: over $\mathbb{F}_2$, the nonzero vector $e_1 + e_2$ pairs to zero with itself. Thus $(V')^{\perp}$ is the space of vectors pairing to zero with every vector in V', and need not be a complementary subspace to V'. The transpose is characterised by $\langle Tv, w\rangle = \langle v, T^{\mathsf{T}}w\rangle$.

The name *top quotient* records what remains after quotienting by the incoming directions from shorter tuple spaces. It does not assert that this quotient is semisimple.

The transpose map has a direct combinatorial meaning. For a $(b-1)$-tuple u with position i missing,

$$\partial_i^{\mathsf{T}} e_u = \sum_{x\notin \operatorname{im} u} e_{u[i\mapsto x]}.$$

It inserts each admissible label once. There is no factor $1/N$ and no division by the number of available labels. This is the incoming subspace V', and it is a sum of images, not an arbitrarily selected copy of P_{b-1}.

The incoming space V' contains directions lifted from shorter tuples. The zero-marginal space contains directions invisible after deleting any one position. The quotient T_b forgets the incoming directions. Label permutations commute with all marginal maps, and position permutations permute the marginal maps. Thus these spaces carry the two appropriate actions.

Example 11.1 (Lengths one and two)**.** For $b = 1$, ∂ sums coordinates, V' is the constant line and $\mathcal{Z}_1(N)$ is the zero-sum space. For $b = 2$, a coefficient vector is an off-diagonal array $f(i, j)$, $i \neq j$. Its marginals are its row and column sums. Thus $\mathcal{Z}_2(N)$ consists of off-diagonal arrays with every row and column sum zero. On three labels it is spanned by

$$e_{12} + e_{23} + e_{31} - e_{21} - e_{32} - e_{13}.$$

In characteristic two the six coefficients are all one, but every line still sums to zero because it contains two ones.

11.1.1 The old error already has a special shape

The annihilator ideal $J_{b-1}(N)$ consists of combinations of label permutations that act as zero on every tuple space of length below b. Suppose w_0 is a witness on all lengths below b, and set

$$\varepsilon = zw_0z - z \quad \text{on } V.$$

This is the action of an element of $J_{b-1}(N)$. For any $x \in J_{b-1}(N)$,

$$\partial_i x = x\partial_i = 0, \qquad x\partial_i^{\mathsf{T}} = \partial_i^{\mathsf{T}} x = 0.$$

In these equations the same group-algebra element x acts on both tuple spaces. The marginal and its transpose commute with label permutations, while x acts as zero on the shorter $(b-1)$-tuple space by the definition of $J_{b-1}(N)$. Therefore x maps V into $\mathcal{Z}_b(N)$ and kills V'. In particular ε factors through

$$V \longrightarrow T_b(N) \longrightarrow \mathcal{Z}_b(N) \longrightarrow V.$$

To *factor through* the quotient means that $\bar{\varepsilon}([v]) = \varepsilon v$ is well defined: replacing v by $v + v'$ with $v' \in V'$ does not change the answer. Its values belong to $\mathcal{Z}_b$, and the final arrow is that space's actual inclusion into V. This turns the error into a map from the quotient by incoming shorter-tuple directions to the zero-marginal output space. It also commutes with position permutations. This is much more special than an arbitrary operator on V. The improved route exploits exactly this special form.

There are two tasks. First, construct a position-compatible correction of this shape with $z\varphi z = -\varepsilon$. Second, realise that correction by an element of the actual label group algebra which kills all shorter tuples. The first task needs retained labels; the second will be solved by a rook-placement theorem without adding labels.

11.1.2 A length-three zero-marginal vector

For three positions and four labels, write ϵ_{ijkl} for the sign of the permutation (i, j, k, l) of $(1, 2, 3, 4)$, and let l be the unique missing label. Then

$$Y = \sum_{i,j,k \text{ distinct in } [4]} \epsilon_{ijkl} e_{ijk}$$

has all three marginals zero. Fix, for example, the second and third entries $2, 3$. The only contributions are e_{123} and e_{423}, with opposite coefficients, so their sum under deletion of the first position is zero. The same argument for any surviving ordered pair uses the transposition between the missing label and the deleted entry. It changes the sign and pairs the two terms. In characteristic two the signs become equal, and the two coefficients still sum to zero. Thus the zero-marginal condition is concrete at lengths one, two and three; it is not a name for an unspecified representation.

The linear witness proof does not require this particular vector to span $\mathcal{Z}_3$. It illustrates the cancellations which the partition and cone maps must respect for *every* vector in that space.

11.2 Partitioning a zero-marginal vector so that one label is avoided

For a finite position set S and label set L, let $P_S(L) = K[\mathrm{Inj}(S, L)]$ and $\mathcal{Z}_S(L) = \bigcap_{i \in S} \ker \partial_i$. If $L' \subseteq L$, regard $\mathcal{Z}_S(L')$ as the vectors of $\mathcal{Z}_S(L)$ supported on tuples using only L'. Write $\mathcal{Z}_c$ when there are c positions.

Suppose a block label β occupies a new position and $y \in \mathcal{Z}_c(L)$ occupies the other positions. The vector $e_\beta \otimes y$ has zero marginals in those other positions, but its new-position marginal is y. We want a retained correction with the same marginal. If

$$y = \sum_{x \in L} \pi_x(y), \qquad \pi_x(y) \in \mathcal{Z}_c(L \setminus \{x\}),$$

then

$$e_\beta \otimes y - \sum_{x \in L} e_x \otimes \pi_x(y) \tag{11.2}$$

has every marginal zero. Avoiding x is essential: it makes each displayed tuple injective. The maps must also commute with permutations of the old positions, so they can be used consistently in all layers. Here $e_\beta \otimes y$ means inserting β into the designated new position of every tuple in y; the other positions retain their names. It is tuple-coordinate notation, not multiplication of polynomials.

Example 11.2 (One old position)**.** Take $L = \{1, 2, 3\}$ and $y = ae_1 + be_2 - (a + b)e_3$. Define

$$\pi_1(y) = b(e_2 - e_3), \quad \pi_2(y) = a(e_1 - e_3), \quad \pi_3(y) = 0.$$

Each summand avoids its subscript and has coordinate sum zero; their sum is y. For $y = e_1 - e_2$, formula (11.2) is the six-term vector

$$e_{\beta 1} - e_{\beta 2} + e_{12} - e_{13} - e_{21} + e_{23}.$$

Check its two marginals by grouping terms with the same surviving label. With only labels $1, 2$, every $\mathcal{Z}_1(L \setminus \{x\})$ is zero, while $e_1 - e_2 \neq 0$. The third label is what permits the construction.

The phrase *partition of unity* below refers to the identity $\sum_x \pi_x = I$ between linear maps. Each map distributes part of the input into a space avoiding its label. The outputs may overlap; the maps are not required to be positive or idempotent projections. Their equivariance concerns permutations of the tuple positions. They can choose distinguished labels and need not commute with all permutations of the label set.

Lemma 11.3 (Partition of unity)**.** *If $|L| \geq 2c + 1$, there are position-equivariant linear maps*

$$\pi_x : \mathcal{Z}_c(L) \longrightarrow \mathcal{Z}_c(L \setminus \{x\}), \qquad \sum_{x \in L} \pi_x = I.$$

This statement and its proof work over any field.

Proof. We induct on c. At $c = 0$ the space is K on the empty tuple; choose one label x, make $\pi_x = I$ and all others zero.

Let $c \geq 1$. Choose $\xi \in L$ and put $L' = L \setminus \{\xi\}$. For $y \in \mathcal{Z}_c(L)$ and position i, let $y^{(i)}$ be the coefficients of those tuples in which ξ appears in position i, with that position then removed. This lies in $\mathcal{Z}_{[c]\setminus\{i\}}(L')$. To see this, fix ξ in position i and sum another position: it is a marginal of y, hence zero. There cannot be a second occurrence of ξ in an injective tuple.

By induction, partition $y^{(i)}$ on L' as $\sum_{x \in L'} \pi'_x(y^{(i)})$, each term avoiding x. For a vector $v \in \mathcal{Z}_{[c]\setminus\{i\}}(L' \setminus \{x\})$, partition once more on $L' \setminus \{x\}$:

$$v = \sum_{x' \in L' \setminus \{x\}} \pi''_{x'}(v).$$

Both applications are justified: $|L'| \geq 2c$ and $|L' \setminus \{x\}| \geq 2c - 1 = 2(c - 1) + 1$.

Construct $F_{i,x}(v)$ by inserting ξ into position i of v, then subtracting the sum obtained by inserting x' into position i of $\pi''_{x'}(v)$. Every term avoids x. All tuples are injective because the lower-dimensional vector avoids the inserted label. A marginal in a position other than i vanishes term by term. The marginal in position i is $v - \sum_{x'} \pi''_{x'} v = 0$. Thus

$$F_{i,x}(v) \in \mathcal{Z}_c(L \setminus \{x\}).$$

Its ξ-part in position i is v, and its ξ-parts in other positions are zero.

For $x \in L'$ set

$$\pi_x(y) = \sum_{i=1}^{c} F_{i,x}(\pi'_x(y^{(i)})), \qquad \pi_\xi(y) = y - \sum_{x \in L'} \pi_x(y).$$

The ξ-part of the last expression in every position is $y^{(i)} - \sum_x \pi'_x(y^{(i)}) = 0$. It therefore uses no label ξ, and belongs to $\mathcal{Z}_c(L')$. The sum identity is built into the definition.

For equivariance, use the same inductively constructed maps on every position set of size $c-1$, transported along a bijection. The transport is independent of the chosen bijection because the inductive maps commute with position permutations. Permuting the c positions simply permutes the terms indexed by i in the displayed sum. No averaging over a symmetric group is used. □

The count $2c+1$ is now visible in an actual induction: reducing one position removes two labels, the label ξ being cleared and the label x being avoided. This is a sufficient bound proved by the construction, not an appeal to a dimension formula.

11.2.1 Why the two recursive partitions have different jobs

In the proof just given, the first partition decomposes the slice where the distinguished label occurs in position i. It chooses which label x the completed vector will avoid. The second partition then chooses a label x' to insert in the correction replacing the distinguished label. Avoiding x' is necessary for injectivity; avoiding x is necessary for the final destination of π_x. One partition cannot simply be reused as the other, since the two labels are excluded for different reasons.

The residual component $\pi_\xi(y)$ is not justified only by the sum identity. Its slice containing ξ must vanish at every position. The inserted terms were designed so their ξ-slice at position i is exactly the first partition of $y^{(i)}$, and their ξ-slices at all other positions are zero. This proves that the residual really avoids ξ. Subtracting zero-marginal vectors then keeps all its marginals zero. These two checks establish the stated codomain of π_ξ, as well as the identity $\sum_x \pi_x = I$.

The distinguished label called ξ in that induction is local bookkeeping; it is unrelated to the proof-degree budget. All the requirements in the induction count retained labels. To avoid any confusion, the subsequent applications describe this chosen label as the distinguished label rather than as a degree parameter.

11.3 Adding positions coherently

Writing ∂_J means forgetting every position in J, by successive marginal maps. Their order does not matter: they are sums over distinct coordinates. Inserting an extra label and then forgetting it resembles the elementary combinatorial operation of adding a vertex to a configuration, often called coning. Here the inserted label must be avoided by the old tuple, so the partition just proved supplies a linear sum of admissible insertions. The construction is a section of a specified marginal map; it does not assume a topological contraction theorem.

Let R_0 and J be disjoint position sets of sizes c and ℓ. Here ℓ counts positions being added; the target-index support parameter t is not involved in this layer construction. We want to add the positions in J while keeping the marginals in R_0 zero. If $|L| \geq 2c + \ell$, the map deleting all positions in J has a position-R_0-equivariant section

$$\mathcal{Z}_{R_0}(L) \longrightarrow \{\Phi \in P_{R_0 \cup J}(L) : \partial_i \Phi = 0 \ (i \in R_0)\}. \tag{11.3}$$

Being a section means that deleting all the new positions J after this insertion returns the original vector. The equivariance required here concerns the old positions R_0, not arbitrary permutations of the label set. Here is the construction. Suppose some positions $J' \subsetneq J$ have already been added. Split the vector according to their injective labelling γ. On that component the remaining zero-marginal vector uses the label set $L \setminus \operatorname{im} \gamma$. Its size is at least

$$2c + \ell - |J'| \geq 2c + 1.$$

Partition that vector by Lemma 11.3. For a new position j, insert label x in the summand π_x. Summing the new position returns the original vector, and the old zero marginals stay zero. Repeat until all ℓ positions have been added. Every step is linear and commutes with the R_0 position action, proving (11.3).

11.3.1 Sorting a zero-marginal vector by its block positions

Split the universe as retained labels $[n]$ and block labels B. Every vector $f \in P_b(n+q)$ has components f_D, for subsets $D \subseteq [b]$, consisting of tuples whose block positions are exactly D:

$$f_D \in K[\mathrm{Inj}(D, B)] \otimes P_{[b]\setminus D}(n).$$

Write ϵ_i for summing a block entry and $\partial_i^{\mathrm{ret}}$ for summing a retained entry. The condition of zero marginal in position $i \notin D$ is

$$\partial_i^{\mathrm{ret}} f_D + \epsilon_i f_{D \cup \{i\}} = 0. \tag{11.4}$$

It simply separates the possible value of the deleted entry into a retained label or a block label.

Let $\mathcal{Z}^{\leq \ell}$ be the zero-marginal vectors with no components using more than ℓ block positions. The first space is $\mathcal{Z}^{\leq 0} = \mathcal{Z}_b(n)$. The top components of a vector in $\mathcal{Z}^{\leq \ell}$ have zero retained marginals by (11.4). We obtain a map onto the prospective layer

$$G_\ell = \bigoplus_{|D|=\ell} K[\mathrm{Inj}(D, B)] \otimes \mathcal{Z}_{[b]\setminus D}(n).$$

Its kernel is $\mathcal{Z}^{\leq \ell - 1}$. We next prove surjectivity with a position-compatible section, so this is an actual quotient description rather than a count of dimensions.

A *filtration* is simply this increasing sequence of subspaces. Its layer at ℓ records what appears after allowing ℓ block positions, modulo everything already allowed at $\ell - 1$. A quotient layer is not yet a chosen subspace of the larger space. The section below will choose representatives, and the subsequent operator argument will make that choice compatible with z. Here G_ℓ names a space of tuple components, not a target polynomial $G^{(n)}$.

In the displayed tensor product, the first factor records which block labels occupy the positions in D, while the second carries the retained-tuple coefficients. An elementary tensor $e_\beta \otimes y$ places the retained vector y at the block pattern β. General vectors are sums of these tensors. The direct sum keeps different sets D of block positions in separate components.

11.3.2 An explicit lift of one top component

Take $J = [\ell]$, an injective block tuple $\beta : J \to B$, and $y \in \mathcal{Z}_{R_0}(n)$ with $R_0 = [b] \setminus J$. If

$$n \geq 2(b - \ell) + \ell = 2b - \ell,$$

the section (11.3) gives Φ on the b retained positions such that $\partial_J \Phi = y$ and $\partial_i \Phi = 0$ for $i \in R_0$. For $J' \subseteq J$, set

$$\phi_{J'} = (-1)^{|J'|} \partial_{J \setminus J'} \Phi.$$

For $D \subseteq J$ define

$$f_D = e_{\beta|D} \otimes \phi_{J \setminus D},$$

and put the other components zero. At $D = J$ this is $e_\beta \otimes y$, the required top component. For $i \in R_0$, the retained marginal vanishes. For $i \in J \setminus D$,

$$\partial_i \phi_{J \setminus D} = -\phi_{J \setminus (D \cup \{i\})},$$

which cancels the block marginal of the adjacent component in (11.4). Thus the whole lift belongs to $\mathcal{Z}^{\leq \ell}$.

To lift all components equivariantly, choose an ordering of each ℓ-element subset of block labels. Use it in the representative positions $[\ell]$. Its stabiliser among position permutations consists exactly of permutations of the remaining $b-\ell$ positions. The constructed lift commutes with this stabiliser, by the cone construction. Transport it along position permutations to every other placement of the same ordered block labels. If two permutations give the same placement, their difference is in the stabiliser, so the transported lift is the same. Extend linearly over all block subsets and retained vectors. This supplies the promised S_b-equivariant section without division by any group order.

For $1 \leq \ell \leq b$, the largest requirement $2b - \ell$ occurs at $\ell = 1$. Therefore the single hypothesis

$$n \geq 2b - 1 \tag{11.5}$$

supplies all these sections.

11.3.3 Two worked lifts at tuple length three

The signs in the lift formula can be checked without abstract notation. First take one block position, a block label β, and

$$y = e_{12} + e_{23} + e_{31} - e_{21} - e_{32} - e_{13}$$

on the other two positions. The vector y uses only retained labels $1, 2, 3$, so the particular vector

$$e_\beta \otimes y - e_4 \otimes y$$

is injective and has every marginal zero. The first marginal is $y - y = 0$; the two other marginals are zero because those of y are zero. This is a single-vector instance of the one-block-position construction. Constructing a linear map for *all* y is the harder task solved by the partition of unity, and it is what needs $n \geq 5$ at $b = 3$.

For two block positions let the retained position be first and use $y = e_1 - e_2$. With retained labels $1, 2, 3, 4$, the cone can use

$$\Phi = e_{134} - e_{234}.$$

It has zero first marginal, and deleting the last two positions returns y. For distinct block labels β, γ, the signed lift is

$$f = e_{1\beta\gamma} - e_{2\beta\gamma} - e_{13\gamma} + e_{23\gamma} - e_{1\beta4} + e_{2\beta4} + e_{134} - e_{234}.$$

Deleting position one cancels each adjacent pair. Deleting position two cancels the terms with β against those with 3, and those with both retained values against the remaining terms. Deleting position three gives the analogous cancellations between γ and 4. Its top component is exactly $e_{\beta\gamma} \otimes (e_1 - e_2)$. The count for this layer is $2b - \ell = 4$; the one-block layer is the more expensive layer, requiring five retained labels. This is why a maximum over the layers, rather than a sum, controls the completed construction.

11.4 Making the splitting compatible with the block operator

The sections just constructed commute with position permutations, but need not commute with z. We now fix that, without adding labels.

The grading used here sorts basis patterns into two classes. It is not polynomial degree. Its diagonal operator D reads the chosen orbit coordinate, whereas z moves patterns by translations. Their *commutator* $[D, z] = Dz - zD$ measures the difference between doing these two operations in opposite orders. The following calculation will turn that difference into an invertible operator W.

A block pattern is an injective partial tuple β with a specified nonempty position domain of size ℓ, and values in B. Its image is a nonempty subset of B of size $\ell < q$. Such a subset cannot be fixed by all block translations: a nonempty translation-invariant subset of the regular block is the whole block. Choose the orbit coordinate π from Section 9.1 on these *image subsets*, and define

$$De_\beta = \pi(\operatorname{im}\beta)e_\beta.$$

Let E be the vector space on all the current block patterns, and let E_0, E_1 be the spans of the patterns with coordinate $0, 1$, respectively. Thus $E = E_0 \oplus E_1$ before any use of z. They are stable under position permutations and split by the position domain, since neither operation changes the image subset. On each subset orbit, let $\lambda : A \to \mathbb{F}_2$ be the nonzero linear functional vanishing on its stabiliser used to define π. The previous calculation gives

$$[D, z] = W = \sum_i c_i\lambda(\varepsilon_i)S_i, \qquad W^2 = \Big(\sum_i c_i\lambda(\varepsilon_i)\Big)^2 I.$$

The scalar is nonzero by independence of the block coefficients, so W is invertible on the whole pattern space.

For either $e = 0, 1$, suppose $v_0 + zv_1 = 0$ with $v_0, v_1 \in E_e$. Apply $D - eI$. The first term vanishes and the second gives Wv_1, so $v_1 = 0$, and then $v_0 = 0$. Thus the map $(v_0, v_1) \mapsto v_0 + zv_1$ from $E_e \oplus E_e$ is injective, giving $2\dim E_e \le \dim E$. Since $E = E_0 \oplus E_1$, both inequalities must be equalities. In particular

$$E = E_0 \oplus zE_0.$$

Tensor each domain component with its retained zero-marginal factor. Because z fixes retained labels, the layer has the analogous decomposition

$$G_\ell = Q_\ell \oplus zQ_\ell$$

with Q_ℓ position-stable and $z : Q_\ell \to zQ_\ell$ injective.

11.4.1 Turning a section into a commuting section

A section of $f : E \to Q$ may choose good representatives but fail to respect z. The two-level decomposition supplies a way to repair it: choose the representative of $q_0 \in Q_0$, and require the representative of zq_0 to be its image under z. Injectivity of z on Q_0 makes the two inputs uniquely identifiable. The next lemma checks that this prescription is linear and compatible with positions as well. Once a section τ is available, the splitting is explicit:

$$v = \big(v - \tau f(v)\big) + \tau f(v), \qquad v - \tau f(v) \in \ker f.$$

When f and τ commute with z, both subspaces are stable under z.

Lemma 11.4 (A section commuting with a square-zero operator)**.** *Let E, Q carry square-zero operators z commuting with position permutations, and let $f : E \to Q$ be a surjection commuting with both actions. Suppose Q_0 is position-stable,*

$$Q = Q_0 \oplus zQ_0, \qquad z : Q_0 \longrightarrow zQ_0 \text{ is injective.}$$

If σ is a position-equivariant section of f, then a section commuting with both actions is given by

$$\tau(q_0 + zq_1) = \sigma(q_0) + z\sigma(q_1).$$

Proof. The direct sum and injectivity make both q_0, q_1 unique. We have $f\tau = I$. Moreover

$$z\tau(q_0 + zq_1) = z\sigma(q_0) = \tau(zq_0),$$

using $z^2 = 0$. Stability of Q_0 and commutation of z, σ with positions show that τ also commutes with positions. □

The construction lifts the first member of each pair q_0, zq_0 freely and obtains the second by applying z. Apply the lemma to each quotient $\mathcal{Z}^{\leq \ell} \to G_\ell$. Its kernel is $\mathcal{Z}^{\leq \ell-1}$, and the commuting section makes the next space the direct sum of that kernel and the image of the section. That image is $\tau(Q_\ell) \oplus z\tau(Q_\ell)$, since the quotient map is injective on the section image. Inducting on ℓ gives the direct sum of the retained layer and these two-level summands. Let C be the direct sum of all $\tau(Q_\ell)$ for $\ell \geq 1$. Every summand is position-stable and the map $z : C \to zC$ is injective. This proves the splitting of the filtration as actual subspaces, not merely an equality of their dimensions. We have proved

$$\mathcal{Z}_b(n + q) = \mathcal{Z}_b(n) \oplus C \oplus zC, \tag{11.6}$$

where C is position-stable, $z : C \to zC$ is an isomorphism, and z kills the retained summand. This is a concrete direct-sum decomposition, valid under $b < q$ and $n \geq 2b - 1$.

11.4.2 The four consequences used for correction

Write $V = P_b(N)$. The next four properties are precisely the interface used in the correction. They do not assert that every invariant subspace of V is free for z. Recall $V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}$, $\mathcal{Z}_b = (V')^\perp$ and $T_b = V/V'$. Their dual pairing is $\langle [v], \zeta \rangle = \langle v, \zeta \rangle$ for $\zeta \in \mathcal{Z}_b$. It is independent of the representative v because every vector in V' is orthogonal to $\mathcal{Z}_b$. This is the pairing used in the transpose argument below. It is a perfect pairing: it identifies T_b with the space of linear functionals on $\mathcal{Z}_b$. Nondegeneracy follows from $((V')^\perp)^\perp = V'$ in the finite-dimensional tuple pairing. The notation $z^{-1}(U)$ in the proof means the *inverse image* $\{v : zv \in U\}$, not an inverse for the nilpotent operator. For subspaces U_1, U_2 of the tuple ambient, the orthogonal-complement calculations use the elementary rules

$$(U_1 + U_2)^\perp = U_1^\perp \cap U_2^\perp, \qquad (U_1 \cap U_2)^\perp = U_1^\perp + U_2^\perp, \qquad (zU_1)^\perp = (z^{\mathsf{T}})^{-1}(U_1^\perp).$$

The last identity is exactly the transpose pairing formula. These rules apply over our field without positivity or semisimplicity.

Lemma 11.5 (The layer correction interface)**.** *Under $b < q$ and $n \geq 2b - 1$, the following hold.*

1. *$zV \cap \mathcal{Z}_b(N) \subseteq z\mathcal{Z}_b(N)$.*
2. *$zV \cap V' \subseteq zV'$.*
3. *There is a position-equivariant w_C on $\mathcal{Z}_b(N)$ with $zw_Cz = z$ there.*
4. *There is a position-equivariant w_T on $T_b(N)$ with $zw_Tz = z$ there.*

Proof. For the third assertion, let w_C send zc back to c and kill C and the retained summand. For the first, the kernel of z in (11.6) is $\mathcal{Z}_b(n) \oplus zC$. An element of $zV \cap \mathcal{Z}_b(N)$ is therefore $s + zc$. But zV consists of vectors on tuples meeting the block; so does zc. The remaining s is supported entirely on retained tuples, forcing $s = 0$.

For the other two assertions use transpose. In characteristic two each σ_i is an involution, so $z^{\mathsf{T}} = z$. The pairing identifies $T_b(N)$ with $\mathcal{Z}_b(N)^*$, giving w_T by duality. To see the second inclusion explicitly, take orthogonal complements. The first inclusion implies

$$z^{-1}(\mathcal{Z}_b) = \ker z + \mathcal{Z}_b.$$

Indeed, if $zv \in \mathcal{Z}_b$, it equals zc for some $c \in \mathcal{Z}_b$, so $v - c \in \ker z$. But

$$(zV')^{\perp} = z^{-1}(\mathcal{Z}_b), \qquad (zV \cap V')^{\perp} = \ker z + \mathcal{Z}_b.$$

The complements agree, hence so do the spaces. This proves the required inclusion, with no appeal to semisimplicity. □

11.5 Correcting the error on one layer

The correction has a reusable algebraic form. Write $\pi : V \to T_b$ for the quotient and $\iota_{\mathcal{Z}} : \mathcal{Z}_b \to V$ for inclusion. Since both spaces are stable under labels, z induces operators $z_T, z_{\mathcal{Z}}$ satisfying

$$\pi z = z_T \pi, \qquad z\iota_{\mathcal{Z}} = \iota_{\mathcal{Z}} z_{\mathcal{Z}}.$$

The distinction between these three domains keeps the construction independent of any choice of representatives. The maps w_C and w_T below are auxiliary inverses on these two spaces, not yet a common group-algebra witness. They construct one correction matrix; realisation by permutations is the separate task of the next chapter.

The auxiliary maps reverse only the relevant arrows in nilpotent chains. On a pair $c \mapsto zc \mapsto 0$, for example, w_C sends zc back to c; it is not an inverse of z on the whole space. The correction will control its output with w_C on $\mathcal{Z}_b$ and its input with w_T on T_b. Composition is read from right to left through these two different spaces. The two inverse conditions in the next lemma are what make the defect cancel; we do not invert the residual $\bar{\varepsilon}$.

Lemma 11.6 (The two-sided correction contract)**.** *Let $1 \leq b \leq N$, and suppose $w_0 \in K[S_N]$ satisfies $zw_0z = z$ on every tuple length below b. On $V = P_b(N)$ write its residual as*

$$\varepsilon = zw_0z - z = \iota_{\mathcal{Z}}\bar{\varepsilon}\pi, \qquad \bar{\varepsilon} : T_b \longrightarrow \mathcal{Z}_b.$$

Suppose position-equivariant maps $w_C : \mathcal{Z}_b \to \mathcal{Z}_b$ and $w_T : T_b \to T_b$ satisfy the two inverse conditions

$$z_{\mathcal{Z}} w_C \bar{\varepsilon} = \bar{\varepsilon}, \qquad \bar{\varepsilon} w_T z_T = \bar{\varepsilon}. \tag{11.7}$$

Then the position-equivariant correction

$$\bar{\varphi} = -w_C \bar{\varepsilon} w_T : T_b \to \mathcal{Z}_b, \qquad \varphi = \iota_{\mathcal{Z}} \bar{\varphi} \pi : V \to V$$

has image in $\mathcal{Z}_b$, kills V', and satisfies $z\varphi z = -\varepsilon$. If an element $j \in J_{b-1}(N)$ realises φ, then $w_0 + j$ is a witness on every length at most b.

Proof. The factorisation through quotient and inclusion gives the two shape conditions. The residual $\bar{\varepsilon}$ commutes with positions because the label group-algebra action does, and hence so does φ. The inverse conditions give the fully typed calculation

$$z\varphi z = -\iota_{\mathcal{Z}} z_{\mathcal{Z}} w_C \bar{\varepsilon} w_T z_T \pi = -\iota_{\mathcal{Z}} \bar{\varepsilon} \pi = -\varepsilon.$$

Thus $z(w_0 + j)z - z = \varepsilon + zjz = 0$ on V. On each shorter tuple module j is zero, so its addition preserves the old witness identity. □

11.5.1 Verifying the two inverse conditions

Return to the residual constructed at the start of the chapter. It kills V' and has image in $\mathcal{Z}_b$, so its factorisation through $\bar{\varepsilon}$ is available. Its image also lies in zV. The first assertion of Lemma 11.5 therefore gives $\operatorname{im}\varepsilon \subseteq z\mathcal{Z}_b$, and the third gives

$$z_{\mathcal{Z}} w_C \bar{\varepsilon} = \bar{\varepsilon}.$$

Here we use the surjectivity of π and injectivity of $\iota_{\mathcal{Z}}$ to express the identity on the indicated domains. This is the first condition of (11.7).

For the other side, if $zv \in V'$, the second interface assertion gives $zv = zv'$ for some $v' \in V'$. Since $\varepsilon = (zw_0 - I)z$,

$$\varepsilon v = (zw_0 - I)zv' = \varepsilon v' = 0.$$

Now $z_T w_T z_T = z_T$ by the fourth assertion. For any representative $v_T \in V$ of $w_T z_T \pi(v)$, it follows that $z(v_T - v) \in V'$. The implication just proved gives $\varepsilon v_T = \varepsilon v$, or

$$\bar{\varepsilon} w_T z_T \pi = \bar{\varepsilon} \pi.$$

Surjectivity of π gives the second inverse condition. This calculation needs no section $T_b \to V$: the value of ε is independent of representatives because it kills V'.

The contract now supplies the correction. Theorem 12.5, proved in the next chapter, realises its shape by an actual $j \in J_{b-1}(N)$ using the existing labels. This is the step that replaces the fresh-label compression of the quadratic construction.

Theorem 11.7 (The characteristic-two linear witness)**.** *Let $q = 2^a > r$, and let $n \geq \max\{0, 2r-1\}$. With the independent block coefficients specified in Chapter 9, one common element $w \in K[S_{n+q}]$ satisfies $zwz = z$ on every $P_b(n+q)$, $0 \leq b \leq r$. Deletion is therefore exact on $\mathcal{C}_r(n+q)$, with the natural basis and tensor maps proved there.*

Proof. At length zero take $w_0 = 0$. Suppose one element w_{b-1} works on lengths $0, \ldots, b-1$. Its residual on length b has the shape proved at the beginning of this chapter. The layer interface verifies the two inverse conditions, and Lemma 11.6 constructs φ with $z\varphi z = -\varepsilon$. Theorem 12.5, whose complete proof is in the next chapter, realises it by $j \in J_{b-1}(n+q)$. Set $w_b = w_{b-1} + j$. Then

$$zw_b z - z = \varepsilon + zjz = 0$$

on length b. On every shorter length j is zero, so the old witness identity is unchanged. Every step uses only the same labels; $n \geq 2r-1$ implies $n \geq 2b-1$ for all $b \leq r$. Induction gives $w = w_r$. The common-projector proof of Chapter 9 gives exactness, while its orbit calculation gives the natural maps. □

Why the cost is a maximum

The filtration sections at length b require $2b - \ell$ retained labels for its layer with ℓ block positions. All such layers use the same retained set, so their total requirement is $2b - 1$. The group-algebra realisation then uses that same board without adding labels. At the next tuple length, the new residual is again special and the same pool can be used. The final requirement is therefore

$$\max_{1 \le b \le r} (2b - 1) = 2r - 1,$$

not $\sum_b b$. The entire improvement is a change in how labels are allocated to correction maps. It changes the onset for a fixed degree budget from quadratic to linear. The later consequence classification inverts that onset to obtain the degree lower bound.

Chapter 12

Realising the lift's corrections by permutations

The property we need

The previous chapter constructed a correction with zero marginals in its outputs and zero incoming shorter layers. To repair a common witness, that correction must be an actual linear combination of label permutations. A different matrix correction for each tuple length would not be sufficient. We now prove the required realisation statement from finite combinatorics, in this main chapter.

Recall the spaces on which that correction acts: $V = P_b(N)$ is the ordered injective-b-tuple space, $V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}$ contains the incoming shorter-tuple directions, and $\mathcal{Z}_b = \bigcap_i \ker \partial_i$ is the zero-marginal space. We want a single element of $J_{b-1}(N)$, the ideal of permutation combinations acting as zero on every tuple length below b, whose action on V is the required correction.

The proof has two parts. First, values on a carefully selected set of rook placements determine any zero-line-sum function. Second, those selected values can be prescribed by triangular elimination in permutation coefficients. The triangular system has diagonal coefficient one, so it can be solved by subtraction over any commutative ring. It needs no averaging, semisimplicity or division by a factorial.

The order of the board is a bookkeeping tool for this construction. It is not a built-in order in the uniform polynomial input. We are free to order the existing finite labels while constructing a group-algebra element; the final element acts on the original uninterpreted labels. No label is added.

12.1 Rook coordinates and realisation

The representation-theoretic term for maps commuting with every permutation of tuple positions is their *centraliser algebra*. Addition and composition preserve that condition. Label permutations commute with position permutations, so their linear combinations belong to this algebra. The realisation problem asks whether the particular correction matrix can be obtained from such a combination, with its required zero action on shorter tuples as well. Commutation alone does not supply the coefficients; the following coordinate construction does.

The correction contract in Lemma 11.6 leaves one finite spanning problem: realise the correction's whole matrix by permutations. Rook coordinates express its two shape conditions

as line sums. The argument is independent of the block and works over any commutative ring.

12.1.1 From a map to a board of non-attacking rooks

A matrix entry of a position-equivariant map $\varphi : P_b(N) \to P_b(N)$ is indexed by an output tuple s and input tuple t. Permuting their positions together leaves the entry unchanged. Thus it depends only on

$$\rho = \{(s_i, t_i) : i = 1, \dots, b\},$$

a set of b cells with distinct rows and distinct columns on an $N \times N$ board. Call this a b-rook. Board rows and columns are *labels*; they are different from the positions of the original tuples.

For instance, output $(3, 4)$ and input $(1, 2)$ give the cells $\{(3, 1), (4, 2)\}$. Reordering both tuples to $(4, 3)$ and $(2, 1)$ gives the same cells, so position equivariance gives the same matrix entry. Reordering only one tuple would give different cells. Passing to rook coordinates forgets the simultaneous ordering, while retaining which input label is paired with which output label.

The matrix therefore defines a function F on b-rooks. For a $(b-1)$-rook σ, choose an unused column y. A column line through σ is the family of rooks $\sigma \cup \{(x, y)\}$ as x ranges over unused rows. Define row lines analogously. The map has image in $\mathcal{Z}_b$ precisely when its column lines sum to zero. It kills V' precisely when its row lines sum to zero. Write $\mathcal{R}_b^0$ for this zero-line-sum space of rook functions.

A label permutation g has graph $\{(g(y), y)\}$. Its matrix entry is one exactly when ρ is contained in that graph. Set

$$\chi_g(\rho) = [\rho \subseteq \operatorname{graph} g], \qquad L_b\Big(\sum_g a_g g\Big) = \sum_g a_g \chi_g$$

on b-rooks. For a fixed unused row or column, exactly one extension of a rook σ lies in the graph of g if σ itself does, and none otherwise. Therefore

$$\text{each line sum of } L_b(x) \text{ through } \sigma = L_{b-1}(x)(\sigma). \tag{12.1}$$

If $F \in \mathcal{R}_b^0$ is realised as $L_b(x)$, this identity forces $L_{b-1}(x) = 0$, then every smaller $L_j(x) = 0$. Thus $x \in J_{b-1}(N)$ automatically. It remains to prove that every $F \in \mathcal{R}_b^0$ is realisable.

Here an indicator $[\rho \subseteq \operatorname{graph} g]$ is the scalar 1 or 0 according to whether the displayed containment holds. The map L_b reads the matrix of a permutation combination in these coordinates; it is unrelated to the lower-degree polynomial space $L_d(n)$. A line sum varies one cell while the other $b-1$ cells are held fixed. It is the rook-coordinate version of a marginal.

12.1.2 A first example: matrices with zero row and column sums

At $b = 1$, $\mathcal{R}_1^0$ is the space of square matrices with zero row and column sums. If row and column N are designated as last, such a matrix is determined by its upper-left $(N-1)\times(N-1)$ entries and is

$$\sum_{i,j<N} F_{ij}(E_{ij} - E_{iN} - E_{Nj} + E_{NN}).$$

Each four-entry matrix is a difference of two permutation matrices: choose a permutation taking column j to row i and column N to row N, and subtract the permutation with these two outputs swapped. The other entries cancel. This proves realisation at length one over every ring. The general rook argument is a higher-length replacement for this elementary construction.

12.1.3 Length two: a four-term correction

Take four labels and disjoint swaps $\tau_1 = (13)$ and $\tau_2 = (24)$. The group-algebra element

$$j = (1 - \tau_1)(1 - \tau_2) = 1 - \tau_1 - \tau_2 + \tau_1\tau_2$$

annihilates every one-tuple. A one-tuple omits at least one whole pair; the corresponding swap fixes it, and the disjoint factors commute. On the two-tuple $(1, 2)$ it gives

$$e_{12} - e_{14} - e_{32} + e_{34}.$$

Deleting either position makes this vector zero. More generally, its matrix has zero output and input line sums because j kills one-tuples and commutes with the marginal maps. This is a concrete correction in $J_1(4)$.

12.1.4 Length three: an eight-term correction

On six labels take $\tau_1 = (14)$, $\tau_2 = (25)$ and $\tau_3 = (36)$. Then $j = (1 - \tau_1)(1 - \tau_2)(1 - \tau_3)$ annihilates every tuple of length at most two: such a tuple leaves one whole pair unused. Its action on $(1, 2, 3)$ is

$$je_{123} = e_{123} - e_{423} - e_{153} - e_{126} + e_{453} + e_{426} + e_{156} - e_{456}.$$

Each marginal pairs terms differing only in the deleted coordinate, with opposite coefficients. In characteristic two the signs coincide and the paired terms still cancel. Thus $j \in J_2(6)$ provides a literal length-three example of the shape required by the correction contract.

These products illustrate some realisable corrections. They do not prove that *every* correction is realisable, and their use of $2b$ displayed labels is not a new bound on the retained-label pool. The forcing theorem below proves the spanning statement on the existing $N \times N$ board, for arbitrary zero-line-sum data.

12.2 A forcing set of coordinates

Work on any two equally large finite sets of integer labels R, C, of size M, with their inherited orders. In this combinatorial proof M is the board size and c is the number of rooks; the application later sets $M = N$ and $c = b$. Numerical thresholds such as $y + 1$ denote the cut strictly above the integer label y. A permutation here is a bijection $g : C \to R$. An increasing chain is a subset of its graph increasing in both coordinates. Let $\mathrm{LIS}(g)$ be the largest chain length. Among the column sets of longest chains choose the lexicographically first one, I_g: at the first differing column, the set containing that column is earlier. Define

$$\rho_g = \{(g(y), y) : y \notin I_g\}.$$

If $\mathrm{LIS}(g) = M - c$, this is a c-rook. Let Ψ_c consist of these rooks.

The longest increasing subsequence can be read directly from the list $g(1), \ldots, g(M)$ when the labels are $1, \ldots, M$: choose increasing column numbers whose listed values also increase. For the values $(2, 1, 3)$ there are two length-two choices, with columns $\{1, 3\}$ and $\{2, 3\}$. The first is lexicographically earlier because it contains the first differing column, 1. Removing it leaves the cell $(1, 2)$ as the canonical complement. This rule fixes a coordinate even when the longest chain is not unique.

To call Ψ_c a *forcing set* means that a zero-line-sum function is determined by its values there. If two such functions agree on Ψ_c, their difference vanishes there, and the theorem below will show that it vanishes everywhere. This is uniqueness under the line equations, rather than a statement about arbitrary functions on the board. Existence of permutation coefficients is a separate task after forcing has been proved.

Theorem 12.1 (Forcing)**.** *A function in $\mathcal{R}_c^0$ that vanishes on Ψ_c vanishes on every c-rook.*

Example 12.2 (The 3×3 board)**.** We write permutations in one-line notation: 312 means $g(1) = 3$, $g(2) = 1$, $g(3) = 2$. The permutations with longest increasing subsequence of length two are $312, 231, 213, 132$. Their canonical complement rooks are respectively

$$(3,1),\ (1,3),\ (1,2),\ (2,3).$$

If a zero-line-sum matrix is zero on these four cells, row one first forces $(1,1)$ to be zero. Column one then forces $(2,1)$; row two forces $(2,2)$; column two forces $(3,2)$; and row three forces $(3,3)$. This is the small model of the forcing argument.

12.2.1 Why a longest chain determines a useful coordinate

A permutation is a complete placement of M rooks. Removing a longest increasing chain leaves its canonical complement ρ_g. The chain can be reconstructed once that complement is fixed: the remaining columns and rows must be paired in increasing order. The forcing coordinates therefore encode a rigid increasing completion, while still having the same number of cells as the corrections we want to realise.

The lexicographic choice controls ties when a corner is added. The two contractions in Example 12.4 show why a minimum-row corner can be allowed a tie, whereas a minimum-column corner must be excluded from every longest chain. To keep that information through induction, we pair a chain bound above a cut with a vanishing condition based on unused labels below it.

12.2.2 Strengthened induction statement

To remove the bottom row and the first column, we must know which canonical longest chains survive contraction. Bounds on longest chains in upper parts of the board keep track of this. These are bookkeeping conditions for the induction, not new mathematical assumptions on F.

Let $L = M - c$. A condition is (s, θ, η), where s is "row" or "column", θ is a label threshold, and $\eta \geq 1$. A bijection satisfies it if the longest increasing chain using only cells with that coordinate at least θ has length at most $L - \eta$. Bounds are interpreted as integer inequalities; a negative upper bound is impossible for a nonnegative chain length.

For a collection $\mathcal{E}$ of conditions let $\Psi(\mathcal{E})$ be the canonical complement rooks of permutations with $\mathrm{LIS} = L$ satisfying every condition. Let $Q(\mathcal{E})$ be the set of c-rooks for which, for at least one condition, at most $\eta - 1$ *unused* labels of the specified side lie below θ.

The two descriptions are paired deliberately. A bound on chains above a threshold will, through a line-sum identity, force a value at a rook with too few unused labels below it.

An induction invariant is the collection of facts carried to each smaller board. Here M is its size; for a c-rook, c is the number of occupied cells, and $L = M - c$ is the number of unused rows and also of unused columns. A condition with deficit η permits at most $L - \eta$ members of an increasing chain above its cut. The set $\Psi(\mathcal{E})$ records canonical coordinates obeying those chain restrictions; $Q(\mathcal{E})$ records additional zeros established through line equations. These are different kinds of information. For example, with row labels $1, 2, 3, 4$, the row condition with cut 3 and $\eta = 2$ puts a one-rook in Q when at most one of rows $1, 2$ is unused, that is, when its occupied row is 1 or 2. The extra conditions below are introduced within the induction; the desired forcing theorem starts with no such conditions.

Lemma 12.3 (Strengthened forcing)**.** *If $F \in \mathcal{R}_c^0$ vanishes on $\Psi(\mathcal{E}) \cup Q(\mathcal{E})$, then $F = 0$.*

Theorem 12.1 is its case of no conditions.

The *induction invariant* is the following data at the current board:

$$L = M - c, \qquad F \in \mathcal{R}_c^0, \qquad F|_{\Psi(\mathcal{E})} = 0, \qquad F|_{Q(\mathcal{E})} = 0.$$

Each use of induction must preserve the three assertions about F with the updated conditions. The first two sweeps replace (M, c) by $(M-1, c-1)$, keeping L fixed; the final restriction replaces it by $(M-1, c)$, decreasing L by one. The conditions record exactly what these contractions change.

12.2.3 Three facts about adding a corner cell

Let $r_0 = \min R$ and $y_0 = \min C$. The following statements describe the only possible new increasing chains.

1. Remove row r_0 and column y, and let $\widetilde{g}$ be a bijection on the smaller board with longest chain length L. Add the cell (r_0, y). Every new chain using it continues with a chain in columns strictly above y. If that region had longest chain at most $L-1$, the new permutation has longest chain L. Its lexicographically first longest chain is still the old one: the old chain must use some column below y, whereas every new chain through the corner uses columns at least y. Therefore $\rho_g = \rho_{\widetilde{g}} \cup \{(r_0, y)\}$. For a column threshold $\theta \le y$, the new regional maximum is

$$\max\{\mathrm{LIS}(\widetilde{g}|_{\mathrm{col}\ge\theta}),\ 1 + \mathrm{LIS}(\widetilde{g}|_{\mathrm{col}>y})\}.$$

2. Remove row x and column y_0, and add (x, y_0). If the old region of rows strictly above x has longest chain at most $L-2$, every chain using the new corner has length at most $L-1$. Thus all longest chains remain old, the canonical one remains unchanged, and $\rho_g = \rho_{\widetilde{g}} \cup \{(x, y_0)\}$. The analogous maximum formula holds for row thresholds at most x. The stronger deficit two eliminates a tie involving the smallest column.
3. Remove both minimal labels and add (r_0, y_0). This corner can be placed before every increasing chain of the smaller board. The longest length increases by one, every new longest chain contains the corner, and $I_g = \{y_0\} \cup I_{\widetilde{g}}$. Thus $\rho_g = \rho_{\widetilde{g}}$.

These facts follow just by splitting chains according to whether they use the added cell. They do not rely on a representation-theoretic result.

Example 12.4 (One board, two contractions)**.** Take $M = 4, c = 1$, so $L = 3$, and the permutation 2134. Rows increase upwards and columns to the right. The vertical rule marks the cut at column 3; the starred cell is the corner $(1, 2)$ to be contracted:

	1	2	3	4
4	·	·	·	●
3	·	·	●	·
2	●	·	·	·
1	·	★	·	·

remove row 1 and column 2 →

	1	3	4
4	·	·	●
3	·	●	·
2	●	·	·

The contracted permutation has the increasing chain with columns $\{1, 3, 4\}$. Above the cut, its chain length is $2 = L - 1$. Adding the corner creates another length-three chain with columns $\{2, 3, 4\}$, but $\{1, 3, 4\}$ is lexicographically earlier. Thus deficit one suffices to preserve the canonical chain in corner fact 1. The removed column 2 is absent from the contracted label set; the labels below the cut are now just $\{1\}$. For a rook on that board, the corresponding Q condition counts whether this remaining label is unused.

Instead contract $(2,1)$ from the same full board. The horizontal rule now marks the row cut at 3:

	2	3	4
4	·	·	●
3	·	●	·
1	●	·	·

The old longest chain uses columns $\{2,3,4\}$, and the region of rows above 2 again has length $L-1=2$. Adding $(2,1)$ creates the competing chain $\{1,3,4\}$, which is now earlier. This explains the stronger requirement in corner fact 2: imposing regional length at most $L-2$ makes every chain through the added corner have length at most $L-1$, so there can be no tie. The example exhibits why the two sweeps require different deficits.

12.2.4 Reading the cuts as integer inequalities

Row labels and column labels may have gaps; they need not be $1,\dots,M$ after a contraction. The notation $y+1$ means the integer cut strictly above the particular label y. On the contracted column set, which excludes y, being below $y+1$ is exactly being strictly below y. This observation justifies the unused-label counts in both sweeps.

The inequalities are ordinary integer inequalities. In particular, $\mathrm{LIS} \le L-\eta$ is impossible if $L-\eta<0$; it does not silently mean a truncated natural-number bound of zero.

12.2.5 Base cases and degenerate conditions

The proof uses two levels of induction. The outer induction reduces the board size M. Within a fixed board, a decreasing sweep remembers which corner choices have already been eliminated. Contracting a specified corner means fixing that cell and deleting its row and column: $G(\sigma)=F(\sigma\cup\{\text{corner}\})$. Thus (M,c) becomes $(M-1,c-1)$ and L stays unchanged. The final restriction has no fixed cell and leaves c unchanged, so L drops by one. Keeping the integer row and column names, including their gaps, makes the threshold conditions unambiguous.

Prove Lemma 12.3 by induction on M. If $c>M$ there are no rooks. If $c=M>0$, each line through an $(M-1)$-rook contains exactly one M-rook, so the zero-line-sum equation forces its value to zero.

A condition is degenerate if the entire board has at most $\eta-1$ labels below its threshold. Then every rook belongs to Q, and the conclusion is immediate. Assume no condition is degenerate. Every column threshold is therefore above y_0 and every row threshold above r_0.

If $c=0$, consider the increasing bijection. It has longest length M. For a column condition, at least η columns lie below θ, so the part above θ has length at most $M-\eta$; the same holds for rows. Thus its empty complement rook belongs to $\Psi(\mathcal{E})$, and the sole value of F is zero. We may now assume $1\le c<M$.

12.2.6 First sweep: eliminate every rook meeting the bottom row

For $y\in C$ in decreasing order, contract the cell (r_0,y) and define

$$G(\sigma)=F(\sigma\cup\{(r_0,y)\})$$

on $(c-1)$-rooks of the board with that row and column removed. Its line sums are line sums of F, so $G\in\mathcal{R}^0_{c-1}$. The smaller board has size $M-1$ and complement length $(M-1)-(c-1)=L$.

Retain the old conditions and add

$$\begin{aligned}&(\text{column}, y+1, 1),\\ &(\text{column}, y+1, \eta+1) \quad \text{for each old } (\text{column}, \theta, \eta) \text{ with } \theta \le y.\end{aligned}$$

The first new condition bounds the right region by $L-1$, giving corner fact 1. The second bounds it by $L-(\eta+1)$ whenever needed. An old column condition with $\theta \le y$ is preserved upon adding the corner because

$$\max\{L-\eta, 1+L-(\eta+1)\} = L-\eta.$$

Old column conditions above y and old row conditions are unaffected by the corner. Thus a canonical rook satisfying the new conditions extends to one in $\Psi(\mathcal{E})$, and G vanishes on its new Ψ set.

We must also verify every case of its new Q set. An old condition passes directly to the enlarged rook: the deleted row and column are occupied by the added cell, so the count of unused labels below an old threshold is unchanged.

For a new condition $(\text{column}, y+1, e)$, let

$$U = \{y' \in C : y' < y,\ y' \text{ unused by } \sigma\}.$$

On the contracted board this is exactly the unused set below $y+1$, so $|U| \le e-1$. If $e=1$, there are no such columns. If the condition came from an old $(\text{column}, \theta, \eta)$, then $e = \eta+1$ and $|U| \le \eta$. For each $y' \in U$, the enlarged rook $\sigma \cup \{(r_0, y')\}$ has at most $\eta-1$ unused columns below θ. Indeed, if $y' < \theta$, its occupation removes one member from a subset of U; if $y' \ge \theta$, y' itself lies outside that subset, again leaving at most $|U|-1$. Thus every such rook lies in the old Q set.

Now use the row line through σ in row r_0. Its terms at columns greater than y are zero by the decreasing inner induction. Its terms below y are either absent ($e=1$) or zero by the old Q condition. The only remaining term is $G(\sigma)$, so it is zero. This proves vanishing on the new Q set in all cases. The outer induction on M now gives $G=0$.

Completing the decreasing sweep shows that F vanishes on every rook meeting row r_0.

12.2.7 A sweep term with its unused labels displayed

Here is one instance of the new-Q verification. Use row and column labels $1, \dots, 7$, take $c=3$ and hence $L=4$, and suppose an old condition is $(\text{column}, 3, 1)$. In the bottom-row sweep at $y=4$, contract row 1 and column 4. The smaller board has $(M, c, L) = (6, 2, 4)$; its column labels are $1, 2, 3, 5, 6, 7$. The new condition $(\text{column}, 5, 2)$ comes from that old one.

Choose the smaller rook $\sigma = \{(2,1), (3,3)\}$. Below the new cut 5, the contracted board has columns $1, 2, 3$, of which only 2 is unused. Thus σ is in the new Q set since $1 \le 2-1$. The row-line equation on the original board is exactly

$$0 = F(\sigma \cup \{(1,2)\}) + F(\sigma \cup \{(1,4)\}) + \sum_{y' \in \{5,6,7\}} F(\sigma \cup \{(1,y')\}).$$

The last three terms were eliminated earlier in the decreasing sweep. In the first term, both columns below the *old* cut 3, namely $1, 2$, are occupied. Its unused count is zero, so it lies in the old Q set and is also zero. The remaining term is the contracted value $G(\sigma)$, which is therefore zero. This trace shows why unused labels are counted after contraction, why the deficit increases by one, and how the new vanishing condition is proved by a line equation.

12.2.8 Second sweep: eliminate every rook meeting the first column

Adjoin the condition $(\mathrm{row}, r_0+1, 1)$ to obtain $\mathcal{E}^+$. Its new Q rooks are exactly those using row r_0: there must be no unused row below r_0+1, and that row is the only candidate. They have just been proved zero. Its Ψ set is a subset of the old one. Thus the vanishing assumptions still hold for $\mathcal{E}^+$.

Sweep $x \in R \setminus \{r_0\}$ in decreasing order. Contract (x, y_0) and keep every old condition in $\mathcal{E}^+$. For each row condition $(\mathrm{row}, \theta, \eta)$ with $\theta \le x$, add $(\mathrm{row}, x+1, \eta+1)$. In particular the recently added condition produces $(\mathrm{row}, x+1, 2)$, because $x > r_0$. Therefore the region above x has maximum at most $L-2$, exactly the hypothesis of corner fact 2.

The verification for Ψ is now explicit: row thresholds at most x use the same maximum calculation as in the first sweep; row thresholds above x are unchanged; every column threshold exceeds y_0, so excludes the new corner. For Q, old conditions pass unchanged. A new condition from deficit η says that there are at most η unused rows below x. When any such row x' is occupied instead of x, the number unused below its source threshold is at most $\eta-1$, by the same two cases $x' < \theta$ and $x' \ge \theta$. These terms lie in $Q(\mathcal{E}^+)$. The column-line equation at y_0 then forces the remaining term to zero, since larger rows were handled earlier. The induction on M proves the contracted function zero.

The case $x = r_0$ was already covered by the first sweep. Thus F now vanishes on every rook meeting either row r_0 or column y_0.

12.2.9 The remaining smaller board

Restrict F to $(R \setminus \{r_0\}) \times (C \setminus \{y_0\})$ without changing c. A line in this board is an old line minus one term using a removed label; that term is zero, so the restriction has zero line sums.

For every condition in $\mathcal{E}^+$ with deficit $\eta \ge 2$, keep its side and threshold and replace its deficit by $\eta-1$. Discard the conditions with deficit one. The smaller complement length is $L-1$.

A permutation on the smaller board satisfying these conditions and having longest length $L-1$ extends by the minimal corner. Corner fact 3 gives longest length L and the same complement rook. Every old threshold excludes that corner. For $\eta \ge 2$ the regional bound is

$$(L-1) - (\eta-1) = L - \eta;$$

for $\eta = 1$ it follows from the whole smaller board having longest length $L-1$. So the restricted function vanishes on the new Ψ set.

If a smaller rook belongs to the new Q set, its unused count below a threshold is at most $\eta-2$. In the original board the removed minimum label is also unused by this rook and lies below that threshold, increasing the count by exactly one. The resulting count is at most $\eta-1$, so the rook lies in the old $Q(\mathcal{E}^+)$. Thus the new Q vanishes too. The induction on M finishes the proof of Lemma 12.3, and hence of the forcing theorem.

12.2.10 What each part of the induction accomplished

The chain conditions ensure that canonical coordinates survive adding a corner. The unused-label conditions ensure, through a line equation, that the other terms of a sweep have already vanished. Once both sweeps are finished, restriction loses only zero terms. The final deficit change preserves the same induction invariant with L decreased by one. These two records let a local contraction prove a global forcing theorem.

12.3 Triangular realisation and the common correction

The forcing theorem says that the selected values on Ψ_b determine a zero-line-sum function. We now prescribe those values by permutation coefficients while simultaneously making every smaller tuple action zero. Recall that $L_k(x)$ records the matrix entries of a permutation combination x on k-rooks.

This second ingredient is simpler. If ρ_g lies on the graph of $g' \neq g$, then g' agrees with g outside I_g and permutes the remaining values $g(I_g)$. On I_g the values of g are increasing. At the first column where g' differs, it must use a larger remaining value. Thus g' is lexicographically larger than g.

This is triangular elimination in the ordinary linear-algebra sense. Equations of the form

$$a_i + \sum_{j>i} t_{ij} a_j = b_i$$

are solved from the last unknown backwards. For example, $a_1 + a_2 = b_1$, $a_2 = b_2$ give $a_2 = b_2$ and $a_1 = b_1 - b_2$. No division is needed because the coefficient of each new unknown is 1. In the present system, t_{ij} is the indicator that the later permutation contains the earlier one's canonical rook. The prescribed right-hand side is F at length b and zero at every shorter length. Solving those lower equations is what preserves the witness identities on all shorter tuple spaces.

Fix $F \in \mathcal{R}_b^0(N)$ and let $\mathcal{T} = \{g : \mathrm{LIS}(g) \geq N - b\}$. Its canonical complement rooks have sizes at most b. Define coefficients a_g in decreasing lexicographic order by

$$a_g = \begin{cases} F(\rho_g) & |\rho_g| = b, \\ 0 & |\rho_g| < b \end{cases} - \sum_{\substack{g' \in \mathcal{T},\ g' \neq g \\ \rho_g \subseteq \mathrm{graph}\, g'}} a_{g'}. \tag{12.2}$$

Every coefficient on the right was already defined by the preceding observation. Put $x = \sum_{g \in \mathcal{T}} a_g g$, and let $H_k = L_k(x)$ for each $k \leq b$. The construction makes H_k zero on Ψ_k for $k < b$ and makes $H_b = F$ on Ψ_b.

At level zero the only coordinate is the empty rook, and it is prescribed zero when $b > 0$. Inductively suppose $H_{k-1} = 0$. Equation (12.1) shows that H_k has zero line sums. For $k < b$, it vanishes on Ψ_k, so forcing gives $H_k = 0$. At $k = b$, $H_b - F$ has zero line sums and vanishes on Ψ_b, so forcing gives $H_b = F$. The case $b = 0$ is immediate by a scalar multiple of any permutation.

We have proved that every zero-line-sum rook function is realised by a group-algebra element, and the realising element annihilates all smaller tuple modules. Translating back to matrices proves:

Theorem 12.5 (Realisation of the layer correction)**.** *For $1 \leq b \leq N$, every position-equivariant map $\varphi : P_b(N) \to P_b(N)$ with image in $\mathcal{Z}_b(N)$ and killing V' is the action of an element $j \in J_{b-1}(N)$. No additional labels are required.*

Example 12.6 (Actual permutation coefficients)**.** On a 3×3 board take

$$F = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

The recursion gives coefficients $-1, -1, 2, 1, -1$ for $312, 231, 213, 132, 123$, respectively. Therefore

$$F = -P_{312} - P_{231} + 2P_{213} + P_{132} - P_{123}.$$

Here P_g has its one in row $g(y)$ of column y. The coefficients sum to zero, so the action on the empty tuple is zero, as required for J_0. Reducing this integer identity modulo two or any odd prime gives the corresponding realisation in that characteristic.

Example 12.7 (The six-term cycle from a two-rook function)**.** Take $b = 2, N = 3$. Every two-rook ρ has a unique completion to a permutation g_ρ: its one unused column must be paired with its one unused row. Define

$$F(\rho) = \operatorname{sign}(g_\rho).$$

Each row or column line has two rooks, and their completions differ by a transposition. Their signs add to zero, so $F \in \mathcal{R}_2^0$. The only permutation with longest chain length $N - b = 1$ is 321. Its chosen chain uses column 1, and hence

$$\Psi_2 = \{\{(2,2),(1,3)\}\}, \qquad F(\rho_{321}) = -1.$$

This is the single forcing coordinate. In the triangular construction, every smaller canonical rook is prescribed zero. The complete decreasing recursion is:

g	ρ_g	a_g
321	$\{(2,2),(1,3)\}$	-1
312	$\{(3,1)\}$	$0 - a_{321} = 1$
231	$\{(1,3)\}$	$0 - a_{321} = 1$
213	$\{(1,2)\}$	$0 - a_{312} = -1$
132	$\{(2,3)\}$	$0 - a_{312} = -1$
123	$\varnothing$	$0 - (a_{321} + a_{312} + a_{231} + a_{213} + a_{132}) = 1$

Thus the realising element is

$$j = 123 - 132 - 213 + 231 + 312 - 321 = \sum_{g \in S_3} \operatorname{sign}(g) g.$$

Here the three-digit strings denote permutations in the group algebra. On a one-tuple, the two permutations taking its label to any fixed output label have opposite signs; on the empty tuple, all six coefficients sum to zero. Consequently $j \in J_1(3)$. Its action on e_{12} is

$$je_{12} = e_{12} + e_{23} + e_{31} - e_{21} - e_{32} - e_{13},$$

the zero-marginal cycle of the preceding chapter. Also $L_2(j) = F$ entry by entry, because only g_ρ contributes at ρ. This example connects the tuple vector, its forcing coordinate and the actual permutation coefficients. Reducing the integer calculation in characteristic two turns every sign into one while preserving all cancellations.

12.4 Why the triangular system has diagonal one

The graph of g contains ρ_g, so its contribution at that coordinate is exactly one. The lexicographic argument above places every other contributing permutation later. Subtraction in decreasing order therefore solves the system without dividing by a dimension, orbit size or group order, even over a ring with zero divisors.

The lower-level prescriptions are essential: they make $H_0 = 0$, then make each H_k a zero-line-sum function by (12.1). Forcing successively gives $H_k = 0$ for $k < b$ and $H_b = F$. Thus the same triangular construction proves both realisation and annihilation of shorter tuples, as the table in Example 12.7 shows.

The stated realisation theorem is for $1 \le b \le N$. If $b > N$, the tuple module is zero and its only endomorphism is realised by the zero group-algebra element. In the witness construction we always have $b \le r < q \le N$, so the nonempty-board case is the one being used.

What this theorem gives the lifting proof

Return to $\bar{\varphi} : T_b \to \mathcal{Z}_b$ in the two-sided correction contract and its composite

$$\varphi = \iota_{\mathcal{Z}} \bar{\varphi} \pi : P_b \to P_b.$$

Position equivariance turns this composite into a rook function. Its image in $\mathcal{Z}_b$ gives zero column lines, and its vanishing on V' gives zero row lines. The realisation theorem supplies an actual $j \in K[S_N]$ with this whole matrix, and with $jP_k = 0$ for every $k < b$. Lemma 11.6 therefore applies: adding that single j to the old witness repairs length b and leaves every shorter length unchanged.

No fresh labels were used. The construction only ordered the existing rows and columns, performed finite forcing, and subtracted already known coefficients. It proves the realisation contract in every characteristic, and that is why the same correction mechanism will survive the transition from characteristic two to arbitrary prime characteristic in the next chapter.

Chapter 13

Lifting in every positive characteristic

The property we need

Recall the *block operator* used to construct the lift:

$$z = \sum_i c_i(\sigma_i - I).$$

Here σ_i translates the added block in its ith coordinate, I is the identity operator, and the c_i are the chosen field coefficients. Thus z is a linear operator on our labelled vector spaces, not a scalar. The equation $z^2 = 0$ says that applying z *twice* gives zero: $z(z(v)) = 0$ for every vector v. It does not require $z(v) = 0$ on the first application.

For a concrete example over $\mathbb{F}_2$, subtract the identity from the operator which swaps two coordinates. The resulting matrix is

$$z = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq 0, \qquad z^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Each entry in its square is $1 + 1 = 0$ in the field. Such an operator is called *square-zero*; it sends its own image to zero.

The characteristic-two proof used two special properties of the general block operator: it was square-zero, and its matrix on the permutation basis equalled its transpose. The latter holds because each coordinate translation is an involution, so its permutation matrix is symmetric. In odd characteristic these properties require different replacements. The combinatorial core, however, used signed sums and triangular elimination without division by a group order, and remains available.

This chapter proves every replacement needed for the same linear retained-label budget. The block has $q = p^a$ labels and $z^p = 0$. The deletion quotient must discard every nilpotent chain longer than one, not just the full length-p chains. A common witness for the top power $u = z^{p-1}$ yields a multi-factor projector for that actual quotient. We first prove the requisite purity on the tuple generators, then obtain identities in their common annihilator algebra, and only then apply those identities to arbitrary subspaces and quotients.

The layer splitting now has p levels. Duality uses the block relabelling $v \mapsto -v$ to handle transpose. Finally, the same rook realisation repairs the top-power residual without new labels. These steps supply the exact-deletion theorem in every prime characteristic.

The conclusion has the same progression form as before: fix d, choose $q = p^a$ larger than its support budget, and lift with degree d at every step. Odd characteristic changes the algebra which builds the lift, not the meaning of repeated lifting or the linear label cost.

13.1 The block operator in prime characteristic

A linear operator is *nilpotent* when some number of successive applications gives zero. A nilpotent chain has independent members $c, zc, \dots, z^{j-1}c$, with $z^j c = 0$; applying z moves one step down this chain. These are the elementary blocks often called Jordan chains. The equation $z^p = 0$ bounds their length by p, but does not by itself exclude lengths between 2 and $p-1$. That exclusion is a property to prove for the actual block action.

Let now $\operatorname{char} K = p$ for any prime. Keep $N = n + q$ for the larger universe, with n retained labels and q added block labels. The support budget r bounds the tuple lengths needed by the proof diagram; it is not the block exponent a or the proof degree d. As in the characteristic-two construction, we temporarily extend the coefficient field if needed and write K for the working field below. The descent in Lemma 3.3 preserves the original degree question. Take a regular block $A = \mathbb{F}_p^a$ of size $q = p^a > r$, choose $c_1, \dots, c_a$ independent over $\mathbb{F}_p$ in K, and put

$$z = \sum_i c_i(\sigma_i - 1).$$

The generators commute and have order p, so the binomial formula in characteristic p gives

$$z^p = \sum_i c_i^p(\sigma_i^p - 1) = 0.$$

For $p > 2$ we cannot replace this by $z^2 = 0$. For example, for a three-cycle σ in characteristic three,

$$z = \sigma - I, \qquad z^2 = \sigma^2 + \sigma + I \neq 0$$

on its three-point permutation space.

The binomial step uses the elementary characteristic-p identity $(U+V)^p = U^p + V^p$ for *commuting* operators: each intermediate binomial coefficient is divisible by p. It is valid for the coordinate translations because they commute. It is not a rule for arbitrary matrix sums, nor a statement that powers of unrelated operators commute.

13.1.1 The deletion quotient

Since $\operatorname{im} z$ need not be contained in $\ker z$, the characteristic-two expression $\ker z / \operatorname{im} z$ is no longer defined. Use

$$\mathcal{D}(M) = \ker z / (\ker z \cap \operatorname{im} z). \tag{13.1}$$

A nilpotent chain has basis $v_0, \dots, v_{j-1}$ with $zv_0 = 0$ and $zv_i = v_{i-1}$ for $i \geq 1$. Its kernel is Kv_0. If $j = 1$, its image is zero and the chain survives in (13.1). If $j \geq 2$, v_0 is in the image and the chain disappears. Thus deletion keeps the one-dimensional chains and discards every longer chain.

One might instead try $\ker z / \operatorname{im} z^{p-1}$. It is a defined quotient, but it is the wrong one: a chain of length two in characteristic three would survive because $z^2 = 0$ there, even though it should be discarded by (13.1). Keeping the quotient precise is important.

13.2 The commutator argument in characteristic p

Work on one moved orbit of a permutation basis. Choose a basis object x_0, and let $H = \{v \in A : vx_0 = x_0\}$ be its stabiliser. Because the orbit is moved, H is a proper subspace of A. Choose a nonzero linear functional $\ell : A \to \mathbb{F}_p$ vanishing on H, and define $\pi(vx_0) = \ell(v)$. This is well defined because two translations taking x_0 to the same object differ by an element of H. Let D multiply each basis vector by its orbit coordinate π, and let S_i act by translation

in coordinate i. With ε_i the ith unit vector, put $e_i = \ell(\varepsilon_i)$ and $\alpha = \sum_i c_i e_i \neq 0$. The choices are made orbit by orbit. Exactly as before,

$$Dz - zD = W = \sum_i c_i e_i S_i, \quad Wz = zW, \quad W^p = \alpha^p I.$$

The commutator $Dz - zD$ compares the order of two operations. Here W names that commutator operator, distinct from a polynomial ambient space W_N or a consequence space $W_d(n)$. Its value W is invertible for a concrete reason: $W^{-1} = \alpha^{-p} W^{p-1}$. Independence of the c_i makes the scalar α nonzero. Thus it is W whose inverse is used here, not the nilpotent operator z. Consequently the auxiliary linear operator $B = W^{-1}D$ satisfies

$$[B, z] = Bz - zB = I. \tag{13.2}$$

Here B names an operator, not the set of block labels used in the earlier chapters; brackets denote the displayed commutator of composed operators. This proof works also on a moved permutation basis tensored with an arbitrary representation: D reads only the basis coordinate, while the S_i act diagonally and still have order p.

There is an elementary model for this commutator. On the space with basis $1, T, \dots, T^{p-1}$, work modulo T^p, let z_0 multiply by T, and let B_0 differentiate. Differentiation is well defined modulo T^p because $(T^p)' = pT^{p-1} = 0$. For $j < p-1$, $(B_0 z_0 - z_0 B_0)T^j = ((j+1)-j)T^j = T^j$. At $j = p-1$ the first term is zero and the second has coefficient $-(p-1) = 1$ in the field. Hence $[B_0, z_0] = I$ on the whole space. The actual orbit construction above produces its commutator directly; this model explains why a finite-dimensional commutator can equal the identity in characteristic p.

If $zm = 0$, commuting z past powers of B yields

$$zB^j m = -jB^{j-1} m, \qquad z^j B^j m = (-1)^j j!\, m.$$

The first identity follows by induction from $zB = Bz - I$; applying it repeatedly gives the second. At $j = p-1$ the scalar $(p-1)!$ is nonzero in characteristic p, because every factor $1, \dots, p-1$ is nonzero. Therefore

$$\ker z = \operatorname{im} z^{p-1} \tag{13.3}$$

on the moved summand. The reverse inclusion uses $z^p = 0$.

The displayed formula starts with a kernel vector m and constructs a preimage at the full height $p-1$ by applying B^{p-1} and dividing by $(-1)^{p-1}(p-1)!$. Thus every vector killed by z on a moved orbit already comes from the top power. A full chain is a copy of the truncated polynomial model just described; a direct sum of such chains is what the earlier phrase "free for z" means here. This is a statement about the block operator, not freeness for the full symmetric-group action. Equivalently, let the variable T in the ring $K[T]/(T^p)$ act as z. A full chain generated by c has basis $c, zc, \dots, z^{p-1}c$ and is one free copy of that ring. A direct sum of full chains is therefore a free module for this truncated polynomial ring. This is the precise module meaning of freeness used in the calculation.

This is the full replacement for the square-zero freeness calculation. It does not divide by p, by $|A|$, or by a symmetric-group order. It divides only by a nonzero orbit coefficient and by $(p-1)!$.

On a bounded-support permutation basis the retained summand has $z = 0$ and the moved summand satisfies (13.3). Therefore the identifications with the retained basis and the tensor identifications from Chapter 9 still hold, with quotient (13.1). Stable matrices are transported by exactly the same retained-component argument.

13.2.1 A three-cycle computation

In characteristic three, let σ cycle e_0, e_1, e_2. The operator $z = \sigma - I$ has

$$z^2 = I + \sigma + \sigma^2, \qquad z^2 e_0 = e_0 + e_1 + e_2, \qquad z^3 = 0.$$

Its kernel is the line spanned by $e_0 + e_1 + e_2$, and this line is exactly $\operatorname{im} z^2$. It lies in $\operatorname{im} z$, so the deletion quotient is zero on this moved orbit. A separate fixed basis vector has zero image and survives. This is the prime-characteristic counterpart of the moved two-point calculation, with a full chain of three levels instead of two.

The commutator argument proves the same conclusion for every moved orbit, including those with nontrivial stabilisers and those tensored with an arbitrary representation. The factorial is $(p-1)!$, not $p!$: each factor being divided by is nonzero. It is this explicit calculation that excludes shorter nontrivial nilpotent chains in the original moved tuple modules.

13.3 The top-power witness

Set $m = p - 1$ and $u = z^m$. Here m records this fixed power exponent, rather than a tuple length or support size. Like z, u is a linear operator; $u^2 = 0$ because $2(p-1) \geq p$ and $z^p = 0$. Recall that $J_r(N)$ consists of the group-algebra elements acting as zero on all ordered-tuple spaces $P_b(N)$ with $b \leq r$. The common certificate becomes

$$uwu - u \in J_r(N), \qquad \text{or } uwu = u \text{ on every } P_b(N),\ b \leq r. \tag{13.4}$$

This says that w takes the bottom of a full length-p chain back to a preimage at its top. It is enough to build a projector for the actual quotient, but more steps are needed than in characteristic two.

13.3.1 Fixed directions, full chains, and purity

The condition called *purity* below is $\ker z \cap \operatorname{im} z \subseteq \operatorname{im} z^{p-1}$. It says that a kernel vector which already has a preimage under z also has a preimage under its top power. On a chain of length $j \geq 2$, the bottom vector is in $\ker z \cap \operatorname{im} z$. It is in $\operatorname{im} z^{p-1}$ only when the chain has the full length p. On a fixed line the intersection is zero. This is why purity distinguishes fixed directions and full chains from shorter nontrivial chains.

Definition 13.1 (Fixed and full-chain decomposition)**.** For an operator z with $z^p = 0$, a fixed and full-chain decomposition is

$$V = F \oplus \bigoplus_{j=0}^{p-1} z^j C, \qquad zF = 0,$$

where the sum is direct and $z^{p-1} : C \to z^{p-1}C$ is injective. Thus each basis vector c of C supplies the full chain $z^{p-1}c, \ldots, zc, c$; F consists of fixed directions.

"Fixed" here means killed by z. It need not mean fixed by every label permutation. The direct-sum decomposition is a statement about this nilpotent operator, not an equivariant decomposition for the full symmetric group.

This definition will describe the layer splitting constructed below. For the original tuple modules, purity follows directly from the orbit calculation, without choosing chain bases:

$$\ker z \cap \operatorname{im} z \subseteq \operatorname{im} z^{p-1}. \tag{13.5}$$

The image has no retained component, and a moved kernel vector is in the top-power image by (13.3). The algebraic contract is therefore: prove purity on these generators; use one top-power witness to obtain projector identities there; transfer the identities through $J_r(N)$ to prove exactness on the whole annihilator class. A chosen chain complement need not be compatible with submodules or quotients.

Example 13.2 (Why the distinction matters)**.** In characteristic three, a lone length-two chain has $u = z^2 = 0$. The relation $uwu = u$ is then vacuous for any w, but $\ker z \cap \operatorname{im} z$ is nonzero and $\operatorname{im} u = 0$. Such a chain does not satisfy (13.5). The argument must establish its projector identities on the actual tuple modules first; the top-power relation alone on an arbitrary vector space is insufficient.

13.3.2 The ordered projector

For $p > 2$, the square-zero operator $u = z^{p-1}$ can have a larger kernel than z. In characteristic three, on the chain $zv_2 = v_1$, $zv_1 = v_0$, $zv_0 = 0$, the top power kills v_1, while z does not. If w sends v_0 to v_2 and kills the other two vectors, the characteristic-two expression $(I - uw)(I - wu)$ would leave v_1 unchanged. It therefore cannot model the deletion quotient for z. The ordered product below uses intermediate powers of z to remove every level, not merely the top and bottom ones.

For a vector satisfying $z^{h+1}v = 0$, call h an upper bound on its chain height. This height is independent of polynomial degree. The proof controls the height of the image after each successive factor. The permitted height decreases; the degree budget of a PC proof is not being decreased. Only the completed product is claimed to be the required projector; its factors are not assumed to commute.

Lemma 13.3 (Projector from purity and a top-power witness)**.** *Let z, w be endomorphisms of a vector space over a field of characteristic p, and put $m = p - 1$. Suppose $z^{m+1} = 0$, $z^m w z^m = z^m$, and (13.5) holds. Define*

$$S_0 = I, \qquad S_{k+1} = (I - z^k w z^{m-k})S_k \quad (0 \le k < m), \qquad e = (I - z^m w)S_m. \tag{13.6}$$

Then $ze = ez = 0$, $e^2 = e$, and $ev = v - z^m wv$ whenever $zv = 0$.

Proof. Operator products act from right to left. Their order is part of the construction; no factor is commuted past w.

Eliminating high levels. At step k put $h = m - k$. Inductively $z^{h+1}S_k = 0$; this is true at $k = 0$ because $z^{m+1} = 0$. For $k < m$ we have $h \ge 1$, so $y = z^h S_k v$ belongs to $\ker z \cap \operatorname{im} z$, hence to $\operatorname{im} z^m$. The witness gives $z^m wy = y$, and therefore

$$z^h S_{k+1} v = z^h (I - z^{m-h} w z^h) S_k v = y - z^m wy = 0.$$

The permitted height drops from h to $h - 1$ at each step. Thus $z^{m+1-k}S_k = 0$ for every k, and finally $zS_m = 0$. As $z^{m+1} = 0$, the last factor also gives $ze = 0$.

Action on a kernel vector. If $zv = 0$, every factor in S_m fixes v, because its rightmost power of z has exponent at least one. Therefore

$$ev = v - z^m wv \quad (zv = 0). \tag{13.7}$$

Killing the image. The first factor $I - wz^m$ fixes $\operatorname{im} z$, since $z^{m+1} = 0$. All later factors preserve $\operatorname{im} z$ because their correction terms have a left factor z^k with $k \ge 1$. Hence $S_m zv \in \ker z \cap \operatorname{im} z \subseteq \operatorname{im} z^m$. The final factor $I - z^m w$ kills this vector by the witness relation. Thus $ez = 0$.

Idempotence. The witness gives

$$z^m we = (z^m w - z^m w z^m w)S_m = 0.$$

Since $ev \in \ker z$, formula (13.7) now gives $e^2 v = ev$. We have proved $ze = ez = 0$ and $e^2 = e$. □

The final factor kills the bottom of each full chain as well. Formula (13.7) is especially useful for lifting: only the final correction acts on a kernel vector. It is the identity needed for $\mathsf{L} = e\iota$ and the retained-coordinate right inverse.

13.3.3 Passing from tuples to every relevant space

Recall that $\mathcal{C}_r(N)$ contains every finite-dimensional label representation killed by $J_r(N)$, including the invariant subspaces and quotients used by the consequence calculation. The point of a common witness is that identities verified on all the tuple generators then hold on every space in this class.

Concretely, $K[S_N]/J_r(N)$ is the algebra in which two permutation combinations are identified when their difference kills every tuple test space of length at most r. The action on each $M \in \mathcal{C}_r(N)$ factors through this quotient. Consequently an identity such as $e^2 = e$ proved on all the tuple spaces is an identity of this algebra and acts on every such M. A chosen chain complement need not restrict to a submodule, but an identity of common operators does. The argument therefore transfers identities rather than separately chosen bases.

Naturality here concerns maps f of label representations: they intertwine every common group-algebra operator, giving $f(ev) = ef(v)$. This does not assert that e commutes with every label permutation. Its image is used as a vector-space model for the deletion quotient, which carries the appropriate residual action.

For the actual block action, purity on every tuple generator has already been proved in (13.5). The proposition below isolates the implication from that property and a common witness to exactness. Its purity premise records the input to this algebraic step; it is not an outstanding assumption of the block construction.

Proposition 13.4 (Common identities give exact deletion)**.** *For the block operator z above, $z^p = 0$. Suppose purity holds on every $P_b(N)$, $b \le r$, and one $w \in K[S_N]$ satisfies (13.4). The group-algebra expression e in (13.6) gives a natural vector-space isomorphism $\mathcal{D}(M) \simeq eM$ for every $M \in \mathcal{C}_r(N)$, and $\mathcal{D}$ is exact on this class.*

Proof. The lemma proves the identities on the tuple generators, so ze, ez and $e^2 - e$ belong to $J_r(N)$. They act as zero on every object of $\mathcal{C}_r(N)$, whether or not that object has a fixed and full-chain decomposition. Formula (13.7) follows there directly from the ordered product, since all factors in S_m fix a kernel vector.

The quotient-to-image map sends $[v]$ to ev. It is well defined because $ez = 0$, and $eM \subseteq \ker z$ because $ze = 0$. For $zv = 0$, the difference $v - ev = z^{p-1}wv$ belongs to both $\operatorname{im} z$ and $\ker z$. Taking the quotient class of ev therefore returns $[v]$. Conversely, idempotence makes every element of eM fixed by e. These maps are inverse for the actual quotient (13.1).

Every module map commutes with the same group-algebra expression e. For an exact sequence $0 \to U \xrightarrow{i} V \xrightarrow{f} W \to 0$, restriction to $eU \to eV$ preserves injectivity. If $y \in eW$, any lift $v \in V$ gives the projected lift ev, since $f(ev) = ef(v) = y$. If $v \in eV$ maps to zero, write $v = i(u)$ and then $v = ev = i(eu)$. Thus the projected sequence is exact. The natural quotient-to-image isomorphisms prove exactness of deletion, without a separate purity assumption on subobjects. As in Chapter 9, eM is a vector-space model; the quotient retains the residual label action. □

Example 13.5 (A full chain in characteristic three)**.** Let $zv_2 = v_1$, $zv_1 = v_0$, $zv_0 = 0$, and let $wv_0 = v_2$, with w zero on the other chain vectors. Then $z^2wz^2 = z^2$. The projector is

$$e = (I - z^2w)(I - zwz)(I - wz^2).$$

The rightmost factor removes v_2, the middle factor removes v_1, and the leftmost removes v_0. If a separate fixed vector v_{fix} is added and $wv_{\mathrm{fix}} = 0$, every factor fixes v_{fix}. Thus the projector removes exactly the full chain and retains the fixed direction. For $p = 2$ the same formula has only the two factors $(I - zw)(I - wz)$ from Chapter 9.

13.4 The linear construction in odd characteristic

We now adapt the characteristic-two construction from Chapter 11, using characteristic-independent combinatorial ingredients and new operator arguments. The combinatorial ingredients are the partition of unity in Section 11.2, the cone and signed component lifts in Section 11.3, and the rook forcing and permutation realisation in Chapter 12. Their insertion and cancellation formulas use 1 and -1 without a characteristic-two cancellation, so these combinatorial proofs remain valid in every field. They supply the same retained-label count $2b - 1$.

The operator arguments do require replacements. The two-level splitting must become a full p-level splitting, and the characteristic-two identity $z^{\mathsf{T}} = z$ must be replaced by conjugacy with the transpose. The following subsections establish these replacements before using the common correction contract with $u = z^{p-1}$.

Fix the current tuple length $b \le r$, and recall $V = P_b(N)$, $V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}$, $\mathcal{Z}_b = \bigcap_i \ker \partial_i$ and $T_b = V/V'$. Thus V' is the incoming shorter-tuple space, $\mathcal{Z}_b$ is the zero-marginal space, and T_b is the quotient by the incoming directions. The filtration $\mathcal{Z}^{\le \ell}$ consists of zero-marginal vectors supported on tuples with at most ℓ block positions. Its prospective quotient layer is

$$G_\ell = \bigoplus_{|J|=\ell} K[\mathrm{Inj}(J, A)] \otimes \mathcal{Z}_{[b]\setminus J}(n),$$

where $J \subseteq [b]$ is a set of positions and A is the block-label set. These are the same layer spaces used in the characteristic-two construction; only the block action and its chain structure change.

13.4.1 Constructing a fixed and full-chain decomposition

A nonempty block pattern has an image set of size $\ell < q$. Give it a grade $e \in \mathbb{F}_p$ using the orbit coordinate of that image set, as in Chapter 11. Let E be the space on all these block patterns, and let E_e span the patterns of grade e, and let D read the grade. The same commutator $W = [D, z]$ is invertible and commutes with z. In this subsection e is a scalar grade; the projector denoted by e in the preceding sections is an operator and plays no role in the grade decomposition.

Fix a grade e and suppose $\sum_{j=0}^{p-1} z^j v_j = 0$ with $v_j \in E_e$. The commutator identity gives

$$(D - eI)z^j v_j = jWz^{j-1}v_j \quad (j \ge 1).$$

Applying $D - eI$ to the relation and cancelling W leaves $\sum_{j=1}^{p-1} jz^{j-1}v_j = 0$. Repeat $p - 1$ times to obtain $(p-1)!v_{p-1} = 0$, hence $v_{p-1} = 0$. Remove that term and repeat with one fewer power. All v_j vanish. This proves both directness of $\sum_j z^j E_e$ and injectivity of each z^j on E_e. It does not require W to preserve the grade: after each cancellation the coefficients are still the original grade-e vectors.

Consequently $p \dim E_e \le \dim E$ for each of the p grades. As they partition the basis, $\sum_e \dim E_e = \dim E$. Summing the inequalities forces equality in every one. In particular

$$E = \bigoplus_{j=0}^{p-1} z^j E_0.$$

The grade depends only on the image set, so it is stable under position permutations and splits by position domain. It ignores the retained tuple. Tensoring each domain component with its retained zero-marginal factor therefore gives $G_\ell = \bigoplus_{j=0}^{p-1} z^j Q_\ell$, with Q_ℓ position-stable and z^{p-1} injective on Q_ℓ.

This is the p-level version of the two-level section construction. The grade $e \in \mathbb{F}_p$ is a coordinate on an image-set orbit; ℓ instead counts block positions. To make a section commute with z, choose preimages of the chain starts $q \in Q_\ell$, then prescribe the preimage of $z^j q$ to be $z^j \sigma(q)$. Directness and injectivity make all the chain coordinates unique. The last step goes to zero because $z^p = 0$, so the prescription is compatible with z at every level without an averaging operation.

For a layer quotient map f and the position-equivariant section σ supplied by the signed lift, define

$$\tau\Big(\sum_{j=0}^{p-1} z^j q_j\Big) = \sum_{j=0}^{p-1} z^j \sigma(q_j).$$

Uniqueness of the decomposition makes the formula well defined; $f\tau = I$ follows from $fz = zf$. Commuting with z is checked by shifting each power up by one; the term at power p is zero. Its image splits that filtration quotient as an actual subspace. Position stability of Q_ℓ and position equivariance of z and σ also make τ position-equivariant. Successively splitting the quotients gives

$$\mathcal{Z}_b(n+q) = \mathcal{Z}_b(n) \oplus \bigoplus_{j=0}^{p-1} z^j C, \tag{13.8}$$

where C is position-stable and z^{p-1} is injective on C. This is a fixed and full-chain decomposition in the sense of Definition 13.1. The hypotheses remain $n \ge 2b-1$ and $b < q$: making the sections commute with z adds no labels to the retained-label cost of the signed lift.

13.4.2 Top and zero-marginal spaces

Write $V = P_b(N)$. We verify the four properties used in Lemma 11.5, now with $u = z^{p-1}$ in place of z. In (13.8), every vector has a unique expression $s + \sum_{j=0}^{p-1} z^j c_j$, with $s \in \mathcal{Z}_b(n)$ and $c_j \in C$. Since $zs = 0$,

$$z\Big(s + \sum_{j=0}^{p-1} z^j c_j\Big) = \sum_{j=0}^{p-2} z^{j+1} c_j.$$

Directness and injectivity of the powers of z on C force $c_0 = \cdots = c_{p-2} = 0$ in a kernel vector. Hence

$$\ker(z|_{\mathcal{Z}_b}) = \mathcal{Z}_b(n) \oplus uC, \qquad u\mathcal{Z}_b = uC.$$

If a vector belongs to $uV \cap \mathcal{Z}_b$, it is therefore $s + uc$. Both that vector and uc have zero retained coordinates: u kills retained tuples and preserves the moved-tuple summand. Thus $s = 0$, and

$$uV \cap \mathcal{Z}_b = u\mathcal{Z}_b.$$

Define w_C on $\mathcal{Z}_b$ by sending uc to c and killing the retained summand and the levels $C, zC, \dots, z^{p-2}C$. Injectivity of u on C makes this well defined; position stability makes it position-equivariant. It satisfies $uw_C u = u$ there.

13.4.3 Duality with the actual transpose operator

The transpose changes which space carries the inverse used by duality. In the tuple pairing, $\langle uv, w\rangle = \langle v, u^{\mathsf{T}}w\rangle$. Thus the operator induced by u on the quotient T_b is paired with u^{T} on $\mathcal{Z}_b$, rather than automatically with u itself. This is the part of the characteristic-two argument that must change.

All transposes here are ordinary bilinear matrix transposes. They do not conjugate the coefficients c_i. A permutation matrix transposes to the matrix of its inverse, giving

$$z^{\mathsf{T}} = \sum_i c_i(\sigma_i^{-1} - 1).$$

The map $\iota(v) = -v$ on the block is a genuine label permutation, fixes retained labels and commutes with position permutations. It therefore conjugates the entire tuple-layer construction for z to that for z^{T}: $\iota\sigma_i\iota^{-1} = \sigma_i^{-1}$ and $\iota u\iota^{-1} = u^{\mathsf{T}}$. It preserves V' and $\mathcal{Z}_b$, since these are defined by label-equivariant marginals. The symbol ι here denotes this block inversion and its action on tuple spaces; it is different from the inclusion $\iota_{\mathcal{Z}} : \mathcal{Z}_b \hookrightarrow V$ used in the correction formulas.

For a three-element block labelled by $\mathbb{F}_3$, inversion fixes 0 and swaps $1, 2$. Conjugating translation by 1 with this swap gives translation by -1. This is the elementary meaning of $\iota z\iota^{-1} = z^{\mathsf{T}}$: the change is an actual relabelling of the block, so the marginal constructions remain valid. The inverse-image notation $(u^{\mathsf{T}})^{-1}(U)$ below means $\{v : u^{\mathsf{T}}v \in U\}$, not an inverse matrix for u^{T}.

In particular, $u^{\mathsf{T}}V \cap \mathcal{Z}_b \subseteq u^{\mathsf{T}}\mathcal{Z}_b$. This gives

$$(u^{\mathsf{T}})^{-1}(\mathcal{Z}_b) = \ker u^{\mathsf{T}} + \mathcal{Z}_b.$$

Taking orthogonal complements in the nondegenerate tuple pairing,

$$(uV')^{\perp} = (u^{\mathsf{T}})^{-1}(\mathcal{Z}_b), \qquad (uV \cap V')^{\perp} = \ker u^{\mathsf{T}} + \mathcal{Z}_b.$$

Their equality proves $uV \cap V' = uV'$.

The perfect pairing $T_b \times \mathcal{Z}_b \to K$ identifies a quotient vector with a linear functional on $\mathcal{Z}_b$. The dual of a map h sends such a functional λ to $\lambda \circ h$. This reverses composition order, as with ordinary matrix transposes. In the following construction, this dual map is w_T; the displayed pairing calculation verifies its inverse condition on the correct space.

To specify the quotient witness, put

$$h = \iota w_C \iota^{-1} : \mathcal{Z}_b \longrightarrow \mathcal{Z}_b, \qquad u^{\mathsf{T}}hu^{\mathsf{T}} = u^{\mathsf{T}}.$$

The tuple dot product induces the perfect pairing $\langle\ ,\ \rangle : T_b \times \mathcal{Z}_b \to K$. Define $w_T : T_b \to T_b$ by $\langle w_T t, \zeta\rangle = \langle t, h\zeta\rangle$. Then

$$\langle uw_T ut, \zeta\rangle = \langle t, u^{\mathsf{T}}hu^{\mathsf{T}}\zeta\rangle = \langle ut, \zeta\rangle,$$

so $uw_T u = u$. The pairing is invariant under position permutations, and h commutes with them; equivalently, their matrices transpose to their inverses. Thus w_T is position-equivariant. This supplies all four correction properties with the actual transpose operator.

13.4.4 A trace from transpose to correction in characteristic three

The following small linear-algebra model displays the maps in the interface. Let $V = Kf \oplus Ke_0 \oplus Ke_1 \oplus Ke_2$ over $K = \mathbb{F}_3$, let σ fix f and cycle the three e_i, and put $z = \sigma - I$, $u = z^2$. Choose the incoming space $V' = Kf$. For the ordinary dot product its orthogonal

complement is $\mathcal{Z} = \operatorname{span}\{e_0, e_1, e_2\}$, and $T = V/V'$ is paired perfectly with $\mathcal{Z}$. These are model spaces for the interface, rather than an assertion about a particular tuple length.

Write $v_\Sigma = e_0 + e_1 + e_2$. On $\mathcal{Z}$ the top power satisfies $ue_i = v_\Sigma$, so its matrix is the all-one 3×3 matrix. The chain-start space can be chosen as $C = Ke_0$: the vectors e_0, ze_0, v_Σ are independent. Define w_C by

$$w_C e_2 = e_0, \qquad w_C e_0 = w_C e_1 = 0.$$

It sends v_Σ back to e_0 and kills C and zC, as the construction requires. Block inversion fixes e_0 and swaps e_1, e_2; hence $h = \iota w_C \iota^{-1}$ sends e_1 to e_0 and kills the other two basis vectors. Taking the dual in the displayed pairing gives

$$w_T[e_0] = [e_1], \qquad w_T[e_1] = w_T[e_2] = 0.$$

For example, $uw_T u[e_2] = uw_T[v_\Sigma] = u[e_1] = [v_\Sigma] = u[e_2]$. The same calculation applies to every basis vector. This shows the transpose reversal explicitly: w_T is the dual of h, with its specified domain T, rather than a reuse of w_C on quotient vectors.

Now start this model's repair with $w_0 = 0$, so $\varepsilon = -u$. The typed correction $\varphi = \iota_{\mathcal{Z}}(-w_C \bar{\varepsilon} w_T)\pi$ sends e_0 to e_0 and kills f, e_1, e_2: indeed $w_T[e_0] = [e_1]$, $\bar{\varepsilon}[e_1] = -v_\Sigma$, and $-w_C(-v_\Sigma) = e_0$. Therefore $u\varphi u e_i = v_\Sigma = -\varepsilon e_i$, while both sides vanish on f. Thus the residual is cancelled on every vector.

Using $w = \varphi$ in the ordered projector makes the three stages equally visible. The factor $I - wu$ kills e_0 and fixes ze_0 and v_Σ; $I - zwz$ then kills ze_0 and fixes v_Σ; $I - uw$ kills v_Σ. Every factor fixes f. The full chain is removed and the fixed line survives. On actual tuple spaces the preceding construction supplies position equivariance and the incoming/zero-marginal conditions; the rook theorem supplies the common permutation combination.

13.4.5 The same correction, with the correct power

The remaining correction is the algebraic contract from Chapter 11, with u in place of z. Its typed composition uses the quotient T_b at the input and the zero-marginal space at the output. It uses neither an arbitrary choice of quotient representatives nor a characteristic-two cancellation. The preceding chain and transpose arguments have supplied its two inverse conditions; the rook theorem then realises its matrix on the same existing label set.

Suppose w_0 works on shorter tuples for the top-power relation. Its error is

$$\varepsilon = uw_0u - u.$$

It maps into $\mathcal{Z}_b$ and kills V', giving the factorisation $\varepsilon = \iota_{\mathcal{Z}} \bar{\varepsilon} \pi$, where $\bar{\varepsilon} : T_b \to \mathcal{Z}_b$. Write u_T and $u_{\mathcal{Z}}$ for the induced operators. We now verify both inverse relations explicitly. The error's image lies in $uV \cap \mathcal{Z}_b = u\mathcal{Z}_b$, so the witness on that space gives $u_{\mathcal{Z}} w_C \bar{\varepsilon} = \bar{\varepsilon}$. If $uv \in V'$, the inclusion $uV \cap V' \subseteq uV'$ gives $uv = uv'$ for some $v' \in V'$. As $\varepsilon = (uw_0 - I)u$ and $\varepsilon V' = 0$,

$$\varepsilon v = (uw_0 - I)uv' = \varepsilon v' = 0.$$

The quotient witness $u_T w_T u_T = u_T$ says that a representative v_T of $w_T u_T \pi(v)$ satisfies $u(v_T - v) \in V'$. The preceding implication therefore gives $\varepsilon v_T = \varepsilon v$. Injectivity of $\iota_{\mathcal{Z}}$ gives $\bar{\varepsilon} w_T u_T \pi = \bar{\varepsilon}\pi$; surjectivity of π then gives $\bar{\varepsilon} w_T u_T = \bar{\varepsilon}$. No compatible choice of representatives is assumed.

With $\pi : V \to T_b$ and $\iota_{\mathcal{Z}} : \mathcal{Z}_b \to V$, define the actual endomorphism

$$\varphi = \iota_{\mathcal{Z}}(-w_C \bar{\varepsilon} w_T)\pi.$$

Lemma 11.6, applied with u in place of z, now gives $u\varphi u = -\varepsilon$. The map is position-compatible, has image in $\mathcal{Z}_b$ and kills V'. The characteristic-independent realisation theorem supplies $j \in J_{b-1}(N)$ with this endomorphism. Hence

$$u(w_0 + j)u - u = \varepsilon + uju = 0$$

on length b. On every shorter tuple module, $j = 0$, so its old witness relation is unchanged. The same label pool is used throughout.

Inducting on b proves (13.4) on the same label pool whenever $n \geq 2r - 1$. The longer projector gives exactness, the moved-orbit calculation gives natural basis and tensor deletion, and the original PC closure transport gives the same period statement.

Completion of the proof of Theorem 7.1. Begin at tuple length zero with $w_0 = 0$, since z acts as zero on constants. At length b, the preceding correction and Theorem 12.5 replace w_{b-1} by w_b satisfying $z^{p-1}w_b z^{p-1} = z^{p-1}$ on every length at most b. The same retained labels are used at every step. The largest requirement is $\max_{1\leq b\leq r}(2b - 1) = 2r - 1$; length zero needs no label. The projector identities then imply exactness on $\mathcal{C}_r(N)$. The orbit commutator calculation supplies the natural retained-basis and tensor identifications, including the actual stable coefficient maps. These arguments prove every assertion of Theorem 7.1. □

Returning to the prescribed-target theorem

The construction has proved the common top-power witness on every tuple length up to an abstract budget r, at retained universe size $n \geq \max\{0, 2r - 1\}$ and block size $q = p^a > r$. The tuple purity calculation and the explicit projector prove exact deletion on the full support class. The moved-orbit commutator proves natural basis and permutation-first-factor tensor deletion. Stable coefficient matrices therefore represent the actual smaller axiom and target maps, and exactness transports the full PC closure.

For a fully invariant target the application uses $r = kd + s$. For the indexed request it uses $r = 2kd + s + t$, covering the certificate source and the doubled polynomial pairing used to reflect target membership. These choices are made by the target-transfer proof; the finite algebra just proved is independent of the equations. The target remains the prescribed full polynomial, including when the degree budget is below its actual degree. No target is added to the axioms and no semantic-entailment inference is introduced.

Finally, extend scalars only while constructing the generic block operator and its common witness. Lemma 3.3 returns actual derivability of each original-coefficient target to the original field. Thus the construction applies to every coefficient field of fixed positive characteristic. It has not asserted that a coefficientwise projection of an arbitrary extension-field proof preserves each proof line: the descent argument uses the finite-dimensional consequence spaces and recovers a legal proof from their bases.

Both the onset and the period are linear in the fixed proof-degree budget. Chapter 14 now assembles the actual target-transfer theorem; the classification argument in Chapter 6 inverts its two quantitative estimates, with the exact easy-residue and finite-peak qualifications. The proof engine is now complete in every prime characteristic.

Part IV

Completing lifting and comparing NS with PC

Chapter 14

Lifting the actual axioms, proof rules and targets

All the finite-algebra obligations have now been supplied. We return to the original question $\Gamma_n \vdash G_b^{(n)}$: can every requested target be derived from the given axioms within degree d? We verify each part of the contract in Chapter 5. The point of doing the assembly slowly is to make clear that a prescribed target, rather than an arbitrary lifted vector, is transported.

14.1 The theorem we can now state without hidden assumptions

Let Γ_n be a fixed uniform axiom template and let $G_b^{(n)}$ be a fixed uniform target family. Put

$$R(d) = 2kd + s + t, \qquad n_* = \max\{n_0, 1, kD_G + t\}.$$

Here k, s, t bound the label supports of a variable, an axiom index and a target index; D_G bounds target degree. The universe size n_0 is the axiom stability threshold. These are fixed data; only d is a proposed proof degree.

Theorem 14.1 (Actual target transfer)**.** *Over any field of fixed prime characteristic p, suppose*

$$q = p^a > R(d), \qquad n \geq \max\{n_*, 2R(d) - 1\}, \qquad N = n + q.$$

For either NS or ordinary PC, every prescribed size-N target has a derivation of degree at most d if and only if every prescribed size-n target does. The assertion holds for every $d \geq 0$, with the separate zero-degree interpretation given below. It also holds for any representative of one fixed global target orbit at each eligible universe size. For a fully invariant target use the sharper budget $kd + s$ and start $\max\{n_0, 1, kD_G\}$.

The theorem concerns original-field polynomials and actual derivations. It does not require invariant certificates, does not assume that the axioms are inconsistent, and does not add the target to them. If a stabilised target degree exceeds d, the threshold predicate is simply false. The empty target request is true at every degree.

14.2 Transporting the entire consequence space

Fix a positive degree budget d and write $W_N = P_{N,\leq d}$ for the ordinary polynomial space of degree at most d. Recall

$$I_d(N) = \operatorname{span}_K\{\mu f_{a,N} : \deg \mu + \deg f_{a,N} \leq d\}, \qquad C_d(N) = V_\infty(N).$$

The first space records NS identities of total-product degree at most d, with μ ranging over monomials; the second is the closure of the usable axioms under the legal ordinary-PC rules (Proposition 3.2). Boolean equations, when present, remain inputs in this polynomial space. We first transport the spaces $I_d(N)$ and $C_d(N)$, without choosing any target. This separates the proof-system calculation from the question of whether a particular uniform expression belongs to it. The operation $\mathcal{D}(M) = \ker z_M/(\ker z_M \cap \operatorname{im} z_M)$ deletes the block through the label action; the common projector represents its classes, and $\mathsf{L} = e\iota$ lifts smaller-universe vectors. Exactness means that kernels, images and quotients are preserved by this operation.

Concretely, a deletion class is represented by a vector v with $z_M v = 0$. Two such vectors represent the same class if their difference lies in $\ker z_M \cap \operatorname{im} z_M$. The projector provides representatives which can be compared with smaller-universe vectors. We use the actual inclusions and maps between these classes, not just equal dimensions of the resulting spaces (Chapter 7).

For NS the certificate map has source basis the pairs (μ, a) with

$$\deg \mu + \deg f_{a,N} \le d, \qquad (\mu, a) \longmapsto \mu f_{a,N} \in W_N.$$

Its source has support at most $kd + s$, and W_N has support at most kd. The matrix entry at a target monomial ν is zero unless $\mu \mid \nu$, and otherwise is $c(a, \nu/\mu)$. These are the actual size-independent invariant entries of the certificate map. The stability threshold n_0 makes each axiom degree independent of universe size. Exactness and matrix naturality therefore identify the embedded image $\mathcal{D}(I_d(N))$ with $I_d(n)$.

For PC let $X_N = K[\mathcal{V}_N]$ and $L_N = P_{N,\le d-1}$. The tensor product records a variable and a lower-degree line together: a pure tensor $x_v \otimes f$ is sent to their polynomial product, and sums of pure tensors record sums of such products. The degree restriction is imposed on f before multiplication. This is the variable-labelled register of Chapter 5. The multiplication map

$$m_N : X_N \otimes L_N \longrightarrow W_N, \qquad x_v \otimes f \longmapsto x_v f$$

has stable entries. Its source is supported on at most $k + k(d-1) = kd$ labels. The usable axioms give the image $V_0(N)$ of their coefficient map. The recurrence is

$$V_{j+1}(N) = V_j(N) + \operatorname{im}\big(m_N|_{X_N \otimes (V_j(N) \cap L_N)}\big).$$

Every stage is a submodule of W_N, and every multiplication domain is a submodule of the fixed support-bounded space $X_N \otimes L_N$. Suppose stage j has been identified at both universe sizes. Put $U_N = V_j(N) \cap L_N$ and $U_n = V_j(n) \cap L_n$. The intersection is the kernel of $(v, l) \mapsto v - l$ on $V_j(N) \oplus L_N$, so exactness identifies $\mathcal{D}(U_N)$ with U_n, including their inclusions $i_N : U_N \hookrightarrow L_N$ and $i_n : U_n \hookrightarrow L_n$.

Write $m'_N = m_N \circ (1 \otimes i_N)$ and $m'_n = m_n \circ (1 \otimes i_n)$ for the restricted multiplication maps. The required compatibility is the commutative square

$$\begin{array}{ccc} \mathcal{D}(X_N \otimes U_N) & \xrightarrow{\mathcal{D}(m'_N)} & \mathcal{D}(W_N) \\ \simeq\downarrow & & \simeq\downarrow \\ X_n \otimes U_n & \xrightarrow{m'_n} & W_n. \end{array} \tag{14.1}$$

The left identification is (7.2), natural in the second factor: it sends $\mathcal{D}(1 \otimes i_N)$ to $1 \otimes i_n$. The ambient multiplication map m_N is sent to m_n by stable matrix naturality. Composing these two statements proves the square for the restriction. Exactness now transports its image and the sum with V_j, giving stage $j + 1$. The initial stage is the actual usable-axiom image, so induction identifies the embedded $\mathcal{D}(V_j(N))$ with $V_j(n)$ for every j.

Both ascending chains stabilise because their ambient spaces are finite dimensional. Choose a stage beyond both stabilisation points. We obtain

$$\mathcal{D}(I_d(N)) = I_d(n), \qquad \mathcal{D}(C_d(N)) = C_d(n), \tag{14.2}$$

where equality means equality after the natural identification $\mathcal{D}(W_N) \simeq W_n$. This transports every cancellation in a PC proof, including a low-degree result subsequently multiplied by a variable. No bound on proof length and no symmetry of individual proof lines was used.

14.3 Why deletion also reflects target membership

Preserving an image is not by itself enough: an exact operation can annihilate a nonzero quotient. The next lemma supplies the reflection needed for a general target source.

The notation S'/S means vectors in S' considered equal when their difference lies in S; it is a *subquotient* of the ambient space. In the lemma, $\mathcal{C}_{2u}(N)$ is the common-annihilator class of Chapter 7, not a PC consequence space. Here u is an integer support bound, unrelated to the top-power operator $u = z^{p-1}$ used in the witness construction. The doubled budget has a concrete purpose: a basis pair in $P \otimes P$ can use the union of two support-u label sets. The annihilator class is not being assumed closed under arbitrary tensor products.

Lemma 14.2 (Nonzero subquotients survive)**.** *Let $P = K[\mathcal{U}_N]$ have support at most u. Suppose deletion is exact on $\mathcal{C}_{2u}(N)$, its block size exceeds $2u$, and the natural identifications (7.1)–(7.2) hold. If $S \subsetneq S' \subseteq P$ are invariant subspaces, then $\mathcal{D}(S'/S) \neq 0$.*

Proof. The permutation basis gives a nondegenerate dot product on P, equivariant under S_N. It induces a pairing

$$S^{\perp} \otimes (S'/S) \longrightarrow K.$$

This pairing is surjective: for $v \in S' \setminus S$, a linear functional vanishing on S can take value one at v; the dot product represents that functional by an element of $S^{\perp}$. Let $Y = S'/S$. Tensoring over a field is exact, so the displayed source lies in $P \otimes Y$, itself a subquotient of $P \otimes P$. The last permutation module has support at most $2u$, justifying every use of exactness. If $\mathcal{D}(Y) = 0$, permutation-first-factor tensor compatibility gives $\mathcal{D}(P \otimes Y) = 0$. Deleting the inclusion $S^{\perp} \otimes Y \hookrightarrow P \otimes Y$ gives an injection

$$\mathcal{D}(S^{\perp} \otimes Y) \hookrightarrow 0,$$

so the pairing source also deletes to zero. Exactness applied to the surjective pairing then gives $0 \twoheadrightarrow \mathcal{D}(K) \simeq K$, which is impossible because $K \neq 0$.

We use this surjective pairing, not a trace or an averaged inner product. In characteristic p, composing a pairing with coevaluation can yield a vanishing dimension scalar and would not prove the lemma. □

In the detector proof, $S^{\perp}$ consists of vectors whose dot product with every vector of S is zero. Thus pairing $h \in S^{\perp}$ with $v + S \in S'/S$ is well-defined: replacing v by $v + s$ changes the value by $\langle h, s\rangle = 0$. The tensor product is the linear source of this bilinear pairing. Surjectivity says that some such pair has value one, rather than that a dimension scalar is nonzero.

Lemma 14.3 (An invariant target needs no doubled budget)**.** *Let $j : K \to W$ and $H \subseteq W$ be equivariant and belong to the exactness class. Then $j(1) \in H$ if and only if its deleted image belongs to $\mathcal{D}(H) \subseteq \mathcal{D}(W)$.*

Proof. Membership is the vanishing of $K \xrightarrow{j} W \to W/H$. Its image is either zero or isomorphic to the simple trivial module K. Exactness preserves this image, and $\mathcal{D}(K) \simeq K \neq 0$. Thus the composite vanishes exactly when its deleted composite does. The vector $j(1)$ may be any invariant target, not just the constant polynomial one. □

14.4 The actual uniform target map

For indexed targets, let $\mathcal{B}$ be the fixed index basis and let $\gamma(b, \mu)$ be its stable coefficient rule. Write δ for the largest stabilised target degree, taking $\delta = 0$ for an empty request. At a universe size where all target degrees are at most d, define

$$J_N : K[\mathcal{B}_N] \longrightarrow W_N, \qquad [b] \longmapsto G_b^{(N)}.$$

Here J_N is a linear target map. It is distinct from the group-algebra annihilator ideal $J_r(N)$ used to construct the common projector. Its entry in row μ, column b is precisely $\gamma(b, \mu)$. Equivariance and independence of universe size are the hypotheses of the natural matrix identification in Theorem 7.1. Consequently $\mathcal{D}(J_N)$ is the actual smaller target map J_n. This statement concerns the prescribed full polynomials $G_b^{(n)}$; we do not replace a target by its degree-d truncation.

Let H_N be either $I_d(N)$ or $C_d(N)$. Every target belongs to H_N exactly when $\operatorname{im} J_N \subseteq H_N$. Put

$$S_N = H_N + \operatorname{im} J_N, \qquad Y_N = S_N/H_N.$$

The condition is $Y_N = 0$. Exactness, target-map naturality and (14.2) identify its deletion with

$$Y_n = (H_n + \operatorname{im} J_n)/H_n.$$

The polynomial permutation basis of W_N has support at most kd. Lemma 14.2, applied with the original monomial support bound, reflects vanishing of Y_N once the budget covers $2kd$. The affine budget

$$R(d) = 2kd + s + t$$

also covers the certificate source $(kd + s)$, the target source (t) and every multiplication domain. It therefore proves membership in both directions. The target source need not be simple; reflection comes from a subquotient of the ambient polynomial target space. Doubling $R(d)$ once again would be unnecessary: the pairing uses $W_N \otimes W_N$, whose support is $2kd$.

Proof of Theorem 14.1. First extend scalars to an algebraic closure, as permitted by Lemma 3.3. The degree of every indexed target stabilises beyond $n_* = \max\{n_0, 1, kD_G + t\}$. If d is smaller than the largest stable target degree, the all-target predicate is false at both universe sizes. Otherwise the full map J just described applies. Take $N = n + q$, $q = p^a > R(d)$ and retained universe size $n \geq \max\{n_*, 2R(d) - 1\}$. Exact deletion transports H_N, J_N and their sum and quotient. The detection lemma yields

$$(\forall b \in \mathcal{B}_N,\ G_b^{(N)} \in H_N) \quad \Longleftrightarrow \quad (\forall b \in \mathcal{B}_n,\ G_b^{(n)} \in H_n).$$

Field descent holds for each actual target separately, so the equivalence holds over the original field. Repeated steps connect any eligible universe sizes in a common residue class and prove the stated periodicity. For one fixed global target orbit, replace $\mathcal{B}$ by that orbit; at eligible universe sizes, equivariance makes every representative equivalent.

At degree zero the stabilised targets are either all zero, or are scalars with at least one nonzero value. In the latter case every target is derivable in degree zero exactly when the

original axioms include a nonzero constant. If a target has positive stable degree, degree-zero all-target membership fails. These are fixed predicates beyond n_* and give the separate zero-level assertion.

For a fully invariant target use $j_N : K \to W_N$, $j_N(1) = G^{(N)}$, in place of J_N. Lemma 14.3 reflects membership without the tensor pairing. The budget reduces to $kd + s$, with stable onset $\max\{n_0, 1, kD_G\}$. Taking $G^{(N)} = 1$ gives the sharper refutation special case. No target has been added to the axiom system. □

14.5 From one deletion step to a residue class

Write $h_S(n)$ for the maximum least degree of the requested targets in system S. Recall the safe onset $N(d)$ and chosen p-power period $Q(d)$ from (6.1); their separate zero-degree values are $N(0) = n_*$ and $Q(0) = 1$. Fix d and an eligible power q. If two sizes at or above the chosen onset have the same residue modulo q, repeatedly subtract q from the larger until reaching the smaller. Every retained size in those steps remains eligible. Applying Theorem 14.1 in both directions proves threshold periodicity.

To preserve an exact minimum degree rather than just the assertion $h(n) \le d$, repeat the argument at every lower threshold. The periods are powers of the same prime and divide the chosen degree-d period; zero degree has a stable separate predicate. The vector of all those threshold answers determines the truncated value $\min\{h(n), d + 1\}$. This completes the input used in Theorem 6.5 for the four profiles.

14.6 The complete lower-bound argument in a paragraph

Fix a minimum finite degree d occurring at a size beyond its own onset. The two-way threshold theorem makes that exact degree recur along a prime-power progression. Therefore a degree which is not allowed to recur must occur before its onset. Since the onset is $O(d)$, those isolated occurrences must have degree $\Omega(n)$. On T3 hard residues the same contradiction excludes every recurring easy level. On T4 we instead use unbounded recurring levels and the short period to bring a representative down to size $O(d)$, producing finite linear peaks. No iteration ever reduces the absolute degree: it is the availability of recurrence and its onset which produce the gap.

Chapter 15

An independent uniform NS–PC comparison

The linear PC lower bounds were proved by transporting the PC closure itself. We now establish a different result: every bounded-degree PC consequence has an NS identity of degree bounded independently of universe size. The comparison applies to arbitrary polynomials, not only invariant targets or refutations. Its qualitative consequence is that NS and PC assign every fixed uniform request the same type. Throughout, Γ_n is a fixed uniform template over a field K of fixed prime characteristic p. Recall that $I_d(n)$ is the span of axiom products of total-product degree at most d, while $C_d(n)$ is the ordinary-PC consequence closure inside $P_{n,\le d}$.

15.1 What comparison can and cannot prove

Suppose you already know that NS degree is large. It is tempting to use $h_{\mathrm{PC}} \le h_{\mathrm{NS}}$ as a PC lower bound. The inequality points in the wrong direction. Even an upper comparison $h_{\mathrm{NS}} \le F(h_{\mathrm{PC}})$ need not preserve a linear degree lower bound if F is large. Our linear PC bound therefore came from its own closure transport, not from inverting F.

The purpose of comparison is instead to say that bounded PC degree can never coexist with unbounded NS degree on the same sequence of sizes. This is its role in the running example of Chapter 4: the elementary induction-chain restriction supplies a growing NS obstruction, and comparison rules out recurring bounded PC degree for the original uniform order family. The linear PC constant then comes from PC's own lifting range, without inverting the possibly large function F. It also controls recurring finite levels well enough to identify the same one of the four types. The proof will make this statement for *every* polynomial vector in the PC consequence space, before specialising to any uniform target.

15.2 Why one plateau makes a space closed

The ideal generated by Γ_n consists of finite sums $\sum_f a_f f$ with arbitrary polynomial multipliers. The spaces $I_e(n)$ impose a product-degree bound on those sums, so they need not be ideals: multiplication may leave the allowed degree window. The argument below finds enough closure inside a fixed polynomial ambient without asserting that a truncated NS space is closed under arbitrary multiplication.

Imagine a chain of NS consequence spaces with increasing allowed degree, but look only at their intersection with the fixed degree-d ambient. Thus $I_{d+j}(n) \cap P_{n,\le d}$ allows larger certificates while retaining only consequences of polynomial degree at most d. An invariant-

chain bound forces two successive intersections to be equal. At that equality, multiplication of a degree-$d-1$ vector cannot create a new degree-d vector: it has only used one extra unit of NS certificate degree, precisely the step covered by the plateau. Thus the plateau is closed under the ordinary-PC rules.

This is the core idea. The first lemma supplies the uniform chain bound; the second argument turns a plateau into whole-consequence containment. Neither step assumes symmetry of the proof lines or the target.

The length being bounded is the number of strict inclusions in an ascending chain. For example, $A_0 \subsetneq A_1 = A_2 \subsetneq A_3$ has two strict inclusions. Invariance means that relabelling sends each subspace to itself. Exact deletion transports the inclusions, and the detector keeps a strict inclusion strict by preserving its nonzero quotient. This is why a dimension bound at small universe size can control chains at every universe size.

Lemma 15.1 (Uniform chain bound)**.** *For any bounded basis system $\mathcal{U}$, strict S_n-invariant subspace chains in $K[\mathcal{U}_n]$ have a length bound independent of n.*

Proof. Let r bound the basis support. Over an algebraic closure, choose a fixed $q = p^a > 2r$. At $N = n + q$ with $n \geq 4r + 1$, deletion is exact at the doubled budget $2r$. Lemma 14.2 shows that each nonzero subquotient B/A of $K[\mathcal{U}_N]$ survives. Thus $A \subsetneq B$ implies $\mathcal{D}(A) \subsetneq \mathcal{D}(B)$: the whole strict chain becomes a strict chain of the same length in $K[\mathcal{U}_n]$.

Repeatedly subtract q while the universe size exceeds $M = 4r + q$. Every retained universe size in such a step exceeds $4r$, so all steps are eligible. The final chain has at most $\dim K[\mathcal{U}_M]$ strict inclusions, because a smaller-universe basis injects into the size-M basis. The same bound covers the initial universe sizes at most M. Each step may choose its own generic coefficients: the detector applies to arbitrary invariant subspaces, without a naturality assumption on the chain across universe sizes. Finally, scalar extension preserves strict inclusions, giving the result over the original coefficient field. □

Only a finite exactness threshold at each fixed support is needed for this argument. In characteristic two the earlier quadratic construction also suffices. For an arbitrary positive characteristic the completed linear construction already provides the needed threshold. The bound may be large; its role is uniformity in n, not a sharp comparison of the two minimum degrees.

Theorem 15.2 (Uniform comparison for all consequences)**.** *For every uniform axiom template over a field of prime characteristic p, there is a nondecreasing $F : \mathbb{N} \to \mathbb{N}$, independent of n and of the target, with $F(d) \geq d$ and*

$$C_d(n) \subseteq I_{F(d)}(n) \qquad (n, d \geq 0). \tag{15.1}$$

Consequently, every polynomial with an ordinary-PC derivation of degree at most d from Γ_n has a total-product NS identity of degree at most $F(d)$ from the same axioms. For the aggregate profiles of any stable uniform target family, and for the profiles of any one fixed target orbit,

$$h_{\mathrm{PC}}(n) \leq h_{\mathrm{NS}}(n) \leq F(h_{\mathrm{PC}}(n)), \qquad F(\infty) = \infty. \tag{15.2}$$

Their NS and PC types in Theorem 6.5 therefore agree.

Proof. Fix d. Monomials of degree at most d form a permutation basis of support at most kd. Let L_d bound strict invariant-subspace chains in $P_{n,\leq d}$, uniformly in n, by Lemma 15.1. Consider the invariant spaces

$$W_j = I_{d+j}(n) \cap P_{n,\leq d}, \qquad 0 \leq j \leq L_d + 1.$$

They form an ascending chain. Some $W_j = W_{j+1}$ with $j \le L_d$, otherwise there would be more than L_d strict inclusions.

This plateau contains every axiom usable within degree d and is closed under linear combinations. If $f \in W_j$ and $\deg f \le d-1$, then multiplication of an NS identity for f by a variable gives $x_v f \in I_{d+j+1}(n)$. Its degree is at most d, hence $x_v f \in W_{j+1} = W_j$. For $d = 0$ there is no permitted multiplication of a nonzero line; the zero line already lies in W_j. Induction on *any* degree-d PC derivation now places every one of its lines in W_j. In particular,

$$C_d(n) \subseteq I_{d+L_d}(n).$$

This induction retains cancellations: a lower-degree linear combination can be multiplied because it lies in the plateau, even if its separate summands have degree d.

Choose

$$F(d) = d + \sum_{u=0}^{d} L_u.$$

It is nondecreasing, satisfies $F(d) \ge d$, and gives (15.1). No symmetry of a target was used. Conversely, expansion of NS identities into monomial–axiom products gives a same-degree PC derivation, so the first inequality in (15.2) holds. If the aggregate PC degree is finite, every requested target belongs to $C_d(n)$ at that budget; the inclusion puts every target in $I_{F(d)}(n)$. Empty requests have degree zero. Infinite PC degree forces infinite NS degree by the first inequality. These observations prove the complete extended-order comparison.

For type agreement write $g = h_{\mathrm{PC}}$ and $h = h_{\mathrm{NS}}$. The inequalities $g \le h \le F(g)$ transfer eventual boundedness in both directions. If g takes one finite value d infinitely often, then on those sizes h lies in the finite interval $[d, F(d)]$; some finite value of h recurs. A recurring finite value of h likewise forces a recurring finite value of g. Thus the recurring sets are empty together. If recurring values of g are unbounded, the same finite-interval argument yields unbounded recurring values of h. Conversely, choose a recurring value of h exceeding $F(B)$. On its infinite level set, $g > B$ and $g \le h$, so some finite value above B recurs for g. Thus infinitude of the recurring sets agrees as well. Boundedness, emptiness and infinitude determine exactly the four alternatives. □

Corollary 15.3 (Common finite peaks)**.** *If a fixed uniform request has common type T4, there is a nondecreasing F such that, beyond every bound B, some size $n \ge B$ has*

$$h_{\mathrm{PC}}(n) = d \ge B, \qquad d \in E(h_{\mathrm{PC}}), \qquad d \le h_{\mathrm{NS}}(n) \le F(d) < \infty,$$

with $n \le (p+2)(\kappa d + \sigma)$. Both degrees satisfy the rational linear lower bound in (6.7) at these same sizes. Here $E(h)$ is the set of recurring finite degrees, and $R(d) = \kappa d + \sigma$ is the applicable transfer budget: $(\kappa, \sigma) = (2k, s+t)$ for indexed targets and (k, s) for an invariant target.

Proof. Take the finite PC peaks from Theorem 6.5. The pointwise lower comparison gives the NS lower bound, and the uniform upper comparison gives its finiteness at those sizes. □

The comparison does not identify exact degrees, easy-class values or recurring degree sets. Its potentially large function F cannot be inverted to obtain a linear PC lower bound from a linear NS lower bound. That quantitative result came directly from the PC transfer theorem.

15.3 Reading the result as three tests of a profile

A profile has a recurring finite level if it takes one finite value at infinitely many sizes. The comparison sends that infinite set into a finite interval of degrees in the other system, so some

value of the other profile repeats infinitely often. It also preserves whether recurring levels are unbounded: choose levels above $F(B)$ and use pigeonhole inside a finite interval. Finally, it preserves eventual boundedness directly. These three tests distinguish T1, T2, T3 and T4.

This is why one does not need equality of exact degrees, equality of recurring levels, or equal easy-class values to obtain same-type agreement. For common finite T4 peaks, take a finite PC peak; pointwise $\mathrm{PC} \leq \mathrm{NS}$ supplies its lower bound, and the upper comparison supplies finiteness of the NS degree at that same size.

Part V

Examples, witnesses and logical consequences

Chapter 16

Examples with prescribed targets

The classification allows both finite proof degrees and infinity. An example should make clear which kind of hardness it illustrates. We already have a finite T2 family from a Boolean first-order sentence: the descending-predecessor example of Chapter 4, with its full lifting proof. Here we begin with a consistent family whose target is derivable on one residue class and underivable on the others. We then obtain finite hard proofs of a nonconstant target. Finally we describe the four first-order refutation families used by the undecidability argument. Recall the test in Theorem 6.5: T1 is eventually bounded; T2 has no recurring finite degree; T3 is not eventually bounded and has a finite nonempty set of recurring finite degrees; T4 has infinitely many. For a request, h_S is its least-degree profile; $d_S(\Gamma)$ denotes the special least degree of a refutation, whose target is 1.

16.1 A consistent family with an exact residue split

Fix characteristic p and, for $n \geq 1$, take

$$\Gamma_n = \{x_i - 1 : i < n\}, \qquad G^{(n)} = \sum_{i<n} x_i.$$

The axioms have the solution $x_i = 1$ for every i, so they are consistent at every size. Both the axioms and the invariant target have stable coefficient descriptions.

Proposition 16.1. *For both NS and PC, the target profile is exactly*

$$h_S(n) = \begin{cases} 1, & p \mid n, \\ \infty, & p \nmid n. \end{cases}$$

In particular, it is T3. With the same axioms but the indexed request $G_i^{(n)} = x_i$ for every $i < n$, the profile is constantly infinity and has type T2.

Proof. Summing the axioms gives $G^{(n)} - n$. If $p \mid n$, this is $G^{(n)}$, so it has an NS certificate, and hence a PC proof, of degree one. The target is nonzero of degree one, proving equality. If $p \nmid n$, the root with all coordinates one evaluates $G^{(n)}$ to the nonzero field element n. Soundness excludes any derivation. The same root evaluates each indexed target x_i to one, proving the final assertion. □

This example also distinguishes an invariant sum from its individual summands. The statement that the sum is a consequence does not say that each summand is a consequence. The aggregate profile is a maximum of separate proof degrees, not the degree of a proof of their sum.

The easy residue class here is zero modulo p; its degree is exactly one. The other classes have infinite degrees. This is a genuine consistent-axiom example of the classification, but it does not yet provide finite proofs requiring degree $\Omega(n)$. The next construction addresses that different question.

16.2 A fresh variable as a nonconstant target

The degree of a target need not reflect the degree needed to derive it. A polynomial of degree one may require arbitrarily high proof degree. We first give an exact construction, then apply it to a uniform family whose hard values are finite.

A simple invariant target illustrates the distinction between finite hardness and failure of derivability. For $G^{(n)} = \sum_{i<n} x_i$, the input $\{1\}$ gives degree exactly one in both systems for $n \geq 1$. With no inputs the same target is underivable and has degree ∞. Both fit the uniform definition, but neither supplies a finite hard example. The next construction does.

Here *fresh* means a new indeterminate in the polynomial ring, independent of every old variable and absent from every old axiom. Any coefficient polynomial has a unique expansion $a(x, y) = \sum_j a_j(x)y^j$; the notation $[y^1]a$ selects $a_1(x)$. This linear operation is different from substituting a value for y. The proof uses substitution for PC and coefficient extraction for NS, because they control the two degree conventions differently.

Proposition 16.2 (Adjoining a fresh target)**.** *Let K be any field, let $\Gamma \subseteq K[x_v : v \in V]$ be any axiom set, and adjoin a fresh variable y. Each certificate uses only finitely many axioms. Let $\Gamma^{\uparrow}$ be the same axioms viewed in $K[x_v : v \in V, y]$, with no additional axiom involving y. Then*

$$\begin{aligned} h_{\mathrm{PC}}(\Gamma^{\uparrow} \vdash y) &= \max\{1, d_{\mathrm{PC}}(\Gamma)\}, \\ h_{\mathrm{NS}}(\Gamma^{\uparrow} \vdash y) &= d_{\mathrm{NS}}(\Gamma) + 1. \end{aligned} \tag{16.1}$$

These equalities include degree zero and infinity, with $\infty + 1 = \infty$.

Proof. For PC, embed a refutation of Γ in the larger ring and multiply its last line 1 by y. If the refutation had degree d, the resulting proof of y has degree $\max\{1, d\}$: only the new last line is added, and its degree is one.

Conversely, substitute $y = 1$ in every line of a PC proof of y. The old axioms are unchanged and linear combinations commute with the substitution. Multiplication by an old variable remains a multiplication step; multiplication by y becomes a repeated line, which the linear combination rule admits. No degree increases, and the last line becomes 1. Every proof of y also has degree at least one. These two lower bounds prove the first equality, including the case when no refutation exists.

For NS, multiplying the coefficients of an identity $1 = \sum_f a_f f$ by y gives an identity for y with degree one larger. For the reverse inequality, suppose

$$y = \sum_f a_f(x, y) f(x)$$

has total-product degree at most e. Take the coefficient of y^1. Since every f is independent of y, this gives the exact identity

$$1 = \sum_f [y^1] a_f(x, y)\, f(x).$$

Every nonzero product on the right has degree at most $e - 1$: it is the coefficient of y^1 in a product of total degree at most e. Zero products may be discarded. Hence $d_{\mathrm{NS}}(\Gamma) \leq e - 1$. In particular a finite proof of y exists exactly when a finite refutation exists. If the original refutation degree is zero, both new degrees are one, as asserted. □

Example 16.3 (Why the two exact identities differ)**.** Take $\Gamma = \{x, 1-x\}$. Both refutation degrees are one. PC derives $1 = x + (1-x)$ and then multiplies that last constant by y; every line still has degree at most one. An NS identity for the fresh target is

$$y = yx + y(1-x).$$

Its products have degree two. No degree-one NS identity can produce y, since its multipliers of these degree-one axioms would have to be constants. Thus the new PC degree is one and the NS degree is two. PC can finish the refutation before multiplying its last line; an NS identity carries the extra variable in its coefficient polynomials.

The absence of a y-axiom is part of the construction; in particular, we do not add $y^2 - y = 0$. For a uniform template, give the new variable empty support and keep all old axiom indices and coefficients. Set all coefficients of axiom monomials involving y to zero. The result is again a uniform template with the same bounds k, s, D, n_0, and $G^{(n)} = y$ is a fully invariant target of degree one at every universe size. Thus (16.1) holds at every universe size.

This operation preserves each of the four types, separately for NS and PC. The NS recurring finite degrees are shifted by one. For PC, the map $d \mapsto \max\{1, d\}$ merges only degrees zero and one; all larger finite degrees and infinity are unchanged. Eventual boundedness, absence or unboundedness of recurring finite levels, and the mixed periodic case are consequently preserved.

The notation $v_p(n)$ is the exponent of p in the factorisation of n: writing $n = p^a u$ with $p \nmid u$ gives $v_p(n) = a$. For instance, $v_3(12) = 1$ and its associated power is $q = 3$. A window such as $[q, q+4]$ bounds a finite proof degree; the condition $n \geq 2q$ must still be checked before using its lower endpoint.

Corollary 16.4 (A finite hard nonconstant consequence family)**.** *Let K have prime characteristic p, and let $\widehat{B}_n$ be the normalised weak-bijection system defined in (16.5), with all of its function and coverage witnesses retained. Adjoin a fresh variable y as above, and request the target $G^{(n)} = y$. Both target-degree profiles are of type T4.*

More precisely, write $n = p^a u$, where $p \nmid u$, and put $q = p^a$. Whenever $n \geq 2q$,

$$q \leq h_{\mathrm{PC}}(n) \leq q+4, \qquad q+1 \leq h_{\mathrm{NS}}(n) \leq q+5,$$
$$h_{\mathrm{PC}}(n) \leq h_{\mathrm{NS}}(n).$$

In particular, at every size $n_a = (p+1)p^a$,

$$\frac{n_a}{p+1} \leq h_{\mathrm{PC}}(n_a) \leq h_{\mathrm{NS}}(n_a) \leq \frac{n_a}{p+1} + 5 < \infty. \tag{16.2}$$

Proof. The refutation bounds (16.6), proved below together with the rectangular obstruction in the next chapter, give $q \leq d_{\mathrm{PC}}(\widehat{B}_n) \leq d_{\mathrm{NS}}(\widehat{B}_n) \leq q+4$. Apply (16.1); since $q \geq 1$, the maximum in the PC identity changes nothing here. At $n_a = (p+1)p^a$ we have $v_p(n_a) = a$ and $n_a \geq 2p^a$, yielding (16.2).

For each fixed a, infinitely many sizes $n = p^a(p(j+1)+1)$ have the same valuation and satisfy the displayed bounds. Some finite degree in each of the bounded windows must therefore recur infinitely often. The windows tend to infinity with a, proving T4 in each system. On the other hand, the old upper bound gives $h_{\mathrm{PC}}(n) \leq 5$ and $h_{\mathrm{NS}}(n) \leq 6$ whenever $p \nmid n$. The linear peaks thus coexist with infinitely many easy sizes, rather than giving an eventual linear lower bound at all sizes. □

This example has finite hard proofs of a nonconstant target, but its axioms are inconsistent: the difficulty is inherited from refuting $\widehat{B}_n$. It does not establish a family of satisfiable axioms with the same hard consequence profile. The small consistent example in Chapter 1 illustrates ordinary-PC consequences in a different setting; no growing lower bound is asserted for it.

16.3 The refutation special case: four first-order examples

We now take the constant target $G^{(n)} = 1$. This is the refutation special case of the uniform-consequence theorem; the sharper invariant-target budget applies. Write $\psi_1, \psi_2, \psi_3, \psi_4$ for the four sentences described in order below. The four cases all occur among fixed first-order sentences with no finite nonempty models. We give examples over a field of fixed prime characteristic p. Each sentence is translated by the graph-and-sum convention: relation and function-graph variables are Boolean, and every function has a totality sum and pairwise functionality equations, as in Chapter 2. These are ordinary polynomial inputs, including the Boolean equations. The parameter $n \geq 1$ is the universe size. For the matrix notation in this section, we relabel the universe as $\{1, \ldots, n\}$; this changes no equation or degree.

T1: a local contradiction. The sentence $\forall x\,(R(x) \wedge \neg R(x))$ contributes r_i and $1 - r_i$. Their sum refutes the system in degree one, independently of n. The contradiction is already visible at one element.

T2: an injection that misses a point. Let η say that a unary function g is injective and that a constant c is outside its image. Write x_{ij} for $g(i) = j$ and c_j for $c = j$. In addition to the Boolean equations, its system Γ_n^η consists of

$$\begin{gathered}\sum_j x_{ij} - 1 = 0, \qquad x_{ij}x_{ik} = 0 \ (j \neq k), \qquad x_{ij}x_{hj} = 0 \ (i \neq h), \\ \sum_j c_j - 1 = 0, \qquad c_j c_k = 0 \ (j \neq k), \qquad x_{ij}c_j = 0.\end{gathered} \tag{16.3}$$

A solution would inject n elements into the $n - 1$ elements other than c, so every instance is unsatisfiable. For either proof system, every degree-d refutation satisfies

$$n < 3pd \qquad (n \geq 3).$$

Here is how the internal matching bound gives this estimate. Choose the largest p-power $q \leq n/3$ and put $r = n - q$. Set $c_n = 1$, all other $c_j = 0$, and restrict the image of g to $[r]$. After transposition, every remaining axiom is an axiom of the bijection system $B_{r,n}$. Here $B_{r,m}$ is the Boolean r-by-m matrix system with every row and column sum equal to one and both row and column collisions equal to zero. If $d < q$, then $2d \leq r$ and $q \mid n - r$, contradicting the rectangle obstruction of Lemma 17.1. The substitutions do not increase ordinary-PC or NS degree. Thus $d \geq q$, while maximality of q gives $n < 3pq \leq 3pd$. This proves T2 without an external pigeonhole lower-bound theorem.

The injection certificate and its linear degree bound. The paper's matching upper bound is useful to inspect directly. Put

$$R_i = \sum_{j=1}^{n} x_{ij}, \qquad C = \sum_{j=1}^{n} c_j, \qquad P = \prod_{i=1}^{n} R_i.$$

Every monomial in CP has the form $c_j \prod_i x_{i,\lambda(i)}$. If λ is not injective, two factors give a column-collision axiom. If it is injective, it is a permutation of $[n]$, and its unique row i with $\lambda(i) = j$ gives the omission axiom $c_j x_{ij}$. In either case the monomial is an axiom multiple of degree $n + 1$. Summing these identities expresses CP in the raw NS span at that degree.

The second certificate is the exact telescoping identity

$$CP - 1 = (C - 1) + C \sum_{i=1}^{n} (R_i - 1) \prod_{h<i} R_h.$$

Its summands use only constant and row totality axioms, with each complete product of degree at most $n+1$. Subtracting this certificate from the one for CP derives 1 at degree at most $n+1$. For a two-element universe the telescoping list is $(C-1)+C(R_1-1)+CR_1(R_2-1)$, with degrees $1, 2, 3$. A term $c_1x_{11}x_{21}$ in CP is a column-collision multiple; $c_1x_{11}x_{22}$ is an omission-axiom multiple. These are the two possibilities in the general expansion.

Thus, over any field and for every $n \geq 1$,

$$d_{\mathrm{PC}}(\Gamma_n^\eta) \leq d_{\mathrm{NS}}(\Gamma_n^\eta) \leq n+1.$$

The PC inequality uses the degree-preserving simulation of an NS certificate by repeated variable multiplication. No Boolean reduction is used. With the preceding $n < 3pd$ lower bound for $n \geq 3$, both degrees are $\Theta(n)$ for each fixed positive characteristic. This family proves sharpness of the general linear degree lower bound; it does not supply an upper bound for every uniform family. This certificate and the sharpness deduction are written mathematics, as in the paper; no additional Lean endpoint is claimed for them.

T3: hardness on one residue class. Let χ_p assert

$$\forall x \qquad f^p(x) = x \quad \text{and} \quad f(x) \neq x.$$

Its finite models are disjoint unions of directed p-cycles. In particular, it has a 0/1 graph assignment exactly when $p \mid n$. An explicit normalisation uses the function equations for f, the equations $f_{ii} = 0$, and

$$f_{i_0 i_1} f_{i_1 i_2} \cdots f_{i_{p-1} i_p} = 0 \qquad (i_p \neq i_0). \tag{16.4}$$

If $p \nmid n$, summing all length-p walks gives an NS refutation of degree at most p. Indeed, totality derives the sum of their monomials minus n within degree p. Nonclosed walks vanish by (16.4). A closed walk with a repeated vertex either contains a functionality collision or follows a shorter deterministic cycle; since p is prime, the latter is a forbidden fixed point. Each remaining cycle contributes its monomial p times, according to its starting vertex, and hence contributes zero. The resulting nonzero scalar n yields the refutation.

Now take $\psi_3 = \eta \wedge \chi_p$ in disjoint nonlogical symbols. For $p \mid n$, substitute a p-cycle assignment into the second block. This preserves proof degree and shows that the conjunction has exactly the degree of η, separately for NS and PC. For $p \nmid n$, the second block refutes in degree at most p. Consequently this all-size unsatisfiable family is bounded off the multiples of p and has linear degree on the multiples: it has type T3. No exact degree is needed on the bounded residue classes.

T4: the valuation of the universe size controls the difficulty. The weak-bijection example is more revealing. Let unary functions f, g be injective with disjoint images, and require their images together to cover the universe. Use a unary witness h for coverage:

$$\forall z \qquad f(h(z)) = z \ \vee \ g(h(z)) = z.$$

Let a_{ij}, b_{ij}, w_{ji} represent $f(i) = j$, $g(i) = j$, and $h(j) = i$. The combined $2n$-by-n matrix $(a; b)$ has row sums equal to one and both row and column collisions equal to zero. The witness block consists of

$$\sum_i w_{ji} - 1 = 0, \qquad w_{ji} w_{ji'} = 0 \ (i \neq i'), \qquad w_{ji}(1 - a_{ij})(1 - b_{ij}) = 0, \tag{16.5}$$

together with the Boolean equations. Call this complete system $\widehat{B}_n$. Its witnesses are retained: it is not identified with the simpler matrix system $B_{n,2n}$.

Put $q = p^{v_p(n)}$. The proved bounds give

$$q \le d_{\mathrm{PC}}(\widehat{B}_n) \le d_{\mathrm{NS}}(\widehat{B}_n) \le q + 4 \qquad \text{whenever } n \ge 2q. \tag{16.6}$$

The upper bound holds at every positive size, without the displayed size condition. We give both degree-preserving reduction and certificate explicitly.

A substitution is used here as a map of proofs, not merely as a map of solutions. An affine substitution replaces a variable by a linear polynomial plus a constant, so it does not increase polynomial degree. One must also check that every substituted input is derivable from the new inputs within the permitted degree. The following identities supply that check for the coverage and Boolean witness equations.

For the lower bound transpose the combined matrix and substitute $w_{ji} = a_{ij} + b_{ij}$ into every witness variable. Its totality becomes the corresponding totality of $B_{n,2n}$. The identities

$$(a+b)^2 - (a+b) = (a^2 - a) + (b^2 - b) + 2ab,$$
$$(a+b)(1-a)(1-b) = -(1-b)(a^2 - a) - (1-a)(b^2 - b)$$

put the substituted Boolean and coverage equations in the raw NS spans at degrees two and three respectively. A substituted witness collision expands into four column collisions of the raw combined matrix. Thus every substituted input has a derivation within its own degree. Substitution is affine: a multiplication by an old variable becomes a sum of permitted multiplications by new variables, with the preceding line still of degree at most $d - 1$. A degree-d ordinary-PC refutation would therefore refute $B_{n,2n}$ within the same degree. The rectangle obstruction excludes this when $2d \le n$ and $d < q$, yielding the lower bound in (16.6).

For the upper bound fix a column j and abbreviate

$$u_i = a_{ij} + b_{ij}, \qquad T = \sum_i u_i, \qquad Z = \sum_i w_{ji} u_i.$$

The coverage axiom and disjointness collision give the exact identity

$$Z - 1 = (\sum_i w_{ji} - 1) - \sum_i w_{ji}(1 - a_{ij})(1 - b_{ij}) + \sum_i w_{ji} a_{ij} b_{ij}.$$

Consequently $Z - 1$ lies in the degree-three NS span. If x is one entry of this column, then

$$Tx - x = (x^2 - x) + \sum_{y \ne x} yx$$

lies in the degree-two span, by the Boolean equations and column-collision equations. Adding this for a_{ij} and b_{ij} gives $Tu_i - u_i$ in that span. Finally

$$T - 1 = \sum_i w_{ji}(Tu_i - u_i) - (T - 1)(Z - 1)$$

puts the missing column totality in the degree-four span. This retains the coverage witnesses throughout the derivation.

For completeness, consider any matrix with row sums R_i and column sums C_j, with their differences from one in the NS span at degree e, and with both collision families in degree two. For a fixed k-element row set S, telescoping the product of its row sums gives $\prod_{i \in S} R_i - 1$ in degree at most $e + k$. Summing over S gives

$$\sum_{|S|=k} \prod_{i \in S} R_i - \binom{\#\text{rows}}{k}$$

in that same span. Expand the products as partial functions from k rows to columns. A noninjective partial function contains a column collision, so its monomial is already in the degree-k span. The remaining monomials are exactly those of k-edge matchings. The column calculation gives the same matching sum, since inversion pairs partial injections with identical monomials. Subtracting derives

$$\binom{\#\text{rows}}{k} - \binom{\#\text{columns}}{k}$$

within degree $e + k$. Applied to the $2n$-by-n matrix with $e = 4$, this is the desired counting certificate.

Write $n = qu$ with $q = p^{v_p(n)}$ and $p \nmid u$. In characteristic p, $(1 + t)^n = (1 + t^q)^u$. The coefficient of t^q is therefore u; for $(1 + t)^{2n}$ it is $2u$. Thus $\binom{2n}{q} - \binom{n}{q} = u \neq 0$ in K. Rescaling the counting certificate gives an NS refutation of degree at most $q + 4$.

For each fixed a, the sizes $n = p^a(p(k + 1) + 1)$, $k \geq 0$, have $v_p(n) = a$ and satisfy (16.6). Hence some finite degree in $[p^a, p^a + 4]$ recurs infinitely often. These windows are unbounded as a varies, proving T4 for both proof systems. On the other hand, $p \nmid n$ implies degree at most five. There are explicit simultaneous linear peaks: at $n_a = (p + 1)p^a$,

$$\frac{n_a}{p+1} \leq d_{\mathrm{PC}}(\widehat{B}_{n_a}) \leq d_{\mathrm{NS}}(\widehat{B}_{n_a}) \leq \frac{n_a}{p+1} + 4.$$

For example, in characteristic two every odd n has degree at most five, whereas at $n = 3 \cdot 2^a$ both degrees lie between 2^a and $2^a + 4$. The peaks are linear in their universe size even though the profile returns to bounded values infinitely often. Thus linear peaks coexist with infinitely many easy sizes. This is the reason the T4 conclusion concerns peaks rather than an eventual linear lower bound at every size.

16.4 Exercises

Exercise 16.5. For the consistent family of Proposition 16.1, compute the two profiles at sizes $1, 2, 3, 4, 5, 6$ in characteristic three. Compare the invariant sum request with the indexed request for every individual variable. Which statements have finite proof degrees?

Model answer (p. 169).

Exercise 16.6. For $\Gamma = \{x, 1 - x\}$ over an arbitrary field, write a degree-one PC proof of the fresh target y and prove that its least NS degree is two. Do not insert an axiom involving y.

Model answer (p. 170).

Exercise 16.7. In characteristic three, apply the fresh-y bounds to the normalised weak-bijection family at $n = 6$ and $n = 36$. What can the stated bounds say at $n = 9$? Distinguish the lower-bound size condition from the all-size upper certificate.

Model answer (p. 171).

Exercise 16.8. Suppose Γ has a root over some extension field of K. Explain why the fresh target y is underivable from $\Gamma^{\uparrow}$. If one instead adds $y^2 - y$ as an axiom, identify the extra term which appears when extracting the coefficient of y^1 from an NS identity. Why does the original coefficient-extraction proof no longer apply unchanged?

Model answer (p. 171).

Chapter 17

Matching designs and the concrete lower bounds

The examples use a lower bound for a concrete rectangular system. Its proof illustrates why a functional excluding a Nullstellensatz certificate does not by itself exclude a Polynomial Calculus derivation. To handle degree falls we construct a whole space of functionals, together with an extension operator between successive degrees. The use of designs as linear obstructions to Nullstellensatz proofs has a direct antecedent in Buss [26]. The argument here also proves the extension property needed to handle ordinary-PC lines. Here the target is 1. The capital D below is the proposed proof-degree budget, rather than the input-degree bound of a general template. The integers N, M describe row and column counts in the auxiliary boards.

17.1 Why this separate example proof is informative

The general linear theorem used an exact operation on whole consequence spaces. The concrete rectangular family gives a second way to see why ordinary-PC degree falls must be handled honestly. Here we will use linear functionals instead of a group-algebra projector.

A single functional annihilating the permitted axiom products excludes an NS certificate. It need not remain valid when a cancelled low-degree polynomial is multiplied in PC. The remedy is not to announce that NS and PC are the same. We construct a whole space of designs and prove that every lower-degree design extends. The PC invariant will quantify over *all* designs. This universal quantifier is what survives a multiplication after cancellation.

The proof has four steps: choose a terminating signed reduction of matchings, prove independence of its integral coordinates in a square, compress complete histories to obtain a binomial identity in rectangles, and use that identity to extend designs. The last step applies the functionals to every legal PC line. We give the complete construction in this chapter, rather than referring to a pigeonhole lower-bound paper.

For positive integers $r \le m$, let $B_{r,m}$ consist of the equations

$$\sum_{j=1}^{m} x_{ij} - 1 = 0 \quad (i \in [r]), \qquad \sum_{i=1}^{r} x_{ij} - 1 = 0 \quad (j \in [m]),$$

the Boolean equations $x_{ij}^2 - x_{ij} = 0$, and the row and column collision equations

$$x_{ij}x_{i\ell} = 0 \quad (j \neq \ell), \qquad x_{ij}x_{hj} = 0 \quad (i \neq h).$$

Here $[r] = \{1, \dots, r\}$. A Boolean solution would describe a bijection between the two sides, so a proper rectangle has no solution. Nevertheless, in positive characteristic the two linear

totality sums can agree because $m - r$ vanishes in the field. The following statement identifies a range in which this obstruction persists through ordinary PC.

Lemma 17.1 (Rectangle obstruction)**.** *Let K be a field of prime characteristic p, and let a, D be nonnegative integers. If*

$$1 \le r \le m, \qquad 2D \le r, \qquad D < p^a, \qquad p^a \mid m - r,$$

then $B_{r,m}$ has no ordinary-PC refutation of degree at most D. Consequently it has no Nullstellensatz refutation of total-product degree at most D.

We give the full construction over the integers before passing to K. An *edge set* is a subset of $[N] \times [M]$. It is a matching if no two distinct edges share a row or a column. Write $\operatorname{rows} P$ and $\operatorname{cols} P$ for its two projections. A degree-d design may take values in a coefficient ring: it is a function ψ on edge sets, zero on nonmatchings, for which

$$\sum_{j=1}^{M} \psi(P \cup \{(i,j)\}) = \psi(P), \qquad \sum_{i=1}^{N} \psi(P \cup \{(i,j)\}) = \psi(P) \quad (|P| < d). \tag{17.1}$$

The indices $i \in [N]$ and $j \in [M]$ are arbitrary. For K-valued designs, denote the resulting vector space by $E_d(N, M)$. The signed coordinate reductions below are constructed integrally and then interpreted in K. Normalisation, $\psi(\varnothing) = 1$, is a separate condition. We will construct a normalised member of $E_D(r, m)$ and prove that, for $D \ge 1$, *every* member of $E_{D-1}(r, m)$ extends to $E_D(r, m)$ while preserving its values through degree $D - 1$.

To interpret the table, associate an edge set P with the squarefree monomial $x_P = \prod_{e \in P} x_e$. Its proposed value is $\psi(P)$. For an actual matching assignment this value would be its containment indicator; a design instead imposes only the stated linear consistency conditions in the tested degree window. A normalised design can be read as a *pseudoexpectation* in the following algebraic sense: a linear assignment of values to monomials satisfying those tests and giving the constant monomial value one, without a probability distribution or a positivity assumption. Its extension to ordinary monomials is spelled out in the last section. Unnormalised designs are also needed for the extension argument.

17.2 A degree-one design which is not a solution

Over $\mathbb{F}_2$, consider four rows and six columns. Give the six columns respectively the selected row labels $1, 1, 1, 2, 3, 4$, and let $\psi(\{(i,j)\})$ be one on those six selected edges and zero elsewhere. Let $\psi(\varnothing) = 1$. Every column sum is one. The four row sums are $3, 1, 1, 1$, all one in the field. Thus these values satisfy the degree-one totality conditions, although the selected edges do not form an actual bijection: three columns use the same row.

This is an honest linear witness, not a Boolean model. Degree one has not yet tested the quadratic collision axioms. Higher-degree designs must also assign compatible values to matching edge sets. The extension construction below supplies these values exactly in the range of the rectangle obstruction, without division by a factorial.

> **Checkpoint.** An actual solution evaluates every polynomial multiplicatively. A design is a consistent linear response only to the tested degree window. It can exist for an inconsistent rectangle and still rule out every proof within that window.

17.3 A terminating signed reduction

The reduction will replace one matching by a formal integer linear combination of others. The word *signed* allows negative coefficients; it does not count matchings with positive weights.

The required identity is that every design gives the same value before and after a replacement. Keeping integer coefficients first lets us use the same identities after reducing modulo any characteristic.

A chain in a matching is a subset whose row coordinates and column coordinates both increase. Let $h(P)$ be the maximum chain length, and write $P_{\le b}$ for the edges in rows at most b. A matching P is *terminal* if, for every positive integer b,

$$h(P_{\le b}) \le b - |\operatorname{rows}(P_{\le b})|,$$

and the corresponding inequality holds for column prefixes. The right side counts unused labels in the prefix. This definition depends only on the labelled matching, not on the ambient rectangle. Prefixes beyond its last occupied row or column cannot introduce a new failure: their left side stays fixed and their right side increases.

For a nonterminal matching, use the least failed row prefix if one exists; otherwise use the least failed column prefix. In the selected prefix choose a longest chain, breaking ties by the lexicographically greatest list of its edges, with the edges themselves in row-first lexicographic order. In a column step use this rule after transposition. Thus all choices are deterministic.

Consider a row step with failed prefix b and chosen chain A. Put

$$P_0 = P \setminus A, \qquad S = \operatorname{cols} A, \qquad B = \{i \le b : i \notin \operatorname{rows} P\},$$

$$U = \operatorname{rows} A \ \cup \ \{b < i \le N : i \notin \operatorname{rows} P\}.$$

The sets B and U are disjoint and partition all rows unused by P_0. Moreover, $|B| < |A|$, because the selected prefix fails. A longest chain in the first failed prefix ends in row b. Indeed, if it ended earlier, the same chain would belong to the preceding prefix, whereas the number of unused labels cannot decrease when one adds the next row. That preceding prefix would already fail.

For sets of columns S and rows V, a full injection from S into V means a matching J with $\operatorname{cols} J = S$ and $\operatorname{rows} J \subseteq V$. A partial injection has only $\operatorname{cols} J \subseteq S$. Replace P by

$$\sum_{\substack{J \text{ partial injection} \\ S \to B}} (-1)^{|J|}[P_0 \cup J] \ - \sum_{\substack{J \text{ full injection} \\ S \to U \\ J \ne A}} [P_0 \cup J],$$

omitting colliding terms. Square brackets here denote formal basis elements, not polynomial equivalence classes. The transposed rule defines a column step.

To verify the identity for a degree-d design with $|P| \le d$, repeatedly apply its column equation to the columns of S. At each application the partial matching has fewer than $|P|$ edges, so the degree restriction is satisfied. An occupied row contributes zero by collision vanishing. For a still unassigned column the sum over available rows in U is therefore the previous value minus the sum over rows in B. Expanding these successive differences gives

$$\sum_{J : S \to U} \psi(P_0 \cup J) = \sum_{A' \subseteq S} (-1)^{|A'|} \sum_{J : A' \to B} \psi(P_0 \cup J).$$

Every surviving term is an injection, because collisions have value zero. The right side contains each partial injection exactly once, indexed by its column set. Isolating the term $J = A$ on the left proves the stated reduction. The column step follows by interchanging rows and columns in the design equations. No assertion that the deterministic reduction itself commutes with transposition is needed.

We next prove termination, including the equal-degree case. For the fixed grid $G = [N] \times [M]$, set $H = N + M$, $c = NM + 1$, and define

$$w(i,j) = c^{H-i-j}, \qquad W(P) = \sum_{(i,j) \in P} w(i,j), \qquad \rho(P) = |P|(W(G) + 1) + W(P).$$

Every lower term has fewer edges, since an injection into B has at most $|B| < |A|$ edges. Thus its ρ is smaller irrespective of its weight. For a full injection $J \neq A$ into U, let e be the first edge of A not belonging to J, in increasing row order. Every earlier chain edge still belongs to J. If $x \in J \setminus A$ had row smaller than that of e, its row would be an earlier chain row already occupied by its retained edge; this contradicts the matching condition. A row from the other part of U is greater than b and hence cannot be smaller. Likewise, a column of x smaller than the column of e belongs to an earlier retained chain edge, again giving a collision. Consequently

$$e_1 \leq x_1, \qquad e_2 \leq x_2, \qquad e_1 + e_2 < x_1 + x_2.$$

At least one inequality is strict because $x \neq e$. Each newly added edge has weight at most $w(e)/c$, and there are at most $NM < c$ such edges. Their total weight is strictly smaller than $w(e)$. Cancelling the common edges of A and J shows $W(P_0 \cup J) < W(P)$. The identical argument applies to a column step, since $i + j$ is unchanged by transposition. Every child therefore has strictly smaller nonnegative integer universe size ρ. The tree is finitely branching, and every path has length at most its initial universe size. Hence the complete signed reduction tree is finite.

For terminal T, let $C_{N,M}(P, T) \in \mathbb{Z}$ be the sum, over paths from P to T, of the products of their step signs; set it to zero for colliding P. The empty path gives

$$C_{N,M}(S, T) = \delta_{S,T} \quad (S \text{ terminal}), \qquad C_{N,M}(P, T) = 0 \quad (|T| > |P|).$$

Induction on $\rho(P)$ using the reduction identity gives, for every degree-d design,

$$\psi(P) = \sum_{\substack{T \text{ terminal} \\ |T| \leq d}} C_{N,M}(P, T)\psi(T) \qquad (|P| \leq d). \tag{17.2}$$

Thus the terminal values determine every design in the relevant range. It remains to show that these coordinates themselves satisfy the required totality equations and can be restricted from a square to a rectangle.

For a small illustration of the preceding reduction, in a 2-by-2 square the matching $\{(1, 2)\}$ fails the first row prefix. Its chosen chain is that one edge, B is empty, and $U = \{1, 2\}$. The rule gives

$$[\{(1, 2)\}] \longmapsto [\varnothing] - [\{(2, 2)\}].$$

Here $\{(2, 2)\}$ is terminal. For a design of degree at least one, the corresponding identity is simply the column sum $\psi(\{(1, 2)\}) + \psi(\{(2, 2)\}) = \psi(\varnothing)$. Thus a terminal coordinate can be negative; it is not a probability.

17.4 Integral independence in a square

An indicator here is a function on actual permutations of the square: it takes value one when the permutation contains the given matching, and zero otherwise. The argument is *integral* because it uses integer coefficients throughout. Here it proves independence after passing to any coefficient ring in the theorem, as well as over the integers. The signed moments below isolate a coefficient with multiplier 1 or -1, both invertible in every ring. This is the reason for proving a unit diagonal rather than a nonzero factorial.

Let $T \subseteq [N] \times [N]$ be terminal. For an edge $e \in T$, let its height be the maximum chain length in the southwest downset $\{f \in T : f_1 \leq e_1,\ f_2 \leq e_2\}$. A longest such chain ends at e: otherwise e can be appended, because distinct edges of a matching have distinct row and column coordinates. If e is southwest of f with both coordinates strictly smaller,

appending f raises the height. It follows that every equal-height level T_t is decreasing: as its row coordinate increases, its column coordinate decreases. The levels partition T.

Let $h = h(T)$. The terminal inequality at prefix N implies that there are at least h unused rows and h unused columns. Let a_t and b_t be the tth smallest unused row and column, respectively, for $1 \le t \le h$. These anchors precede every coordinate in T_t. For example, if $e \in T_t$, a chain of length t ending at e belongs to its row prefix. Terminality requires at least t unused rows in that prefix; row e_1 is occupied, so the tth unused row is strictly less than e_1. The column argument is identical. Form the paired blocks

$$A_t = \operatorname{rows} T_t \cup \{a_t\}, \qquad B_t = \operatorname{cols} T_t \cup \{b_t\}.$$

They are disjoint on each side, their paired sizes agree, and their minima are a_t, b_t. Their complements also have equal sizes. Fix the increasing bijection on the complements, and an increasing base bijection $A_t \to B_t$ in each block. Vary each base bijection by a permutation σ_t of A_t. The union of all these bijections is an actual permutation π_σ of $[N]$.

For any edge set P, define the integer signed moment

$$M_T(P) = \sum_{\sigma} \left(\prod_{t=1}^{h} \operatorname{sgn} \sigma_t \right) \mathbf{1}_{P \subseteq \pi_\sigma}.$$

If a block has two rows absent from rows P, interchange their images. This is a fixed-point-free involution on the choices, changes the sign, and preserves the containment condition. Its terms cancel over $\mathbb{Z}$. In particular, $M_T(P) = 0$ when $|P| < |T|$: otherwise every block would use at least $|A_t| - 1 = |T_t|$ rows, forcing $|P| \ge \sum_t |T_t| = |T|$.

The diagonal is a unit. In each block the edges of T_t prescribe the bijection on every row except a_t; its remaining value is necessarily b_t. Thus exactly one choice σ contains T, and

$$M_T(T) \in \{1, -1\}.$$

This is a single signed contribution; no factorial or positive integer multiplicity must be inverted.

For equal-degree triangularity, suppose $|P| = |T|$ and $M_T(P) \neq 0$. There is a choice containing P, so P is a matching. Cancellation shows that at most one row in each block is unused. Counting over the disjoint blocks forces exactly $|T_t|$ edges in block $A_t \times B_t$, one unused row in every block, and no edge in the complement. Write P_t for this block matching.

Among all matchings of size $|T_t|$ in $A_t \times B_t$, the decreasing matching T_t has uniquely least weight W. Here is a direct proof. Choose a weight minimiser Q from this finite set. If it uses the minimum row a_t, move that edge to the unique unused row. This is a legal matching, has the same cardinality, and strictly smaller weight, because its row coordinate has increased. Thus a_t is unused. The same argument shows that the minimum column b_t is unused. If Q contains an increasing pair $(i, j), (i', j')$ with $i < i'$ and $j < j'$, replace it by the crossed pair $(i, j'), (i', j)$. Both added edges have coordinate sum strictly larger than that of (i, j), and their combined weight is smaller than its weight, since $c = N^2 + 1 > 2$ when such a pair exists. The unchanged context is disjoint from the crossed pair. This again contradicts minimality. Therefore Q is decreasing, uses precisely $A_t \setminus \{a_t\}$ and $B_t \setminus \{b_t\}$, and must equal T_t: on two fixed ordered sets there is exactly one order-reversing bijection. Any distinct matching with the same minimum weight would also be a minimiser and would give the same conclusion. Hence

$$W(P_t) \ge W(T_t), \quad \text{with equality only if } P_t = T_t.$$

Summing over the disjoint blocks proves that

$$|P| = |T|, \quad P \neq T, \quad M_T(P) \neq 0 \quad \Longrightarrow \quad W(P) > W(T).$$

Together with the lower-degree cancellation, this says that $M_T(P) = 0$ for every $P \neq T$ with $\rho(P) \leq \rho(T)$.

Now suppose a finite linear combination of terminal matching indicators vanishes at every permutation of $[N]$, with coefficients in any commutative ring R. Among its nonzero coefficients choose a terminal T of greatest universe size ρ. Apply the integer signed moment M_T to this relation, then map the integers to R. All other nonzero terms vanish by triangularity; the result is the coefficient of T multiplied by 1 or -1. This contradicts its being nonzero. The terminal indicators are therefore independent over every commutative ring, including fields of characteristic two and rings with zero divisors.

Each square permutation satisfies the design equations in every degree. Equation (17.2), applied to its containment indicator, expresses every matching indicator in terminal coordinates. For a fixed row i, the permutation identity

$$\sum_{j=1}^{N} \mathbf{1}_{P\cup\{(i,j)\}\subseteq\pi} = \mathbf{1}_{P\subseteq\pi}$$

holds because π has exactly one edge in that row. Expand all terms in terminal coordinates and use the integral independence just proved. For each terminal T this yields

$$\sum_{j=1}^{N} C_{N,N}(P \cup \{(i,j)\}, T) = C_{N,N}(P, T).$$

The column identity follows in the same way from the unique edge in a column. Colliding sets have zero coordinates and zero permutation indicators, so they are included. Thus every square coordinate is an unnormalised design, in every degree, over the integers.

17.5 Whole-history compression and the marked binomial identity

The last matching of a path need not remember every row visited by that path. We must group paths using their *entire* histories. Once the total signed weight is the same for every chosen k-element set of appended rows, summing over those sets produces $\binom{L}{k}$. The mark ensures that k is positive, even if its initial edge is later deleted. This will remove the unwanted appended-row contribution when the binomial coefficients vanish in K.

Fix $P, T \subseteq [r]\times[m]$, with P a matching and T terminal, and a column $j \notin \operatorname{cols} P$. Append L rows, and consider all pairs consisting of a marked initial row $i \in \{r+1, \ldots, r+L\}$ and a signed reduction path from $P \cup \{(i,j)\}$ to T. The weight of a pair is the product of its path signs. Its *history support* is the set of appended rows appearing anywhere on the path, including the initial state and edges that are subsequently deleted.

If $|P| + 1 \leq D$ and $2D \leq r$, every such support is nonempty and has size at most $|P| + 1$. To prove the latter assertion, give one token to each initial edge. In a row rewrite, transport a token by its column to the unique child edge in that column, or delete it if that column is absent. The child uses only columns of the parent, so this describes every child edge exactly once. In a column rewrite transport by row, with the analogous deletion rule. No deleted token reappears, and no two surviving tokens represent the same edge.

For any matching Q of at most D edges, failure at row prefix b implies

$$b < h(Q_{\leq b}) + |\operatorname{rows}(Q_{\leq b})| \leq 2|Q| \leq 2D \leq r.$$

Thus every chosen row chain lies in old rows. An edge already in an appended row is part of the unchanged context of a row rewrite and is retained verbatim. A column rewrite preserves

its token's row or deletes the token. It follows that a token can encounter at most one appended row throughout its lifetime. There are $|P| + 1$ initial tokens, so the full history has at most that many appended rows. The initial mark belongs to it even when its token later disappears.

We next verify that histories of the same size carry the same total integer weight. Let S be a prescribed k-element subset of the L appended rows, and let f be the increasing injection fixing $[r]$ and sending $r + 1, \dots, r + k$ to the elements of S in order. For $t \geq 1$, set $f(r + k + t) = r + L + t$. This extends f increasingly and makes the inverse image of the large row interval $[r + L]$ exactly $[r + k]$. Rename the row coordinate of every edge by f and retain its column and step sign.

First, this renaming preserves the deterministic selector. Every failed row prefix lies below r, as just proved; those prefixes and their unused-row counts are unchanged. Their chains are unchanged as labelled edge sets. For a column prefix, the number of occupied columns and all chain lengths are preserved by the increasing row renaming. The first failed column prefix is therefore unchanged. The lexicographic tie choice is also preserved: a strictly increasing map preserves the lexicographic order of sorted edge lists, including the transposed lists used in a column step. Orientation, first failure and chosen chain all agree under renaming, and terminality is preserved.

Second, renaming gives a bijection on the branches whose child edges remain in $[r] \cup S$. In a row step, the inverse images of the sets B and U are exactly the corresponding sets in the compressed rectangle: order comparisons with the old endpoint and occupied-row membership are preserved. An image-supported replacement therefore has a unique inverse injection. Its column set, cardinality, exclusion of the original chain, collision status and sign are all preserved. For a column step the permissible column sets are unchanged; the same argument, using row membership, applies. A child cannot encode two different replacement injections, since its replacement edges are recovered by removing the fixed context: their columns are disjoint from the context columns in a row step, and their rows are disjoint in a column step. The lower and upper branch families are disjoint because their cardinalities differ. Thus no multiplicity is lost. There may be other branches using rows outside S, but these belong to different history supports and are not part of this correspondence.

Applying the branch bijection successively gives a two-way, sign-preserving bijection between marked paths with history exactly $\{r + 1, \dots, r + k\}$ in the compressed rectangle and marked paths with history exactly S in the large rectangle. Its inverse simply renames every state and the initial mark back. The endpoint T is fixed because it uses only old rows. All intermediate states have at most D edges, so the selector argument applies at every step.

Define $c_k(P, T, j)$ to be the total weight of the marked paths in the rectangle $[r + k] \times [m]$ whose full history is all k appended rows. Partitioning the marked paths by their history support, and then by its cardinality, proves the integer identity

$$\sum_{i=r+1}^{r+L} C_{r+L,m}(P \cup \{(i, j)\}, T) = \sum_{k=1}^{D} \binom{L}{k} c_k(P, T, j). \tag{17.3}$$

The coefficients are independent of L. Its constant term is absent because each path retains its initial mark in the *history*, not necessarily in its final state. This distinction is essential.

17.6 Rectangular designs and their extensions

We now pass from integer reduction coordinates to the fixed field K. The two goals are a design with $\psi(\varnothing) = 1$ and an extension of every degree-$(D - 1)$ design; the first will detect 1, and the second will make the obstruction survive ordinary-PC multiplication. Assume

now the hypotheses of Lemma 17.1, with $D \geq 1$, and put $q = p^a$ and $L = m - r$. For $1 \leq k \leq D < q$ the integer $\binom{L}{k}$ is zero in K. Indeed, $q \mid L$ and in characteristic p,

$$(1+z)^L = (1+z^q)^{L/q};$$

the right side has no coefficient in these degrees. Take the square of size m and an old terminal matching $T \subseteq [r] \times [m]$. The function

$$\theta_T(P) = C_{m,m}(P,T) \quad \text{in } K, \qquad P \subseteq [r] \times [m],$$

is a degree-D rectangular design. Its row equations are the square row equations, with the same column set. Its column equations follow from the square column equation after deleting the appended-row sum, which vanishes by (17.3) for $|P| < D$ and an unused column. If the column is occupied, every appended insertion collides, so the omitted sum is already zero. If P collides, every insertion still collides and both sides are zero. This covers all design equations.

On old terminal inputs, the square reduction is still terminal and

$$\theta_T(S) = \delta_{S,T}.$$

In particular, $\theta_\varnothing$ is normalised. For an arbitrary $\psi \in E_{D-1}(r,m)$ define explicitly

$$\widetilde{\psi}(P) = \sum_{\substack{T \text{ terminal in } [r]\times[m] \\ |T| \leq D-1}} \psi(T)\theta_T(P).$$

This is a member of $E_D(r,m)$ and agrees with ψ on all terminal inputs of size at most $D-1$. Apply (17.2) in the old rectangle to both designs, after restricting the new design to degree $D-1$. They agree on every edge set of size at most $D-1$. Thus the extension exists for every unnormalised design, with no choice of an unspecified lift.

17.7 Why extension is precisely the multiplication rule

The polynomial spaces $P_{\leq 0} \subseteq P_{\leq 1} \subseteq \cdots$ form a *filtration*: nested spaces indexed by the degree allowed. The corresponding design problem asks for compatible values on each window. Extending a design means keeping every tested value on edge sets of size at most $D-1$ and supplying the extra degree-D values. It is stronger than constructing one normalised design at degree D; the multiplication argument needs an extension for whichever lower-degree table a given edge produces.

Fix an edge e. To test a new line $x_e f$ of degree at most D, start with any degree-D design ψ. Force the edge into the design by defining $s_e\psi(P) = \psi(P \cup \{e\})$. This is a degree-$(D-1)$ design. Extend it back to degree D. The extension evaluates every monomial of f exactly as the original design evaluates that monomial multiplied by x_e, because f has degree at most $D-1$.

The induction hypothesis says that *every* degree-D design kills earlier lines. It therefore kills f under this chosen extension, which proves that the original ψ kills $x_e f$. This is the exact point at which a space of designs succeeds where one isolated witness would be insufficient.

17.8 The invariant through ordinary-PC degree falls

For a design ψ define a linear functional on the original polynomial ring by

$$\Lambda_\psi\left(\sum_\alpha b_\alpha x^\alpha\right) = \sum_\alpha b_\alpha \psi(\operatorname{supp}\alpha),$$

where $\operatorname{supp}\alpha$ is the set of edges with positive exponent. This formula is a functional on ordinary polynomials; it does not replace a PC proof by a proof in a Boolean quotient. Every degree-D design annihilates each axiom usable in a degree-D proof. Totalities use (17.1) at $P = \varnothing$; collisions use vanishing on nonmatchings; and the Boolean equations use the equality of the supports of x_e^2 and x_e.

For an edge e, put $(s_e\psi)(P) = \psi(P \cup \{e\})$. If $\psi \in E_D(r, m)$, then $s_e\psi \in E_{D-1}(r, m)$. Collision vanishing is immediate. In either totality equation at $|P| < D - 1$, apply the degree-D equation to $P \cup \{e\}$, whose cardinality is at most $D - 1$. The insertion operations commute, including repeated insertion of the same edge. Also, directly from the monomial formula,

$$\Lambda_\psi(x_e f) = \Lambda_{s_e\psi}(f).$$

Induct through a degree-D ordinary-PC derivation, asserting that every line is annihilated by *every* member of $E_D(r, m)$. Input axioms were handled above, and linear combinations preserve the assertion even when their highest-degree terms cancel. Consider multiplication of an earlier line f by x_e. If $f = 0$ there is nothing to prove. Otherwise ordinary polynomial degree gives $\deg(x_e f) = \deg f + 1$, so $\deg f \le D - 1$. For a given $\psi \in E_D(r, m)$, extend $s_e\psi$ to a design $\nu \in E_D(r, m)$ using the preceding construction. The earlier line is annihilated by ν by the inductive assertion. Since the two designs agree on all edge sets of size at most $D - 1$, and every monomial of f has support of that size or smaller,

$$\Lambda_\psi(x_e f) = \Lambda_{s_e\psi}(f) = \Lambda_\nu(f) = 0.$$

This proves the invariant for multiplication as well. Finally the normalised design $\theta_\varnothing$ sends the constant 1 to 1, so that constant cannot be the last line of the derivation.

For $D = 0$, all displayed axioms have positive degree because both sides of the rectangle are nonempty. No nonzero constant axiom is available, and multiplication of a nonzero polynomial cannot produce a degree-zero line. Linear combinations of the available zero lines remain zero. Thus a degree-zero refutation is impossible directly. This completes the ordinary-PC proof in every case. The NS conclusion follows from the degree-preserving simulation of a total-product NS certificate by ordinary PC.

Chapter 18

The topological classification and algebraic witnesses

This chapter expands Section 11 of the paper. The main description of T1–T4 is the topological classification in Section 18.1: empty, full, nonempty proper clopen, or closed but not open hardness spectrum. Its proof uses eventual prime-power periodicity and compactness, and applies to all the uniform requests covered by the theorem. We then distinguish algebraic witnesses from actual models. The final Euler-structure discussion is optional historical background; it is not needed for this classification and supplies no converse criterion. A uniform target invites a semantic question: does every solution of Γ_n satisfy $G^{(n)} = 0$? Our theorem classifies syntactic algebraic derivation, not that semantic question. The two are related through linear witnesses, but the relation is not an identification of the four types with four classes of ordinary models. Keep a fixed uniform axiom family Γ_n and its fixed target request over a field K of prime characteristic p. Its profile $h_S(n)$ is the maximum least degree needed for those targets in NS or ordinary PC.

18.1 The four types as shapes of a hardness spectrum

The current paper gives a useful picture of the classification. This picture applies to *every* aggregate or fixed-orbit uniform request in the theorem, over an arbitrary field of characteristic p. It is not restricted to Boolean equations or to a particular translation of first-order logic. Its input is eventual prime-power periodicity of bounded-degree derivability.

The residue-class structure is already implicit in Riis's 2009 report [8, Section 4, pp. 18–19]. The following reformulates that structure and extends its reading to the uniform consequences considered here; it is not a claim to have newly discovered the four cases. No converse to the Euler-model criterion is needed.

18.1.1 A p-adic point is a compatible address in a residue tree

At level one record a residue modulo p; at level two record one modulo p^2 which reduces to the first residue; continue in this way. A compatible infinite sequence is a p-adic integer. For example, in characteristic two, the successive addresses of the ordinary integer 5 are 1 mod 2, 1 mod 4, 5 mod 8, 5 mod 16, and so on. Not every infinite address represents a nonnegative ordinary integer. This distinction is precisely what makes the extra points useful. For further background see Gouvêa [28]; the argument needed here is supplied below.

A neighbourhood specifies a finite part of the address: $a + p^L\mathbb{Z}_p$ contains all points with the chosen residue modulo p^L. Such a neighbourhood is both open and closed, or *clopen*: its complement is the finite union of the other residue classes. The whole space $\mathbb{Z}_p$ is compact.

One way to see why is to view its points as infinite paths in the finitely branching residue tree. If a cover by residue neighbourhoods had no finite subcover, one could choose a residue at each level whose descendants still had no finite subcover. The resulting infinite path would be uncovered, a contradiction. General open covers reduce to these basic ones.

18.1.2 Colour the residues which are eventually easy

Use exactly $N(d)$ and $Q(d)$ from (6.1), including the separate convention $N(0) = n_*$ and $Q(0) = 1$. Define

$$A_d(h) = \{a \in \{0, \dots, Q(d)-1\} : \ h(n) \le d \text{ for all } n \ge N(d),\ n \equiv a \pmod{Q(d)}\},$$
$$\mathcal{E}_d(h) = \bigcup_{a \in A_d(h)} (a + Q(d)\mathbb{Z}_p), \qquad \mathcal{H}_p(h) = \mathbb{Z}_p \setminus \bigcup_{d \ge 0} \mathcal{E}_d(h).$$

Here $A_d(h)$ is a *set of residues*, whereas $N(d)$ is a *size threshold*. The calligraphic $\mathcal{E}_d(h)$ is a subset of $\mathbb{Z}_p$; it is not the set $E(h)$ of recurring degree values. The sets $\mathcal{E}_d(h)$ increase with d. To check this, pass to the common finer period and compare sufficiently late positive integer representatives. Each set is clopen, so the hardness spectrum $\mathcal{H}_p(h)$ is closed.

A point lies *outside* this spectrum precisely when it has a residue neighbourhood on which all sufficiently large integer sizes have one uniform finite degree bound. This equivalent description shows that choosing later valid onsets or finer valid prime-power periods does not change the spectrum. Finitely many exceptional sizes also make no difference. We have not defined a polynomial system at a noninteger p-adic size: the point records compatible information about ordinary finite sizes.

Proposition 18.1 (The four shapes)**.** *For the profiles in the four-way theorem,*

Type	*Shape of* $\mathcal{H}_p(h)$
T_1	*Empty*
T_2	*All of* $\mathbb{Z}_p$
T_3	*Nonempty, proper and clopen*
T_4	*Closed but not open*

For the same uniform request, NS and ordinary PC have the same hardness spectrum, even if their exact degree profiles differ.

Proof. If the spectrum is empty, the increasing open sets $\mathcal{E}_d$ cover compact $\mathbb{Z}_p$. A finite subcover is contained in one $\mathcal{E}_D$, so that set is the whole space. This gives an eventual degree bound D, exactly T1. Conversely such a bound makes the spectrum empty. The spectrum is the whole space exactly when every $\mathcal{E}_d$ is empty. This means that no finite degree recurs, exactly T2.

In T3 the classification theorem gives a fixed nonempty proper union of easy residue classes, with a uniform bound D. No bounded-degree residue neighbourhood can intersect the linearly hard region: a common refinement would contain arbitrarily large integers satisfying both incompatible bounds. Thus $\bigcup_d \mathcal{E}_d = \mathcal{E}_D$, and the spectrum is nonempty, proper and clopen. Conversely, for a nonempty proper clopen spectrum its complement is compact. The same finite-subcover argument makes the complement equal to some $\mathcal{E}_D$. Any recurring finite value e occupies a residue neighbourhood on a tail, by truncated periodicity. That neighbourhood belongs to $\mathcal{E}_e \subseteq \mathcal{E}_D$, whence $e \le D$. At least one finite value recurs, and the nonempty spectrum excludes T1. This is T3. The remaining closed-but-not-open possibility is therefore exactly T4.

Finally $h_{\mathrm{PC}}(n) \le h_{\mathrm{NS}}(n) \le F(h_{\mathrm{PC}}(n))$, with F independent of n. A residue neighbourhood is eventually bounded for one system if and only if it is eventually bounded for the other. This proves equality of their spectra; it does not prove equality of $\mathcal{E}_d$ for a fixed d. □

18.1.3 A worked binary example: hard at the limit, finite at every size

Take the abstract profile $h(n) = 2^{v_2(n)}$ for positive integers n. Thus odd sizes have value one, sizes twice an odd number have value two, and sizes four times an odd number have value four. This is the toy profile already used in Chapter 6, not an assertion that a particular polynomial family has exactly these degrees. The easy regions are

Budget d	$\mathcal{E}_d(h)$	Still outside this easy region
0	$\varnothing$	$\mathbb{Z}_2$
1	$1 + 2\mathbb{Z}_2$	$2\mathbb{Z}_2$
$2, 3$	$\mathbb{Z}_2 \setminus 4\mathbb{Z}_2$	$4\mathbb{Z}_2$
$4, \ldots, 7$	$\mathbb{Z}_2 \setminus 8\mathbb{Z}_2$	$8\mathbb{Z}_2$
$8, \ldots, 15$	$\mathbb{Z}_2 \setminus 16\mathbb{Z}_2$	$16\mathbb{Z}_2$

Consequently $\mathcal{H}_2(h) = \bigcap_L 2^L\mathbb{Z}_2 = \{0\}$. Every positive size has finite value, yet there is no bound valid on any neighbourhood of this limiting address. For every L, the neighbourhood $2^L\mathbb{Z}_2$ contains the arbitrarily large positive sizes 2^j with values 2^j. It also contains the arbitrarily large sizes $2^L(2k + 1)$ of the fixed finite value 2^L. Thus each neighbourhood of zero meets the easy region but is not uniformly easy. The singleton $\{0\}$ is closed and not open. This is the T4 picture. The point $0 \in \mathbb{Z}_2$ is an address, not a new instance with an empty underlying universe.

For comparison, the abstract profiles $h(n) = 1$ and $h(n) = n$ give empty and full spectra. The profile which is 1 on odd sizes and n on even sizes gives $2\mathbb{Z}_2$, the T3 shape. These examples separate the four pictures without claiming a realisation theorem for arbitrary closed subsets of $\mathbb{Z}_p$.

18.1.4 What topology explains, and what still needs the quantitative proof

In T3, looking at a single finite number of residue digits suffices to separate the easy and hard regions. In T4 it never suffices near a boundary point. Compactness is the step turning local finite bounds into one global bound in T1, or one bound on the entire easy part in T3. It does not compute that bound.

Topology alone does not give linear lower bounds. For example, $g(n) = 1 + v_2(n)$ has the same singleton spectrum as the preceding toy profile, but $g(n) \leq 1 + \log_2 n$. Its threshold periods grow exponentially with the degree budget, rather than obeying the linear period bound proved here. Thus it is not a counterexample to the paper's theorem. The estimates $N(d) = O(d)$ and $Q(d) = O(d)$ are what produce the linear gap and finite linear peaks. Keep the qualitative picture and that quantitative input distinct.

18.2 A quotient remembers what remains unproved

For a fixed degree budget d, the ambient polynomial space must also contain the target. We therefore allow polynomials up to degree $\max\{d, D_G\}$, where D_G is the fixed target-degree bound. Form its quotient by $I_d(n)$ for NS or by $C_d(n)$ for PC, the consequence spaces of Proposition 3.2. A target is derivable exactly when its quotient class is zero. If the class is nonzero, elementary linear algebra separates it from zero by a functional. This is a completely abstract description of the failure to prove a consequence.

For target one the separating functional is normalised by $\lambda(1) = 1$. For a general target it must instead satisfy $\lambda(G) = 1$. That small change matters: a witness for failure to derive x need not be normalised on one at all. Nor is it generally evaluation at a model. The two examples below make the distinction explicit.

18.3 Dual witnesses for failure to derive a target

For refutations, the design viewpoint of Buss [26] organises such linear witnesses. We state the elementary separation criterion for a prescribed target, whose normalisation is on that target rather than necessarily on the constant 1. For either system write $W_d(n)$ for its degree-d consequence space: $W_d(n) = C_d(n)$ for PC and $W_d(n) = I_d(n)$ for NS. Let $\delta = D_G$ be the fixed upper bound on target degree. We view $W_d(n)$ inside $P_{\leq \max\{d,\delta\}}(n)$, so the ambient space also contains every requested target even when its degree exceeds d.

The *dual* of a vector space is its space of linear functionals. A functional *annihilates* a subspace when it is zero on every vector of that subspace. Thus the two conditions below say that the functional ignores all proved consequences but detects the chosen target. The condition on the target fixes the scalar normalisation of the functional; it does not make the functional multiplicative.

Definition 18.2 (A degree-bounded target witness)**.** For a requested target $G_b^{(n)}$, a degree-d witness is a K-linear functional

$$\lambda : P_{\leq \max\{d,D_G\}}(n) \longrightarrow K, \qquad \lambda(G_b^{(n)}) = 1, \qquad \lambda(W_d(n)) = 0.$$

For an all-target request, a witness consists of an index b and such a functional. For a single fixed orbit, one may use any available representative. With target 1, this reduces to the usual normalisation $\lambda(1) = 1$ for a bounded-degree non-refutability witness.

Proposition 18.3 (Dual form of nonderivability)**.** *A degree-d target witness exists exactly when the target has no derivation of degree at most d in the chosen system. For an all-target request it exists exactly when $h(n) > d$.*

Proof. If $G_b^{(n)} \in W_d(n)$, a functional annihilating $W_d(n)$ cannot take value one on it. Conversely its class in the finite-dimensional quotient $P_{\leq \max\{d,D_G\}}(n)/W_d(n)$ is nonzero whenever it is not in $W_d(n)$. Extend that class to a basis, assign it value one and assign the other basis vectors zero. Composing with the quotient map gives the required functional. If the target degree is greater than d, it automatically lies outside $W_d(n)$; the enlarged ambient space also covers this degree-floor obstruction. Failure of all-target derivability is failure for at least one target. □

A K-valued solution a of Γ_n which violates $G_b^{(n)} = 0$ gives a witness by $\lambda(f) = f(a)/G_b^{(n)}(a)$. For target one this is ordinary evaluation, with its multiplicative property. A general witness need not be multiplicative, positive or normalised on one. It records a linear separation from all permitted bounded-degree consequences, not necessarily a truth assignment to the original variables. If the solution is instead defined over an extension field E/K, the scaled evaluation initially takes values in E. Choose a K-linear map $\pi : E \to K$ with $\pi(1) = 1$, by extending 1 to a K-basis of E. Composing the scaled evaluation with π gives a K-valued separating functional: it still vanishes on every consequence and takes value one on the target. This descent uses linearity, not multiplicativity of π.

Example 18.4 (Semantic entailment need not be derivation)**.** The equation $x^2 = 0$ over a field forces $x = 0$ at every solution. Nevertheless, neither NS nor PC derives x from this axiom: every consequence belongs to the ideal (x^2). For $d \geq 2$, the functional on $P_{\leq d}$ taking value one on x and zero on all other monomials annihilates the degree-d consequence space. This is a target witness although no actual solution violates $x = 0$. The example explains why our minimum degree is a syntactic ideal-membership resource.

Example 18.5 (Linear consistency without truth-value evaluation)**.** Over $\mathbb{F}_3$, take $x^2-x=0$, target one and degree budget two. The PC consequence space is $K(x^2-x)$: no nonzero vector in that span has degree at most one, so no multiplication is usable. The functional

$$\lambda(1)=1, \qquad \lambda(x)=\lambda(x^2)=2$$

is a witness but is not evaluation: $\lambda(x^2)=2$ whereas $\lambda(x)^2=1$. Boolean equations impose the intended truth-value semantics on actual solutions, without making every linear witness an actual evaluation.

Annihilating only axiom multiples gives an NS witness. For PC the functional must annihilate the full closure, including polynomials obtained by cancellation followed by new multiplication. This is the dual form of the distinction between I_d and C_d.

18.4 Four patterns of bounded-degree algebraic information

Let $h(n)$ be a fixed-orbit or all-target degree profile in the chosen system. By Proposition 18.3, a degree-d witness exists precisely when $h(n) > d$. Recall that $E(h)$ consists of the finite values occurring at arbitrarily large universe sizes. The four alternatives classify eventual boundedness and whether $E(h)$ is empty, finite nonempty or infinite; they do not classify the sets of roots of Γ_n.

Proposition 18.6 (A dual reading of the classification)**.** *The four types have the following mutually exclusive interpretations.*

T1. *Some fixed D suffices to exclude degree-D witnesses at every sufficiently large size.*
T2. *For every fixed d, witnesses exist at every sufficiently large size.*
T3. *There are a fixed D, a p-power period and a nonempty proper easy residue set. No degree-D witness exists on easy residues at large sizes. On hard residues, witnesses exist at every fixed degree for all sufficiently large sizes in those residues.*
T4. *Infinitely many distinct finite degrees e recur, each at arbitrarily large sizes. At an occurrence with $e \geq 1$, a degree-$(e-1)$ witness exists while a degree-e witness does not. There is also a fixed degree at which witnesses fail at arbitrarily large sizes. No fixed periodic partition separates a uniformly bounded-degree part from a part with positive linear hardness.*

Proof. T1 follows directly from $h(n) > d$ being equivalent to witness existence. In T2 the linear bound eventually exceeds any fixed d; conversely eventual witness existence at every fixed degree excludes every recurring finite value. For T3 use the easy/hard residue split of Theorem 6.5. Conversely the displayed split makes the recurring values a nonempty subset of $\{0,\dots,D\}$ and leaves the profile unbounded. In T4 use each recurring e for the two adjacent-degree assertions, and use the least recurring value for infinitely many failures at a fixed degree. The impossibility of a fixed periodic bounded/linear split is Proposition 6.6. Infinitely many recurring finite values characterise T4, completing the converse. □

For each fixed d, the existence of a witness is eventually determined by a residue modulo a p-power. T3 needs only one finite degree window to distinguish all easy residues. In T4, larger windows keep revealing new recurring finite levels. The linear onset bounds the last possible isolated low degree, while the short period locates finite linear peaks. For indexed targets, the separating index may vary with size within the fixed requested system; this is not a change to the request itself.

The *finite-model spectrum* of a sentence is the set of positive integers n for which it has a model with n elements. It records existence of solutions, without recording a polynomial proof budget.

The refutation examples in Section 16.3 all have no finite nonempty models yet exhibit all four types. Their identical empty finite-model spectra cannot distinguish the cases. The fresh-target construction in Section 16.2 also shows that a nonconstant consequence can retain these algebraic distinctions. The classification is a property of uniform algebraic reasoning about the chosen equations and targets, rather than a satisfiability classification.

18.5 Optional background: the Euler criterion

This section explains the separate sufficient criterion recalled at the end of Section 11 of the paper. It is not another description of the four types. Readers interested in the topological classification can skip it without losing any step of the argument.

Krajíček's Euler structures [13] equip the permitted parameter-definable sets with a strong counting characteristic χ. It assigns one to a point, is additive on disjoint unions, multiplicative on products and invariant under definable bijections. It also satisfies the fibre rule $\chi(X) = r\chi(Y)$ for a definable map $X \to Y$ whose fibres all have characteristic r. On finite sets this extends ordinary counting; the permitted definable sets may be infinite. The characteristic on definable sets is distinct from the field characteristic p.

For the degree criterion the values lie in $\mathbb{Z}/p^\nu\mathbb{Z}$, with precision ν depending on the degree budget and indexing bounds. Reduction modulo p interprets a definable $\mathbb{F}_p$-valued sum as

$$\sum_{x\in X}^{\chi} a(x) = \sum_{c\in\mathbb{F}_p} c\,\overline{\chi}(a^{-1}(c)).$$

Under the full definability and index-sort hypotheses of [13, Theorem 6.1], including the characteristic on the required index sorts, a definable solution violating the target $g = 0$ excludes derivations within the specified degree budget at sufficiently large sizes in the residue class $\chi(M) \bmod p^\nu$. The refutation case is $g = 1$. A merely mod-p counting rule or an arbitrary infinite model does not supply these hypotheses. The Grothendieck-ring viewpoint [14] organises the additive and multiplicative identities; the strong fibre rule remains additional structure.

This is a *sufficient obstruction* to bounded-degree derivations when the required structure exists. We do not prove that every hard uniform family admits such a structure, nor identify T1–T4 with four classes of Euler structures. A general converse remains a separate question. The hardness-spectrum classification above needs no Euler structure, no Boolean encoding and no model-existence assumption.

18.6 What the model-theoretic language does not replace

The finite-dimensional target theorem requires neither an infinite model nor an Euler structure. Such structures can explain particular residue obstructions, but the proof of the four types works directly with finite consequence spaces and their actual maps. Boolean equations are essential for the intended truth-value interpretation of the first-order encoding, not for the general support-bounded transfer theorem.

A useful test of understanding is to say which statement is being proved: that a target vanishes at all roots, that its polynomial lies in an ideal, or that it has a derivation within one particular degree budget. These are three different questions. The book's profile concerns the last.

Chapter 19

No algorithm can distinguish the types

Already in the special case of the fixed target $G^{(n)} = 1$, the classification cannot be turned into an algorithm which determines the case from a sentence. This remains true under the promise that every finite instance is inconsistent. Throughout this section, the input is a well-formed normalised sentence and the request is a refutation of its translated axioms, with the translation of Chapter 2; the characteristic p is fixed. For a sentence ψ, write $d_S(\psi, n)$ for the least degree of deriving 1 from its size-n translation. This is its refutation profile. Its type is the alternative in Theorem 6.5: eventually bounded (T1), no recurring finite level (T2), not eventually bounded with a finite nonempty set of recurring levels (T3), or infinitely many recurring finite levels (T4).

19.1 Knowing the alternatives need not identify the alternative

The classification gives four exhaustive structural patterns. It does not give an algorithm deciding which pattern belongs to a presented uniform family. The reduction below makes this difference sharp even for the constant target one, where every finite instance is inconsistent.

Think of a program whose halting is unknown. A uniform computation-prefix sentence has models of every finite size if the program never halts. If it does halt, a fixed finite prefix supplies a bounded-degree contradiction at all large sizes. We use this switch inside a controlled join of two example families. The join has one type in the nonhalting case and the other type in the halting case. An algorithm distinguishing those types would therefore decide halting.

Three algebraic points need proof, and each is supplied below. The halting prefix yields an actual bounded-degree NS certificate, not just a semantic contradiction. Adjoining a satisfiable block preserves the other branch's degree. The guarded disjunction keeps both branches visible with at most an additive degree-one overhead. These are the connections between the computational reduction and our exact proof systems.

An algorithm receives a finite string encoding a sentence: its symbols, clauses and function arities can all be read effectively. *Partial computable* allows the algorithm to run forever on some inputs. The theorem demands termination and a correct answer only on the *promised* inputs, namely sentences with no finite nonempty model and type in $A \cup B$. It does not demand a procedure for recognising that promise. The reduction will produce promised inputs in both the halting and the nonhalting case, so a separator could not evade them by failing to halt.

Theorem 19.1. *Let A, B be disjoint nonempty subsets of $\{\mathrm{T}_1, \mathrm{T}_2, \mathrm{T}_3, \mathrm{T}_4\}$. There is no partial computable algorithm which, on every normalised sentence having no finite nonempty model and having type in $A \cup B$, halts and correctly distinguishes membership in A from membership in B. This holds for NS and for ordinary PC, over every field of characteristic p.*

In particular, each of the six pairs is undecidable, and so is membership in any nonempty proper union of types. The last conclusion is the scope of the historical every-union theorem. The pairwise statement requires the additional construction below; it does not follow merely by listing all the undecidable union problems.

19.2 A sentence that records a computation prefix

A *reduction* transforms each finite program description into a finite sentence description by a fixed effective procedure. It must not first decide whether the program halts. Calling the transformation *primitive recursive* states a stronger effectiveness property: it is built using composition and recursion whose iteration counts are given as natural-number arguments. The later unbounded search is performed by the program, not by the procedure producing its sentence.

A counter program has finitely many natural-number registers and instructions for incrementing a register, branching on zero and decrementing, and halting. All registers initially contain zero. The number of registers is finite but need not be two. Write c_M for one more than the largest register index occurring in M, so $c_M \geq 1$.

From a program M, form a sentence θ_M with a strict linear order, least and greatest elements o, e, a successor function S, counter-value functions C_r, and state predicates Q_ℓ. Require $S(e) = e$ and require $S(t)$ to be the immediate successor of t when $t \neq e$. At o, all counters equal o and the state is the initial one. Exactly one state holds at each element, and halting states are forbidden.

At $t \neq e$, the state determines the transition to $S(t)$. Increment replaces $C_r(t)$ by $S(C_r(t))$, with the guard $C_r(t) \neq e$. Decrement branches according as $C_r(t) = o$; on the positive branch, the new value v satisfies $v < C_r(t)$ and $S(v) = C_r(t)$. Other counters are unchanged. These are finite universal clauses with flattened function graphs, and $M \mapsto \theta_M$ is primitive recursive.

Lemma 19.2 (Computation switch)**.** *If M never halts, θ_M has a model of every size $n \geq 1$. If the run of M first reaches a halting state at time T, the translated system has an NS refutation of degree bounded by a constant C_M at every size $n \geq T + 1$.*

Proof. In the first case, use $[n] = \{0, \ldots, n-1\}$ in its natural order and record the first n configurations. Every counter value at time j is at most j, so no guarded increment overflows before the last element. This includes $n = 1$.

For the second case put $a_j = S^j(o)$. If the structure has $T+1$ distinct elements, then $a_j \neq e$ for $j < T$: otherwise immediacy of successor and linearity of the order leave only $a_0, \ldots, a_j$. Induction along this finite prefix forces the actual state at a_j and forces $C_r(a_j) = a_{c_r(j)}$, where $c_r(j)$ is the numerical counter value. At a_T this gives a forbidden halting state. The argument needs neither well-ordering nor an induction axiom in the structure.

This local contradiction gives a degree bound by the finite-term certificate in Lemma 19.3. More explicitly, with c_M counters one may take

$$m = T + 2 + 2c_M(T+1), \qquad C_M = m + \sum_R (m + T + 1)^{\mathrm{arity}(R)} + \max_{C \text{ a clause}} |C|.$$

The term nodes are $o, e, a_1, \ldots, a_T, C_r(a_j), S(C_r(a_j))$; $T + 1$ distinct additional labels supply the size condition. The constant depends on the program and its halting time, never on n. □

For completeness, the reduction needs effective universality, not a decision procedure for the halting time. Every primitive recursive function can be compiled into finite counter code: composition uses fresh registers, and primitive recursion uses a counted loop, preserving the input registers. Let $H(k, c)$ equal one when the program with code c, on its fixed encoded input, halts within k simulated steps, and zero otherwise. Bounded simulation computes H by a fixed number of primitive-recursive transition updates, so this predicate is primitive recursive. A counter program that successively tests $H(0, c), H(1, c), \ldots$ therefore halts exactly when the coded program does. Loading c into this fixed search program is primitive recursive. This reduces the usual undecidable halting problem to the programs just used. Only effective compilation and the standard undecidability of halting are used; the number of counters is unrestricted but finite.

19.3 Finite local contradictions and their certificates

The halting switch uses a fixed finite computation prefix even though function totality is expressed by sums with n summands. The following elementary certificate construction separates these two sizes.

An acyclic evaluation list is an ordered list of term evaluations whose arguments have already been named. This is a dependency condition on the list, not a prohibition on a function taking the same value twice; constants can be treated as evaluations with no arguments. The proof sums over every labelling of that finite list. A *falsity indicator* for a Boolean literal is $1 - x$ for a positive literal and x for a negative literal. The translated clause is the product of its literal falsity indicators: it vanishes exactly when at least one literal is true. These products are the actual input polynomials used below.

Lemma 19.3 (Finite-term certificate)**.** *Let $n \geq 1$ and let a finite acyclic list of m function evaluations be given, with every argument supplied by an earlier node. Fix b distinct labels. Suppose that every assignment of labels to the nodes, and every Boolean assignment to the relation atoms on the named and fixed labels, either has inconsistent function values or falsifies a clause instance whose variables lie among the named and fixed labels and all of whose function evaluations occur in the list. If clauses have length at most L, the graph-and-sum translation has an NS refutation of degree at most*

$$m + \sum_{R} (m + b)^{\mathrm{arity}(R)} + L.$$

The hypothesis and conclusion concern sizes for which the b distinct labels exist.

Proof. For a labelling α of the nodes, let q_α be the product of the m graph variables asserting those function evaluations. Expanding the totality sum for each node, in the order of the acyclic list, gives

$$\sum_{\alpha} q_\alpha - 1$$

in the degree-m NS span. This is a telescoping identity: at step j, the product for the first $j - 1$ evaluations multiplies the totality equation for evaluation j. Summing over the possible earlier values gives the next partial sum. Repeated graph variables cause no problem.

If α gives different values to the same function on the same input, q_α contains a functionality collision and already belongs to the span within degree m. Otherwise, list the distinct relation variables on the at most $m + b$ named labels. Write this actual set as $Y_\alpha = \{y_1, \ldots, y_{a_\alpha}\}$, where $a_\alpha \leq a_{\mathrm{rel}} = \sum_R (m + b)^{\mathrm{arity}(R)}$. Expand over these distinct variables:

$$1 = \prod_{i=1}^{a_\alpha} (y_i + (1 - y_i)) = \sum_{\beta} u_\beta.$$

Each u_β is the Boolean assignment monomial for one partial relational structure. By hypothesis that structure falsifies an available clause. Its clause polynomial is a product of at most L falsity indicators. Write $P = q_\alpha u_\beta$ and the clause polynomial as $F_1 \cdots F_\ell$, where each F_j is the falsity indicator of a literal false in the chosen partial structure. For each literal, $P(1 - F_j)$ has an explicit expression in the structural NS span:

- For a positive relation literal, $F_j = 1 - x$ and $u_\beta = (1 - x)u'$. Thus $P(1 - F_j) = q_\alpha u' x(1 - x) = -q_\alpha u'(x^2 - x)$. For a negative relation literal, $F_j = x$ and $u_\beta = xu'$; the same identity with $x(1 - x)$ applies.
- For a negative graph literal false in the partial structure, its graph variable x is one of the chosen evaluations, so $q_\alpha = xq'$. Its falsity indicator is $F_j = x$ and $P(1 - F_j) = -q' u_\beta(x^2 - x)$.
- For a positive graph literal false there, the chosen evaluation has a different output variable x' on the same function input. Here $q_\alpha = x'q'$, $F_j = 1 - x$, and $P(1 - F_j) = q' u_\beta x' x$, an explicit functionality-axiom multiple.
- A false equality or disequality has the constant falsity indicator $F_j = 1$, so $P(1 - F_j) = 0$.

These are all cases of the normalised literals. Each identity has product degree at most $m + a_{\mathrm{rel}} + 1$. The exact telescoping identity

$$P(1 - F_1 \cdots F_\ell) = \sum_{j=1}^{\ell} P(1 - F_j) \prod_{i<j} F_i$$

therefore puts its left side in the NS span within degree $m + a_{\mathrm{rel}} + \ell$. Since $PF_1 \cdots F_\ell$ is a multiple of the falsified clause axiom, P is in that same span. Sum first over β, then over α, and use the totality identity to obtain one. □

A *partial structure* for θ_M has a domain of labels, all relations defined on it, and function values defined only on some arguments. It is *locally consistent* if every clause instance whose variables take values in the domain, and whose function atoms are all defined, has a true literal.

Lemma 19.4 (Local switch)**.** *Let the run of M first reach a halting state at time T, and let M have c_M counters. Suppose a locally consistent partial structure defines o, e, $a_{j+1} = S(a_j)$ for $j < T$ (with $a_0 = o$), $C_r(a_j)$ and $S(C_r(a_j))$ for $j \le T$, $r < c_M$, and that all these elements lie in its domain. Then its domain has at most T distinct elements.*

Proof. Suppose it contains $T + 1$ distinct elements. If $a_j = e$ for some $j < T$, take j least. Every domain element x satisfies $x = a_0$ or $a_0 < x$ (least element), and, if $a_i < x$ with $i < j$, then $x = a_{i+1}$ or $a_{i+1} < x$ (no element strictly between a_i and $S(a_i)$, and linearity). As $a_j = e$ is greatest, the domain is $\{a_0, \ldots, a_j\}$, too small. So $a_j \neq e$ for $j < T$, and the order clauses make $a_0 < a_1 < \cdots < a_T$. For the counter induction, the initial clauses give the initial state and $C_r(a_0) = a_0$ for every counter. A counter value at time j is at most j, so the notation a_v uses only this named prefix. Suppose its states and counters agree through time $j < T$. An increment of value v gives $S(a_v) = a_{v+1}$; because $v \le j < T$, this successor is among the defined evaluations and its guard is valid. The zero branch of a decrement is selected exactly when $v = 0$. On a positive decrement, the transition specifies a value $y < a_v$ with $S(y) = a_v$. The immediate-predecessor property forces $y = a_{v-1}$: if $y < a_{v-1}$ then a_{v-1} lies strictly between y and $S(y)$, and if $a_{v-1} < y < a_v$ then y lies strictly between a_{v-1} and $S(a_{v-1})$. Both alternatives contradict immediacy. Unchanged counters retain their named values. State uniqueness and the chosen transition identify the next state. Every clause used has its variables in the domain and its function atoms in the specified finite evaluation list. This proves the induction through time T. At $j = T$ the state is halting, which a clause forbids. □

Corollary 19.5. *In that case, for every $n \geq T+1$ the translation of θ_M at size n has an NS refutation of degree at most*

$$C_M = m + \sum_R (m+T+1)^{\mathrm{arity}(R)} + \max_C |C|, \qquad m = T+2+2c_M(T+1).$$

Proof. Apply Lemma 19.3 to the m evaluations o, e, $a_1, \ldots, a_T$, $C_r(a_j)$, $S(C_r(a_j))$ ($j \leq T$, $r < c_M$), each argument supplied by an earlier node, with $b = T+1$ fixed labels. A consistent labelling and a Boolean assignment give a partial structure of the kind in Lemma 19.4, whose domain contains the $T+1$ fixed labels; so it is not locally consistent, which is the hypothesis of Lemma 19.3. □

Here c_M is one more than the largest counter index occurring in M (so $c_M \geq 1$), and T must be the first halting time. Sizes $n \leq T$ are genuine exceptions: θ_M can have models there.

19.4 A disjunction with controlled degree

The notation $d_*(\alpha, n)$ below always refers to the original graph-and-sum translation of α, with target 1; the symbol $*$ stands for either NS or ordinary PC. A signature is the collection of relation and function symbols. Disjoint signatures are obtained by renaming the symbols of one branch, so the two polynomial variable families are independent even though they use the same universe. *Structural equations* below mean the Boolean, function-totality and function-functionality equations of the compiler. The clause equations impose the sentence's further demands. This distinction lets the inactive branch retain a structural root while its clause equations are guarded away.

Conjunction is easy to control when one block is satisfiable. In disjoint signatures, if β has a root at size n, then

$$d_*(\alpha \wedge \beta, n) = d_*(\alpha, n), \qquad * = \mathrm{NS}, \mathrm{PC}. \tag{19.1}$$

One inequality follows by using the refutation of α; the other follows by substituting the root of β into every proof line.

We also need a disjunction which does not accidentally simplify either branch. Introduce a fresh unary relation Z. Keep both branches' Boolean, totality and functionality equations. Replace every clause polynomial f of α by $z_t f$, and every clause polynomial g of β by $(1-z_t)g$, for a fresh universally quantified label t. Denote this normalised sentence by $\alpha \vee_* \beta$. Thus setting every z_t to one exposes the α clauses, while setting them to zero exposes the β clauses. Keeping the structural equations makes either specialisation an actual copy of the selected branch's polynomial system.

The guard z_t is itself a Boolean relation variable. Multiplying a clause by it makes the clause active when $z_t = 1$; multiplying by $1 - z_t$ selects the other value. The lower-bound specialisations set *all* guards to the same value. For the upper bound one label t_0 suffices, since the universally instantiated inputs include the clauses guarded by that label. The hypothesis $n \geq 1$ guarantees that such a label exists.

Lemma 19.6 (Controlled join)**.** *For $n \geq 1$, with ∞ allowed, each of NS and PC satisfies*

$$\max\{d_*(\alpha, n), d_*(\beta, n)\} \leq d_*(\alpha \vee_* \beta, n) \leq \max\{d_*(\alpha, n), d_*(\beta, n)\} + 1.$$

Proof. Set all selectors to one, and give the inactive block a structural root: all its functions take one fixed value and all its relations are false. Its clauses are guarded away, while its

structural equations vanish. The resulting proof refutes α without increasing degree. Selectors zero give the other lower inequality.

For the upper bound fix a label t_0. Simulate each branch refutation with its fixed guard $c = z_{t_0}$ or $c = 1 - z_{t_0}$, representing a branch line f by cf. A guarded clause is already an axiom. A structural input h is retained unguarded, so PC can form $z_{t_0}h$ and, for the other guard, $h - z_{t_0}h$. Thereafter a linear combination of branch lines is simulated by the same combination of their guarded lines, and a step $f \mapsto x_v f$ is simulated by the legal variable multiplication $cf \mapsto x_v(cf)$. This uses no unguarded intermediate branch line and remains valid after cancellations. Every line has degree at most one above the branch budget. The two simulations end in z_{t_0} and $1 - z_{t_0}$, whose sum is one. For NS the same construction multiplies the two certificates by their guards. □

19.5 The reduction

Let $\psi_1, \psi_2, \psi_3, \psi_4$ be the four sentences of Section 16.3. Each has no finite nonempty model. For these particular witnesses, the dominant type under the maximum operation is ordered by

$$\mathrm{T}_1 \prec \mathrm{T}_4 \prec \mathrm{T}_3 \prec \mathrm{T}_2.$$

This is a property of the chosen witnesses, not an ordering assertion about arbitrary profiles. A bounded profile has no effect on the type of a maximum; the all-size linear lower bound for ψ_2 makes it dominant; and the maximum of ψ_4 and ψ_3 is bounded off the multiples of p and linear on them. An additive error of at most one preserves these conclusions. More generally, adjoining an eventually bounded profile and allowing this error preserves each of the four intrinsic types: recurrence transfers through finite intervals of degrees, and unboundedness transfers directly.

Choose indices a, b with $\mathrm{T}_a \prec \mathrm{T}_b$ in this order and define, using disjoint signatures,

$$\Phi_M = \psi_a \vee_* (\psi_b \wedge \theta_M).$$

If M does not halt, (19.1) makes the second branch have exactly the profile of ψ_b. The join has type T_b. If M halts, the second branch is eventually bounded by Lemma 19.2, and the join has type T_a. Every Φ_M is within the all-finite-inconsistent promise, whether or not M halts. The map is primitive recursive and uses no information about that alternative. A separator of types $\mathrm{T}_a, \mathrm{T}_b$ would decide halting. Choosing a and b on opposite sides of any disjoint nonempty A, B proves Theorem 19.1.

19.6 Where the textbook argument ends

The general consequence theorem has reduced the target question to a support-bounded closure calculation, fulfilled its transfer contract by one common projector, and read off the four profiles from fixed-degree recurrence. The refutation examples and this reduction show that the four alternatives are substantial and cannot be algorithmically separated in general. The exercise solutions which follow return to small calculations and help check that each logical distinction in this route has been retained.

Part VI

Model answers and a reference map

Chapter 20

Model answers and a notation guide

This chapter gives a model answer to every numbered exercise in the book. Each question is repeated, followed by the calculation or proof needed to justify the conclusion. Upper bounds and lower bounds are checked separately when an exact minimum degree is requested.

The index links to all seventeen answers; the exercise number links back to its original question. There is also a direct model-answer link beside every original exercise. The notation guide follows the answers. These are written pedagogical arguments, not additional Lean-verified theorem endpoints.

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20.1 Proof rules and minimum degrees

Model answer to Exercise 1.8

Question. Find the least NS and PC degrees of the targets 0, 1, and x from $\Gamma = \varnothing$ and from $\Gamma = \{1\}$. Explain the role of the last-line degree floor.

Model answer. The answer is the same in NS and ordinary PC:

Γ	$h_S(\Gamma \vdash 0)$	$h_S(\Gamma \vdash 1)$	$h_S(\Gamma \vdash x)$
$\varnothing$	0	∞	∞
$\{1\}$	0	0	1

$(S \in \{\mathrm{NS}, \mathrm{PC}\})$.

From empty input, an NS identity has no nonzero summands. In PC the only initial line available is zero; linear combinations and variable multiplication keep it zero. Thus neither system can produce 1 or x. Zero has the empty NS certificate and a one-line PC proof, both of degree zero.

From $\{1\}$, the input itself proves 1 in degree zero. For x, NS uses the identity $x = x \cdot 1$; PC lists 1 and then multiplies by x. These proofs have degree one. The last-line degree floor gives $h_{\mathrm{PC}} \geq \deg x = 1$, and $h_{\mathrm{NS}} \geq h_{\mathrm{PC}}$ gives the same lower bound for NS. Thus the displayed upper bounds are exact. In particular, inconsistency of the inputs does not make a nonconstant target a degree-zero consequence.

Return to Exercise 1.8.

Model answer to Exercise 1.9

Question. For Example 1.4, prove $h_{\mathrm{PC}}(\Gamma \vdash z) = 2$ and $h_{\mathrm{NS}}(\Gamma \vdash z) = 3$. For the NS lower bound, list the monomials which can occur in a degree-two certificate.

Model answer. Work over $\mathbb{F}_2$, with

$$f_1 = xy + x, \qquad f_2 = xy + 1, \qquad f_3 = xz.$$

A complete PC derivation is

line	polynomial	justification
1	$xy + x$	f_1
2	$xy + 1$	f_2
3	xz	f_3
4	$x + 1$	$1 + 2$
5	$xz + z$	z times line 4
6	z	$5 + 3$

Here the notation $1 + 2$ means addition of the polynomials on those two numbered lines. Every line has degree at most two. Multiplication on line 5 is legal within that budget because line 4 has actual degree one after cancellation.

No nonzero input is usable at degree at most one: all three inputs have degree two. Starting from zero alone produces only zero. Therefore $h_{\mathrm{PC}}(\Gamma \vdash z) = 2$.

For NS, expand the identity

$$z = zf_1 + zf_2 + f_3 = (xyz + xz) + (xyz + z) + xz.$$

Its product degree is three. In a degree-two certificate, each nonzero multiplier must be constant, since each input has degree two and nonzero product degrees add over a field. Such a certificate has form

$$af_1 + bf_2 + cf_3 = (a+b)xy + ax + b + cxz.$$

The only monomials which can occur are $xy, x, 1, xz$. The coefficient of the monomial z is always zero, so this expression cannot equal the target. Hence $h_{\mathrm{NS}}(\Gamma \vdash z) = 3$.
Return to Exercise 1.9.

Model answer to Exercise 1.10

> **Question.** A proposed degree-two PC calculation multiplies the derived line xy by x and replaces x^2y by xy, using a Boolean equation. Explain why this is not an ordinary degree-two step, even if $x^2 - x$ is an input.

Model answer. Ordinary PC works with actual polynomials before reduction. The proposed multiplication produces

$$x \cdot xy = x^2y,$$

which has degree three. That intermediate line already violates the degree-two budget.

The Boolean input would permit the correction

$$y(x^2 - x) = x^2y - xy,$$

but multiplying that input by y also produces a degree-three line. Subtracting the correction would eventually give xy, but the degree of the proof counts both degree-three intermediate lines.

In the quotient by the Boolean equation the classes of x^2y and xy agree. Ordinary PC has no rule which performs multiplication and quotient reduction together as one degree-two inference. This identifies the error in the proposed step; it does not exclude some different legal degree-two derivation in a system with additional axioms.
Return to Exercise 1.10.

20.2 Uniformity and support

Model answer to Exercise 2.14

> **Question.** For variables x_{ij}, find the label supports and total degrees of x_{ij}^3 and $x_{ij}x_{jk}$ when i, j, k are distinct. Compare the actual supports with the general bound $2d$. Explain why the row sum in Example 2.2 has an index of support one.

Model answer. Each pair-indexed variable uses at most two labels. With i, j, k distinct, the requested counts are

monomial	degree d	label support	$2d$
x_{ij}^3	3	$\{i, j\}$ (size 2)	6
$x_{ij}x_{jk}$	2	$\{i, j, k\}$ (size 3)	4

A power repeats the same variable and introduces no extra labels. In the second monomial, the two variable supports overlap at j. These are the reasons the general union bound $2d$ is not an equality.

For the row axiom $f_{i,n} = \sum_{j<n} x_{ij} - 1$, the axiom index is i, with support $\{i\}$. If a permutation π fixes i, then

$$\pi f_{i,n} = \sum_{j<n} x_{i,\pi(j)} - 1 = f_{i,n},$$

because π merely reorders the summands. The whole row sum can mention every label while still having a one-label axiom index. Its individual monomials and its axiom index are different basis objects.

Return to Exercise 2.14.

Model answer to Exercise 2.15

> **Question.** Take no axioms, a single variable x of empty support, and two target indices of empty support requesting 0 and x. Determine the two fixed-orbit profiles and the aggregate profile. What happens if one chooses the first target at sizes in an arbitrary set and the second elsewhere?

Model answer. Let h_0 be the profile for the fixed target 0, h_x that for the fixed target x, and h_{all} the profile for the two-target request. At every universe size,

$$h_0(n) = 0, \qquad h_x(n) = \infty, \qquad h_{\mathrm{all}}(n) = \max\{0, \infty\} = \infty.$$

These statements hold separately for NS and PC. The zero certificate or zero line proves 0, whereas empty input cannot derive any nonzero polynomial.

If the selected target is 0 at universe sizes in A and x elsewhere, the selected profile becomes

$$h_A(n) = \begin{cases} 0, & n \in A, \\ \infty, & n \notin A. \end{cases}$$

Thus its degree-zero threshold can encode an arbitrary set of universe sizes. The two available target sorts have stable coefficient rules, but this size-dependent selection is neither one fixed global orbit nor the fixed all-target request. It therefore falls outside the theorem's request convention. In particular, arbitrary orbit switching cannot be used to infer another case of the four-profile classification.

Return to Exercise 2.15.

Model answer to Exercise 2.16

> **Question.** Take variables x_i, no axioms, and targets $x_i - x_j$ indexed by ordered distinct pairs. Use $k = 1, s = 0, t = 2, D_G = 1, n_0 = 1$ to compute n_*. At which positive universe sizes does this request actually contain a target? Why is the stated stability threshold allowed to be larger than necessary?

Model answer. The prescribed parameters give

$$n_* = \max\{n_0, 1, kD_G + t\} = \max\{1, 1, 1 + 2\} = 3.$$

At universe size 1 there is no ordered pair of distinct labels, so the request is empty and its degree is zero in both systems. At every universe size $n \geq 2$, such pairs exist and every target $x_i - x_j$ is a nonzero degree-one polynomial, including in characteristic two. There are no axioms from which to derive it. Consequently

$$h_S(1) = 0, \qquad h_S(n) = \infty \quad (n \geq 2).$$

The actual nonempty request and its degree floor stabilise at universe size two. The theorem's universe size three is a sufficient uniform bound obtained by adding a monomial-

support bound and an index-support bound. Here those supports overlap, so the general bound is conservative. A safe theorem onset need not be the smallest onset for each example.
Return to Exercise 2.16.

20.3 The closure and its coordinates

Model answer to Exercise 3.6

> **Question.** Describe I_0 and C_0 for an arbitrary system. Prove directly that a nonzero constant target has a degree-zero proof exactly when an input is a nonzero constant.

Model answer. Let $K \cdot 1$ denote the one-dimensional space of constants. Then

$$I_0 = C_0 = \begin{cases} \{0\}, & \Gamma \text{ contains no nonzero constant,} \\ K \cdot 1, & \Gamma \text{ contains a nonzero constant.} \end{cases}$$

At degree zero, no nonconstant input can be listed in PC. In NS, a nonzero product has degree zero only when both its multiplier and its input are constants. Zero inputs contribute nothing. Thus both systems start with the span of the constant inputs.

PC multiplication of a nonzero constant by a variable raises its degree to one, so it contributes no new degree-zero vector. Multiplying zero still gives zero. Hence the starting span is the whole closure at degree zero, and it equals the NS span.

If $c \in \Gamma$ is a nonzero constant and the target is a constant $a \neq 0$, the identity $a = (a/c)c$ is an NS certificate of degree zero and also a legal PC scalar combination. Conversely, a nonzero degree-zero consequence requires a nonzero constant in that initial span, so at least one such input must be present.
Return to Exercise 3.6.

Model answer to Exercise 3.7

> **Question.** With input $\Gamma = \{x^2\}$, show that every vector in every closure C_d has zero coefficient of x. Deduce that x is underivable, although it vanishes at every field-valued solution of Γ.

Model answer. Every input belongs to the ideal (x^2). Linear combinations and variable multiplication preserve that ideal, so an induction on PC lines shows that every derived polynomial is divisible by x^2. Its coefficient of the monomial x is therefore zero.

In the one-variable ambient ring $K[x]$, the closure can be written explicitly:

$$C_d = \begin{cases} \{0\}, & d < 2, \\ \operatorname{span}_K\{x^2, x^3, \ldots, x^d\}, & d \geq 2. \end{cases}$$

The forward inclusion follows from divisibility and the degree cutoff. The reverse inclusion follows by repeatedly multiplying the input x^2 by x and then taking linear combinations. If the ambient ring has additional variables, the divisibility argument still applies, with more multiples of x^2 in the closure.

The target x belongs to none of these spaces; its least PC degree is infinity. It has no NS identity either, since $x \notin (x^2)$. On the other hand, a field has no nonzero element whose square is zero. Every field-valued solution of $x^2 = 0$ consequently satisfies $x = 0$. Vanishing at all roots and algebraic derivability are distinct.

Return to Exercise 3.7.

Model answer to Exercise 3.8

Question. List the monomials counted by $B_2(2)$, and compare the full coefficient list of x^2 with its sparse representation. For $v \geq d$, identify explicitly the 2^d entries used in the final lower bound of Lemma 3.5.

Model answer. With the variable order x, y, one ordering of the degree-at-most-two monomials is

$$1,\ x,\ y,\ x^2,\ xy,\ y^2.$$

Thus $B_2(2) = \binom{2+2}{2} = 6$. In that ordering the full coefficient list of x^2 is $(0,0,0,1,0,0)$. A sparse representation instead records only the coefficient 1 and the exponent pair $(2,0)$. These representations charge for different data.

For $v \geq d$, choose distinct variables $x_1, \ldots, x_d$. Every subset $S \subseteq \{1, \ldots, d\}$ gives a squarefree monomial

$$\prod_{i \in S} x_i.$$

Its degree is $|S| \leq d$, so it has an entry in the full coefficient list. Distinct subsets have different exponent vectors and give distinct monomials. There are 2^d subsets, including the empty subset, whose monomial is 1. This proves $B_v(d) \geq 2^d$; the argument also includes $d = 0$, where there is one entry.

Return to Exercise 3.8.

20.4 Periodicity, recurrence, and peaks

Model answer to Exercise 6.8

Question. Let $p = 2$, $\kappa = 4$, $\sigma = 4$, and $n_* = 7$. Compute $N(0), Q(0)$, $N(3), Q(3)$ and $\beta(62)$. Check $N(\beta(62)) \leq 62$ and explain why the strict inequality defining Q matters.

Model answer. Here $R(d) = 4d + 4$. Degree zero uses its separate definition:

$$N(0) = 7, \qquad Q(0) = 1.$$

It is not obtained by substituting zero into the positive-degree period formula.

At degree three,

$$\begin{aligned} R(3) &= 4 \cdot 3 + 4 = 16, \\ N(3) &= \max\{7, 2 \cdot 16 + 1\} = 33, \\ Q(3) &= \min\{2^a : 2^a > 16\} = 32. \end{aligned}$$

Although 16 is a power of two, it equals the support budget and does not satisfy the strict inequality. Taking it as $Q(3)$ would use an unproved transfer step.

For the inverse onset, the numerator is positive, so

$$\begin{aligned} \beta(62) &= \left\lfloor \frac{(62 - 2 \cdot 4 - 1)_+}{2 \cdot 4} \right\rfloor = \left\lfloor \frac{53}{8} \right\rfloor = 6, \\ N(\beta(62)) &= N(6) = \max\{7, 2(4 \cdot 6 + 4) + 1\} = 57 \leq 62. \end{aligned}$$

For comparison, $N(7) = 65 > 62$. The value $\beta(62) = 6$ locates the largest integer threshold guaranteed by this displayed onset formula. It makes no claim that a sharper theorem could not give a larger admissible threshold.

Return to Exercise 6.8.

Model answer to Exercise 6.9

Question. For $g(1) = 2$ and $g(n) = 1$ for $n \geq 2$, verify forward persistence of every threshold under $n \mapsto n + 1$. Identify a reverse threshold implication that fails. Explain why exact recurrence of the starting value cannot be deduced from forward persistence alone.

Model answer. For nonnegative integer thresholds d, all cases are

d	$\{n \geq 1 : g(n) \leq d\}$
0	$\varnothing$
1	$\{2, 3, 4, \ldots\}$
$d \geq 2$	$\{1, 2, 3, \ldots\}$.

Each set is closed under $n \mapsto n + 1$, proving forward persistence of every threshold. A predicate can become true at the next universe size while still having this property.

At threshold one the reverse implication fails:

$$g(2) = 1 \leq 1, \qquad g(1) = 2 > 1.$$

The starting exact value 2 occurs only once, whereas value 1 occurs at every later universe size. Forward persistence preserves the upper bound $g(n) \leq 2$ but does not exclude a drop to a smaller minimum. To preserve the exact value two, one would also need reverse comparison at threshold one. This is why the recurrence argument uses two-way transfer at lower thresholds as well as at the chosen budget.

Return to Exercise 6.9.

Model answer to Exercise 6.10

Question. For $h(n) = p^{v_p(n)}$, determine every recurring value, its liminf, and a sequence of finite linear peaks. Explain why no eventual linear lower bound holds at all sizes.

Model answer. Every value of $h(n) = p^{v_p(n)}$ is a power p^a, with $a \geq 0$. For each fixed a, the universe sizes

$$n_j = p^a(1 + pj), \qquad j = 0, 1, 2, \ldots$$

have valuation a, since $p \nmid 1 + pj$. They tend to infinity and all satisfy $h(n_j) = p^a$. Thus the recurring finite values are exactly

$$E(h) = \{1, p, p^2, p^3, \ldots\}.$$

Every value is at least one. At the unbounded sequence of universe sizes $n = 1 + pj$ the value is one, giving

$$\liminf_{n \to \infty} h(n) = 1.$$

For finite linear peaks take $n_a = p^a$. Then

$$h(n_a) = p^a = n_a \longrightarrow \infty.$$

Each peak is finite and lies exactly on the line $h = n$.

There is no eventual all-size lower bound $h(n) \geq cn$ with $c > 0$. Given c and any proposed starting universe size, choose $n = 1 + pj$ above that universe size and above $1/c$. Then

$h(n) = 1 < cn$. This exercise concerns the displayed abstract sequence. It does not assert that the normalised weak-bijection system has these exact degrees or that its rectangle lower bound applies at $n = p^a$.
Return to Exercise 6.10.

Model answer to Exercise 6.11

> **Question.** Show that T3 cannot have $E(h) = \{0\}$. State exactly where stability of the degree-zero predicate is used.

Model answer. If $E(h) = \{0\}$, zero recurs at arbitrarily large universe sizes, in particular at some $n \geq n_*$. The degree-zero threshold is constant beyond n_*. Since it is true at that universe size, it must be true at every universe size $m \geq n_*$:

$$h(m) \leq 0.$$

All degrees are nonnegative, so $h(m) = 0$ throughout that range. The profile is eventually bounded and is T1, contrary to the unboundedness required in T3.

The essential use of zero-degree stability is the implication that truth at one eligible universe size gives truth at every eligible universe size. Mere periodicity with a larger period would not suffice. For example, the arbitrary sequence which is zero at even universe sizes and equals n at odd universe sizes is unbounded and has only zero as an infinitely recurring finite value. Its zero predicate is not eventually constant, so it is excluded by the theorem's zero-degree input.

Equivalently, the actual zero-degree period is $Q(0) = 1$. There is no nonempty proper easy-residue set modulo one from which to build a T3 split with only the easy value zero.
Return to Exercise 6.11.

20.5 Consequences and concrete examples

Model answer to Exercise 16.5

> **Question.** For the consistent family of Proposition 16.1, compute the two profiles at sizes $1, 2, 3, 4, 5, 6$ in characteristic three. Compare the invariant sum request with the indexed request for every individual variable. Which statements have finite proof degrees?

Model answer. In characteristic three, set

$$\Gamma_n = \{x_i - 1 : i < n\}, \qquad G^{(n)} = \sum_{i<n} x_i.$$

Write h_{sum} for the invariant target profile and h_{ind} for the request for every target x_i. The table applies to both NS and ordinary PC:

n	1	2	3	4	5	6
$h_{\mathrm{sum}}(n)$	∞	∞	1	∞	∞	1
$h_{\mathrm{ind}}(n)$	∞	∞	∞	∞	∞	∞

Summing the inputs gives $G^{(n)} - n$. At $n = 3, 6$, the scalar n is zero in the field, so

$$G^{(n)} = \sum_{i<n} (x_i - 1).$$

This is an NS certificate of degree one and gives a PC proof of the same degree. The target is a nonzero linear polynomial, so its degree floor proves that the minimum is exactly one.

For the other four universe sizes, the assignment $x_i = 1$ satisfies all inputs but has $G^{(n)} = n \neq 0$ in the field. Soundness excludes any derivation. The same assignment has $x_i = 1 \neq 0$ for every individual target at every positive universe size, excluding the indexed request in all six cases. The finite proofs in this table are precisely those for the sum at universe sizes three and six. A sum being derivable does not make its summands derivable. Return to Exercise 16.5.

Model answer to Exercise 16.6

> **Question.** For $\Gamma = \{x, 1 - x\}$ over an arbitrary field, write a degree-one PC proof of the fresh target y and prove that its least NS degree is two. Do not insert an axiom involving y.

Model answer. A legal PC proof over any field is

1	x	input
2	$1 - x$	input
3	1	$1 + 2$
4	y	y times line 3.

All lines have degree at most one. The target degree is one, so $h_{\mathrm{PC}}(\Gamma^{\uparrow} \vdash y) = 1$.

For NS,

$$y = yx + y(1 - x).$$

Both nonzero products have degree two, giving an upper bound two. In a degree-one certificate, each multiplier of the degree-one inputs must be constant. Any such sum is

$$ax + b(1 - x) = (a - b)x + b \qquad (a, b \in K),$$

which has no y-term. It cannot equal y. Thus $h_{\mathrm{NS}}(\Gamma^{\uparrow} \vdash y) = 2$.

PC first derives the constant 1, then multiplies only that last line. In NS the extra y is carried in the multipliers of the original inputs, increasing each product degree. No input involving y, including $y^2 - y$, has been introduced.

Return to Exercise 16.6.

Model answer to Exercise 16.7

Question. In characteristic three, apply the fresh-y bounds to the normalised weak-bijection family at $n = 6$ and $n = 36$. What can the stated bounds say at $n = 9$? Distinguish the lower-bound size condition from the all-size upper certificate.

Model answer. Use $q = 3^{v_3(n)}$. The numerical data are

n	q	$n \geq 2q$?	h_{PC} bounds	h_{NS} bounds
6	3	yes	$3 \leq h_{\mathrm{PC}} \leq 7$	$4 \leq h_{\mathrm{NS}} \leq 8$
36	9	yes	$9 \leq h_{\mathrm{PC}} \leq 13$	$10 \leq h_{\mathrm{NS}} \leq 14$
9	9	no	$h_{\mathrm{PC}} \leq 13$	$h_{\mathrm{NS}} \leq 14$

Indeed $6 = 3 \cdot 2$ and $36 = 3^2 \cdot 4$, so their valuations are one and two, respectively. Both satisfy the rectangle size condition. The original refutation bounds and the fresh-target identities give

$$q \leq d_{\mathrm{PC}}(\widehat{B}_n) \leq q + 4 \implies q \leq h_{\mathrm{PC}} \leq q + 4,$$
$$q \leq d_{\mathrm{NS}}(\widehat{B}_n) \leq q + 4 \implies q + 1 \leq h_{\mathrm{NS}} \leq q + 5.$$

Here $q \geq 1$, so the PC maximum with one changes nothing.

At $n = 9$, the valuation is still two, but $9 < 2q = 18$. The quoted rectangle lower window supplies neither $h_{\mathrm{PC}} \geq 9$ nor $h_{\mathrm{NS}} \geq 10$. The all-size NS upper certificate, followed by the fresh-target construction, still supplies the displayed upper bounds. The target floor gives $h_{\mathrm{PC}}, h_{\mathrm{NS}} \geq 1$, and $h_{\mathrm{PC}} \leq h_{\mathrm{NS}}$ remains valid at every size. The displayed windows are bounds, not assertions of exact minimum degrees.

Return to Exercise 16.7.

Model answer to Exercise 16.8

Question. Suppose Γ has a root over some extension field of K. Explain why the fresh target y is underivable from $\Gamma^{\uparrow}$. If one instead adds $y^2 - y$ as an axiom, identify the extra term which appears when extracting the coefficient of y^1 from an NS identity. Why does the original coefficient-extraction proof no longer apply unchanged?

Model answer. Let a be a root of the old inputs over $E \supseteq K$. Extend it to the fresh variable by assigning $y = 1$. Every old axiom is independent of y, so still evaluates to zero, whereas the target evaluates to one.

An NS identity for y, even with coefficients in E, would evaluate to the impossible equality $1 = 0$. In PC, each input, linear combination and variable multiple evaluates to zero by induction, so a last line evaluating to one is impossible. Thus y is underivable over K and over E.

If $y^2 - y$ is instead added as an axiom, a proposed NS identity has the form

$$y = \sum_f a_f(x, y) f(x) + b(x, y)(y^2 - y).$$

For each old input, $[y^1](a_f f) = ([y^1]a_f) f$. For the new summand,

$$[y^1](b(x, y)y^2) = 0, \qquad [y^1](-b(x, y)y) = -b(x, 0).$$

Taking the coefficient of y therefore gives

$$1 = \sum_f ([y^1]a_f)f - b(x, 0).$$

The extra term $-b(x, 0)$ need not be a multiple of an old input. Hence this is not the old refutation certificate obtained when no y-axiom was present. The original one-degree-saving extraction argument cannot be reused unchanged.

The soundness obstruction above does survive addition of the Boolean axiom: the extended root with $y = 1$ satisfies $y^2 - y = 0$ as well. The obstruction and the exact-degree certificate argument have different hypotheses. This calculation does not assert a new degree identity for inconsistent inputs after adding that axiom.

Return to Exercise 16.8.

20.6 Notation guide

The entries below give the local meaning of frequently reused symbols and a few terms needed in the later chapters. A degree budget bounds actual proof lines or NS products, while a support budget bounds labels in one basis object. Neither is the universe size. Terms such as design and witness refer to linear tests; they need not describe a solution of the equations.

Γ_n
: The input polynomials at universe size n, all interpreted as equations equal to zero.

$G_b^{(n)}$
: The requested target polynomial at universe size n, with target index b. The superscript is not exponentiation. A fully invariant target is written $G^{(n)}$. The target is not an additional axiom.

$h_S(n)$
: The maximum of the least degrees for the requested targets; infinity means at least one target is underivable. An empty request has profile zero.

$d_S(\Gamma_n)$
: The special minimum degree for deriving 1. For a translated sentence ψ, this is also written $d_S(\psi, n)$.

k, s, t
: Bounds on supports of one variable, one axiom index, and one target index. They do not bound the labels in a whole sum.

d, D, D_G
: A proposed proof-degree bound, a fixed input-degree bound in the template, and a fixed target-degree bound, respectively. In the T3 argument, the locally specified $D = \max E(h)$ instead denotes the largest recurring degree; in the matching chapter, D is a proof budget.

$P_{n,\le d}, P_b(N)$
: The ordinary polynomial space of degree at most d, and the tuple space $K[\mathrm{Inj}([b], [N])]$, respectively. The integer b counts tuple positions, while d bounds polynomial degree.

I_d, C_d
: The NS span of axiom products of degree at most d, and the stable ordinary-PC consequence closure in $P_{n,\le d}$.

$W, W_N, W_d(n)$
: W is a locally specified ambient space or, in an exact sequence, its quotient term. W_N denotes a degree-cutoff polynomial ambient, whereas $W_d(n)$ in the model-theory chapter is the chosen NS or PC consequence subspace. In commutator calculations, $W = [D, z]$ is instead an invertible operator. The displayed type fixes which meaning is intended.

C
A chosen complement or chain-start space in the layer construction; also a fresh label set in the quadratic construction and a column set on a rook board. In the injection certificate, $C = \sum_j c_j$ is a polynomial. These local meanings are distinct from the PC closure C_d and the annihilator class $\mathcal{C}_r$.

$r = R(d), \kappa, \sigma$
The support budget $R(d) = \kappa d + \sigma$. For indexed targets $(\kappa, \sigma) = (2k, s + t)$; for invariant targets they are (k, s). In the rectangular matching system $B_{r,m}$, r instead counts rows; the rook-forcing proof specifies its row set R separately.

$n, N = n + q$
A retained universe size and the larger size before a block of labels is deleted.

z, u, w, e
The block translation operator, its top power $u = z^{p-1}$, a common group-algebra witness with $uwu = u$, and the resulting projector. The witness identity holds on tuple spaces of length at most r. Products and powers of these operators mean composition; $z^p = 0$ means that p successive applications give zero. In Lemma 14.2, u instead names an integer support bound. In $n = p^a u$ it is the factor not divisible by p. The scalar grade $e \in \mathbb{F}_p$ in the layer splitting is also distinct from the projector e.

$\mathcal{D}, \mathsf{L}$
Deletion is the quotient $\ker z/(\ker z \cap \operatorname{im} z)$. The lift $\mathsf{L} = e\iota$ first includes a smaller-universe polynomial and then applies the common projector.

$V', \mathcal{Z}_b, T_b$
In $V = P_b(N)$, marginal maps ∂_i sum over one tuple position. The incoming space is $V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}$, the zero-marginal space is $\mathcal{Z}_b = \bigcap_i \ker \partial_i = (V')^{\perp}$, and the top quotient is $T_b = V/V'$.

$J_r(N), \mathcal{C}_r(N)$
The common group-algebra annihilator of all tuple spaces of length at most r, and the modules killed by that ideal. The latter class is also called $\mathcal{M}_r$; it is distinct from the PC consequence space C_d.

J_N
The actual target map $[b] \mapsto G_b^{(N)}$, distinct from the annihilator ideal $J_r(N)$.

$p, q, Q(d)$
The prime characteristic, an admissible block size $p^a > R(d)$, and, for $d \geq 1$, the chosen least such power for the periodicity bound. The separate zero-degree convention is $Q(0) = 1$. The actual period may be smaller than $Q(d)$.

$n_0, n_*, N(d)$
The axiom stability threshold, the stability threshold including the target-degree floor, and the periodicity onset. For $d \geq 1$, $N(d) = \max\{n_*, 2R(d) + 1\}$; separately, $N(0) = n_*$.

$E(h)$
The finite degrees recurring at arbitrarily large universe sizes.

$A_d(h), \mathcal{E}_d(h), \mathcal{H}_p(h)$
The eventually easy residues, their clopen union in $\mathbb{Z}_p$, and the closed complement of all such easy regions. These are not the matching-design space $E_d(N, M)$ or the finite-degree set $E(h)$.

Subquotient; annihilator
A quotient S'/S of an invariant subspace by a smaller invariant subspace. A linear functional annihilates S when it vanishes on S; $S^{\perp}$ expresses this condition using the dot product. The group-algebra annihilator $J_r(N)$ instead consists of operators killing the specified tuple spaces.

Plateau
An equality of consecutive intersections $I_{d+j}(n) \cap P_{n,\le d} = I_{d+j+1}(n) \cap P_{n,\le d}$. It supplies closure under the legal degree-d PC multiplications; it does not say that a truncated consequence space is an ideal.

$[y^1]a, v_p(n)$
The coefficient of y in a polynomial a, and the exponent of p in the positive integer n, respectively. A fresh y is an independent indeterminate absent from the old axioms.

$B_{r,m}, E_d(N, M), \Lambda_\psi$
The rectangular bijection system, its degree-d space of matching designs, and the associated linear functional on ordinary polynomials. Here E_d is unrelated to the recurring-degree set $E(h)$.

Filtration; extension
Nested degree windows. A design extension supplies the next window's values while preserving all values in the previous window.

Definable; χ
Described by a formula in the permitted structure and language; χ is the Euler counting characteristic of its definable sets, not the prime characteristic of the field.

Promise; reduction
The hypotheses on inputs for which an algorithm must halt correctly; an effective transformation of finite input descriptions which transfers a decision problem to another.

20.7 Connections worth following in the literature

The following connections have different roles. Some are historical antecedents, some supply a useful language, and some suggest analogies. They are not interchangeable external proofs of the present theorem.

Proof complexity and the earlier four-way classification. The NS and PC systems originate in the work of Beame, Impagliazzo, Krajíček, Pitassi and Pudlák [21], and of Clegg, Edmonds and Impagliazzo [22]. Riis's LICS 2008 paper and 2009 report [7, 8] provide the four-way refutation classification. The present linear quantitative estimates and uniform consequence treatment should be read against those earlier results. The tree-resolution gap [1] is a related classification phenomenon, not a theorem about polynomial degree. Its infinite-model criterion should not be silently transferred to NS or PC.

Representation stability: similar objects, stronger transport needs. The characteristic-zero uniform permutation-module work of Riis and Sitharam [24], the FI-module framework of Church, Ellenberg and Farb [15], and Harman's modular periodicity [10] place fixed-support families in a broader setting. Section 5.8 explains the actual comparisons, including why the uniform sum $\sum_{i<n} x_i$ does not behave as an FI-submodule under naive inclusion. Hone and coauthors [11] provide an explicit categorical-functor perspective. Compatibility with natural operations already appears in Harman's results. What must be checked here is preservation and reflection of the specific polynomial closure and target maps, with the stated numerical range.

Homological algebra: exactness is a substantive step. The characteristic-two quotient $\ker z / \operatorname{im} z$ resembles homology. The short nilpotent chains in Chapter 13 explain why that formula must change for general p. The exact-functor vocabulary in the Stacks Project [17] is helpful: a kernel, image or quotient is information about a map, not just the dimension of an object. Our deletion is proved exact on the specified class of representations; arbitrary

homology is not exact. Nor does an exact operation automatically detect every nonzero target obstruction. The separate survival argument is indispensable.

Combinatorial linear algebra: rook coordinates. Bowman, Doty and Martin [12] study integral Schur–Weyl duality for partition algebras. Their setting gives background for encoding equivariant maps through finite equality patterns and permutation actions. Our zero-marginal correction is developed explicitly in the rook chapter, with a signed calculation and a triangular elimination. The reference is mathematical context; it does not replace the required correction identity or its coefficient-by-coefficient check.

Duality and designs: a witness need not be a point. Buss's designs [26] make the linear-dual approach to NS lower bounds concrete. Here the target class in a quotient is separated by a linear functional. For PC, that functional must annihilate the *whole* consequence closure, not just axiom multiples. The example $x^2 = 0$ with target x in Chapter 18 distinguishes ordinary ideal membership from vanishing on solutions. This is the commutative-algebra distinction between an ideal and its radical; see Lang [18] for the general algebraic background.

Topology and logic: two distinct meanings of compactness. The residue-tree construction in Section 18.1 is the inverse-limit description $\mathbb{Z}_p = \varprojlim_L \mathbb{Z}/p^L\mathbb{Z}$. Gouvêa [28] is an accessible introduction to the surrounding p-adic mathematics. Compactness here glues local *degree bounds* on congruence neighbourhoods. It does not assert a solution of the original equations, nor require a nonstandard model or a Boolean encoding. Krajíček's Euler structures [13] and the Grothendieck-ring viewpoint of Krajíček–Scanlon [14] are a separate semantic route: they replace finite counting by a characteristic on definable sets, with essential fibre and precision hypotheses. No general converse to that criterion is asserted here.

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